

DIGITAL TWIN MARKET

GLOBAL FORECAST TO 2027

BY ENTERPRISE, APPLICATION (PREDICTIVE MAINTENANCE, BUSINESS OPTIMIZATION),
INDUSTRY (AEROSPACE, AUTOMOTIVE & TRANSPORTATION, HEALTHCARE,
INFRASTRUCTURE, ENERGY & UTILITIES), AND GEOGRAPHY

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LIST OF ABBREVIATIONS

ABBREVIATION	FULL FORM
ADMS	Advanced Distribution Management Solution
AI	Artificial Intelligence
AMHS	Automated Material Handling System
AR	Augmented Reality
AVG	Automated Guided Vehicle
CAD	Computer-aided Design
CAGR	Compounded Annual Growth Rate
CEO	Chief Executive Officer
CIA	Central Intelligence Agency
CPS	Cyber-Physical Systems
DER	Distributed Energy Resources
DMDII	Digital Manufacturing and Design Innovation Institute
DOE	Department of Energy
EFPIA	European Federation of Pharmaceutical Industries and Associations
EIT	Institute of Innovation & Technology
ERP	Enterprise Resource Planning
EU	European Union
EV	Electric Vehicle
GAC	Global Affairs Canada
GIS	Geographical Information System
GSMA	Groupe Speciale Mobile Association
HEAT	Harmonized Enterprise Analytical Tool
HITL	Human-in-the-loop
ICTC	Information and Communications Technology Council
IoT	Internet of Things
IIoT	Industrial Internet of Things
IoET	Internet of Every Things
IRENA	International Renewable Energy Agency

ISO	International Organization for Standardization
JKPDD	Jammu & Kashmir Power Development Department
MCTIC	Ministry of Science, Technology, Innovation, and Communication
MEC	Multi-access Edge Computing
MBSE	Model-based System Engineering
ML	Machine Learning
MOM	Manufacturing Operations Management
MRO	Maintenance, Repair and Operations
OICA	Organisation Internationale des Constructeurs d'Automobiles
PLC	Programmable Logic Controller
PLM	Product Lifecycle Management
PTMT	Patent Technology Monitoring Team
R&D	Research and Development
ROE	Rest of Europe
ROI	Return on Investment
SCADA	Supervisory Control and Data Acquisition
SMEs	Small & Medium Enterprises
UAE	United Arab Emirates
UK	United Kingdom
US	United States
USD	United States Dollar
USPTO	US Patent and Trademark Office
V2X	Vehicle to Everything
VR	Virtual Reality
WTO	World Trade Organization
XR	Extended Reality

1 INTRODUCTION

1.1 STUDY OBJECTIVES

- To define, describe, and forecast the digital twin market size, by industry, application, and enterprise, in terms of value
- To describe and forecast the size of the digital twin market across four regions, namely, North America, Europe, Asia Pacific, and the Rest of the World, along with their respective country-level market sizes, in terms of value
- To provide detailed information regarding the drivers, restraints, opportunities, and challenges influencing the market growth
- To provide a detailed overview of the process flow of the digital twin market
- To strategically analyze micromarkets¹ with respect to individual growth trends, prospects, and contributions to the overall market size
- To study the complete value chain of the digital twin market
- To analyze opportunities for stakeholders by identifying high-growth segments of the digital twin market
- To strategically profile the key players and comprehensively analyze their market positions in terms of their ranking and core competencies²
- To analyze competitive developments, such as product launches, deals (mergers, acquisitions, partnerships, cooperation, alliances, collaborations, agreements, contracts, and investments), and others (expansions) in the digital twin market

1.2 DEFINITION

According to the Digital Twin Consortium, a digital twin is a virtual representation of real-world entities and processes, synchronized at a specified frequency and fidelity. This allows end users to monitor what is happening inside the physical asset in real time. By providing a holistic view of real-time behavior in a real-world environment mapped to a constantly updated virtual model, digital twins make it possible to anticipate maintenance needs, optimize performance, and avoid costly failures.

¹ Micromarkets are defined as the segments and subsegments of the digital twin market included in the scope of the report.

² Core competencies of companies are captured in terms of their key developments and strategies adopted to sustain their positions in the market.

A digital twin accurately replicates the real-time performance of an equipment, system, person, or process. It functions in the present, mimicking the physical thing but with complete knowledge of its past performance as well as an exact assessment of its future potential. A digital twin's ability to test scenarios is a vital feature. Data consumers can utilize a digital twin to test various scenarios and examine the outcomes and implications without putting themselves in danger in the real world. Users can run "what if" scenarios all the way up and down the supply chain and production process to see which tactics are the most profitable. The digital twin technology enables to create a virtual replica of a physical asset, component, product, process, or system/unit. It utilizes the data obtained from several sensors in the systems or asset, integrates various technologies such as IoT, IIoT, AI, AR/VR, ML, and big data analytics for analysis, and makes reasonable projections about the future process. A digital twin is used in the design and development, simulation, and operational & maintenance phase of any product or system. It helps in reducing the operating cost and extending the life of equipment and assets used in manufacturing industries.

The digital twin ecosystem comprises software/solution providers and system providers, associated technology providers, and distributors. Additionally, it represents the digital twin providers offering a comprehensive range of products/solutions that are utilized by the aerospace, automotive & transportation, energy & utilities, oil & gas, infrastructure, healthcare, agriculture, retail, telecommunications, and other industries. The digital twin market is competitive, and most of the key providers of digital twin products/solutions are from North America and Europe.

1.2.1 INCLUSIONS AND EXCLUSIONS

PARTICULARS	INCLUSION	EXCLUSION
Enterprise	Large enterprises, small & medium enterprises (SMEs)	NA
Application	Product design & development, performance monitoring, predictive maintenance, inventory management, business optimization, and other applications (environmental analysis, customized services, instore planning & security implementation, and energy management)	Any of the supporting hardware (IoT, IIoT, and others) or software
Industry	Automotive & transportation, energy & utilities, infrastructure, healthcare, aerospace, oil & gas, telecommunications, agriculture, retail, other industries (semiconductors, chemicals, paper & pulp, and food & beverages)	NA
Geography	North America, Europe, Asia Pacific, and the Rest of the World	NA

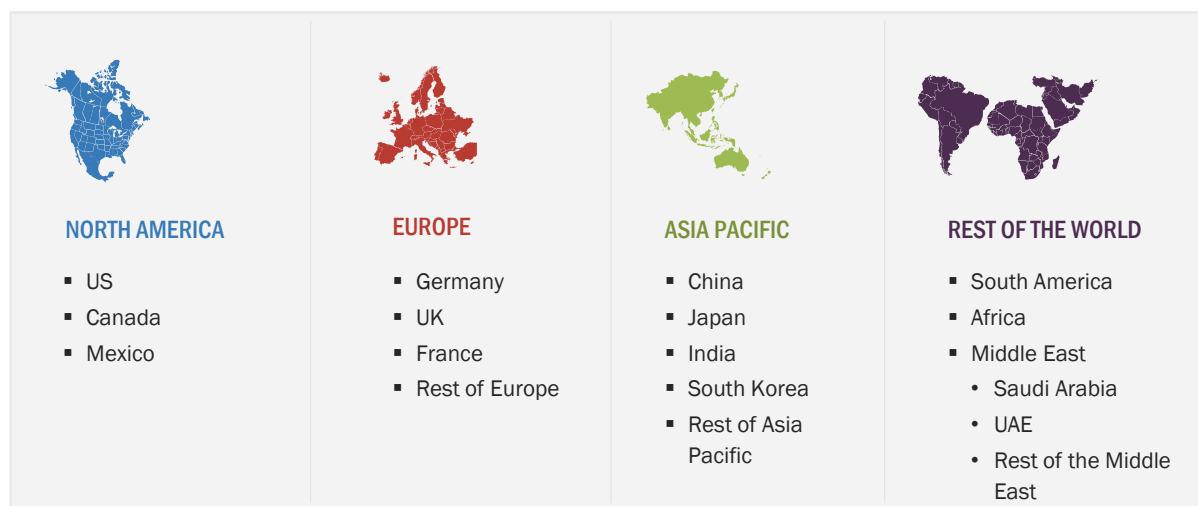
1.3 STUDY SCOPE

1.3.1 MARKETS COVERED

FIGURE 1 DIGITAL TWIN MARKET: SEGMENTATION



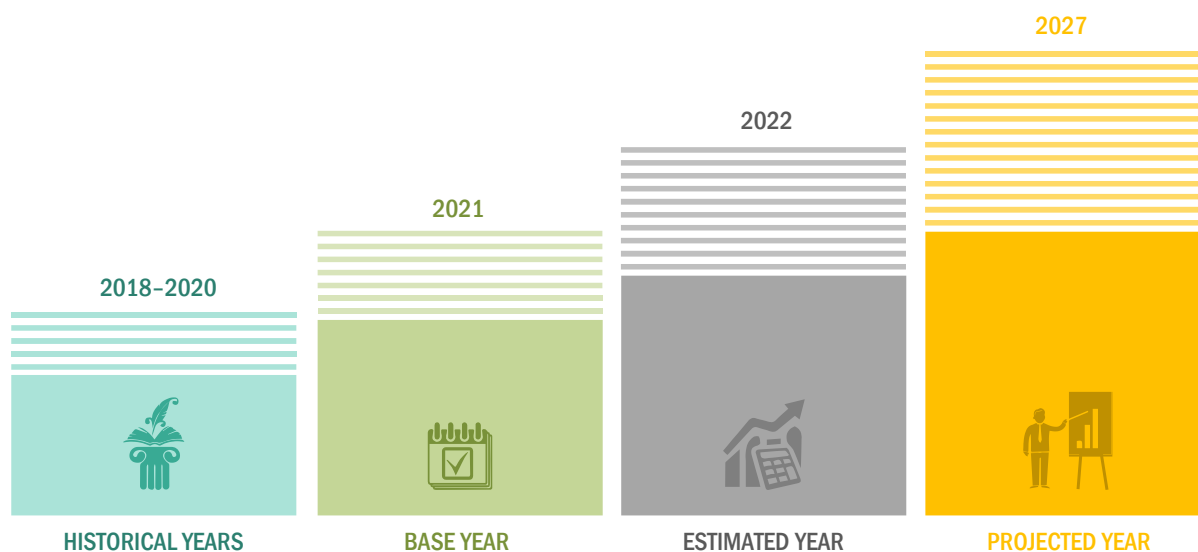
1.3.2 REGIONAL SCOPE



Note: Rest of Europe includes the Netherlands, Norway, Spain, Italy, Sweden, Belgium, and Scandinavian countries. Rest of Asia Pacific includes Taiwan, Australia, New Zealand, Singapore, Malaysia, and the Philippines.

Source: MarketsandMarkets Analysis

1.3.3 YEARS CONSIDERED



Note: The forecast period considered is from 2022 to 2027. The base year considered for the company profiles is 2021. Wherever the information for the base year was unavailable, information from the previous year has been considered.

1.4 CURRENCY

The currency used in the report is the US dollar (USD), with market size indicated in USD million and billion.

- Company revenues have been sourced from the annual reports of respective companies.
- For companies reporting their revenues in currency other than USD, the average annual currency exchange rate for that particular year has been used to convert the values to USD.

1.5 LIMITATIONS

- Industry players have kept quantitative information for some of the market segments confidential. For such segments, the qualitative insights gathered during the study have been used to arrive at the market size in terms of value.
- The size of the digital twin market in the Rest of the World is derived from the size of markets in the Middle East, Africa, and South America. However, the country-wise market sizes for Africa and South America regions have not been included in the study owing to data granularity.

1.6 STAKEHOLDERS

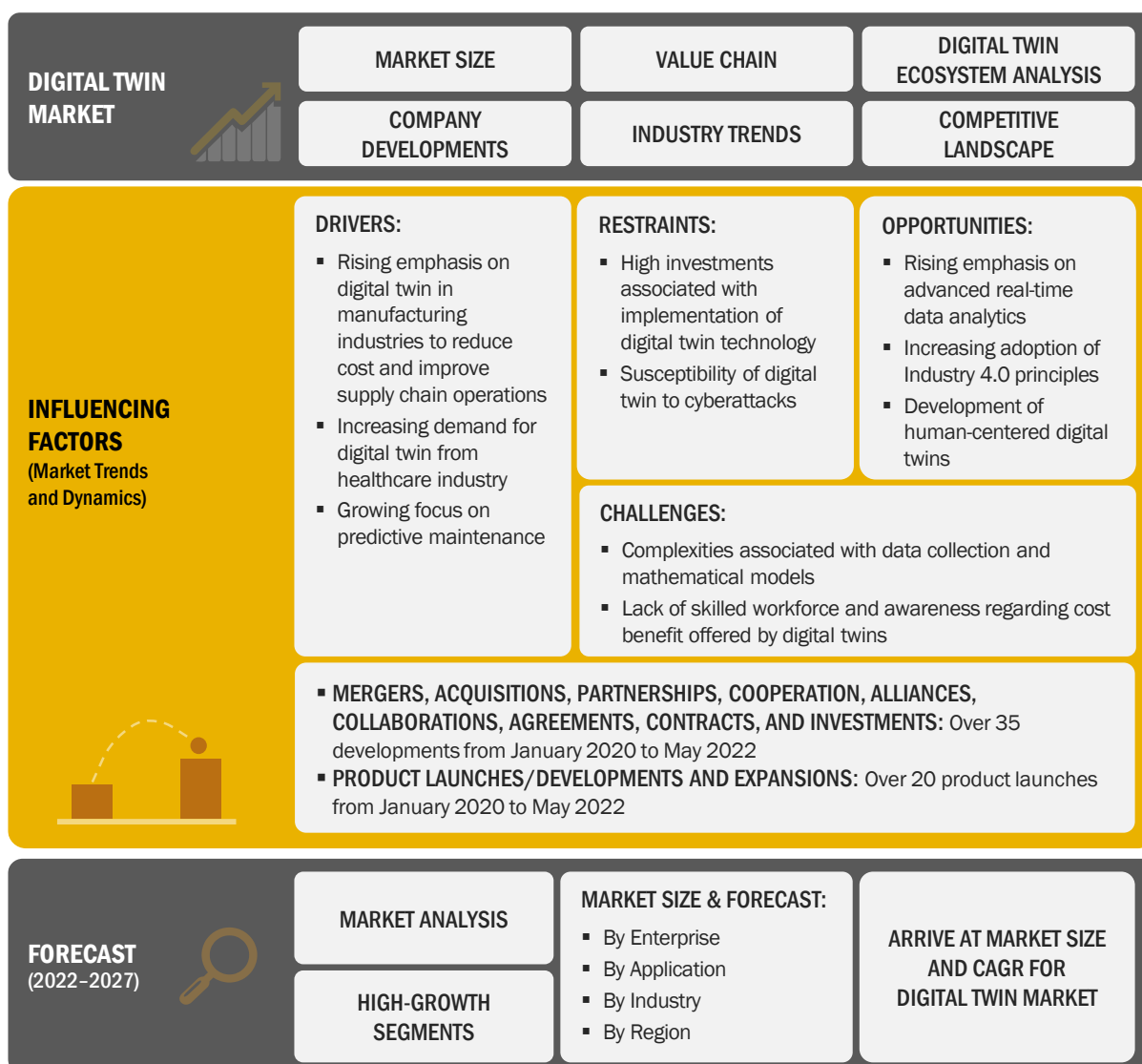
- Government bodies and policymakers
- Industry organizations, forums, alliances, and associations
- Market research and consulting firms
- Raw material suppliers and distributors
- Research institutes and organizations
- Analysts and strategic business planners
- End users of digital twins across various industries such as the aerospace, automotive & transportation, energy & utilities, oil & gas, infrastructure, healthcare, agriculture, retail, telecommunications, and other industries (semiconductors, chemicals, paper & pulp, and food & beverages)

2 RESEARCH METHODOLOGY

2.1 RESEARCH DATA

The research process for this study included the systematic gathering, recording, and analysis of data about customers and companies operating in the digital twin market. This research study involved the extensive use of secondary sources, directories, and databases (Factiva, Oanda, and OneSource) for identifying and collecting information useful for the engineering of the digital twin market. In-depth interviews have been conducted with various primary respondents, including experts from core and related industries and preferred manufacturers, to obtain and verify critical qualitative and quantitative information, as well as to assess growth prospects. Key players in the digital twin market have been identified through secondary research, and their market rankings have been determined through primary and secondary research. This research included studying annual reports of top players and interviewing key industry experts, such as CEOs, directors, and marketing executives.

In the primary research process, various key correspondents from the supply and demand sides have been interviewed to obtain the qualitative and quantitative information relevant to this report. Primary sources from the supply side include CEOs, vice presidents, marketing directors, business development executives, end users, and related executives from various key companies and organizations operating in the digital twin ecosystem.

FIGURE 2 DIGITAL TWIN MARKET: RESEARCH DESIGN

Source: MarketsandMarkets Analysis

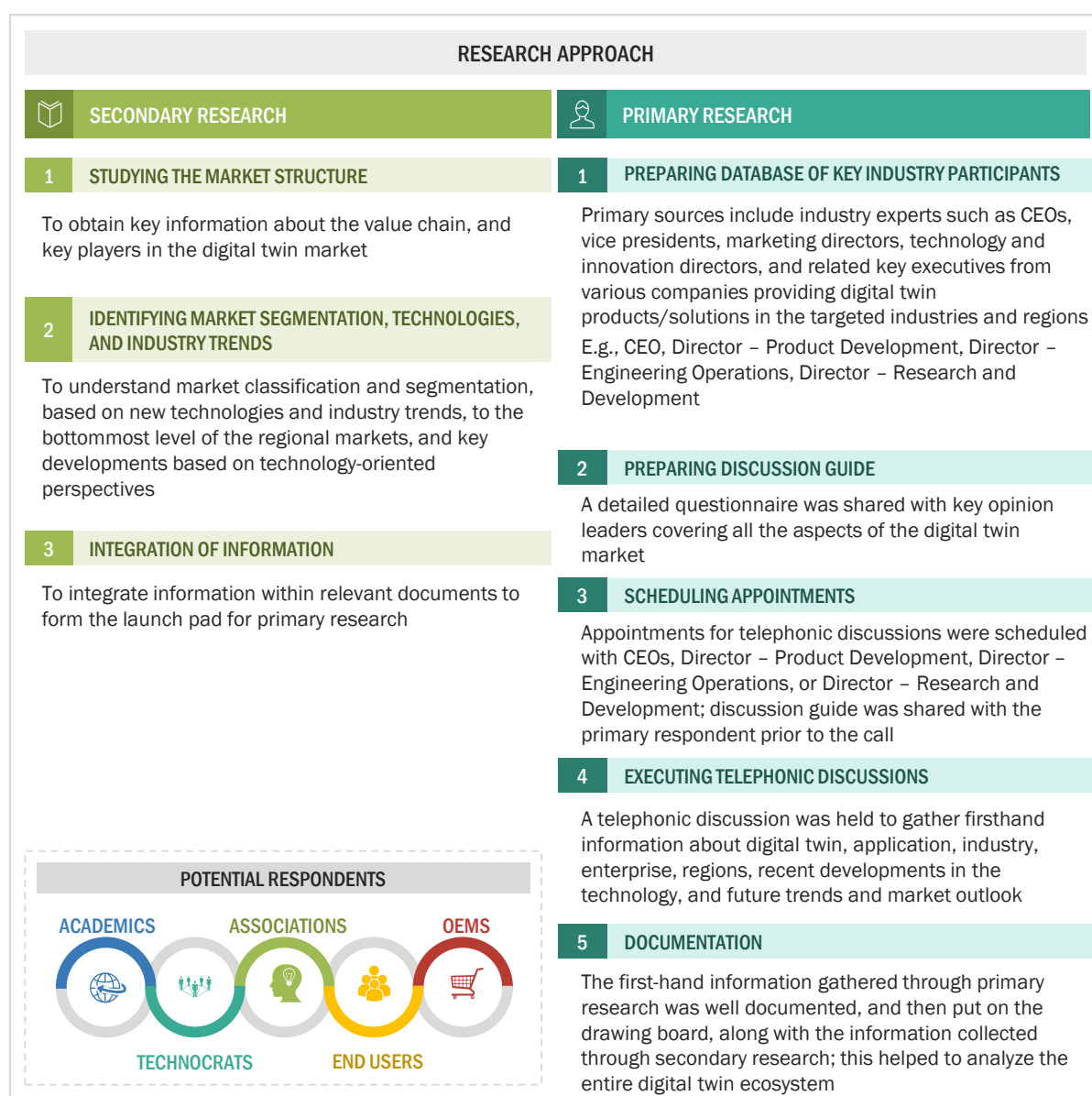
2.1.1 SECONDARY DATA AND PRIMARY RESEARCH

In the secondary research process, various secondary sources have been referred to for identifying and collecting information important for this study on the digital twin market. The secondary sources include annual reports, press releases, and investor presentations of companies; white papers, certified publications, and articles by recognized authors; and directories and databases. Secondary research has been conducted to gather key information regarding the value chain of the digital twin market, the number of key players, market classification and segmentation according to industry trends to the bottom-most level, geographic markets, and key developments from the market- and technology-oriented perspectives.

After the complete market engineering process (which included calculations for market statistics, market breakdown, market size estimations, market forecasting, and data triangulation), extensive primary research has been conducted to gather information and verify and validate the critical numbers obtained.

Primary research has been conducted to identify segmentation types, industry trends, and key players in the market, as well as to analyze the competitive landscape, key market dynamics such as drivers, restraints, challenges, and opportunities, and key strategies adopted by the market players. In the complete market engineering process, both top-down and bottom-up approaches have been extensively used, along with several data triangulation methods, to estimate and forecast the sizes of the market and its segments and subsegments listed in this report. Extensive qualitative and quantitative analyses have been conducted on the complete market engineering process to list the key information/insights. The data gathered through secondary research has been also verified and validated through primary interviews. The figure below illustrates the way primary and secondary research has been conducted for gathering and analyzing the data collected for the digital twin market.

FIGURE 3 DIGITAL TWIN MARKET: RESEARCH APPROACH



Source: MarketsandMarkets Analysis

2.1.2 SECONDARY DATA

In the secondary research, various sources have been referred to for identifying and collecting information important for this study. Secondary sources included corporate filings (such as annual reports, investor presentations, and financial statements); trade, business, and professional associations; white papers, digital twin and capacitor-related journals, and certified publications; articles by recognized authors; gold and silver standard websites; directories; and databases.

Secondary research has been conducted to obtain key information about the industry supply chain, market value chain, the total pool of key players, market classification and segmentation according to industry trends to the bottom-most level, geographic markets, and key developments from both market- and technology-oriented perspectives. Data from secondary research has been collected and analyzed to arrive at the overall market size, which has been further validated by primary research.

2.1.2.1 Major secondary sources

SOURCE	WEBLINK
Digital Twin Consortium	https://www.digitaltwinconsortium.org/
Industrial Digital Twin Association (IDTA)	https://industrialdigitaltwin.org/en/
Association for the Advancement of Artificial Intelligence (AAAI)	https://www.aaai.org/
Association of Asset Management Professionals (AMP)	https://www.maintenance.org/
IEEE Sensors Council	https://ieee-sensors.org/
5G Automotive Association (5GAA)	https://5gaa.org/

2.1.2.2 Key data from secondary sources

PARAMETER	SOURCE
 MARKET SIZE	▪ Company Financials ▪ Magazines ▪ Journals ▪ Press Releases ▪ Paid Databases ▪ MarketsandMarkets Data Repository
 REVENUE OF COMPANIES	▪ Annual Reports ▪ Company Websites ▪ Public Databases ▪ MarketsandMarkets Data Repository
 QUALITATIVE INFORMATION	▪ Company Websites ▪ Annual Reports ▪ Press Releases ▪ MarketsandMarkets Data Repository

2.1.3 PRIMARY DATA

In the primary research process, various primary sources from the supply and demand sides have been interviewed to obtain the qualitative and quantitative information for this report. Primary sources from the supply side included industry experts such as CEOs, VPs, marketing directors, technology and innovation directors, and key executives from major companies and organizations operating in the digital twin market.

After going through the entire market engineering (which includes calculations for market statistics, market breakdown, market size estimations, market forecasting, and data triangulation), extensive primary research has been conducted to gather information and verify and validate the obtained critical numbers. Primary research has been conducted to identify segmentation types, industry trends, key players, competitive landscape, and key market dynamics such as drivers, restraints, opportunities, and challenges, along with the key strategies adopted by players operating in the market.

Extensive primary research has been conducted after obtaining information about the digital twin market through secondary research. Several primary interviews have been conducted with market experts from the demand and supply sides across four major regions—North America, Europe, Asia Pacific, and Rest of the World. Approximately 60% and 40% of the primary interviews have been conducted with parties from the demand and supply sides, respectively. This primary data has been collected through questionnaires, e-mails, and telephonic interviews.

2.1.3.1 Primary interviews with experts

PARTICULARS	PRIMARIES	INTENDED PARTICIPANTS AND KEY OPINION LEADERS
Primary Interview Breakup	- Amazon Web Services - Pratiti Technologies - Siemens - Kalam Technology LLP - Georgia Tech - Enterprise Innovation Institute	- Digital Twin Software/Solution Provider and System Provider - Technology Providers - Government and Regulatory Bodies (External Influencers) - End-user Industry Based on Region

2.1.3.2 Primary sources

TYPE	PARAMETER	KEY DATA
GEOGRAPHIC SPLIT	- Overall size of the digital twin market in 2021 - CAGR of each region during the forecast period (2022–2027)	Digital twin market, by region: North America, Europe, Asia Pacific, and Rest of the World
GLOBAL MARKET SIZE	- Overall size of the digital twin market and its segments - CAGR for the forecast period (2022–2027)	Global digital twin market, by application, industry, enterprise, and region
MARKET SPLIT	- By application - By industry - By enterprise	- Product design & development, performance monitoring, predictive maintenance, inventory management, business optimization, and other applications (environmental analysis, customized services, instore planning & security implementation, and energy management) - Automotive & transportation, energy & utilities, infrastructure, healthcare, aerospace, oil & gas, telecommunications, agriculture, retail, other industries (semiconductors, chemicals, paper & pulp, and food & beverages) - Large enterprises, and small & medium enterprises (SMEs)

2.1.3.3 Key industry insights



Increasing focus of the manufacturing sector to improve bottom line and supply chain is expected to drive the digital twin market.

- Senior Marketing Manager,
Digital Twin Provider



North America currently holds the largest share of the digital twin market. It is further expected to provide immense opportunities for the market players by growing at the highest CAGR during the forecast period.

- Senior Marketing Manager,
Digital Twin Provider



The human-centered design of digital twin is expected to not only improve the overall business optimization but also ensure the wellness and safety of employees.

- Director of Business Development,
Leading Digital Twin Startup

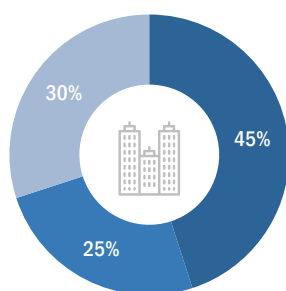


The scarcity of unified data structure and lack of knowledge on mathematical models are major challenges for the digital twin market. On the other hand, the advent of IoT, AI, ML, and blockchain will support the growth and adoption of digital twin technology.

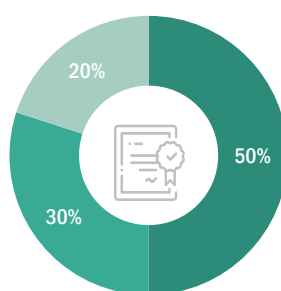
- Head of Autonomous Computing,
Digital twin Provider

2.1.3.4 Breakdown of primary interviews

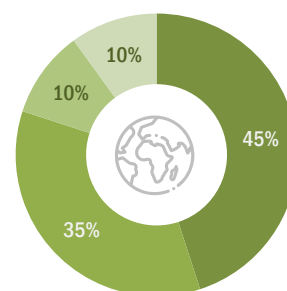
BY COMPANY TYPE



BY DESIGNATION



BY REGION



■ Tier 1 ■ Tier 2 ■ Tier 3 ■ C-level Executives ■ Directors ■ Others ■ North America ■ Europe
■ Asia Pacific ■ Rest of the World

Note: Other designations include sales managers, marketing managers, and product managers. Rest of the World mainly includes the Middle East, Africa, and South America. The three tiers of the companies have been defined based on their total revenue in 2021; Tier 1: Revenue greater than or equal to USD 1 billion, Tier 2: Revenue between USD 500 million and USD 1 billion, and Tier 3: Revenue lesser than or equal to USD 500 million.

2.2 MARKET SIZE ESTIMATION

In the complete market engineering process, both top-down and bottom-up approaches have been used, along with several data triangulation methods, to estimate and forecast the size of the market and its segments and subsegments listed in the report. Extensive qualitative and quantitative analyses have been carried out on the complete market engineering process to list the key information/insights pertaining to the digital twin market.

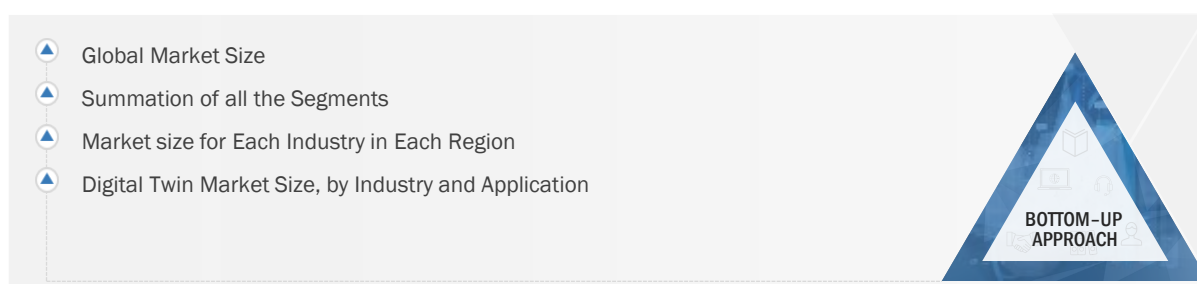
The key players in the market have been identified through secondary research, and their rankings in the respective regions have been determined through primary and secondary research. This entire procedure involved the study of the annual and financial reports of top players, as well as interviews with industry experts such as chief executive officers, vice presidents, directors, and marketing executives for both quantitative and qualitative key insights. All percentage shares, splits, and breakdowns have been determined using secondary sources and verified through primary sources. All parameters that affect the markets covered in this research study have been accounted for, viewed in extensive detail, verified through primary research, and analyzed to obtain the final quantitative and qualitative data. This data has been consolidated and enhanced with detailed inputs and analysis from MarketsandMarkets and presented in this report.

2.2.1 BOTTOM-UP APPROACH

The bottom-up approach has been employed to arrive at the overall size of the digital twin market. The summation of the regional revenues has been carried out to arrive at the market's overall size.

2.2.1.1 Approach to arrive at market size using bottom-up analysis (demand side)

- Identifying the players in the digital twin market influencing the entire market, along with their offerings
- Analyzing major providers/vendors of digital twin, studying their portfolios, and understanding several types of products based on their features and functions
- Analyzing trends pertaining to the usage of digital twin in end-user industries such as automotive & transportation, energy & utilities, infrastructure, healthcare, aerospace, oil & gas, telecommunications, agriculture, retail, and other industries (semiconductors, chemicals, paper & pulp, and food & beverages)
- Tracking the ongoing and upcoming developments in the market, such as investments made, R&D activities, product launches, mergers, acquisitions, partnerships, cooperation, alliances, collaborations, agreements, contracts, expansions, and forecasting the market based on these developments and other critical parameters
- Conducting multiple discussions with key opinion leaders to understand different types of digital twin, applications, industries, and current trends in the market, thereby analyzing the breakup of the scope of work carried out by major system manufacturing companies
- Arriving at the market estimates by analyzing revenues of these companies generated from different types of equipment, and then combining the same to get the market estimate by region
- Verifying and cross-checking the estimates at every level by discussing with key opinion leaders, such as CXOs, directors, and operations managers, and finally with the domain experts at MarketsandMarkets
- Studying various paid and unpaid sources of information, such as annual reports, press releases, white papers, and databases

FIGURE 4 BOTTOM-UP APPROACH

2.2.2 TOP-DOWN APPROACH

2.2.2.1 Approach to arrive at market size using bottom-up analysis (supply side)

- In the top-down approach, the overall market size has been used to estimate the size of individual markets (mentioned in the market segmentation) through percentage splits obtained from secondary and primary research.
- For the calculation of the size of specific market segments, the size of the parent market has been considered to implement the top-down approach. The bottom-up approach has also been implemented for data extracted from secondary research to validate the market size of different segments.
- The market share of each company has been estimated to verify the revenue share used in the bottom-up approach earlier. With data triangulation and data validation through primaries, the size of the parent market and each segment has been determined and confirmed in this study.

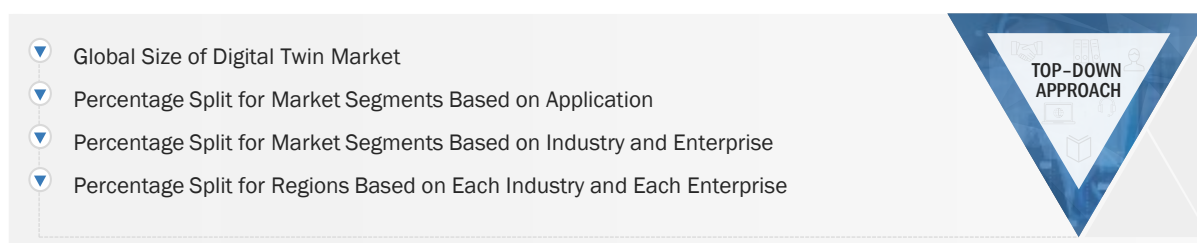
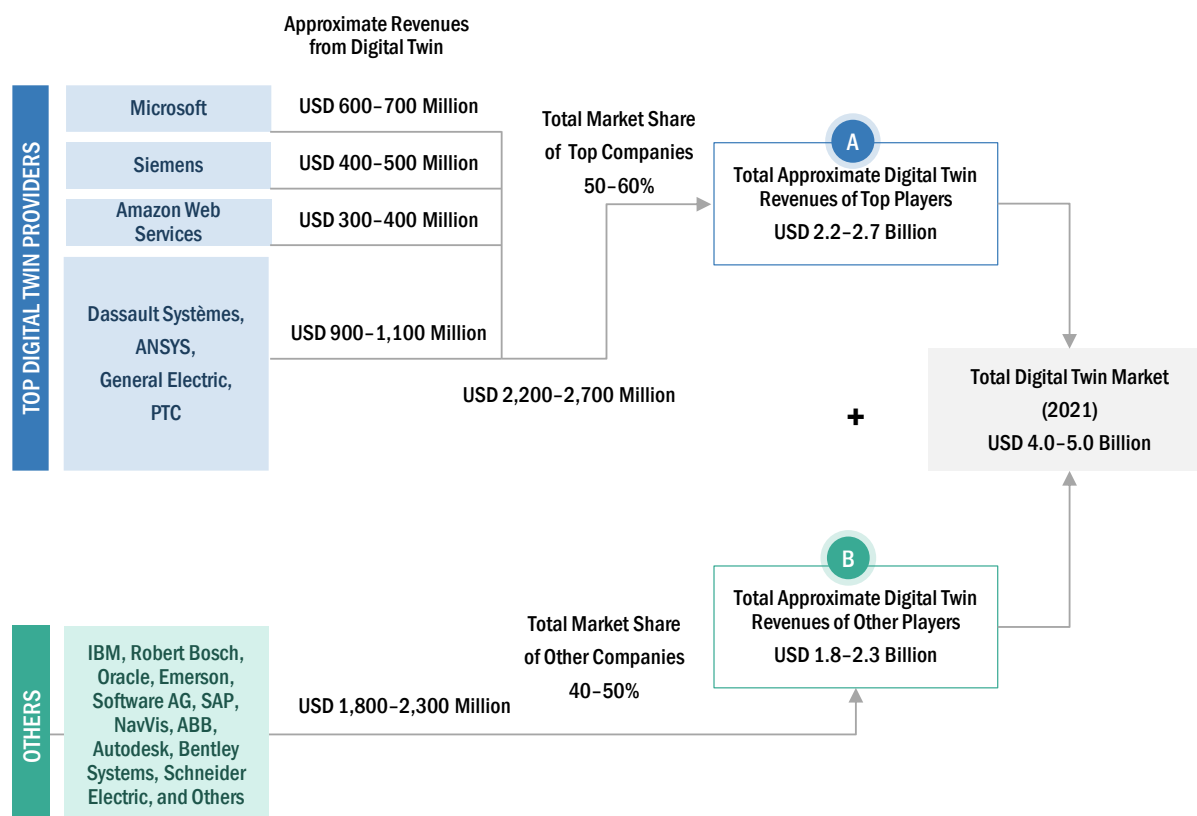
FIGURE 5 TOP-DOWN APPROACH

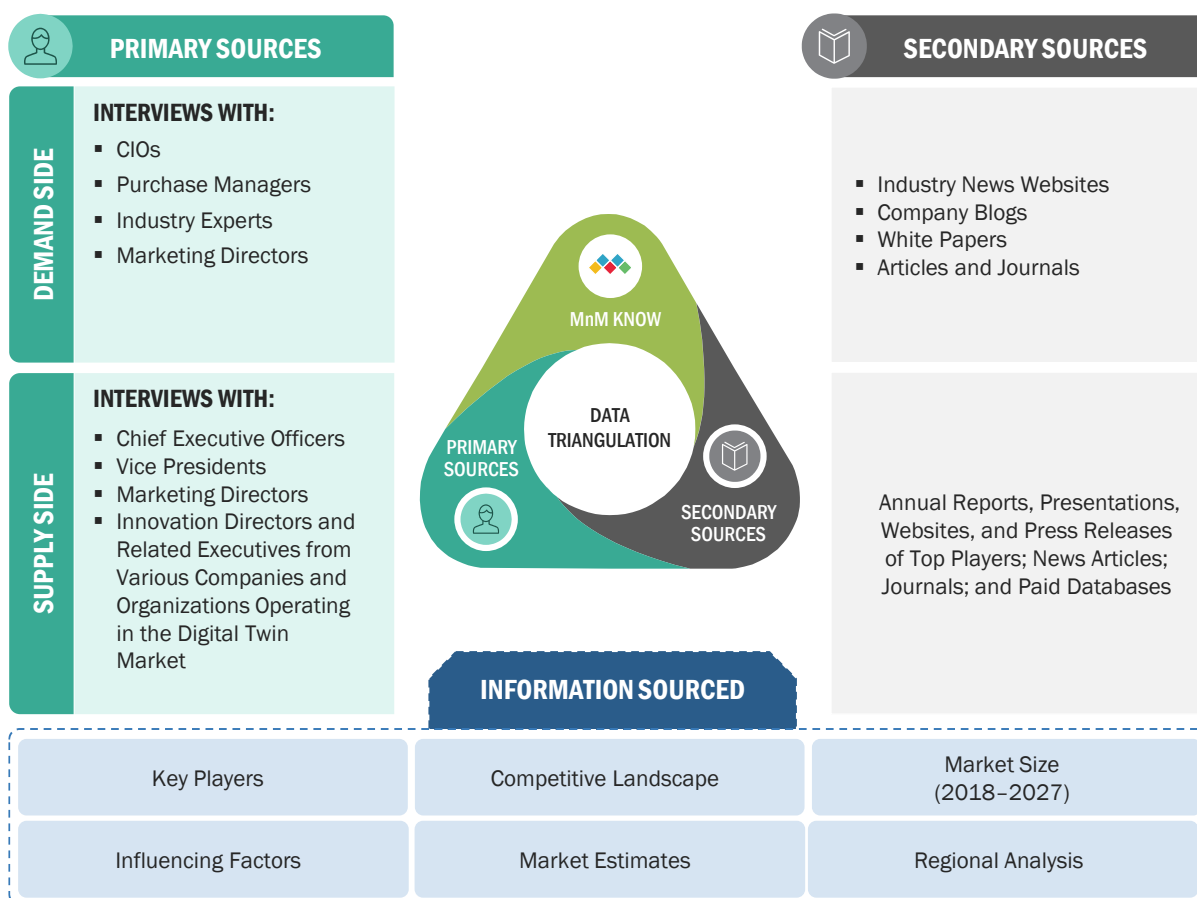
FIGURE 6 MARKET SIZE ESTIMATION METHODOLOGY: (DEMAND SIDE) REVENUE GENERATED BY COMPANIES IN DIGITAL TWIN MARKET



Source: Annual Reports, SEC Filings, Investor Presentations, Secondary Research, Interviews with Experts, and MarketsandMarkets Analysis

2.3 DATA TRIANGULATION

After arriving at the overall size of the digital twin market from the estimation process explained above, the total market has been split into several segments and subsegments. The market breakdown and data triangulation procedures have been employed, wherever applicable, to complete the overall market engineering process and arrive at the exact statistics for all segments and subsegments. The data has been triangulated by studying numerous factors and trends from the demand and supply sides. Along with this, the market size has been validated using both the top-down and bottom-up approaches.

FIGURE 7 DIGITAL TWIN MARKET: DATA TRIANGULATION

Note: MnM KNOW stands for MarketsandMarkets' Knowledge Asset Management framework. In this context, it stands for the existing market research knowledge repository of over 5,000 granular markets, our flagship competitive intelligence and market research platform Knowledge Store, subject matter experts, and independent consultants. MnM KNOW acts as an independent source that helps us validate the information gathered from primary and secondary sources.

2.4 ASSUMPTIONS

The following assumptions have been considered to complete the overall market engineering process for the digital twin market.

PARAMETER	ASSUMPTION
Global Economic Outlook	The global economy has a direct impact on any market. The global economic landscape has been taken into consideration while forecasting the size of the digital twin market in this study.
Exchange Rates and Currency Conversion	For the conversion of various currencies to US dollars, average historical exchange rates have been used according to the year specified. For all historical and current exchange rates required for calculations and currency conversions, the Oanda website has been used.
Currency Value	Forecasts have been done with values calculated under the standard assumption that the value of USD would remain constant during the forecast period.
Data Authenticity	Company revenues and segment-specific information have been derived from the annual reports of the respective companies. The information provided in the annual reports has been assumed to be authentic.
Classification of Segments	All market segments and sub-segments listed in this report through various forms of classification have been considered to be mutually exclusive of each other, with no overlap among them. In addition, for each type of classification, all submarket segments listed under the market are exhaustive in nature, that is, they cover the entire market. In cases where other fields may be present, a category called "others" has been mentioned in the market size tables.
Qualitative Analysis	The qualitative analysis carried out from the quantitative data is based solely on the understanding of the market and its trends by a team of experts involved in preparing this report.

2.5 LIMITATIONS

- The quantitative information for a few of the market segments has been kept confidential by the industry players. Hence, the qualitative insights gathered during the study have been used to arrive at the market size, in terms of value, for such segments.
- The market size in China has been estimated based on historical trends and by analyzing the revenue from the supply side owing to limited access to the domestic data related to the adoption of digital twin.
- As digital twins are unique in their applications as per the clients' requirements, and many SMEs are also a part of the ecosystem, service-/solution-specific revenues have been estimated by taking a holistic view of the product portfolio of the respective players.
- The revenues of vendors offering supporting technologies (artificial intelligence, machine learning, blockchain, 5G, cloud/edge computing, or others), supporting software, or hardware components (IoT, IIoT, and other sensors) and add-ons for digital twin have not been considered in the research study.

- The regional revenues have been derived based on digital twin adoption in different regions listed in the report. The report focuses on the regional revenue of digital twin based on the final usage of end services/solutions in specific regions and not based on where these solutions/services have been developed.

2.6 RISK ASSESSMENT

PARAMETER	DESCRIPTION	RISK
Market in China for 2021	The market size in China has been estimated based on the historical trends and by analyzing the revenue from the supply side owing to limited access to the domestic data related to the adoption of digital twin in China.	The risk has been minimized as much as possible by performing a thorough supply-side analysis specific to China and by discussing and validating the findings with industry experts.
Global Political Conditions	According to primary respondents and secondary research, there might be a slight impact on the respective country's imports and exports owing to the elections held in 2019 and 2020 in various countries worldwide. The impact is unpredictable and, hence, has not been considered while analyzing the market size.	The risk has been minimized as much as possible by discussing and validating the forecast data with industry experts from several countries worldwide.
Adoption of Digital Twin for Various Applications in Different Industries	According to industry trends and inputs from various industry experts, the adoption of digital twin is expected to be higher in the automotive & transport industry due to the increasing adoption of digital twin in product design & development, and predictive maintenance applications in automobiles. The adoption is low in the retail, agriculture, and telecommunications industries as the development and application of digital twin for these industries is yet at a very nascent stage.	Even though the adoption of digital twin in retail, agriculture, and telecommunication is low, the segments are expected to create a significant demand for digital twin during the forecast period. Players catering to the agriculture industry in developing countries are expected to be exposed to low risks due to the involvement of government bodies.

Source: Annual Reports, SEC Filings, Investor Presentations, Secondary Research, Interviews with Experts, and MarketsandMarkets Analysis

3 EXECUTIVE SUMMARY

Digital twins are used for various applications such as predictive maintenance, business optimization, performance monitoring, inventory management, product design & development, and others. Initially, end users were mainly using digital twins for performance monitoring. However, according to present trends, it is expected that end-user industry players will use digital twin solutions more for predictive maintenance and business optimization.

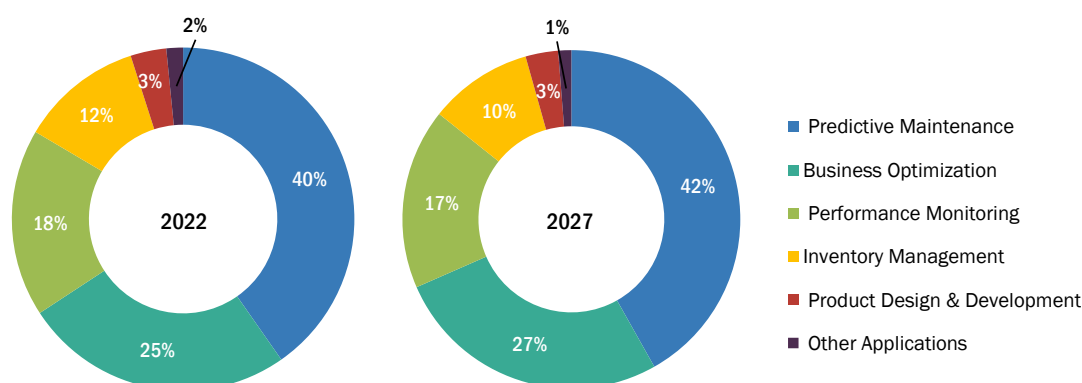
The global digital twin market was estimated at USD 4,500 million in 2021 and is projected to reach USD 73,500 million by 2027, registering a CAGR of 60.6% from 2022 to 2027. The market has been segmented based on application, industry, enterprise, and region. Rising emphasis on digital twin in manufacturing industries to reduce cost and improve supply chain; and increasing demand for digital twins from the healthcare and pharmaceuticals industries are the major drivers for the market. The growing importance of advanced real-time data analytics and increasing adoption of smart manufacturing/industry 4.0 principles on a global scale creates future scope for the digital twin market. On the other hand, the high investment cost involved in implementing digital twin technology is a major factor restraining the growth of the market. In addition, the system complexities, availability of accurate mathematical models, and unified data structure are major challenges for the market players.

Product launches, deals (mergers, acquisitions, partnerships, cooperation, alliances, collaborations, agreements, contracts, and investments), and others (expansions) were the major growth strategies adopted by the players in the digital twin market. These strategies enabled the players to efficiently cater to the growing demand for digital twin from end users and expand their global footprint by offering their products in all the major regions.

Key players in the digital twin market are General Electric (US), Microsoft (US), Siemens (Germany), Amazon Web Services (US), ANSYS (US), Dassault Systèmes (France), PTC (US), Robert Bosch (Germany), Oracle (US), DNV (Norway), Autodesk (US), SAP (Germany), Emerson (US), ABB (Switzerland), Honeywell (US), IBM (US), Johnson Controls (Ireland), Schneider Electric (France), Software AG (Germany), NavVis (Germany), Bentley Systems (US), Altair (US), SWIM AI (US), River Logic (US), and ANDRITZ (Austria).

SMEs/startups covered in the study are Sight Machine (US), Sensat (UK), Cosmo Tech (France), Artemis Technologies (Ireland), FEops (Belgium), Pratiti Technologies (India), QiO Technologies (UK), VEERUM (Canada), Datumix (US), synavision (Germany), Petra Sata Science (Australia), MEIK LABS (US), DigitalTwin Technology (Germany), TWAICE Technologies (Germany), Allvision Io (US), and HASH (UK).

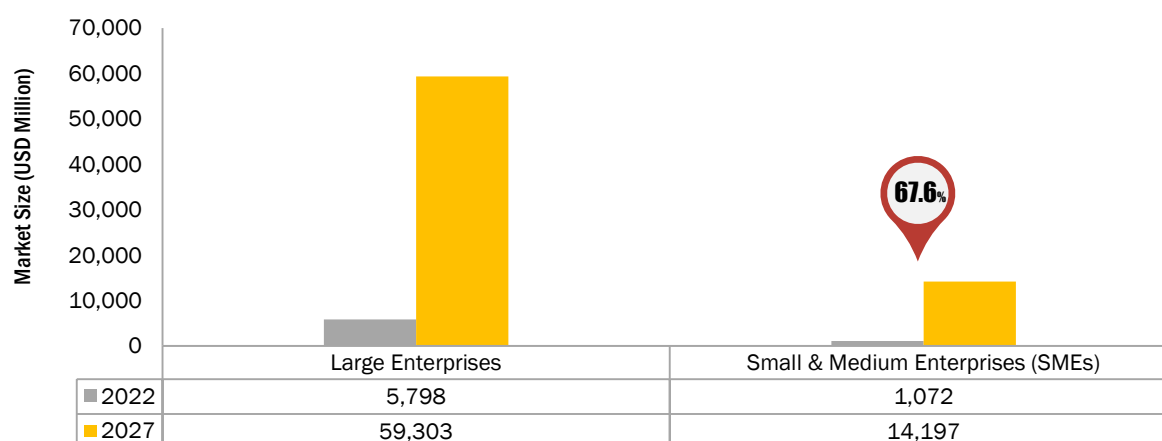
FIGURE 8 PREDICTIVE MAINTENANCE SEGMENT TO HOLD LARGEST SHARE OF DIGITAL TWIN MARKET, BY APPLICATION, DURING FORECAST PERIOD



Source: Secondary Research, Interviews with Experts, and MarketsandMarkets Analysis

The predictive maintenance application is expected to account for the largest size of the digital twin market during the forecast period. It is further expected to grow at the second-highest CAGR during the forecast period. Every organization needs their key equipment to operate at peak efficiency and utilization to realize their return on capital expenditures. This is the fundamental cause for the emergence of predictive maintenance applications of digital twin. With digital twin, companies are focusing on developing a predictive maintenance model that optimizes the maintenance cycle and strikes a balance between corrective and preventative maintenance.

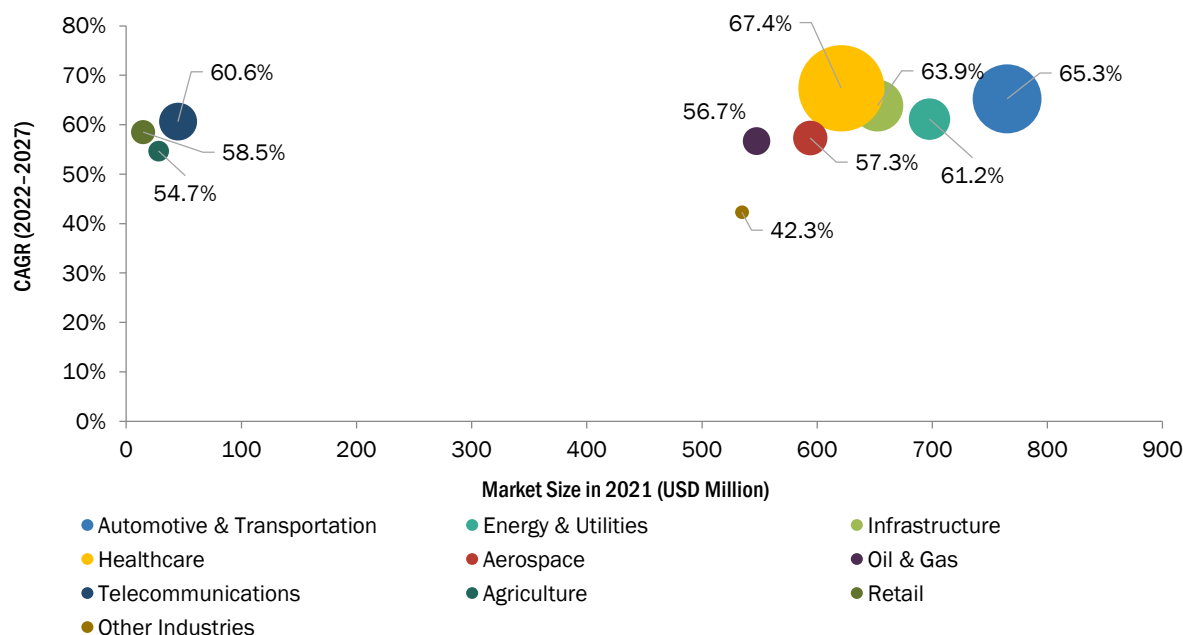
FIGURE 9 SMALL & MEDIUM ENTERPRISES SEGMENT TO REGISTER HIGHEST CAGR DURING FORECAST PERIOD



Source: Secondary Research, Interviews with Experts, and MarketsandMarkets Analysis

The small & medium enterprises segment is expected to dominate the market throughout the forecast period. Many startups are offering digital twin products/solutions that are cost-efficient to some extent and, therefore, a perfect fit for SMEs. For instance, Pratiti Technologies (India) offers cost-efficient digital twin solutions for the renewable energy industry. QiO Technologies (UK) offers PARCs and Digital Health Record algorithms to businesses to measure and benchmark their asset performance, and continuously rank the next best actions. The firm provides multi-platform compatibility as well as the ability to install on any cloud or internet of things (IoT) platform. Interoperability with legacy information technology (IT) and operational technology (OT) systems is a key feature of the solutions.

FIGURE 10 HEALTHCARE INDUSTRY TO REGISTER HIGHEST CAGR IN DIGITAL TWIN MARKET FROM 2022 TO 2027

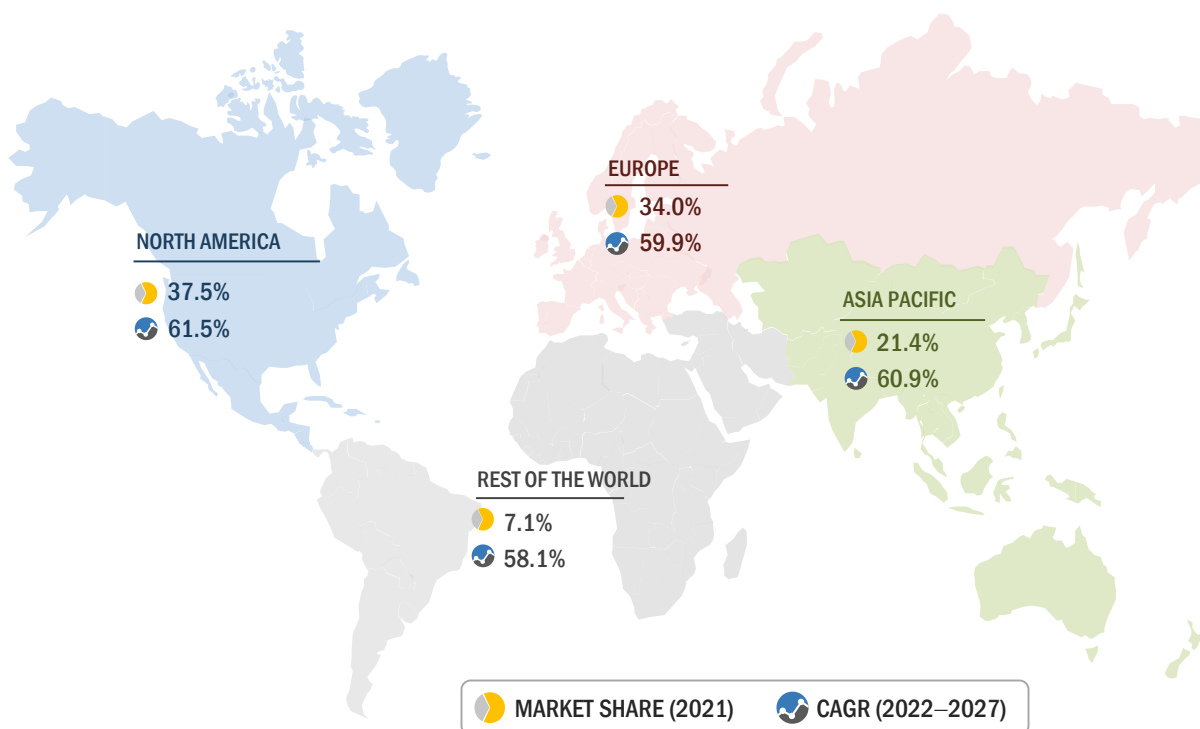


Note: Other industries include semiconductors, chemicals, paper & pulp, and food & beverages.

Source: Secondary Research, Interviews with Experts, and MarketsandMarkets Analysis

The digital twin market for healthcare industry is expected to grow at the highest CAGR during the forecast period. The rapid growth of the healthcare sector is due to the increasing value of patient safety and advancements in the patient care systems with the integration of technologies, such as internet of things (IoT), artificial intelligence (AI), and extended reality (XR). In addition, the use of digital twins is also growing for monitoring, diagnostics, and prognostics; for example, digital twins of organs such as the heart, lung, and brain are developed that help surgeons and doctors to test various parameters on the twin and then implement it in the real-life situation.

FIGURE 11 NORTH AMERICA ACCOUNTED FOR LARGEST SHARE OF DIGITAL TWIN MARKET IN 2021



Note: Rest of the World mainly includes the Middle East, Africa, and South America.

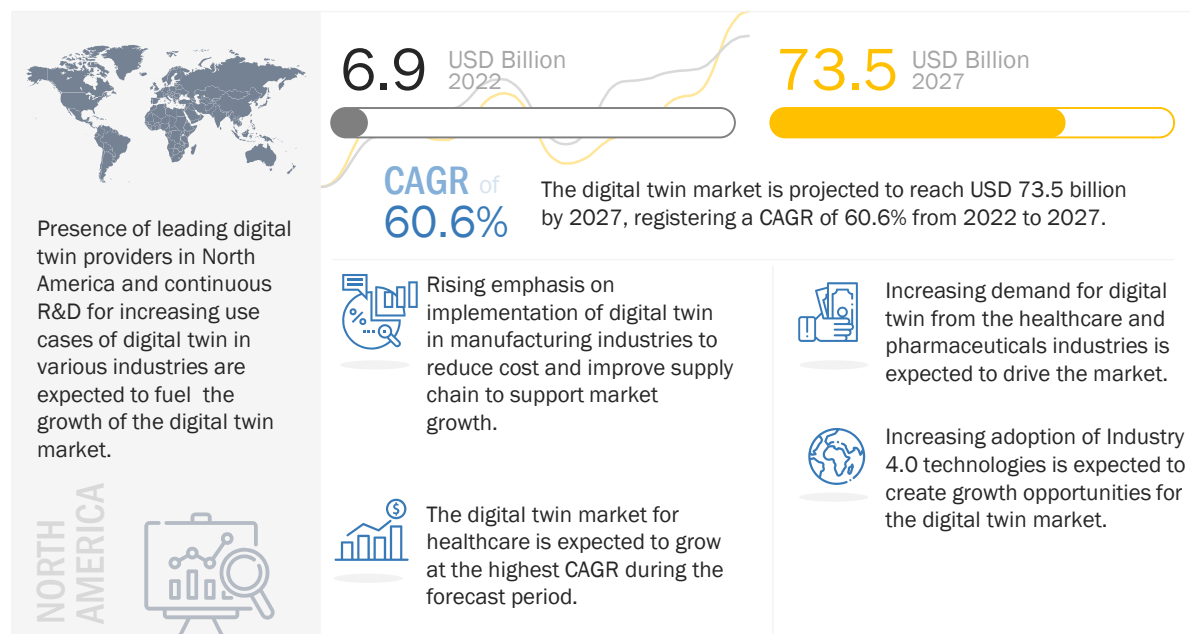
Source: Secondary Research, Interviews with Experts, and MarketsandMarkets Analysis

The North America region accounted for the largest share of the digital twin market in 2021 and is also expected to grow at the highest CAGR during the forecast period. The major factors driving the growth of the digital twin market in North America are the rising demand for smart sensors due to increasing automation in industries; growing adoption of technologies such as IoT and Industrial IoT (IIoT); and the increasing need to optimize productivity and reduce operational and maintenance costs. North America is one of the early adopters of digital twin technology. The region leads in terms of advanced automation technology and the use of digital twin and related/complementary technologies. All these factors contribute to the growth of the digital twin market in North America.

4 PREMIUM INSIGHTS

4.1 DIGITAL TWIN MARKET OVERVIEW

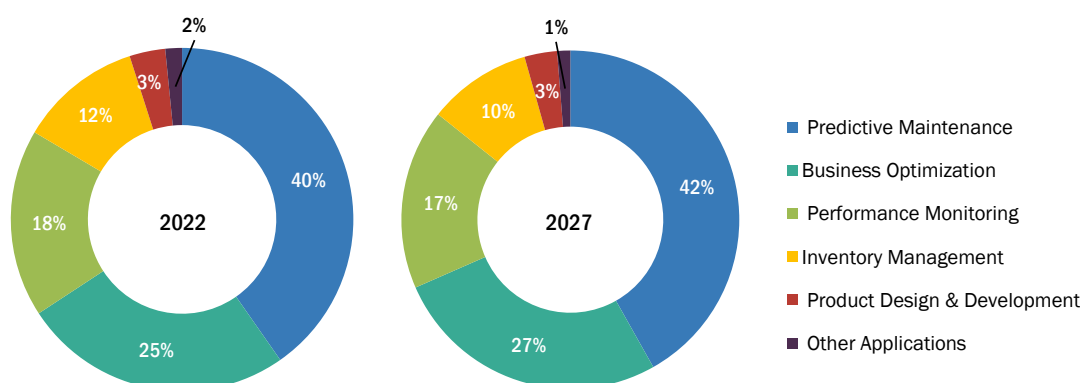
FIGURE 12 RISING EMPHASIS ON DIGITAL TWIN IN MANUFACTURING TO FUEL MARKET GROWTH



Source: Secondary Research, Interviews with Experts, and MarketsandMarkets Analysis

4.2 DIGITAL TWIN MARKET, BY APPLICATION

FIGURE 13 PREDICTIVE MAINTENANCE APPLICATION TO HOLD LARGEST SHARE OF DIGITAL TWIN MARKET DURING FORECAST PERIOD

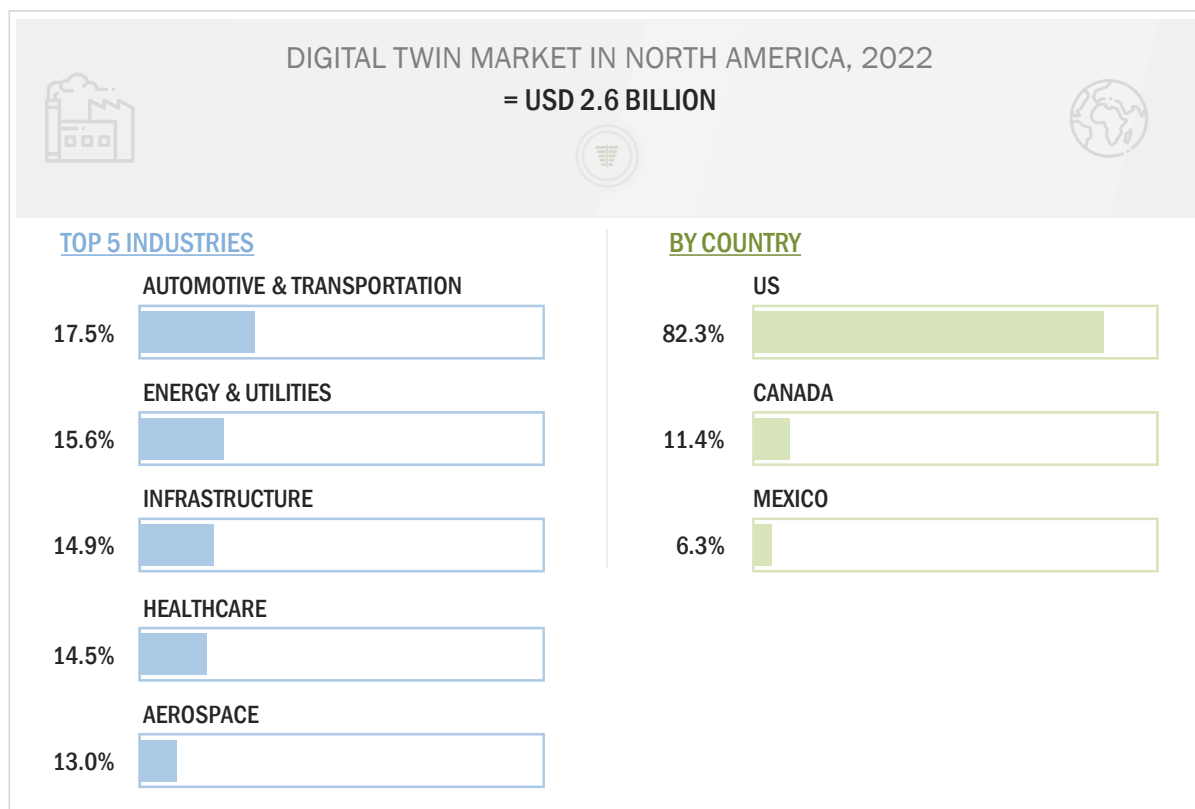


Note: Other applications include environmental analysis, customized services, instore planning & security implementation, and energy management.

Source: Secondary Research, Interviews with Experts, and MarketsandMarkets Analysis

4.3 NORTH AMERICA: DIGITAL TWIN MARKET, BY INDUSTRY AND COUNTRY

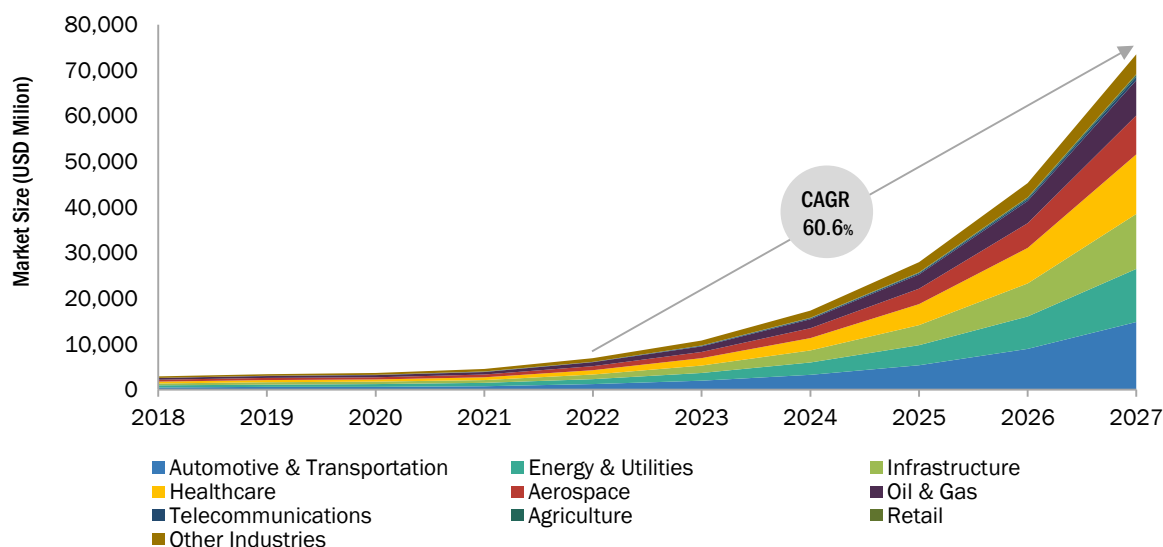
FIGURE 14 AUTOMOTIVE & TRANSPORTATION INDUSTRY AND US TO ACCOUNT FOR LARGEST SHARE OF NORTH AMERICA DIGITAL TWIN MARKET IN 2022



Source: Secondary Research, Interviews with Experts, and MarketsandMarkets Analysis

4.4 DIGITAL TWIN MARKET, BY INDUSTRY

FIGURE 15 HEALTHCARE INDUSTRY TO REGISTER HIGHEST CAGR DURING FORECAST PERIOD

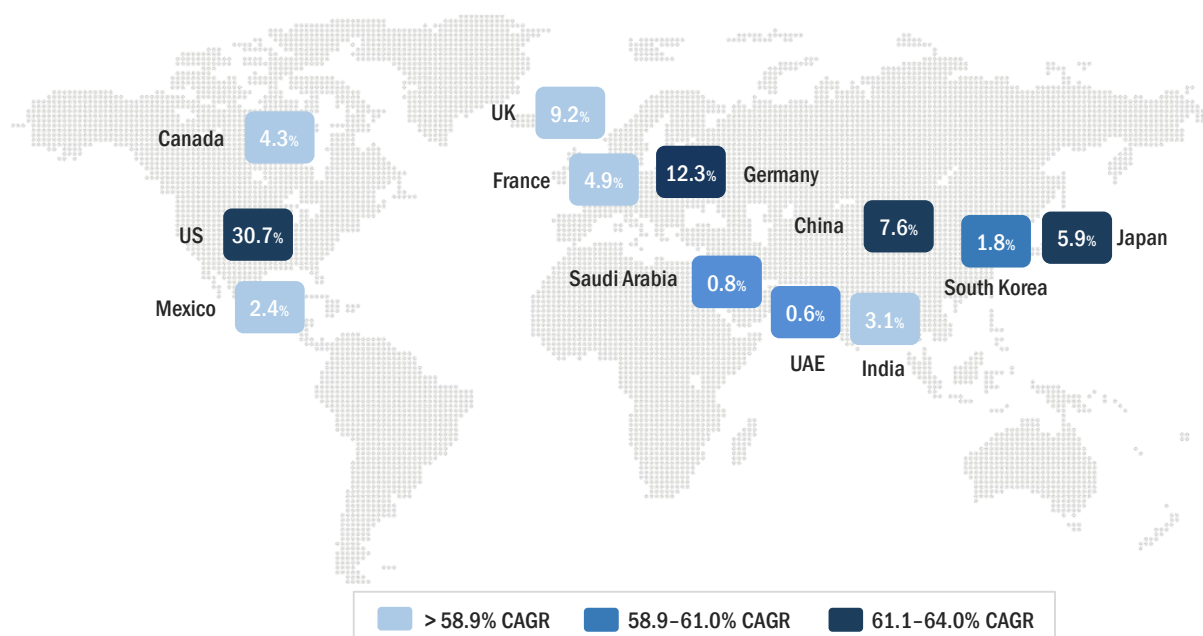


Note: Other industries include semiconductors, chemicals, paper & pulp, and food & beverages.

Source: Secondary Research, Interviews with Experts, and MarketsandMarkets Analysis

4.5 DIGITAL TWIN MARKET, BY COUNTRY

FIGURE 16 US TO DOMINATE GLOBAL DIGITAL TWIN MARKET DURING FORECAST PERIOD



Note: The numbers inside the boxes represent the estimated market shares in 2022, and the color of the boxes represents the CAGR range.

The sum of the market shares does not add up to 100% as Rest of Europe, Rest of Asia Pacific, and Rest of the Middle East have not been considered in the figure.

Source: Secondary Research, Interviews with Experts, and MarketsandMarkets Analysis

5 MARKET OVERVIEW

5.1 INTRODUCTION

This chapter focuses on the analysis of the digital twin market dynamics—drivers, restraints, opportunities, challenges, the supply chain, ecosystem, Porter’s Five Forces analysis, technology trends, average selling price, trade analysis, patent analysis, and tariffs and regulations. Digital twins optimize the product performance by modifying the physical prototype and, hence, reduce the time and cost required for product development. Digital twin technology also facilitates the linking of the physical environment with the virtual environment of the product using technologies such as cloud computing, machine learning, and industrial internet of things (IIoT). Digital twin technology allows for the development/creation of a digital model of a physical system in order to examine and assess its performance precisely. Computer-aided design tools, bill of materials, simulation models, time-series data, IoT sensors, maintenance records, and any other data a user/manufacturer desires to bring together are used to construct digital twins.

Rising emphasis on digital twins in manufacturing industries to reduce cost and improve supply chain operations; increasing demand for digital twins from the healthcare industry; and growing focus on predictive maintenance are the driving factors for the growth of digital twin market. The susceptibility of digital twin to cyberattacks due to the use of IoT and cloud platforms hinders the growth in the market. Presently, digital twin providers have a high growth opportunity in manufacturing industries. Further, the rising emphasis on advanced real-time data analytics, and increasing adoption of Industry 4.0 principles create future scope for the adoption of digital twin.

5.2 MARKET DYNAMICS

This section discusses the drivers, restraints, opportunities, and challenges influencing the digital twin market. These factors help market players design strategies and devise action plans to grow and improve their performance in the market.

FIGURE 17 **DIGITAL TWIN MARKET: DRIVERS, RESTRAINTS, OPPORTUNITIES, AND CHALLENGES**

 DRIVERS	▪ Rising emphasis on digital twin in manufacturing industries to reduce cost and improve supply chain operations ▪ Increasing demand for digital twin from healthcare industry ▪ Growing focus on predictive maintenance
 RESTRAINTS	▪ High investments associated with implementation of digital twin technology ▪ Susceptibility of digital twin to cyberattacks
 OPPORTUNITIES	▪ Rising emphasis on advanced real-time data analytics ▪ Increasing adoption of Industry 4.0 principles ▪ Development of human-centered digital twins
 CHALLENGES	▪ Complexities associated with data collection and mathematical models ▪ Lack of skilled workforce and awareness regarding cost benefit offered by digital twins

Source: Secondary Research, Interviews with Experts, and MarketsandMarkets Analysis

5.2.1 DRIVERS

FIGURE 18 IMPACT OF DRIVERS ON DIGITAL TWIN MARKET

DRIVERS	2 YEARS	5 YEARS	REASON	CONTRIBUTION TOWARD GROWTH
Rising emphasis on digital twin in manufacturing industries to reduce cost and improve supply chain operations			The use of digital twin enables to reduce the excess time and cost required in manufacturing. By implementing digital twins, engineers can optimize a product's performance by modifying the physical prototype that changes at every instance during the design phase. Hence, a digital prototype created with a digital twin can be used to run simulations and can be modified at any instance at a lesser time and with minimal costs.	(+)3.0%–5.0%
Increasing demand for digital twin from healthcare industry			Digital twin plays an important role in monitoring patient health, impact of medicines and therapies, and other parameters. Digital twins are also used to construct personalized medicine and treatment regimens by modeling organs and single cells, as well as an individual's genetic makeup, physiological traits, and lifestyle patterns.	(+) 4.0%–5.0%
Growing focus on predictive maintenance			Digital twin is used widely for predictive maintenance across various industries. Digital twin-based predictive maintenance uses real-time sensor records about the health and working conditions of a product, process, or system/unit and analyzes it against historical data about its failure modes and their criticality. This analyzed data is then used for predictive maintenance.	(+) 3.0%–5.0%
			Cumulative Growth from Drivers	(+) 10.0%–15.0%
IMPACT LEVEL:  HIGH  MEDIUM  LOW  VERY LOW				

Source: Secondary Research, Interviews with Experts, and MarketsandMarkets Analysis

5.2.1.1 Rising emphasis on digital twin in manufacturing industries to reduce cost and improve supply chain operations

Unplanned downtime and production waste severely affect manufacturers. Hence, they are looking for a system that can determine possible defects and failures in advance to avoid future losses. The use of digital twin enables to reduce the excess time and cost required in manufacturing. By implementing digital twins, engineers can optimize a product's performance by modifying the physical prototype that changes at every instance during the design phase. Hence, a digital prototype created with a digital twin can be used to run simulations and can be modified at any instance at a lesser time and with minimal costs. A digital twin has numerous applications in manufacturing, such as configuration management, asset management, process control, performance management, and simulation modeling.

The data that is used in digital twin technology is historical data. In few instances, the digital twin operates with real-time data. Analysts can use digital twins to better understand a supply chain's behavior, foresee uncommon circumstances, and create an action plan to cut down costs and increase process efficiency. A digital twin can aid firms in recognizing trends and simulating the effects of changes to various processes in order to improve supply chain design tests, as well as monitor risk and test probabilities. Digital twin offers a continuous, end-to-end picture of all the processes and bottlenecks associated with the supply chain, allowing manufacturers to solve problems faster and with minimal human intervention. These twins collect data to assist uncover potential flaws in all phases of delivery. For instance, a shipment digital twin will rely on data collected from sensors that communicate updated data throughout shipment, which may be evaluated to identify performance and bottlenecks during transportation and delivery journeys.

Digital twins help supply chain management to better arrange transportation and resources by exploiting real-time data. Furthermore, the technology allows for supply chain optimization across the entire network. River Logic, a US-based company involved in prescriptive analytics (optimization), offers supply chain digital twin to its customer to tackle short-, medium-, and long-term decision-making challenges. Similarly, SAP (Germany), in partnership with Siemens (Germany), co-sells digital twin focused on production and supply chains. As firms venture into new territory, it is critical to embrace digital twins in order to meet the market's changing demands. It will boost the potential to take businesses to new heights by improving data flow. Furthermore, when digital twins are combined with artificial intelligence (AI), machine learning (ML), and digital reality services (virtual reality/mixed reality), firms have new opportunities to incorporate lower-level workers with greater visibility and efficiency.

5.2.1.2 Increasing demand for digital twin from healthcare industry

Digital twin plays an important role in monitoring patient health, impact of medicines and therapies, and other parameters. Amid COVID-19, few digital twin industry players created or deployed solutions to fight the deadly coronavirus. Exactcure (France) has developed a digital twin solution to reduce the impact of inaccurately dosed medication and help patients avoid underdose and overdoses. Through this solution, the efficacy and interactions of medications in the body of a patient based on their personal characteristics such as age, gender, and kidney status are monitored. With the rapid spread of the coronavirus, the healthcare and pharmaceuticals industries resorted to digital twin technology to combat COVID-19. According to NavVis, a Germany-based digital twin provider, the use of NavVis Digital Twin has surged by nearly 300% post the implementation of COVID-19 physical distancing measures. With the help of digital twin in healthcare, many environmental, health, and safety (EHS) responsibilities, as well as layout planning and operations, can be done from home.

Healthcare industry players and digital twin providers are increasingly investing in research and development pertaining to new daily life changes that occurred due to the COVID-19 pandemic. Industry players are trying to develop a digital twin solution to correctly identify people who have symptoms, are infected, or have recovered with antibodies and people who have come in contact with an infected person. For instance, in October 2021, Dell (US) partnered with the i2b2 tranSMART Foundation (US), an open-source open-date community for enabling collaboration for precision medicine to create privacy-preserving digital twins to treat the long-haul symptoms of COVID-19 patients. The study aims to enhance care for the 5% of COVID-19 patients who experience chronic health problems. By using such a solution, it will be easy for state or local governments to monitor green and red zones. Such solutions can be deployed in smart cities effectively. In addition, digital twin technology can also be used to create a virtual twin of a hospital in order to analyze operational strategies, capacity, staffing, and care models. This will help hospitals to identify the areas for improvement, predict future issues, and optimize organizational strategies. Hospital digital twins can be utilized not only to optimize resource allocation but also for risk management. With risk management, hospitals can create a safe environment by implementing data-driven strategic decisions in a complex and sensitive environment.

Digital twins are also used to construct personalized medicine and treatment regimens by modeling organs and single cells, as well as an individual's genetic makeup, physiological traits, and lifestyle patterns. For instance, in February 2020, Dassault Systèmes (France), a digital twin provider, unveiled its digital twin, which aims to revolutionize the healthcare for the treatment of patients by enabling virtual twin experiences of the human body. As the next step in this revolution, in January 2022, at CES 2022 (Las Vegas), the company showcased the virtual twin of the human brain. By providing tailored diagnosis and treatment planning, these reproductions of the human body's internal systems improve medical care and patient treatment. Similarly, Siemens Healthineers has developed a digital twin of the heart, which mimics the electrical and physical properties of the cells in a patient's heart. Cardiologist Benjamin Meder ran simulations to see if the pacemaker can keep a person with congestive heart failure alive before inserting a knife for surgery. This is helping doctors to make more precise diagnoses and optimize decisions to avoid surgery failures.

5.2.1.3 Growing focus of industries on predictive maintenance

Digital twin is used widely for predictive maintenance across various industries. Digital twin-based predictive maintenance uses real-time sensor records about the health and working conditions of a product, process, or system/unit and analyzes it against historical data about its failure modes and their criticality. This analyzed data is then used for predictive maintenance. Along with the prediction of failures, the technology provides various benefits that help in optimizing the complete maintenance process. Digital twin offers the ability to calculate maintenance-related key performance indicators. In addition, digital twin also forecasts the behavior of machines under different circumstances. Being an accurate real-time model for a product's, process's, and system's/unit's condition and performance, a digital twin is used to run simulations and predict how a product, process, and system/unit will "behave" under certain factors, e.g., runtime, exposure to severe operating conditions, temperature, etc.

Digital twins, combined with augmented and virtual reality, help companies in better involvement of customers at the earlier stage in the maintenance process, thereby minimizing the need for expensive late-stage changes and saving time and money. In the aerospace industry, digital twins help minimize maintenance costs, optimize processing, and reduce downtime. General Electric (US) has built digital twins for airplane landing gears. The sensors placed at the landing gear failure points will help in predicting the malfunction or diagnosing the lifecycle of the landing gear. GE Aviation is also producing a digital twin for each engine being manufactured. This gives the engine manufacturer a digital trial for each engine, which helps GE predict the failure of parts.

Digital twin technology offers the ability to timely recognize disruptions in asset performance, forecast potential problems, and simulate various maintenance scenarios. It helps companies eliminate machine downtime, reduce equipment maintenance costs, improve equipment reliability, and extend lifespan. With the introduction of IoT and digital twins, companies are focusing on developing a predictive maintenance model that optimizes the maintenance cycle and strikes a balance between corrective and preventative maintenance. Components are replaced only as needed in predictive maintenance. This method not only replaces components on the verge of failure but also extends component lifetime by lowering unscheduled maintenance and labor expenses. This helps businesses to save money and gain a competitive advantage. Hence, in the near future, end-user industry players will opt for digital twin solutions to avoid losses occurring due to unavailable remote facilities.

5.2.2 RESTRAINTS

FIGURE 19 IMPACT OF RESTRAINTS ON DIGITAL TWIN MARKET

RESTRAINTS	2 YEARS	5 YEARS	REASON	CONTRIBUTION TOWARD GROWTH
High investments associated with implementation of digital twin technology			The requirement for high investments for the deployment of digital twin solutions acts as a restraint for the growth of the market globally. Huge capital is required to set up and install digital twins and the technologies related to it, such as smart automation and IIoT-connected control, product lifecycle management (PLM), and others. If an organization lacks the necessary building blocks and accompanying technology, the cost of establishing a digital twin grows even higher, making it much more costly. This can discourage manufacturers to adopt the technology.	(-) 2.0%–3.0%
Susceptibility of digital twin to cyberattacks			Increasing occurrences of viruses and cyber-attacks have raised concerns regarding data security. Important information can be stolen by malicious viruses in computer systems, resulting in a major loss to the companies. Failure of IoT infrastructure providers and cloud platform providers in implementing proper security measures might lead to disruptions in the flow of information.	(-) 1.0%–2.0%
			Cumulative Decline due to Restraints	(-) 3.0%–5.0%
IMPACT LEVEL:  HIGH  MEDIUM  LOW  VERY LOW				

Source: Secondary Research, Interviews with Experts, and MarketsandMarkets Analysis

5.2.2.1 High investments associated with implementation of digital twin technology

The requirement for high investments for the deployment of digital twin solutions acts as a restraint for the growth of the market globally. Huge capital is required to set up and install digital twins and the technologies related to it, such as smart automation and IIoT-connected control, product lifecycle management (PLM), computer-aided design (CAD) and 3D CAD, manufacturing process management (MPM), manufacturing operations management (MOM), model-based system engineering (MBSE), enterprise resource planning (ERP), and augmented reality (AR)/virtual reality (VR)/extended reality (XR). If an organization lacks the necessary building blocks and accompanying technology, the cost of establishing a digital twin grows even higher, making it much more costly. This can discourage manufacturers from adopting the technology.

The high installation cost of information and enabling technology systems is due to the requirement for consultation, acquisition, and implementation of these systems and services. The implementation of digital twin requires the adoption of advanced smart sensors, IoT, IIoT devices, and several software solutions. The communication between these technologies can be done with the help of more advanced and latest functions such as voice recognition, gesture recognition, or multitouch responsive screens. This further increases the cost of the system. These technologies are still quite expensive and may lead to an additional financial burden on companies operating in price-sensitive economies. Also, it is difficult for small and medium enterprises (SMEs) to invest in digital twin solutions due to the high capital spending and a long payback period.

To improve the efficiency of the primary disciplines of such systems, digital twins are increasingly being used in smart manufacturing systems. For instance, the cost associated with the overall setup of digital twin for electrical vehicles (EVs) and automated guided vehicles (AGVs) systems is high as these vehicles are programmed and require navigation and communication systems to interact with other material handling equipment. The average payback period for AGVs usually ranges from three to five years. Therefore, it is difficult for many companies to invest in digital twin technology for AGVs due to high capital spending and a long payback period.

5.2.2.2 Susceptibility of digital twin to cyberattacks

The implementation of digital technologies, such as the cloud, big data, IoT, and artificial intelligence, is increasing in various facets of businesses. Digital twin involves the integration of various IoT sensors and other digital technologies for virtualizing the physical twin. With growing connectivity arises the risk of security, compliance, data protection, and regulations. Increasing occurrences of viruses and cyber-attacks have raised concerns regarding data security. Important information can be stolen by malicious viruses in computer systems, resulting in a major loss to the companies. Failure of IoT infrastructure providers and cloud platform providers in implementing proper security measures might lead to disruptions in the flow of information. The industries that have implemented digital twins with such platforms are prone to malware attacks that are targeted at industrial systems due to the increasing dependence on web-based data interchange and off-the-shelf IT solutions.

A large volume of data is more prone to the risk of security. Digital twins gather sensitive, business-critical information from many aspects of the business into one place. Business planning needs to include data security, data integrity, privacy, and compliance considerations. Therefore, it is essential to ensure that good security practices are already in place while deploying digital twins, be it for a product, process, or a complete system/unit. With the rise in cyberattacks on critical infrastructure over the past decade, cybersecurity has become a major concern among users and vendors of industrial automation. Thus, the increasing threat pertaining to data security is expected to be a major restraint for the growth of the digital twin market.

Moreover, cybersecurity concerns are increasing, and the industrial sector is finding it difficult to tackle these issues. As the frequency of cybersecurity breaches has increased over the past years, organizations have increased their IT security investments to protect their systems against advanced threats, zero-day malware, and other targeted threats. However, there is a lack of framework and governance due to the high costs of innovation pertaining to the development of advanced cybersecurity solutions. Several organizations consider this a financial constraint, limiting the growth of the smart manufacturing market. Thus, the deployment and maintenance of a digital twin is an expensive proposition. The lack of budget continues to be a concern for cybersecurity professionals to effectively carry out their security operations.

5.2.3 OPPORTUNITIES

FIGURE 20 IMPACT OF OPPORTUNITIES ON DIGITAL TWIN MARKET

OPPORTUNITIES	2 YEARS	5 YEARS	REASON	CONTRIBUTION TOWARD GROWTH
Rising emphasis on advanced real-time data analytics			A digital twin can be used to create twins of components, assemblies, people, or an entire manufacturing plant and can be combined in multiple ways to create a solution with numerous sources of data and information. The foundation of digital twin technology is data and analytics. Digital twins and analytics, once established, allow for more precise diagnostic, predictive, and optimized operations. Businesses have begun to grasp the potential of digital twins after seeing the benefits of AI and data analytics	(+) 3.0%–5.0%
Increasing adoption of Industry 4.0 principles			The trends of Industrial Internet of Things (IIoT) and Industry 4.0/smart manufacturing are ramping up, especially with the rising demand for digitalization.). IIoT will create enormous opportunities for digital twin providers and their customers in the coming years. IIoT brings together the machines, advanced analytics, and people involved in the working of a particular process or product.	(+) 4.0%–5.0%
Development of human-centered digital twins			Digital twins based on human-in-the-loop (HITL) allow artificial intelligence to leverage both human and machine intelligence to create machine learning models. This methodology allows human, product, and system data to be linked to computational toolkits, allowing for ergonomic decision-making to be incorporated into digital twin-driven design. When building a model, it is important to think about human factors such as taking breaks between work sessions, taking fatigue into account, considering the working hours, and so on.	(+) 3.0%–5.0%
			Cumulative Growth from Opportunities	(+) 10.0%–15.0%
IMPACT LEVEL:  HIGH  MEDIUM  LOW  VERY LOW				

Source: Secondary Research, Interviews with Experts, and MarketsandMarkets Analysis

5.2.3.1 Rising emphasis on advanced real-time data analytics

A digital twin can be used to create twins of components, assemblies, people, or an entire manufacturing plant and can be combined in multiple ways to create a solution with numerous sources of data and information. The foundation of digital twin technology is data and analytics. The key element of digital twin includes the ability to monitor a real-world object and model a real-world object. Further, it is possible to inquire for or get alerts about particular events or changes when an item may be monitored in the real world. Whereas the ability to model a real-world object is associated with data such as identity, time, context, and events and, thus, corresponds to a unique physical thing. All of these elements of digital twin represent real-time data analysis that drives the operation of digital twin in any organization.

Digital twins and analytics, once established, allow for more precise diagnostic, predictive, and optimized operations. Businesses have begun to grasp the potential of digital twins after seeing the benefits of AI and data analytics. Currently, multiple AI and machine learning models are built to handle specific challenges, but with a digital twin, enterprises can have access to broader business insights across the entire organization. And due to this, digital twins are becoming a source of truth for facility insights as they take their place in smart workplaces, at the heart of the IIoT, and other network systems. Digital twin analytics is a new standard for decision-making on how to coordinate and operate the workplace because of their ability to contextualize data within a physical model.

Another reason for the potential growth of the digital twin market is the fact digital twins offer a sharper vision to organizations. Predictive analytics alone may envision a simpler model that tests a set of limited variables. However, a digital twin can be considered a more exact and complex form of a predictive model. Digital twins use dynamic, real-time data streams instead of stored data. Furthermore, a digital twin can take many aspects into consideration and generate a picture of the entire situation rather than simply focusing on a small number of outcomes in a certain area. For instance, logistic company United Parcel Service (US) developed harmonized enterprise analytical tool (HEAT), a digital twin that operates in real-time and predicts the resourcing needs at its facilities. Using the combination of digital twin and powerful real-time analytics technologies, UPS manages its extremely complex logistics network by predicting and advising actions in real time. With the help of digital twin and real-time analytics, UPS is delivering COVID-19 vaccines to over 100 countries with 99.99% reliability.

5.2.3.2 Increasing adoption of Industry 4.0 principles

The trends of Industrial Internet of Things (IIoT) and Industry 4.0/smart manufacturing are ramping up, especially with the rising demand for digitalization. According to MarketsandMarkets, the smart manufacturing market was estimated at USD 88.7 billion in 2021 and is expected to reach USD 228.2 billion by 2027, at a CAGR of 18.5% from 2022 to 2027. It is expected that between 2019 and 2025, the number of global IoT connections will grow to be more than double, nearly 25 billion, while global IoT revenue will triple to cross the USD 1.1 trillion mark (GSMA data). IoT will create enormous opportunities for digital twin providers and their customers in the coming years.

IIoT brings together the machines, advanced analytics, and people involved in the working of a particular process or product. This builds up a network of a multitude of industrial devices connected by communication technologies that help to develop systems that can monitor, collect, exchange, analyze, and deliver valuable insights for better decision-making capabilities. With the technology having the ability to cover the entire life cycle of a physical system, process, or product, it provides businesses with a powerful analytical tool that helps to thoroughly assess key performance indicators and provide insights as to where enhancements/upgrades can be made. In the long run, the lessons and suggestions taken from digital twins are expected to create immense opportunities for innovation and growth.

In addition, monitoring the field performance of the manufactured products via a digital twin along with IIoT provides a window with an ongoing view, comparing actual usage with the anticipated usage of manufactured products as reflected in the design parameters of the product. This process provides valuable real-time feedback for improving the design process. The technology also increases the efficiency of product development as the manufacturing and testing of prototypes in a physical environment consumes not only time but also requires high investments. The virtual simulation models of the products or processes or units make it easier for experts from various fields to participate in the product development process. For example, software developers, mechanical engineers, and user interface designers can all simultaneously work on the same digital twin. The deployment of IIoT and digital twins optimizes the efficiency by predicting failures in production and helping engineering teams to fix them before they affect manufacturing targets. Enhancements can be made and simulated by adjusting the parameters of the production line in the digital twin itself, without causing any harm to the actual production. Later, only the successful simulations/enhancements can then be applied to the real-life system. For example, by using a digital twin of turbines for a wind farm, engineers can study how turbines interact with different landscapes and wind sources, then mix and match different turbines to optimize the configuration for each pad on a farm.

5.2.3.3 Development of human-centered digital twins

Digital twins based on human-in-the-loop (HITL) allow artificial intelligence to leverage both human and machine intelligence to create machine learning models. This methodology allows human, product, and system data to be linked to computational toolkits, allowing for ergonomic decision-making to be incorporated into digital twin-driven design. When building a model, it is important to think about human factors such as taking breaks between work sessions, taking fatigue into account, considering the working hours, and so on. The human-centric digital twin design allows for realistic output optimization rather than focusing solely on boosting production and efficiency without taking into account human variables. For instance, in 2022, Kalam Innovation Hub created a human-centric digital twin for its clients, which includes break and leisure time for the employees in the model. This aided the client in optimizing output while keeping employees' basic needs in mind and making the workplace more human-friendly.

This type of human-centered paradigm for digital twin also allows for employee safety and medical fitness at work. It won't just be a case of re-establishing operating systems to meet new requirements; in the present scenario as well as in future, the safety of each individual employee in terms of social distancing will be a far more essential factor. Siemens developed a novel approach to address the issues that businesses are facing as a result of the COVID-19 epidemic. Companies can use Siemens's Simatic RTLS Real-time Locating System to measure, monitor, and maintain social distance among employees. Companies can use the solution to control how personnel interact with one another, with the production line, and with the plant setup in a quick and efficient manner. The integrated digital twin assists the organization in simulating employee safety, testing, and optimizing workspace setup, validating safety measures, and designing future-proof manufacturing lines. Hence, such human-centric digital twins act as a lucrative opportunity for companies to deploy digital twin not just for optimizing their setup and output but also for the safety and wellness of their employees as well.

5.2.4 CHALLENGES

FIGURE 21 IMPACT OF CHALLENGES ON DIGITAL TWIN MARKET

CHALLENGES	2 YEARS	5 YEARS	REASON	CONTRIBUTION TOWARD GROWTH
Complexities associated with data collection and mathematical models			Obtaining the asset data from the supplier may pose a challenge, as it requires close collaboration with multiple tiers of the supply chain. Moreover, establishing the required level of communication in a supply chain to retrieve asset data may be challenging to maintain.	(-) 1.5%–2.5%
Lack of skilled workforce and awareness regarding cost benefit offered by digital twins			As the technology is new for many companies and requires significant changes, end users are unclear about the economic benefits, investments involved, and the cost benefits in the near future. The determination of the potential of a digital twin is considered to be very complex and diverse, which further complicates the adoption of this technology.	(-) 1.5%–2.5%
Cumulative Decline due to Challenges				(-) 3.0%–5.0%
Cumulative Growth from Drivers				(+) 10.0%–15.0%
Cumulative Growth from Opportunities				(+) 10.0%–15.0%
Cumulative Decline due to Challenges				(-) 3.0%–5.0%
Cumulative Decline due to Restraints				(-) 3.0%–5.0%
Cumulative Growth				(+) 14.0%–20.0%
IMPACT LEVEL:  HIGH  MEDIUM  LOW  VERY LOW				

Source: Secondary Research, Interviews with Experts, and MarketsandMarkets Analysis

5.2.4.1 Complexities associated with data collection and mathematical models

Despite the obvious benefits, a major portion of organizations are yet to implement a digital twin model for asset management and business and operation optimization. A digital twin should be able to model simple as well as complex objects and their relationships by accurately capturing their physical properties and simulating behaviors. For instance, the development of an electrical digital twin requires multiple inputs from operators such as facility managers, design engineers, electrical engineers, equipment vendors, and other parties, which adds complexity to the deployment. Obtaining the asset data from the supplier may pose a challenge, as it requires close collaboration with multiple tiers of the supply chain. Moreover, establishing the required level of communication in a supply chain to retrieve asset data may be challenging to maintain. Another challenge comes from the complexity of the physical item that is to be replicated. For instance, processes such as customer relationship management, prospect marketing, or service delivery funnels are significantly easier to duplicate, as they involve a flow of scenarios, ideas, and software inputs instead of moving parts that must be recorded.

In addition, detailed blueprints of historical systems, equipment failures, and other mathematical systems are required to create a digital twin. For instance, a digital twin should be fed with data about equipment failure modes to predict failures. Any change in equipment or asset configuration or operational state would require remodeling of the digital twin, which increases the complexity of maintaining the digital replica. Any modification made to the physical system would require similar changes to its digital model and algorithms. The risk of errors may escalate with the increasing levels of complexity, as the simulation of certain scenarios, such as the addition of new parts at the machine level or operational level changes, may not be precise.

The main problem is obtaining data in the correct format. Classic mathematical techniques such as Bayesian estimation or Bayesian statistics are required to construct the digital model. One of the most significant challenges is that the majority of the world is inexperienced with Bayesian statistics. Furthermore, the lack of unified data from the organizations worldwide is a key challenge. In a company, data is stored in a variety of silos. The data (historical or current) required to develop a digital twin model is accessible from clients in a variety of formats. For example, some data in a supply chain is in pdf format, while others are in engineering models or CAD models. As a result, organizing that type of data from various sources is a difficulty for digital twin developers.

Adopting digital twins entails considerable investments, collaboration, and sustained commitment from the operator. Operators need to establish the required technology infrastructure such as robust IoT deployments and simulation platforms and integrate data sources across departments and IoT contextual systems to maintain a digital twin. These factors restrict the widespread adoption of electrical digital twins by various stakeholders.

5.2.4.2 Shortage of skilled workforce and limited awareness regarding cost-saving benefits offered by digital twins

Companies do not have any specific plan for the implementation of digital twins for product management and have not made any significant investments yet. As the technology is new for many companies and requires significant changes, end users are unclear about the economic benefits, investments involved, and the cost benefits in the near future. The determination of the potential of a digital twin is considered to be very complex and diverse, which further complicates the adoption of this technology.

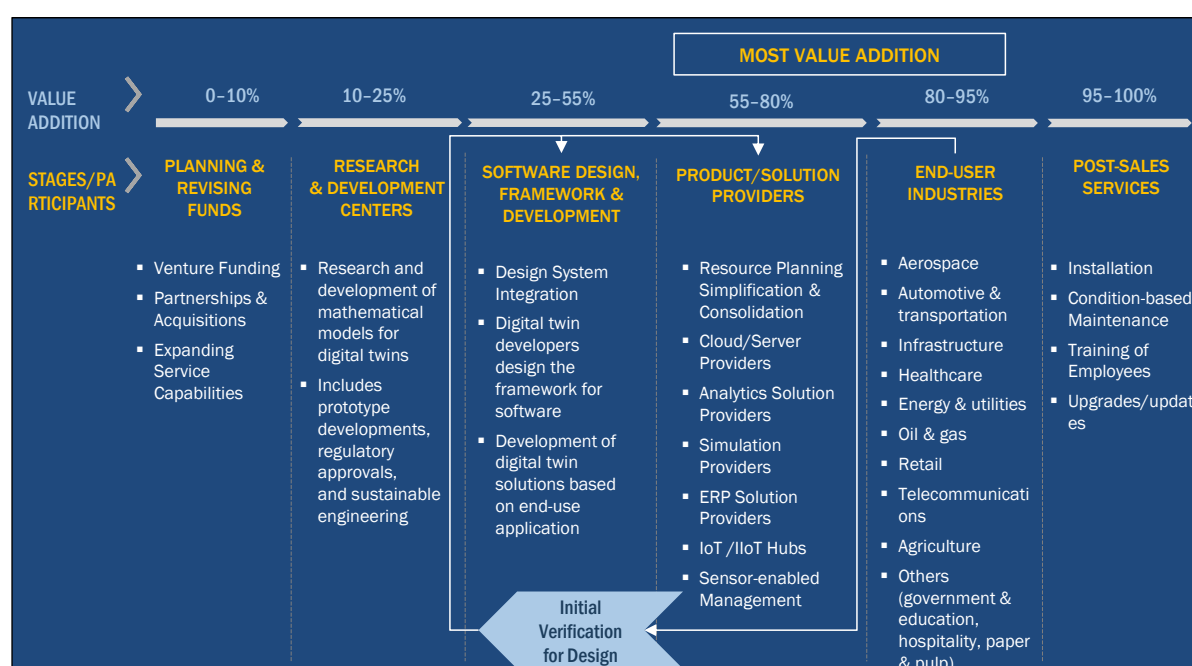
The creation of a digital twin of an asset/process requires various technologies and skillsets, along with trained workers, to handle the latest equipment and software systems. The digital change will lead to a change in the specifications of skills required for employees at different stages of the value chain, starting from development to sales and marketing. The expected increase in working efficiency and data-dependency of processes would lead to the requirement of new employee skill sets and high qualifications. This could create a skill gap between the new workers and trained workers. Moreover, industries are very

dynamic toward adopting new technologies and, at the same time, are facing a shortage of highly skilled and proficient workforce.

Continued support from educational institutions and scientific facilities in generating new talent needs to be encouraged and carefully cultivated with industry tie-ups. This requires a stronger collaboration between various industry stakeholders, such as utilities and grid operators, governments, and universities, to help define and develop digital leaders of the future. Also, despite the increasing popularity of digitization, most of the industries are unaware of the capabilities of digital twin technology due to their complex structure and lack of process understanding for the implementation of this technology.

5.3 VALUE CHAIN ANALYSIS

FIGURE 22 VALUE CHAIN ANALYSIS: DIGITAL TWIN MARKET



Source: Secondary Literature, Interviews with Experts, and MarketsandMarkets Analysis

The value chain analysis section portrays the development of digital twin solutions, beginning from the planning phase to manufacturing and after-sales services; value is added to the product at each stage.

5.3.1 PLANNING AND REVISING FUNDS

The digital twin value chain starts with planning and revising funds, adding significant value to the final products/solutions. This phase involves activities such as bidding and contracting, identifying, and developing partners/suppliers, performing risk benefit and cost benefit analyses, and identifying the key technologies required for digital twin products or solutions. VEERUM, a digital twin provider startup based in Canada, raised a total of USD 16.4 million to boost its growth and expand its digital twin portfolio. The startup raised the fund over the course of eight investment rounds, with the latest funding raised in February 2021. Similarly, TWAICE Technologies (Germany) raised a total of USD 74.4 million in funding over six rounds, with the latest funding raised in April 2022. In December 2021, Software AG (Germany), a major digital solutions provider, raised a total of USD 388.2 million in two rounds of investment.

In February 2022, Bosch Building Technologies (division of Bosch) officially completed the acquisition of Hoerbuerger (Germany). The acquisition of Hoerbuerger is aligned with the Bosch initiative for creating digital building twin centered around building automation and energy management. Digitalization, networking, and artificial intelligence are expanding Bosch's capacity to transform commercial buildings into smart

living and working spaces that boost people's security, safety, and comfort while also ensuring higher sustainability.

5.3.2 RESEARCH AND DEVELOPMENT

The planning and revising funds phase is followed by the R&D phase, which involves prototype and process development, getting regulatory approvals, and sustainable engineering. Digital twin providers are constantly improving existing designs and configurations of products/solutions for greater reliability and better optimization; integration of powerful artificial intelligence (AI) and machine learning (ML) capabilities to analyze the data; and development of mathematical models. At this stage, a higher value is added to the final offering. At this stage, technology experts and researchers identify all the factors that are responsible for upgrading the digital twin technology. Technology experts and researchers add 10% value to the digital twin market value chain.

5.3.3 SOFTWARE DESIGN, FRAMEWORK, AND DEVELOPMENT

This phase includes the development of software according to client requirements. Companies provide customized software to ensure that the clients achieve the desired outputs. In case of digital twin, the entire solution—starting from building blocks, sensors, layout of design (both software and hardware), type of products/solutions—are fully customized for the client. The level of customization is so high that for two different organizations of same size operating within the same industry, the digital twin solution will be different for both the firms. Due such higher level of customization, a higher value is added at this stage of the value chain.

5.3.4 PRODUCTS/SOLUTIONS PROVIDERS

This phase comprises cloud/server providers, analytics solution providers, simulation providers, and ERP solution providers. These providers play a significant role in the entire value chain, and the total value added during this phase is similar to that of software design, framework, and development phase. The digital twin itself cannot operate in isolation. It needs complementary and related technologies to function and provide the desired benefits to the clients. GE (US), IBM (US), Siemens (Germany), Schneider Electric (France), SAP (Germany), Oracle (US), ABB (Switzerland), Microsoft (US), PTC (US), and ANSYS (US) are the major digital twin providers as well as providers of supporting products/solutions for digital twin.

5.3.5 END-USER INDUSTRIES AND AFTER-SALES SERVICE PROVIDERS

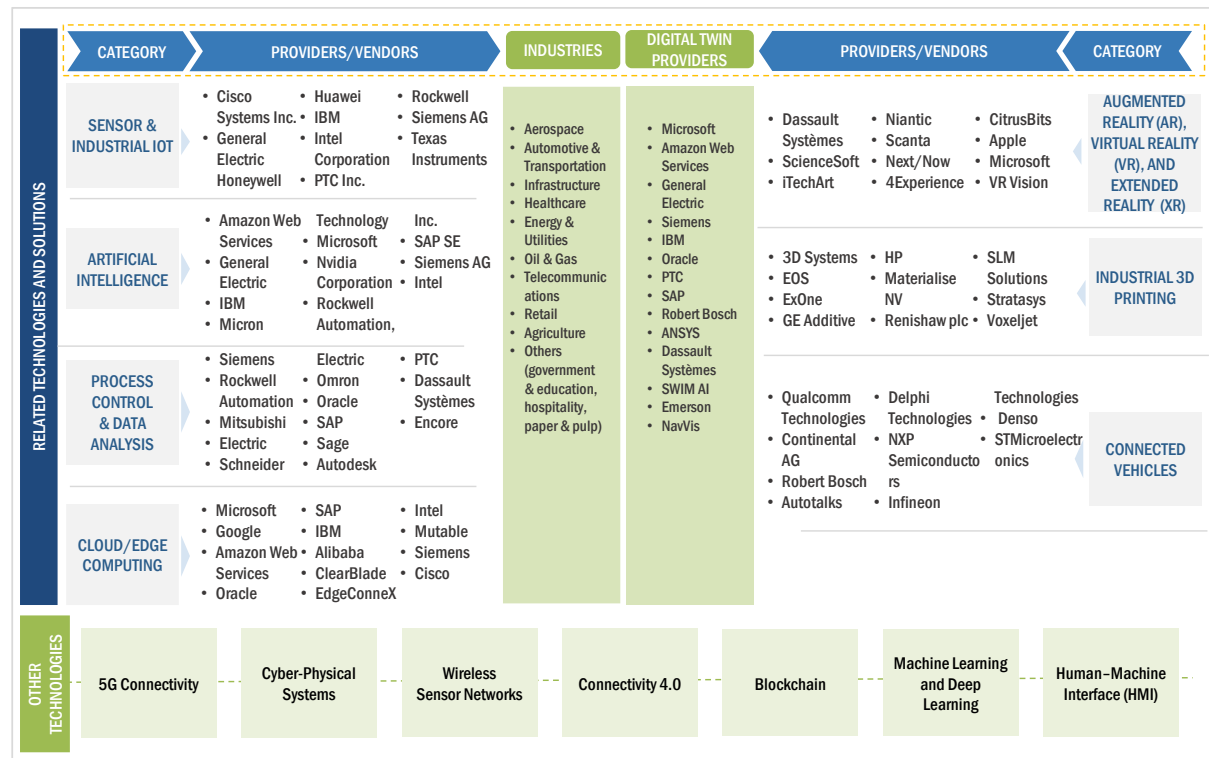
End-user industries for digital twin include aerospace, automotive & transportation, infrastructure, healthcare, energy & utilities, oil & gas, retail, telecommunications, and others. It is the last phase and accounts for ~20% of the value chain. Most of the digital twin providers open their own subsidiaries, sales channels, and after-sales centers worldwide or have close links with third-party providers. Consulting, upgrading, and troubleshooting services are standard across all industries. Microsoft (US) delivers products/solutions in nearly 250 countries with the help of a strong network of partners worldwide. Similarly, ANSYS (US) owns more than 75 strategic sales locations worldwide with a network of channel partners in more than 40 countries. In addition, the company has over 250 global partners that provide hardware, connectivity, and software solutions that host or integrate with ANSYS software solutions.

The after-sales services for digital twin includes the successful installation and integration of the solution with the end-user organization's legacy system. It also includes the training the employees of clients for operating and controlling the dashboards and simulations for real-time performance measurement, and predictive analysis. The clients also avail the services of software upgrades and updates from the digital twin providers. The frequency of updates and upgrades may be subject to the terms of the contract or agreement between the providers and the end users.

5.4 ECOSYSTEM

This section describes the digital twin ecosystem and the deployment of digital twin solutions in different end-user industries, along with the impact of associated technologies on the digital twin market.

FIGURE 23 DIGITAL TWIN ECOSYSTEM



Source: MarketsandMarkets Analysis

The diagram above depicts the schematic analysis of the digital twin ecosystem. Different types of technologies together make up the digital twin ecosystem; these include artificial intelligence (AI), sensors and industrial IoT, process control and data analysis, and cloud/edge computing, industrial 3D printing, blockchain, machine learning and deep learning algorithms, vehicle to everything (V2X), AR/VR/XR, and human-machine interface. Some of the top companies that provide sensors & IoT, software, and data analysis support for the digital twin ecosystem are General Electric (US), Siemens (Germany), IBM (US), Honeywell (US), ABB (Switzerland), Amazon (US), Intel (US), Nvidia (US), and Dassault Systèmes (France). With these emerging technologies, the existing applications of digital twin are expected to expand drastically, creating significant growth opportunities for players in the digital twin market.

The digital twin market is highly competitive. It is marked by the presence of some tier 1 companies, such as Microsoft (US), Amazon (US) – AWS, General Electric (US), Siemens (Germany), PTC (US), ANSYS (US), SAP (Germany), Dassault Systèmes (France), IBM (US), Oracle (US), and Robert Bosch (Germany). These companies have created a competitive ecosystem by investing in research and development activities to launch highly efficient and reliable solutions and software for digital twins. Digital twin technology and solution providers offer a wide range of services that cater to the requirements of various industries. They have been focusing on the development of advanced systems and play a significant role in managing different operational aspects of smart factories, smart buildings, and smart workspace across end-use industries.

TABLE 1 LIST OF COMPANIES AND THEIR ROLE IN DIGITAL TWIN ECOSYSTEM

COMPANY	ROLE IN ECOSYSTEM
General Electric	Digital Twin Product/Solution Provider
IBM	Digital Twin Product/Solution Provider
Microsoft	Digital Twin Product/Solution Provider
Amazon Web Services	Digital Twin Product/Solution Provider
Siemens	Digital Twin Product/Solution Provider
PTC	Digital Twin Product/Solution Provider
ANSYS	Digital Twin Product/Solution Provider
SAP	Digital Twin Product/Solution Provider
Oracle	Digital Twin Product/Solution Provider
Robert Bosch	Digital Twin Product/Solution Provider
Dassault Systèmes	Digital Twin Product/Solution Provider
SWIM AI	Digital Twin Product/Solution Provider
Emerson	Digital Twin Product/Solution Provider
ABB	Digital Twin Product/Solution Provider
Honeywell	Digital Twin Product/Solution Provider
Schneider Electric	Digital Twin Product/Solution Provider
NavVis	Digital Twin Product/Solution Provider
DNV	Digital Twin Product/Solution Provider
Autodesk	Digital Twin Product/Solution Provider
ANDRITZ	Digital Twin Product/Solution Provider
Software AG	Digital Twin Product/Solution Provider
Bentley Systems	Digital Twin Product/Solution Provider
River Logic	Digital Twin Product/Solution Provider
Altair	Digital Twin Product/Solution Provider
Johnson Controls	Digital Twin Product/Solution Provider
Pratiti Technologies	Digital Twin Product/Solution Provider
QiO Technologies	Digital Twin Product/Solution Provider
Petra Sata Science	Digital Twin Product/Solution Provider
Datumix	Digital Twin Product/Solution Provider
synavision	Digital Twin Product/Solution Provider

Sight Machine	Digital Twin Product/Solution Provider
Sensat	Digital Twin Product/Solution Provider
TWAICE Technologies	Digital Twin Product/Solution Provider
MEIK LABS Spa	Digital Twin Product/Solution Provider
Allvision Io	Digital Twin Product/Solution Provider
DigitalTwin Technology	Digital Twin Product/Solution Provider
HASH	Digital Twin Product/Solution Provider
Artemis Technologies	Digital Twin Product/Solution Provider
VEERUM	Digital Twin Product/Solution Provider
Cosmo Tech	Digital Twin Product/Solution Provider
FEops	Digital Twin Product/Solution Provider
Q Bio	Digital Twin Product/Solution Provider

Source: MarketsandMarkets Analysis

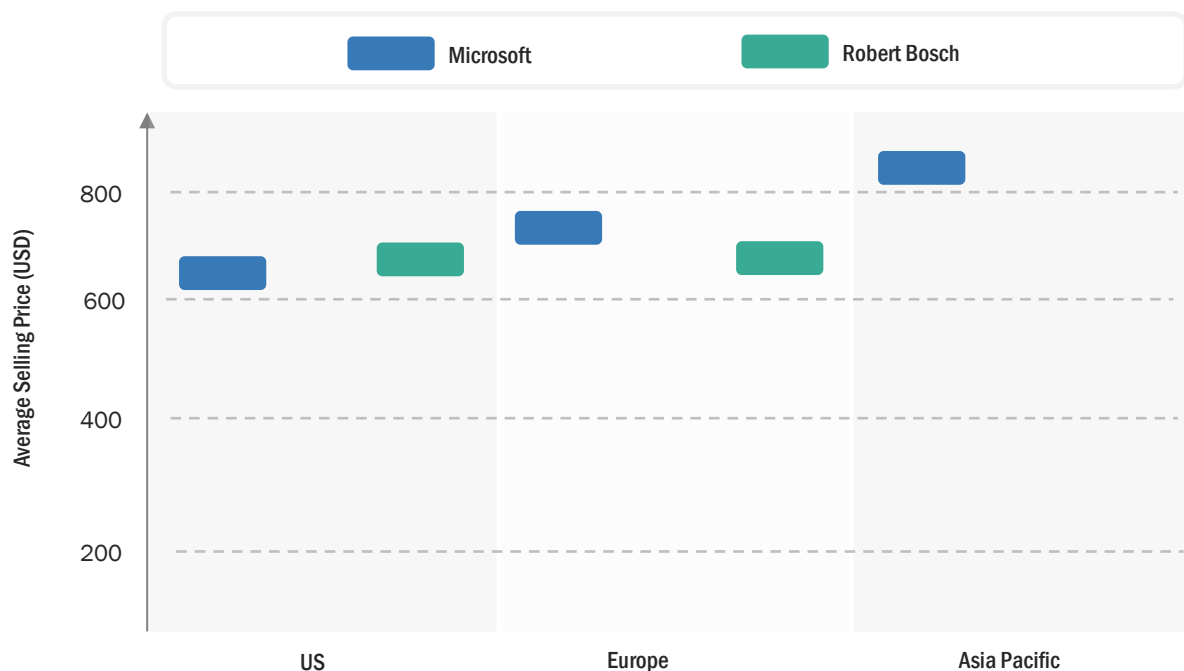
5.5 PRICING ANALYSIS

5.5.1 AVERAGE SELLING PRICE, BY APPLICATION (KEY PLAYERS)

This section includes price ranges of digital twin offered by key players in the top three regional markets based on live messages with message retention (device to cloud) (per device /per month).

- The average selling price of Microsoft Azure Digital twin and Bosch IoT Device Management Digital Twin are represented in the figure below. It is to be noted that the prices are subject to change with time.
- The average selling price of digital twin considered is according to the pricing managed by Microsoft Robert Bosch – based on per 100 million live messages with message retention (device to cloud) (per device/per month) given that the average size of a live message is 10 kB (10 kB/message). It is to be noted that this is not the only factor that decides the price of digital twins. Multiple factors such as number of operations, query units, number of devices/device digital twins, size of digital twins, and others decide the final price of digital twin on a monthly subscription basis or yearly subscription.
- Digital twin is not only available as consumption-based service; a few of the bigger players develop digital twins in-house. The intention here is to indicate how price ranges differ across different cohorts of customers on the basis of the number of messages with an average size of 10 kB.

FIGURE 24 AVERAGE SELLING PRICES OF PRODUCTS/SOLUTIONS, BY TOP THREE REGION (KEY PLAYERS)



Note: Prices mentioned above are estimated prices and not the final prices. These are estimated prices for a million live messages from 1 device (or IoT) to the digital twin provider server/cloud (Microsoft & Robert Bosch). In general, the implementation of digital twin solution in any organization involves multiple devices and systems and the price keeps on cascading with the number of devices and operations. The prices considered here are only for the software and not the hardware components. The price considered here is only for data transfer. The prices for compute power are not considered in this case.

Source: Press Releases, Company Websites, Interviews with Experts, Secondary Journals, and MarketsandMarkets Analysis

TABLE 2 AVERAGE SELLING PRICES OF PRODUCTS/SOLUTIONS, BY TOP THREE REGION (KEY PLAYERS)

Region	Microsoft	Robert Bosch
US	USD 671.00	USD 688.84
Europe	USD 744.16	USD 688.84
Asia Pacific	USD 838.23	-

Note: Prices mentioned above are estimated prices and not the final prices. These are estimated prices for a million live messages from 1 device (or IoT) to the digital twin provider server/cloud (Microsoft & Robert Bosch). It is to be noted that this estimated price is only for 1 device. In general, the implementation of digital twin solution in any organization involves multiple devices and systems and the price keeps on cascading with the number of devices and operations. The prices considered here are only for the software and not the hardware components. The price considered here is only for data transfer. The prices for compute power are not considered in this case.

The estimated prices vary from provider to provider, client to client, industry to industry, and region to region, depending upon the client's requirement.

Source: Press Releases, Company Websites, Interviews with Experts, Secondary Journals, and MarketsandMarkets Analysis

The price of digital twin products/solutions used in the aerospace, automotive & transportation, energy & utilities, and other industries is expected to decrease at a slow pace. As the adoption of digital twin will increase in the next 8 to 10 years, the prices will normalize. The price for digital twin varies from provider to provider, client to client, industry to industry, and region to region, depending upon the client's requirement. In case of digital twin for infrastructure, the price varies on the total area to be covered. For instance, the estimated digital twin budget for a typical A grade commercial office building with 600,000

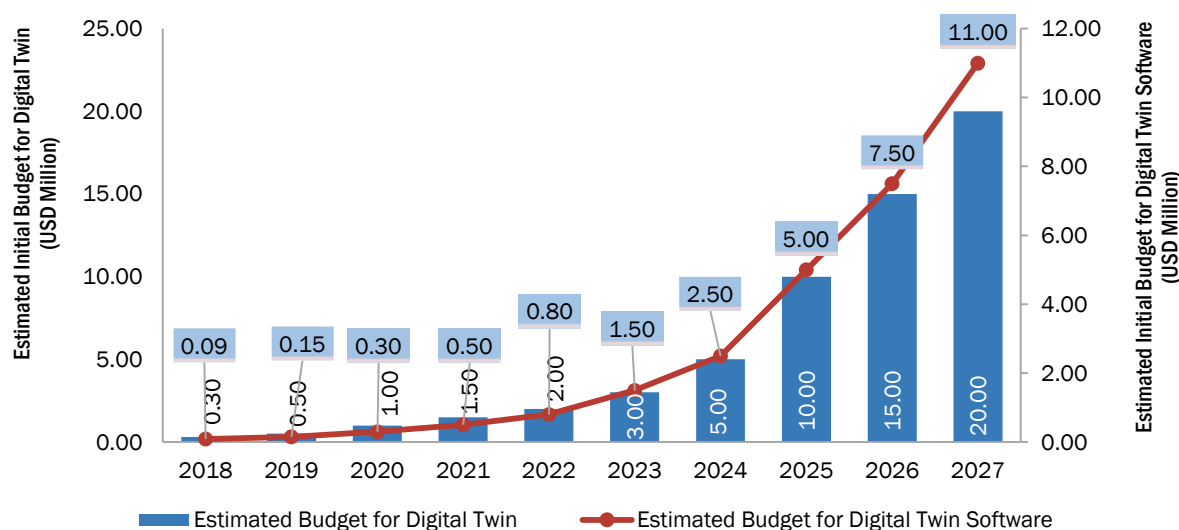
square feet (60,000 square meters) area is typically between USD 1 million to USD 2 million (including hardware and software). In this example, the budget for the digital twin and supporting software is estimated to be near about USD 300,000 to USD 600,000. Similarly, for a group of typical university campus buildings with an area of 1,000,000 square feet (90,000 square meters), the budget for digital twin setup would be between USD 1.5 million to USD 2.5 million (including hardware and software). However, the budget for just the software part of digital twin would be about USD 400,000 to USD 700,000. The selling prices considered for the study were mainly retail prices and not wholesale prices. These costs were triangulated after referring to various secondary sources and discussing the same with industry experts. It can be observed that setup of digital twin is a costly preposition; however, it provides better optimization in operations and offers a long-term benefit in the future. As the organization uses more data sources to improve twin fidelity and completeness, the potential ROI on digital twin also grows.

5.5.2 AVERAGE SELLING PRICE TREND

The pricing of digital twins is different as compared to standard products or services. A digital twin is a combination of physical assets, components, various technologies, software, and others. The cost of building a digital twin includes digital twin software, IoT, integrated software, and training and education. In an industry, there is no standard way to calculate the pricing of the digital twin. As a result, industry players have different ways to charge their digital twin solutions. The cost involved to implement everything from hardware to software is highly customizable and completely depends upon the end use case. Few industry players adopt the node pricing strategy, whereas some charge on a monthly basis.

Node pricing does not involve upfront cost or termination fees. End users are charged on the basis of per node and message. Node indicates a single component in the overall system or solution. For instance, the Azure digital twin solution offered by Microsoft (US) uses the node pricing strategy. Azure digital twin is a consumption-based service, which is based on per node (each object connected to the graph) and per message (API calls to the Azure Digital Twins APIs, incoming messages sent to device and sensor nodes in Azure digital twins or messages sent from Azure digital twins to other systems) pricing system. For example, in the ecosystem of housing areas, all the nodes, such as sensors, devices, are as per Microsoft's Azure. Each communication sent to any of the nodes is considered as a message and is chargeable. Few industry players are charging on a monthly basis. Based on the application and product/process/system, the monthly charges are calculated.

According to a survey by Microsoft IoT Signals report edition 3 (published in October 2021), 24% of the organizations (end users for digital twins) are implementing digital twin, 29% have developed but are not implementing right now, and 24% are still developing it. Whereas 23% of the organizations do not have a strategy for implementing digital twin. According to the same report, the adoption rate of digital twin (implementing at present time) is estimated to be 37% in manufacturing; 38% in energy; 18% in automotive & transportation; and 29% in smart places (smart infrastructures). As mentioned in section 5.5.1 of this chapter, the average budget for digital twin (software only) at present time is between USD 300,000 to USD 700,000 depending upon the number of devices, operations, and area of the space. The estimated budget for digital twin is expected to increase as the adoption of digital twin grows in the next five to six years. The estimated budget trend for digital twin over the next few years is mentioned in the figure below.

FIGURE 25 DIGITAL TWIN MARKET: DIGITAL TWIN SOLUTION BUDGET TREND

Note: Budgets mentioned above are estimated and not the final budgets. The figure above represents that the budget for digital twins in various industries will gradually increase over time as companies invest more in digital twin and data analytics owing to the several benefits offered by them. The figure represents digital twin budget for large enterprises only.

Source: Press Releases, Company Websites, Interviews with Experts, Secondary Journals, and MarketsandMarkets Analysis

Above mentioned historic and forecast budgets were compiled from various sources such as tier 1 supply portals, press releases, company presentations, industry experts, white papers, and MnM data repository. Upon analysis, the projected price was derived based on the OEM announcements, tier 1 new software/solution developments, future demand, historical price trends, and penetration of IoT, IIoT, and analytics in various industries. As the cost for implementing digital twin varies widely due to numerous factors as mentioned in section 5.5.1, only the estimated budget for digital twin has been provided. In 2022, the budget for digital twin has been observed to be between USD 300,000 to USD 2 million. As the adoption in end-user industries grows in the next four to five years, the initial budget for digital twin will also increase. The main reason for the increase in the budget is due to the increasing number of data points throughout the supply chain. In order to handle more data sources or devices, a higher budget is required.

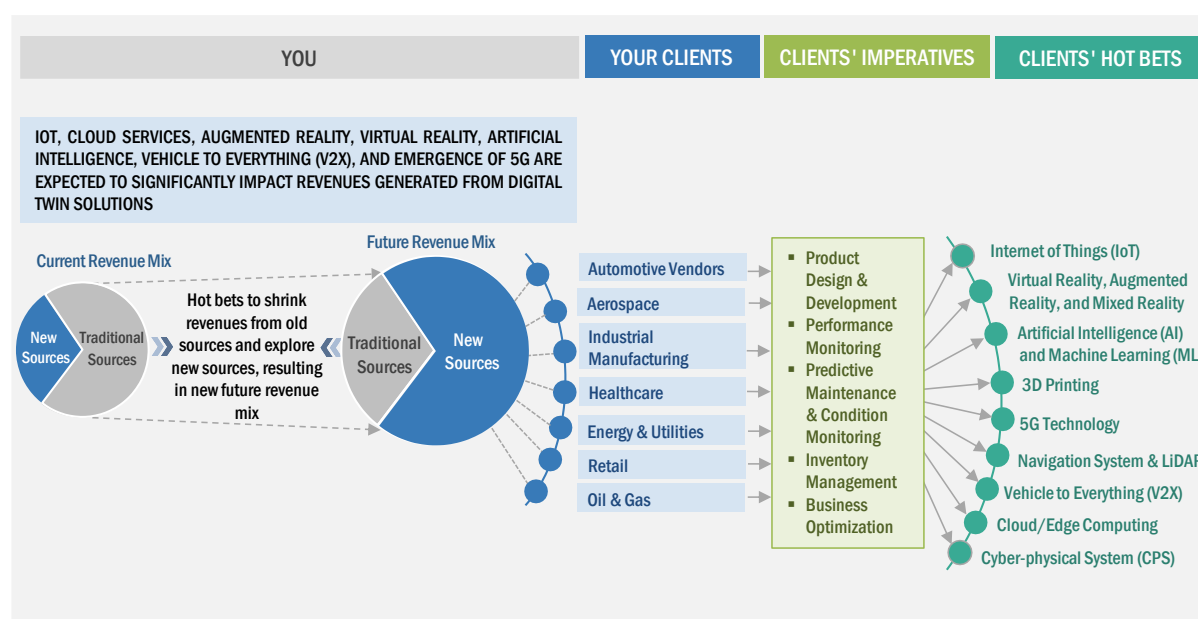
5.6 TRENDS IMPACTING CUSTOMERS' BUSINESSES

The digital twin market is projected to grow at a CAGR of 60.6% from 2022 to 2027. The organizations are shifting toward digital infrastructure and the rising emphasis on digital twin in manufacturing industries is reducing the cost and improving the supply chain which is accelerating the growth of the digital twin market. For instance, utilities and grid infrastructure operators are shifting their focus from traditional power generation toward renewables and low carbon services in the power sector. ENEL (Italy), Tata Power (India), Jibei Electric Power (China), TenneT (Netherlands), SVEA Solar (Sweden), Exelon (US), and Invenery (US) are examples of some of them. In the next 5–10 years, the prime focus of utilities and grid infrastructure operators will be on the digitalization, decentralization, and decarbonization of the processes carried out in the energy industry. This will increase investments in renewable energy integration, distributed energy resources, smart grids, asset performance management, remote monitoring and predictive analytics.

The increasing penetration of AI, ML, IoT and AR/VR/XR technologies in the process and discrete industries is boosting the growth of the digital twin market and will continue to do so in next 4 to 5 years and ahead. The introduction of these technologies in an otherwise traditional factory setting has revolutionized the industries (aerospace, automotive & transportation, retail, infrastructure, and others) and opened up new

avenues for sustainable and optimized production. While many companies in the developing regions of Middle East, Africa, Asia Pacific, and Eastern Europe are still lagging in leveraging opportunities created by the introduction of these technologies, several companies in the developed economies are reaping the benefits of implementing digital twin technologies in their buildings, manufacturing, and operations. For example, IBM (US), Microsoft (US), and General Electric (US) are already carrying out significant research and development in the implementation of digital twin across various industries. Dassault Systèmes (France), a digital twin provider, on the other hand, is focusing on the implementation of digital twin in healthcare for the treatment of patients by enabling virtual twin experiences of the human body. In January 2022, the company showcased the virtual twin of the human brain at CES 2022 (Las Vegas) with the help of AR/VR/XR. Such technological developments are expected to create new revenue streams for companies in the digital twin market.

FIGURE 26 REVENUE SHIFT IN DIGITAL TWIN MARKET



Source: Annual Reports, Press Releases, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

5.7 TECHNOLOGY TRENDS

The idea of digital twin technology revolves around the utilization of data. The major focus is on real-time data monitoring and performing the control actions accordingly. However, it also utilizes the past or historical data for analyzing and taking/giving predictive decisions based on previous patterns and scenarios. For some companies/manufacturers, the implementation of a digital twin may mean investing in the expansion of their data foundation to include things such as IoT-based sensors, a revamped IT infrastructure, and digital solutions that allow better data management and storage or even completely building a newer infrastructure for the digital twin of the products/processes/systems. For instance, a data integration platform that engages an organization's entire digital ecosystem can serve as a single critical environment, which allows digital twin-centric data to reside, converge, and be managed.

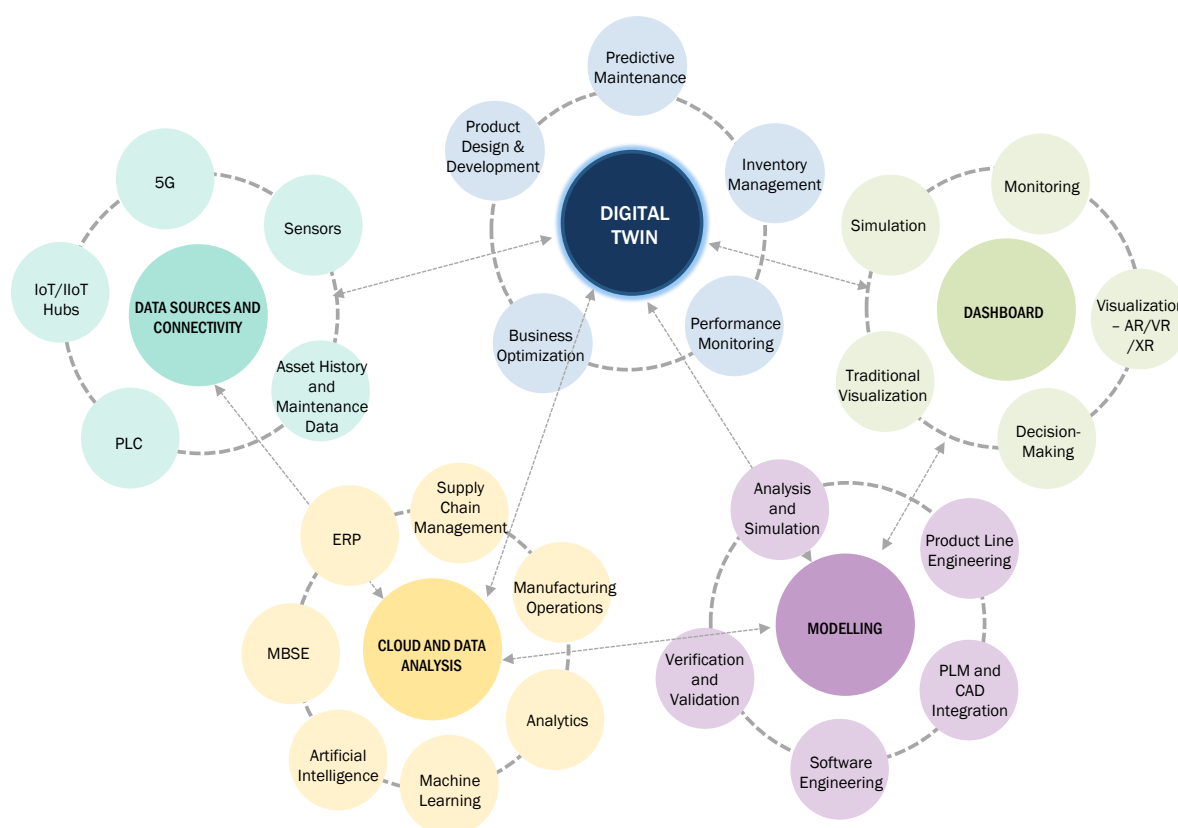
The digital twin is based on massive, cumulative, real-time, and real-world data measurements across an array of dimensions. These measurements can create an evolving profile of the object or process in the digital world and provide important insights on system performance, which will help to take actions in the physical world, such as a change in product design or manufacturing process.

Digital twin represents the convergence of the physical and the virtual world where every industrial product, process, and/or system will get a dynamic digital representation. Throughout the product development life cycle, right from the design phase to the deployment phase, organizations can have a complete digital

footprint of their products, processes, and systems. These “connected digital things” generate data in real time, which helps businesses to better analyze and predict the problems in advance or give early warnings, prevent downtime, develop new opportunities, and even plan better products for the future at lower costs by using simulations. All these factors will help in enhancing the customer experience.

The digital twins incorporated with cloud/edge computing, artificial intelligence (AI), machine learning (ML), and internet of things (IoT) play a key role in the Industry 4.0 and are predominantly used in the Industrial Internet of Things (IIoT), engineering, and manufacturing business space. The widespread usage of the IIoT has made digital twins more cost-effective and accessible for the business world. The figure below represents the related technologies with digital twin, and how they are interconnected with each other.

FIGURE 27 DIGITAL TWIN AND RELATED TECHNOLOGIES



Source: Secondary Research, Interviews with Experts, and MarketsandMarkets Analysis

5.7.1 IOT AND IIOT

The Internet of Things (IoT) is an environment where complex systems are networked together in such a way that the sensor data is transmitted over the Internet for remote monitoring and performance analysis. Digital twin uses an IIoT platform for the virtual representation of a physical asset, product, process, machine, vehicle, device, or system. It digitally represents the data, processes, state of operation, and lifecycle of the asset. Data is fetched from the IIoT sensor mesh and further processed using various technologies according to the desired results. Implementing IIoT with digital twin offers numerous benefits or characteristics such as better visibility, accurate prediction, what-if analysis, documentation and communication, and integration of disparate systems.

Digital twins created in the IIoT space are increasingly used for digitalizing the physical world. For instance, GE (US) has invested in the Industrial IIoT to increase focus on digital twins, creating virtual versions of billions of physical assets that make a global industrial infrastructure. Digital twins are key to exploring the impact of these insights and realizing full cost-saving benefits. Due to the increasing usage of IIoT

sensors, a digital twin receives continuous, real-time data/information from the twin's real-world object or asset. Such one-to-one correspondence makes it possible to test future scenarios, including potential performance enhancements, and proactively anticipate maintenance faults. Few other players such as Microsoft (US), Amazon Web Service (US), Robert Bosch (Germany), and Siemens (Germany) are involved in rapid development of IoT for digital twin.

By linking IIoT to digital twins, businesses can begin to collect data about the operation of the physical system to which it is linked. Further, by gathering environmental data (such as location and configuration) and general information (such as service records, financial models, and so on) IIoT are more than capable to enrich the digital twin. IoT and IIoT sensors collect the data from physical sources and send them over to the server or cloud for further analysis. The Industry 4.0 revolution has had a high impact on the manufacturing industries; it is considerably changing the traditional process of designing industrial machines. The digital twin is one of the aspects of the Industry 4.0 strategy that supports the easy implementation of technologies in the manufacturing and process industries and enables the companies to enhance the efficiency of their processes. A digital twin is at the core of this industrial revolution bringing in unlimited possibilities. It changes the traditional approach of “the first build and then tweak” in the industrial world and brings in a more virtual system-based design process that adds much more efficient role out of any equipment or system by understanding its unique features, performance, and potential issues, if any. Moreover, with a digital twin, an operator can be trained on a virtual machine without the help of a dedicated trainer or simulator.

5.7.2 ARTIFICIAL INTELLIGENCE (AI) AND MACHINE LEARNING (ML)

Machine learning and artificial intelligence are the two approaches that have been leveraging the concept of big data and the IoT. A digital twin is created with the help of data collected from various sensors attached to the assets/products, processes, and/or systems regarding their attributes/desired parameters. This data is collected on a cloud platform and organized into a model called “twin,” which creates the digital model that has the capability to analyze the physical asset and replicate the machine performance in real time.

Machine learning is an application of artificial intelligence (AI) that provides systems with the ability to automatically learn and improve from experience without being explicitly programmed. In addition to rapidly analyzing disparate pieces of data, machine learning can automate and improve complex analytical tasks. It can evaluate data in real time, adjust behavior with minimal need for supervision, and increase outcome efficiency and accuracy. These features of AI & ML are used for implementing digital twin and gaining accurate/precise analysis, monitoring, predictive measure, and for real-time detection and control of defects or faults.

Additionally, machine learning is used to identify anomalies, signatures, and trends in machine performance and develop an understanding of patterns of behavior. It can be applied to help identify the performance of any asset/product, process, or system and use this as the best practice for others in the near future. For instance, digital twins can be implemented by the energy & utilities industry with the help of AI and ML technologies as AI has the potential to manage the accelerating influx of renewables, distributed energy resources. On the other hand, ML can be learned based on the actual real-world behavior of the power plant equipment under all conditions, including seasonal variations, different operating campaigns, start-up/shutdown, and changing duty cycles. It can also consider the deteriorating process and mechanical performance. ML mines the process and asset data for early warning.

5.7.3 EXTENDED REALITY: AUGMENTED REALITY, VIRTUAL REALITY, AND MIXED REALITY

Augmented reality (AR) is a technology that integrates digital information with the user's environment in real time, uses the existing environment, and creates a virtual model of it. To develop and design the digital twin of a physical asset or process, an augmented reality interface interacts simultaneously with the physical asset in real time. Any appropriate augmented reality interface such as a headset, smart glasses, and tablets or smartphones, could be used for creating a digital twin model. Through these interfaces, users can gain access to design data, service data, and simulation analysis as well as to the results obtained from big data analytics. Further, the data can be projected directly on the augmented reality-based model of an asset or a system. In the near future, augmented reality will have the potential to create on-site interaction of digital twin, along with a virtual product linked to its product knowledge base. Augmented reality, virtual reality (VR), extended reality (XR), and mixed reality (or hybrid – including both AR & VR) basically add value to the user interface of that particular digital twin. Extended reality (XR) refers to all real-and-virtual mixed settings and human-machine interactions enabled by computer technology and wearables. VR, AR, and MR are all examples of extended reality.

By using the data provided by sensors for physical assets, a 3D model is created and projected in a virtual environment. The virtual reality supports the digital twin technology in creating a virtual representation of the process and asset. Data deficiencies, such as anomalies or faults, can be traced immediately by analyzing this asset information. This allows the user to interact and control/monitor the twin in a better manner.

Furthermore, by combining an AR-based digital twin with a Geographical Information System (GIS), utilities can streamline field operations such as finding specific locations for digging, monitoring assets, and navigating rugged terrain. In the healthcare industry, AR, VR, and XR can be utilized to study the human organs via their 3D models. AR, VR & XR technologies are extremely useful in visualizing 3D models of automobiles and human organs. The 3D digital twin or digital replicas of automobiles and human organs can be studied to comprehend the fundamental operation of the overall structure. Virtual, augmented, and mixed extended reality are used to approximate the real-world equivalent in digital twin visualization. Using genuine sensor data and models, the software digitalizes and mimics a real process, machine, or thing, allowing people to participate virtually. General Electric (US), Dassault Systèmes (France), and Siemens (Germany) are a few examples of the solution/software provider that are participating in extended reality-based digital twin.

5.7.4 5G

5G technology would be a key enabler for the Internet of Things (IoT) by providing a platform to connect a large number of sensors, devices, and actuators with strict energy and transmission constraints. The digital twin is an integration of various technologies, such as IoT, IIoT, machine learning, and cloud/edge computing, according to the desired outcome or requirement. 5G plays a vital role in all these technologies; data transfer plays a major and important role in its performance. In addition, the 5G infrastructure enabled the support for mission-critical services that require high reliability, global coverage, and/or low latency features.

The continuous increase in the number of IoT-connected devices is expected to be an opportunity for 5G technology. To provide high bandwidth to several use cases, the 5G technology is expected to operate at higher frequency bands, leading to the adoption of small cell network concept. However, the multi-cell interference is one of the challenges in the ultra-dense small-cell network, which the 5G infrastructure is still dealing with. For developing a responsive, near real-time digital twin necessitates low latency and enormous data volumes, both of which 5G ultra-wideband and multi-access edge computing (MEC) can provide. Verizon (US) is one such telecom operator that is providing 5G connectivity for the digital twin. Verizon 5G solutions enable consumers to receive rich insights from a digital twin powered by 5G. Verizon's 5G ultra-wideband network is designed to enable transformational business opportunities. The 5G digital

twin is a novel approach to testing and assurance that provides an emulated, software clone of the 5G physical network, enabling continuous prototyping, testing, assurance, and self-optimization of the living network. Verizon also offers 5G Edge which is a business-focused mobile edge computing platform. It allows developers to create low-latency apps for mobile end-users and wireless edge devices.

The countries that are the early adopters of 5G technology are the US, Japan, South Korea, and China. In North America, the US is a leading country in the advancements in mobile technologies, ranging from analog to 4G, and now in the 5G technology. Various telecom players have started the implementation of 5G across the world. For instance, a European telecommunication company, Swisscom, launched the country's first 5G network across 102 locations of Switzerland in April 2019. Players such as AT&T (US), Verizon (US), Sprint (US), and T-Mobile (US) have already deployed 5G services in the US. 5G is now widely available on some mobile networks in Qatar, China, and Kuwait, and average mobile internet connection speeds in all three countries more than doubled in 2021 (DataReportal). 5G is an important part of discussions about connectivity and digital transformation. It promises a set of features that could alter industry use cases, including low latency, high bandwidth, high capacity, high dependability, enhanced mobility, and longer battery life. 5G could be critical in the context of digital twins for use cases that rely on data being transported from one location to another rapidly, securely, and reliably.

5.7.5 CLOUD COMPUTING AND EDGE COMPUTING

The growing tessellation of data-generating sensors and devices, the ability to process data and create meaningful information quickly is becoming increasingly important. Hence it becomes crucial to handle and manage the enormous amount of data that is to be processed and analyzed. Here, cloud computing plays a vital role. It leverages the power of the Internet to eliminate the need for purchasing, installing, and maintaining independent data centers and networks. The utilization of cloud with IIoT infrastructure offers manufacturers several benefits, such as data monitoring, increasing profit margin, reducing overheads, and enhancing the scalability for discrete manufacturing. Companies optimize maintenance, predict important maintenance events, and stimulate innovation through actual performance feedback by constructing a virtual model of a physical product and mimicking its real-time operation. Because simulation necessitates vast amounts of computer resources and data outputs, cloud computing has long been the technology environment of choice for supporting digital twins due to its scalability and low cost. Cloud computing also helps in better decision-making with real-time asset tracking.

Edge computing, on the other hand extends the centralized cloud paradigm by bringing compute capacity closer to the data source via regionalized or local edge nodes. As a result, the users enjoy multiple benefits with their digital twin such as better speed, privacy & regulations, and uptime in the local area. Edge computing has emerged as a viable option in the present scenarios. Edge computing takes advantage of local resources close to the actual product's location, resulting in lower latency and increased responsiveness, agility, and privacy. According to ANSYS (US), a leading digital twin provider, these benefits of edge computing can act as driver for the firms interested in developing digital twins and begin capturing useful data about their operations. Microsoft (US), Amazon Web Services (US), ANSYS (US), and IBM (US) are few companies that offer cloud computing and edge computing for digital twin solutions.

5.7.6 BLOCKCHAIN

Blockchain is a distributed ledger system that records all transactions in a digital ledger, which is publicly available and evenly distributed to all the participants in the network. In the blockchain, a block consists of a hash of the previous block, a time-stamp, usually both included in the header, and a payload, which is limited in size. This payload can carry information, such as texts, data, or references to other data sources. The blockchain is a decentralized & peer-to-peer nodes network, where each node individually holds the entire transaction history, i.e., the entire blockchain. Hence, there is a dispersed and redundant database for which no intermediary or central authority is required to guarantee the authenticity of transactions. A consensus-based validation of transactions secures the uniqueness of a transaction.

Blockchain plays an important role in digital twin technology when the authenticity and identity of the product, process, or system are of utmost importance. Creating a digital twin on the blockchain means that all information regarding a product, process, and/or system can be saved immutably. Platform security of the digital twin platform and data is crucial. The blockchain technology offers security from hackers due to its encryption features and provides data history transparency. Blockchain technology allows IoT ecosystems to move away from the traditional intermediary-based networking paradigm, in which devices rely on a centralized cloud server to identify and authenticate individual devices. It allows the establishment of secure networks in which IoT devices exchange commercial information in order to complete each transaction. Further, blockchain technology supports the establishment of a scalable network capable of supporting billions of devices without the need for extra resources. Digital twins will benefit from these features, as it can transmit data securely. Both SMEs and large companies have started utilizing blockchain technology to store physical objects and connecting digital twin technology for their products, processes, and systems.

Digital twins on a blockchain will also help in brand protection and counterfeit items. Creating a digital twin of a product on the blockchain will help to view its saved transaction records and information regarding the place of manufacturing and previous owners. In addition, companies that will combine blockchain with virtual simulations will be able to track their product globally with extreme accuracy. For instance, in August 2020, Honeywell (US) deployed blockchain technology for label printers to enhance the security of the supply chains. The iTRACE 2DMI solution can be printed, authenticated, and registered to the blockchain directly on the Honeywell PX940v Industrial Printer, allowing secure product authentication and track & trace along the supply chain. This integration connects the physical printed item to the digital ledger, creating a digital twin for online authentication. This provides an additional layer of security for industries such as transportation & logistics, pharmaceuticals & healthcare, shipping & distribution centers, industrial manufacturing, automotive parts suppliers, and applications that require high precision and error-free printing.

5.8 DIGITAL TWIN MARKET: PORTER'S FIVE FORCES ANALYSIS

The Porter's Five Forces model describes the current scenario of the digital twin market. This model provides exhaustive information on profitability and guides companies in making decisions regarding the entry and exit strategies in a particular segment of an industry. Any change in these forces can have a significant impact on the market.

The competitive environment prevailing in the digital twin market can be analyzed through Porter's Five Forces analytical framework.

FIGURE 28 PORTER'S FIVE FORCES ANALYSIS



Source: Annual Reports, SEC Filings, Press Releases, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 3 DIGITAL TWIN MARKET: PORTER'S FIVE FORCES ANALYSIS

PORTER'S FIVE FORCES	IMPACT
Threat of new entrants	Low
Threat of substitutes	Low
Bargaining power of buyers	Moderate
Bargaining power of suppliers	Moderate
Intensity of competitive rivalry	High

Source: Annual Reports, Press Releases, Interviews with Experts, Secondary Journals, and MarketsandMarkets Analysis

5.8.1 THREAT OF NEW ENTRANTS

The threat of new entrants in the digital twin market is low. Well-established software/solution providers such as Microsoft (US), AWS (US), Siemens (Germany), General Electric (US), ANSYS (US), and Dassault Systèmes (France) bring innovation and continuous R&D in the digital twin market to reduce the operating and manufacturing costs for their clients and increase the optimization and reliability. The industry requires huge capital investment, along with patented technologies and critical expertise in mathematical modelling, and, hence, the threat of new entrants is low. Also, the need for precise and complex algorithms (artificial intelligence, machine learning, and deep learning), mathematical models, and supporting technologies poses high barriers, making it extremely difficult for new entrants to enter and sustain in the market. Substantial investment must be made in the R&D of systems and solutions, as well as data analytics and related technologies, to adhere to the standards regularly. This reduces the window for profits for new firms and can reduce the threat of new entrants in the industry.

5.8.2 THREAT OF SUBSTITUTES

The threat of substitutes is relatively low in the digital twin market. The digital twin technology itself is a substitute to old and traditional methods of performance monitoring and maintenance. More substitutes in the industry result in reduced cost and profitability. Digital twins enhance operational decisions with real-time insights derived from asset and business operations data. Furthermore, digital twin offers more advanced features with real-time insights using Industry 4.0 technologies such as IIoT, AI, ML, and big data analytics and is a complete packaged solution as compared to the traditional systems. The threat of substitutes remains high if the substitutes provide more price-to-performance tradeoff and the buyer's switching cost is low. There are fewer traditional substitute products and fewer substitute providers in this industry, so the threat of substitutes and the chances of customers switching to substitute competitors are low. Customers switch to substitutes when:

- Many players exist that offer better performing substitutes for digital twin in the industry
- The physical assets associated with digital twin reach the end of their life cycle
- End users are looking for lower lead times and improved reliability with better quality products/solutions

In case of the digital twin market, the digital twin technology has a competitive edge in terms of providing better optimization and reliability with real-time asset monitoring and predictive maintenance as compared to traditional substitutes.

5.8.3 BARGAINING POWER OF BUYERS

The bargaining power of buyers refers to the pressure that end users of digital twins can exert on the developers and providers of digital twin products/solutions. The bargaining power of buyers is moderate in the digital twin market. The more substitutes or competitors in the market, the more is the bargaining power of the buyer. When the buyers are in large proportion and the product is slightly differentiated, the bargaining power of the buyer becomes significant because the cost of switching can be high due to customization requirements. Customers always get to choose from whom they want the services. In the case of the digital twin market, the dynamics of aerospace, automotive & transportation, energy & utilities, infrastructure, and other end-user industries are changing rapidly, which fuels the demand for digital twin solutions. However, there are several providers of digital twin products/solutions in the market and offer high level of customization at variable price in the market to stay competitive. The buyers have several options to choose from a large number of providers who offer similar digital twins at a different price range. The buyers of digital twin solutions in emerging countries are price-sensitive; however, the high-cost investment reduces their bargaining power. On the other hand, there are several digital twin startups that offers digital twin at reasonable prices to small and medium enterprises (SMEs) who are more price-sensitive as compared to large enterprises. Hence, due to the availability of more providers, the

buyers have increased bargaining power in such cases. Thus, the overall bargaining power of buyers in the digital twin market is somewhat moderate.

The buying power of the customers can be reduced by:

- Building a large customer base
- Innovating new and easy-to-understand products/solutions for digital twin
- Making the products/solutions easy to use and access so that the customers do not switch to their competitors

5.8.4 BARGAINING POWER OF SUPPLIERS

The bargaining power of suppliers refers to the pressure that suppliers of digital twin products/solutions can exert on end users such as utilities & grid infrastructure operators, automotive manufacturers, pharmaceutical companies, aircraft manufacturers, smart infrastructure developers, and retailers. It also refers to the capability of suppliers/solution providers to dictate the supply/pricing trends leveraging the unique proposition of their digital twin solutions. The switching cost for buyers in the electrical digital twin market is moderate. There are several global vendors offering digital twin solutions, which reduces the bargaining power of the digital twin providers. But at the same time, digital twins are fully customized and there is a considerable switching cost associated. These factors increase the bargaining power of digital twin providers. Hence, the bargaining power of suppliers is moderate in the digital twin market.

5.8.5 INTENSITY OF COMPETITIVE RIVALRY

The intensity of competitive rivalry refers to the number of players competing in an industry and the extent to which market players exert pressure on each other. The presence of various well-established players, such as Microsoft (US), AWS (US), Siemens (Germany), General Electric (US), ANSYS (US), Dassault Systèmes (France), Bentley Systems (US), SAP (Germany), and IBM (US) has intensified the competitive rivalry in the digital twin market. These players have a strong foothold in the local and global markets due to their broad product portfolio and long-term strategic collaborations and alliances. These players are actively adopting different strategies, such as new products/solutions launches, agreements, collaborations, partnerships, acquisitions, and contracts, to boost their growth. This leads to an increase in rivalry among firms to capture a larger market share. The competitive rivalry can be analyzed by:

- Building sustainable differentiation
- Strategic growth—organic and inorganic
- Technological advancements in software offerings

5.9 KEY STAKEHOLDERS & BUYING CRITERIA

This section expands on the role of buyers (customers) introduced in the earlier section and gives insights into the buying cycle.

5.9.1 KEY STAKEHOLDERS IN BUYING PROCESS

FIGURE 29 INFLUENCE OF STAKEHOLDERS IN BUYING PROCESS FOR TOP THREE INDUSTRIES

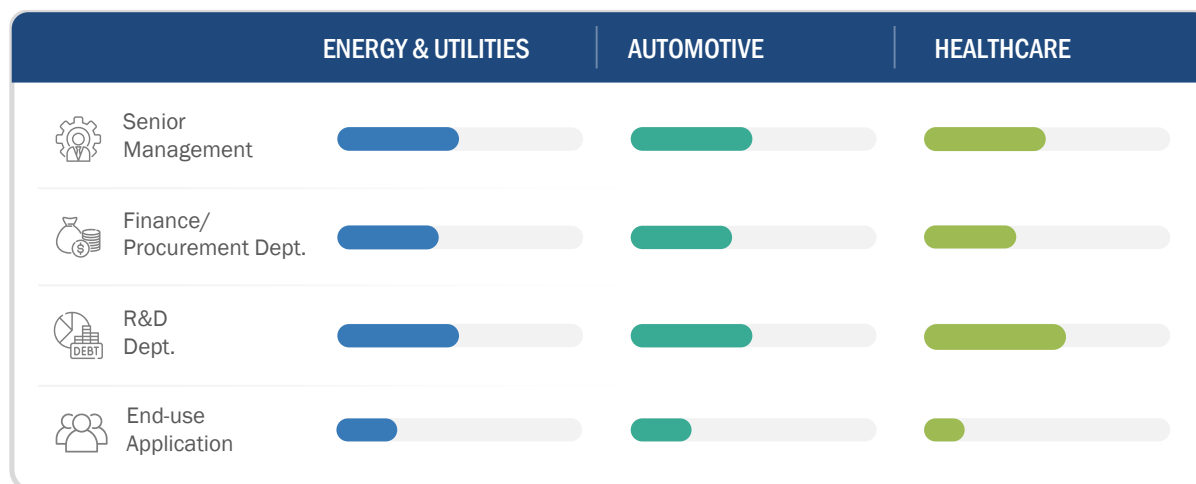


TABLE 4 INFLUENCE OF STAKEHOLDERS IN BUYING PROCESS FOR TOP THREE INDUSTRIES (%)

Particulars	Energy & Utilities	Automotive	Healthcare
Senior Management	30	30	30
Finance/Procurement Dept.	25	25	20
R&D Dept.	30	30	35
End-use Application	15	15	10
Total	100	100	100

This section identifies the key stakeholders and the extent of influence they have in deciding what to buy and who to buy from for each of the top three industries. The senior management in all the three industries has almost equal influence in making the purchasing decision. The influence of financial/procurement department is on the higher side for the automotive and energy & utilities industries due to the high price associated with implementation of sensors at various moving parts. The continuous monitoring generates a massive amount of data from various sources, right from production to operations. In the healthcare industry, the R&D department has a higher influence on decision-making as the risk associated is high with end users and patients. Additional care and expertise are required while dealing with the implementation of digital twin in healthcare as the lives of the patients cannot be risked at any cost. The end-use applications of digital twin are enormous, and the industries are yet to explore the full potential of digital twins. The influence of end-use applications on the automotive & transportation and energy & utilities industries is on higher side as these industries have multiple use cases of digital twin. However, in the healthcare industry, the use cases of digital twin are limited for the time being as the industry is still exploring and optimizing digital twin use cases. For instance, the virtual representation of human organs via digital twin and AR/VR/XR is still in progress and not fully optimized.

5.9.2 BUYING CRITERIA

This sub-section identifies the key criteria for selecting suppliers or vendors and their weights for each of the top three end-user industries.

FIGURE 30 KEY BUYING CRITERIA FOR TOP THREE END-USER INDUSTRIES

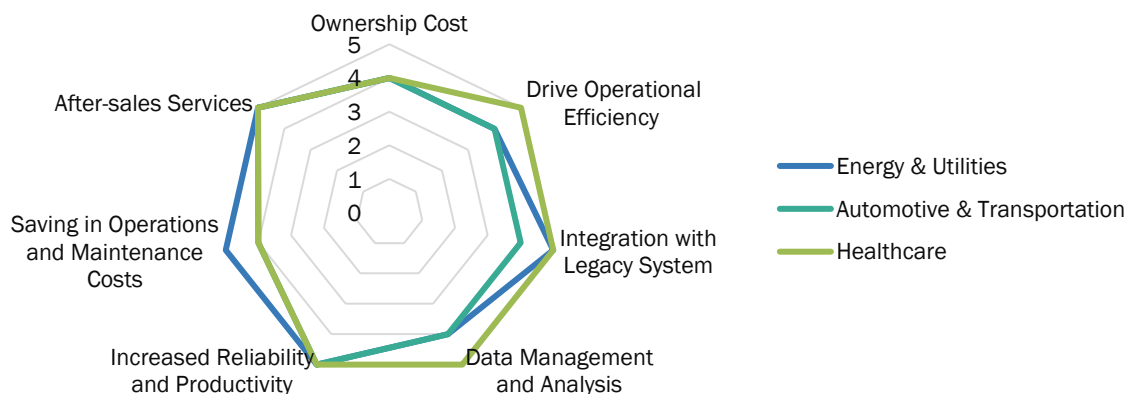


TABLE 5 KEY BUYING CRITERIA FOR TOP THREE END-USER INDUSTRIES

Buying Criteria	Energy & Utilities	Automotive & Transportation	Healthcare
Ownership Cost	4	4	4
Drive Operational Efficiency	4	4	5
Integration with Legacy System	5	4	5
Data Management and Analysis	4	4	5
Increased Reliability and Productivity	5	5	5
Saving in Operations and Maintenance Costs	5	4	4
After-sales Services	5	5	5

Note: Respondents rated each criterion on a scale of 1–5 where 1 indicates least important, and 5 indicates most important or critical.

5.10 CASE STUDIES

5.10.1 USE CASE: AEROSPACE

TABLE 6 INTEGRATING DIGITAL TWIN WITH AI-BASED PREDICTIVE MAINTENANCE INTO AIRCRAFT FLEETS

Customer	Bell Flight, a US-based aerospace manufacturer, headquartered in Fort Worth, Texas
Digital Twin Provider	IBM (US)
Challenge	Even though a piece of military gear appears to be combat ready, it may contain parts or components that are about to fail, posing a risk that is hidden in plain sight. The company needed a setup to detect—and even predict—risks before they have an impact on performance and battle preparedness.
Solution Offered	IBM (US) used deep-learning AI algorithms and IoT technology like digital twin to forecast which vehicles needed preventative maintenance. The algorithm was then able to tie each failure prediction back to a root cause, down to the part or component level, by feeding a variety of technical documents into that pool of data. As a result, detailed recommendations for army aircraft maintainers may be generated automatically. The study demonstrated how forecasting the failure of a very low-cost element can save well over USD 12 million by avoiding the need to replace an entire system—such as an engine. By integrating predictive insights into their ERP systems, Bell Flight can greatly improve the way component inventories are managed. For example, in case of component flaws, early detection is crucial. With digital twin, Bell Flight was able to detect the flaws and avoid the overestimation of the number of available parts, which could jeopardize manufacturing and delivery schedules.
Result	The army can better manage its spare parts inventory and integrate those insights into stronger budgeting and maintenance management procedures in the future with greater transparency into underlying part failure trends.

Source: Company Website

5.10.2 USE CASE: AUTOMOTIVE

TABLE 7 OPTIMIZATION OF AUTOMATED GUIDED VEHICLE LOGISTICS WITH DIGITAL TWIN PLATFORM

Customer	Faurecia (France), a subsidiary of the Group Forvia, one of the largest automotive suppliers in the world
Digital Twin Provider	Dassault Systèmes (France)
Challenge	Manufacturing environments for automotive suppliers are becoming increasingly complicated, with factories managing last-minute engineering changes while producing a higher number of product variants for parallel programmes under narrower timelines.
Solution Offered	In April 2022, Faurecia expanded its usage of Dassault Systèmes' 3DEXPERIENCE digital twin platform to optimize its automated guided vehicle (AGV) inbound logistics. Faurecia is addressing the growing complexity of the supplier production environment by optimizing logistics flow, increasing efficiency, lowering costs, and more sustainably delivering products to customers on time. Optimizing AGV routing is a very critical step in Faurecia's digital factory programme. With the 3DEXPERIENCE platform, Faurecia can simulate and visualize AGV routing in terms of workstations, resource availability, assembly limitations and variabilities, line balancing and flow in one virtual twin experience even before starting production for a customer. It is utilized to change seat, interior, and electronics production processes for connected and personalized vehicles.
Result	Digital twin platform 3DEXPERIENCE is enabling Faurecia to determine and validate the best logistics scenarios for make-to-order assembly lines for its AGVs. Faurecia can synchronize machines, optimize supplies, minimize shop floor layout footprint, shorten cycle times, and enhance energy and manufacturing efficiency with the 3DEXPERIENCE platform, all while maintaining full digital continuity across products, processes, and plants.

Source: Company Website

5.10.3 USE CASE: ENERGY & UTILITIES

TABLE 8 DOOSAN OPTIMIZED ENERGY OUTPUT IN WIND FARMS WITH AZURE DIGITAL TWINS

Customer	Doosan (South Korea), a global leader in energy & utilities industry (power & water –power generation, energy and desalination plants)
Digital Twin Provider	Microsoft (US) and Bentley Systems (US)
Challenge	Doosan wanted to create a digital twin of its wind farms. The key objective was to optimize energy output while lowering maintenance costs for existing facilities and improving the design of its next-generation wind turbines.
Solution Offered	Doosan approached Microsoft and Bentley Systems to build digital twins for its wind farm. Bentley Systems delivers expertise in infrastructure modeling as well as its iTwin technology for creating 3D industrial digital twins. Bentley's iTwin service is supported by Azure Digital Twins and Azure IoT Hub. This solution rapidly ingests and models large amounts of complex operational data from wind turbine IoT sensors and other sources. Doosan employed Azure Digital Twins' flexible modeling capabilities to develop comprehensive digital models that combine data from IoT sensors on wind turbines, weather data, and predicted power output. The Bentley iTwin programme then combines this data with 3D and 4D CAD models, as well as reality models, to build entire solutions that will transform the way infrastructure projects are conceived, built, and operated. Doosan implemented two solutions: Performance Watchdog and Power Prediction using its 3D model in iTwin in conjunction with Azure Digital Twins and IoT Hub. Performance Watchdog provides real-time power generation data and compares it to two data models. One data model is based on physics and current weather conditions, and the other on a machine learning technique based on accessible sensor data. When wind farm operators compare actual output to expected output, they can fine-tune the controls to optimize production and plan necessary repairs. Doosan employed the Power Prediction system to forecast future power generation based on weather forecasts and optimize operational planning.
Result	Doosan developed solutions using Azure Digital Twins, IoT Hub, and Bentley iTwin to optimize turbine operations and collect feedback in order to develop and evaluate advances in wind turbine development. The technology allowed Doosan to maintain its competitive advantage and boost energy efficiency by merging data from different sources into actionable insights.

Source: Company Website

5.10.4 USE CASE: HEALTHCARE

TABLE 9 DIGITAL TWIN OF HUMAN BODY AND BODY ORGANS IMPROVES DRUG TREATMENT

Customer	Patients all across the world
Digital Twin Providers	Dassault Systèmes (France), Q Bio (US), Siemens (Germany), FEops (Belgium), ANSYS (US)
Challenge	To enable a virtual representation of human organs that can be personalized to particular patients and updated in order to better understand disease development over time or understand the response to new medications, therapies, or surgical interventions.
Solution Offered/ in progress	Siemens Healthineers has been developing a digital twin of the heart to help with pharmacological therapy and cardiac catheterization simulations. Similarly, FEops, a European firm, has already commercialized the FEops Heartguide technology and had also received regulatory permission. It improves the investigation and treatment of structural heart disorders by combining a patient-specific copy of the heart with AI-enabled anatomical analysis. In 2014, Dassault Systèmes launched the Living Heart Project to create a virtual twin of the human heart. Medical researchers, surgeons, medical device makers, and medication companies have collaborated on the initiative as an open source partnership. Meanwhile in 2022, the company's Living Brain project and is guiding epilepsy therapy and tracking the evolution of neurodegenerative illnesses. Similar projects have been conducted for the lungs, knees, eyes, and other systems by the company. Most digital twin systems rely on current technology to collect the necessary data, but Q Bio's innovative Gemini Digital Twin platform begins with a full-body scan. Q Bio, an US based startup claims that it can capture a full-body scan in 15 minutes without utilizing radiation or breath holds, and that it can do so using new computational physics models that are more precise than traditional MRI for many diagnoses. Andreessen Horowitz, Kaiser Foundation Hospitals, and others have invested over USD 80 million in the startup. Q Bio is also working on integrating data from genetics, chemistry, anatomy, lifestyle, and medical history to improve these models. Further, ANSYS (US), one of the leading digital twin providers, is working on the development of a digital twin to improve drug delivery using models of simulated lungs as part of the Virtual Human System project.
Result	Digital twins make it possible to pool cohorts of similar patients that would otherwise be impossible or prohibitively expensive to put together in real life. The technology also avoids the hazards and concerns that come with using real-life subjects. Digital-twin technology in the healthcare sector, when applied correctly, allows practitioners to establish optimal medicines, enhance patient outcomes, and maximize efficiency, resulting in hospital cost savings.

Source: Company Website

5.10.5 USE CASE: INFRASTRUCTURE

TABLE 10 TEMPLE UNIVERSITY USED SMART BUILDING DIGITAL TWIN FOR OPTIMIZING OPERATIONS ACROSS UNIVERSITY

Customer	Temple University, Broad Street, Philadelphia, Pennsylvania, US
Digital Twin Providers	Microsoft
Challenge	Temple University sought to create a framework for rolling out an integrated financial management system (IFMS) over the entire campus. In addition, Temple university needed to create a smart-building plan that would lay the groundwork for streamlining operations across the university's 214 buildings, lowering costs, and improving services for its schools, businesses, employees, and students.
Solution Offered	In the first phase of the project, e-Magic established a reference architecture and implemented a working IFMS solution in five campus buildings using TwinWorX, which is built on Azure native services. The buildings' automation, lighting, and utility metering were connected at the edge and incorporated into the Azure cloud using Azure IoT Hub. Temple may now oversee building operations from a single pane of glass display by leveraging assets that were modelled using Azure Digital Twins. The university can also draw insights from fault detection and diagnosis algorithms, Azure Time-Series Insights analytics, and utility reporting. Temple wanted an integrated controls strategy that was future-proof, vendor-neutral, and aligned with the IT agenda of the institution. Temple University adopted digital twin technologies with its partner e-Magic, which leapfrogged old solutions from embedded vendors who supplied restricted offerings at growing costs. As a result of this, Temple University saves money on IT costs, and it will save even more money once the platform allows for open-bid selection of building systems.
Result	Temple University is able to develop a scalable digital twins ecosystem using Azure IoT and Azure Digital Twins, allowing the university to decrease costs and enhance operations across the campus today and in the future as the campus grows.

Source: Company Website

5.11 TRADE ANALYSIS

IoT & IIoT are the most frequently utilized sensors for communicating data between the physical assets and the servers for creating digital twin of the physical assets. IoT & IIoT are an important part of digital twin ecosystem. This section describes the import and export trend of IoT-based tracking and monitoring, including apparatus for the transmission or reception of voice, image or data, including apparatus for communication in a wired or wireless network (excluding telephone sets or base stations). The figures and tables showcased in this section are for the import and export data for products included under the HS code 851769.

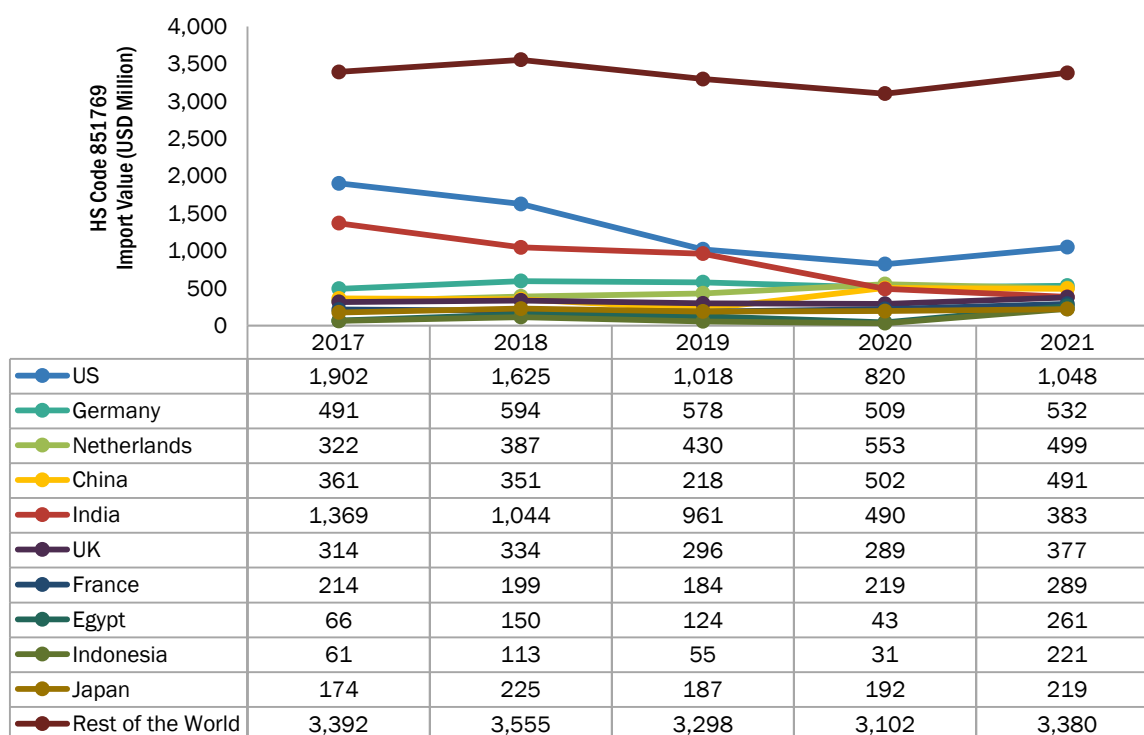
5.11.1 IMPORT SCENARIO

5.11.1.1 Import scenario for digital twin ecosystem

The imports of products having the HS code 851769 worldwide were valued at ~USD 7,699 million in 2021, a rise of 9% from 2020. According to various industry experts, the key reason for this gain is the increasing adopting of IoT & IIoT sensors across various industries such as aerospace, healthcare, and automotive & transportation.

In FY 2021, the US, Germany, the Netherlands, China, and India were the top five countries in terms of global import value for products with the HS code 851769 (including IoT & IIoT). The majority of these countries are key manufacturing hubs that are rapidly moving toward digitization. More than 70% of the top 10 countries increased imports of IoT & IIoT during 2020–2021, except for the Netherlands, China, and India. The table below provides import-related data for products with the HS code 851769 (including IoT & IIoT) in various key countries and the rest of the world during 2017–2021.

FIGURE 31 IMPORT DATA FOR HS CODE 851769 FOR TOP COUNTRIES, 2017–2021 (USD MILLION)



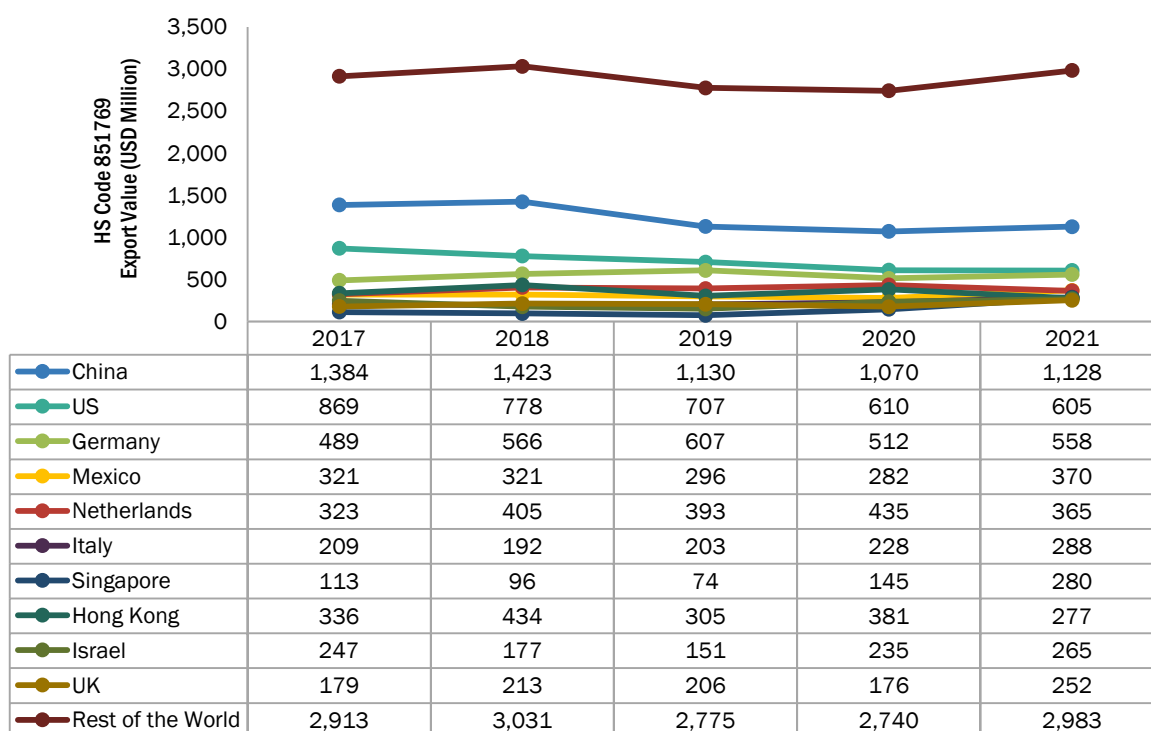
Source: ITC Trade Map

5.11.2 EXPORT SCENARIO

5.11.2.1 Export scenario for digital twin market

The exports of products having the HS code 851769 worldwide were valued at ~USD 7,372 million in 2021, a rise of 8.9% than in 2020. In FY 2020, China was the leader in terms of global export value for products with HS code 851769 (including IoT & IIoT), followed by the US, Germany, Mexico, and the Netherlands. The table below provides export data for products with HS code 851769 (including IoT & IIoT) in various key countries and the rest of the world during 2017–2021.

FIGURE 32 EXPORT DATA FOR HS CODE 851769 FOR TOP COUNTRIES, 2017–2021 (USD MILLION)



Source: ITC Trade Map

5.12 PATENT ANALYSIS

This section provides a brief overview and description of various patents associated with the Digital twin market. This section reveals the patents of various digital twin providers in the ecosystem. Qualcomm (US), General Electric (US), and IBM (US) has received more than 300 patents for digital twin from January 2012 to December 2021. Given below is a table depicting the patents by the top 15 companies in the last 10 years. In the US, the Patent Technology Monitoring Team (PTMT) periodically issues general statistics and miscellaneous reports on patenting activities at the US Patent and Trademark Office (USPTO).

TABLE 11 TOP 15 PATENT OWNERS IN LAST 10 YEARS (2012–2021)

PATENT OWNER	NO. OF PATENTS
Qualcomm	439
IBM	429
General Electric	378
Samsung Electronics	377
Canon Kabushiki Kaisha	306
United Parcel Service of America	302
Acushnet Company	281
Asml Netherlands Bv	226
Sabic Global Technologies Bv	184
Pure Storage	173
Fujifilm Corporation	161
Sumitomo Chemical Company	157
Dow Global Technologies	150
Sony Corporation	150
Monolithic 3d	148

Source: Patentscope-Wipo, Espacenet, USPTO Patent Database, and lens.org

TABLE 12 PATENTS FILED FOR DIGITAL TWIN, 2021

YEAR (APPLICATION GRANTED)	PUBLICATION/ APPLICATION NUMBER	APPLICANT	DESCRIPTION	LOCATION
December 2021	US 202017031970 A	IBM	Real-time market activity of one or more physical assets are monitored via a plurality of Internet of Things sensors and the operational history of each physical asset of the one or more physical assets. One or more digital twin resources associated with each physical asset of the one or more physical assets are also monitored. At least one market change that affects at least one of the monitored one or more physical assets and the respectively associated monitored one or more digital twin resources associated with each physical asset of the one or more physical assets are identified. Based on at least one identified market change, a price for a digital twin resource available for purchase in a digital twin content store is dynamically adjusted.	US
December 2021	US 202017071253 A	IBM	In an approach to digital twin enabled asset performance and upgrade management, upcoming changes for assets are received, where the upcoming changes include at least one of hardware requirements, firmware requirements, software requirements, hardware upgrades, firmware upgrades, and software upgrades. The upcoming changes for the assets are stored into repositories, where the appropriate repository is determined using topic modeling. The digital twins are updated, where each digital twin is continuously updated with the upcoming changes. Continued usage metrics are calculated for each asset based on the output of the digital twins, where the continued usage metrics are specific to each asset. A report is created based on the upcoming changes and the continued usage metrics, where the report includes at least one of hardware dependencies, firmware dependencies, software dependencies, and upgrade dependencies.	US
November 2021	US 201916723014 A	ANSYS	A method and apparatus for extending, customizing and validating a simulation-based digital twin model is described. In an exemplary embodiment, the device transmits a model to a client, where the model is a simulation-based digital twin model. In addition, the device receives a customization to the model, the where the customization adds a functionality to	US

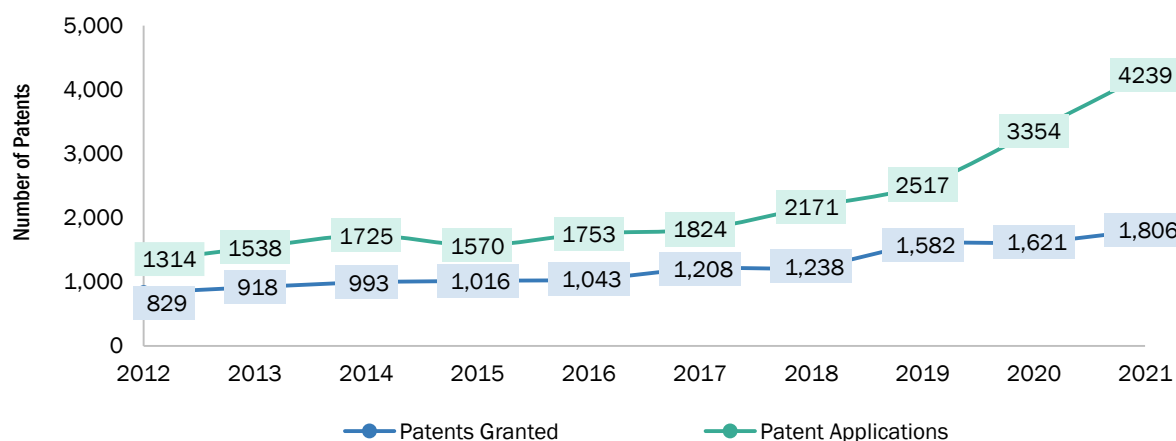
			the model. Furthermore, the device deploys the model in a model platform, where the model is used in a simulation with the model platform and the model is coupled with the model platform.	
October 2021	EP 19158513 A	Siemens	Systems, methods and computer program products for simulating workflows and activities of physical assets using digital twin models. User-defined simulations are performed by selecting digital twin components being analyzed during the simulation, concentrating the analysis on the selectively defined components and bypassing components that will not be simulated. Users can design the digital twin simulation using one or more available digital twin models. The model can be the most current digital twin model, a previous version of a model or a hybridized model comprising components or portions from multiple versions of the available digital twins. Users can further customize simulations by selecting components or sections of the digital twin model to selectively bypass during the simulation or provide overriding values for non-simulated portions of the digital twin which can be used as entry criteria inputted into the next simulated section or component of the digital twin, to complete the simulation.	Europe
September 2021	US 202016752990 A	General Electric	The example embodiments are directed systems and methods for managing a contextual digital twin. In one example, the method may include receiving a request message configured to trigger an action to a digital twin, the digital twin comprising a virtual representation of an asset, determining behavior of the digital twin that is linked to the request message based on capabilities of the behavior stored with respect to the digital twin, determining input parameters of the behavior from the request message, and generating an executable script for performing the determined behavior in association with the digital twin based on the input parameters. The context can provide information about operations of the asset as well as insight into how to handle the operations based on similar occurrences to the asset or other related assets.	US

September 2021	US 201815941067 A	General Electric	A method for controlling operation of an electric power generation system includes receiving power generator data corresponding to the electric power generation system. The method further includes receiving, using a digital prime mover unit, a set-point parameter corresponding to a prime mover unit and generating one or more prime mover parameter estimates corresponding to the plurality of prime mover parameters. Further, the method includes receiving, using a digital generator unit, one or more prime mover parameter estimates and generating one or more generator parameter estimates corresponding to the plurality of generator parameters. The digital prime mover unit and the digital generator unit are real-time operational models of the prime mover unit and the generator unit. The method also includes controlling the operation of the electric power generation system based on at least one or more of the power generator data, the prime mover parameter estimates, and the generator parameter estimates.	US
August 2021	US 201715603082 A	General Electric	A control system and method utilizing one or more processors that are configured to determine contaminant loading of blades of a turbomachinery compressor based on one or more environmental conditions to which the turbomachinery compressor is exposed and one or more atmospheric air inlet conditions of the turbomachinery compressor. The one or more processors then determine a corrosion contaminant concentration on the blades of the turbomachinery compressor based on the contaminant loading that is determined and determine an upper limit on or a distribution of potential corrosion of the blades of the turbomachinery based on the corrosion contaminant concentration, at least one of the environmental conditions to which the turbomachinery compressor is exposed, and the corrosion contaminant concentration that is determined.	US

August 2021	US 201916720186 A	SAP SE	US	Methods, systems, and computer-readable storage media for receiving IoT data from an IoT device of an IoT network, the IoT data including identification data and sensor data, determining that a digital twin of the IoT device is stored in a cloud platform based on at least a portion of the identification data, the digital twin including a digital representation of the IoT device that is provided as an instance of a model of the IoT device, and enriching the digital twin with the sensor data.
June 2021	US 201816103385 A	Microsoft	US	A digital twin may be configured to perform blockchain transactions on behalf of a device with limited memory, limited compute power, and/or limited internet connectivity. A method for performing such a blockchain transaction includes hosting a digital twin for the capability limited device that has a unique identifier matching a unique identifier of the digital twin such that actions performed by the digital twin are attributable to the capability limited device. The method further includes receiving input for a set of blockchain operations at the system capable of performing the blockchain operations, from the capability limited device. The method further includes performing the set of blockchain operations at the system capable of performing the blockchain operations using the digital twin and the unique device identifier, such that performance of the set of blockchain operations is attributed to the capability-limited device.

Source: Secondary Literature, Interviews with Experts, Lens.org, and MarketsandMarkets Analysis

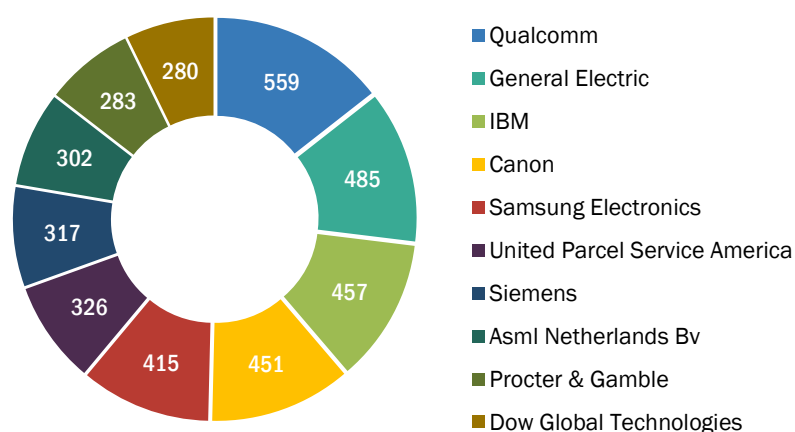
FIGURE 33 DIGITAL TWIN PATENTS FROM 2012 TO 2021



Source: Lens.org

The global patenting trend analysis demonstrates a growing patenting activity for digital twin. Qualcomm (US), General Electric (US), and IBM (US) are at the top of the leadership board in terms of patent applications for digital twins with cumulative 1,501 patents. Most of the patents granted for digital twin belongs to the US jurisdiction as the US is one of the early adopters as well as a prime user of digital twin technology. US-based companies outlay huge amounts as R&D investment in digital twin and account for a larger share of patents all over the world. Few startups also own some unique patents for digital twins. For example, an Indian startup Pratiti Technologies owns a unique patented solar digital twin – in renewable energy. Apart from Pratiti Technologies, only General Electric (US) is the only digital twin provider that owns such type of patent for solar digital twin in renewable energy.

FIGURE 34 TOP 10 COMPANIES WITH LARGEST NUMBER OF PATENT APPLICATIONS FOR DIGITAL TWINS, 2012–2021



Source: Patentscope-Wipo, Espacenet, USPTO Patent Database, and lens.org

5.13 KEY CONFERENCES & EVENTS TO BE HELD DURING 2022–2023

This section includes key conferences and events related to the digital twin ecosystem in the next six quarters.

TABLE 13 DIGITAL TWIN MARKET: DETAILED LIST OF CONFERENCES & EVENTS

YEAR/ QUARTER	NORTH AMERICA	EUROPE	ASIA PACIFIC	REST OF THE WORLD
Q3'22	[August] International Conference on Digital Architecture and Digital Technology	[July] LoRAWAN World Expo [September] DTW 2022	[August] International Conference on Digital Sociology and Digital Technologies	[July] DIGITAL TRANSFORMATION SUMMIT
Q4'22	[October] IoT Tech Expo North America [October] USER Event 2022	[October] Smart Grid Forums — IEC 61850 Week 2022 [December] DRDHum2022	[December] ADIP 2022	[November] CIMOUSEF'19 [December] ICDAICSITIV 2022
Q1'23	[January] International Conference on Internet Computing and Internet of Things	[January] International Conference on Digital Architecture, Digital Design and Digital Fabrication	[February] ICDADDF 2023	[April] International Conference on Internet of Things Data Analytics
Q2'23	[April] ICDAICSITIV 2024	[April] IoTBDs 2023	[May] International Conference on Internet of Things and Blockchain	[April] International Conference on Blockchain for Internet of Things
Q3'23	[August] International Conference on Internet of Nano Things [September] ICTAILTITD 2023	[July] International Conference on Internet Computing and Internet of Things	[July] International Conference on Internet of Medical Things	[August] International Conference on Cyber Security for Internet of Things
Q4'23	[October] International Conference on Internet of Things	[October] ICAIDT 202	[December] ICDIADA 2023 [October] ICDAICSITIV 2023	[December] ICDAICSITIV 2023

Note: * indicates conferences and events that may be more important because of wider reach, sponsorship from leading organizations, thought leadership, etc.

5.14 TARIFFS AND REGULATIONS

5.14.1 TARIFFS

This section represents the MFN tariffs for products with the HS code 851769 (including IoT & IIoT), which are an integral part of the digital twin ecosystem. The most favored nation (MFN) tariff is implemented in a country with the most favored nation status. The MFN tariff is the lowest possible tariff that a country can impose on the products imported from another country. For example, if a country's lowest tariff is 2% of the value of a good, it is its MFN tariff, and the respective country charges this percentage on an import from another country with the MFN status. Members of the World Trade Organization must extend the MFN status to other countries that are not yet a part of the MFN list. In the US, the MFN status is formally called permanent normal trade relations. The table below represents the data for MFN tariffs levied on products with HS code 851769 (including IoT & IIoT).

TABLE 14 US: MFN TARIFFS FOR PRODUCTS WITH HS CODE 851769 (INCLUDING IOT & IIOT), BY KEY COUNTRY

COUNTRY	MFN TARIFF	YEAR
China	0.75%	2021
India	2.86%	2021
Iran	34.44%	2020
Maldives	20%	2020
Mexico	4.29%	2018
Turkey	1.03%	2021
UK	0%	2022

Source: ITC Trade Map Data for HS Code 853224

On products with HS code 851769 (including IoT & IIoT) exported by the US to European countries in 2021, the MFN tariff levied was ~0%. The MFN for exports to APAC countries in 2020 on products with HS code 851769 (IoT & IIoT) exported by the US to India and China were 2.86% and 0.75%, respectively. However, the exports from APAC countries were also levied with similar tariffs. The table below provides the data for MFN tariffs levied on apparatus exported by China to a few key countries.

TABLE 15 CHINA: FOR PRODUCTS WITH HS CODE 851769 (INCLUDING IOT & IIOT), BY KEY COUNTRY

COUNTRY	MFN TARIFF	YEAR
India	2.86%	2021
Brazil	14%	2021
Iran	34.44%	2020
Germany	0%	2022
Mexico	4.29%	2018
Yemen	5%	2017
Turkey	1.03%	2021
Israel	4.56%	2017

Source: ITC Trade Map Data for HS Code 853224

The MFN tariff levied on products with HS code 851769 (including IoT & IIoT) exported by China in 2021 was almost similar to that of the US. The MFN for exports by China to countries with bigger economies in 2021 for the products (considered HS code 851769) was 0%. However, tariffs for exports to low-income economies was between ~18% to 2% in 2021. Due to such drastically varying rates worldwide, tariffs have few negative, as well as positive, impacts on the revenue generated by IoT & IIoT providers.

5.14.1.1 Positive impact of tariffs on digital twin ecosystem

The recent implementation of tariffs by the US due to the ongoing US-China trade war provides insights into the effect of tariffs on the apparatus for transmission or reception. High tariffs are forcing various manufacturers in the US to produce goods at a lower cost, which can be achieved through automation. Products with HS code 851769 (including IoT & IIoT) will help manufacturers that depend on exports subsidize these tariffs through domestic manufacturing. Companies that invest in networking, data transmission, and network upgrade before the implementation of tariffs are expected to be in an advantageous position compared with their competitors.

5.14.1.2 Negative impact of tariffs on digital twin ecosystem

Tariffs often have a strong political influence, and their intended outcome may not ensure growth for fiber optics companies and transmission or reception apparatus (including IoT & IIoT) providers. If higher tariffs continue to prevail, the growth and development of the IoT & IIoT sensors could stagnate, which, in turn, will considerably reduce productivity and affect the digital connectivity and transformation of many industries globally.

5.14.2 REGULATIONS AND STANDARDS

5.14.2.1 Regulatory bodies, government agencies & other organizations

This section provides an overview of codes, regulations, and standards related to the digital twin market. These codes, regulations, and standards are introduced by institutions such as the International Organization for Standardization (ISO), Institute of Electrical and Electronics Engineers (IEEE), National Institute of Standards and Technology (NIST), International Electrotechnical Commission (IEC), European Telecommunications Standards Institute (ETSI), German Automotive Industry (VDA), British Standards Institute (BSI), and China Communications Standards Association (CCSA). As digital twin is an emerging technology, regulatory bodies and organizations are yet to develop specific codes and regulations.

5.14.2.2 Major standards and regulations

FIGURE 35 VARIOUS STANDARDS & REGULATIONS FOR DIGITAL TWIN



Source: MarketsandMarkets Analysis

TABLE 16 DIGITAL TWIN: STANDARDS AND REGULATIONS

SR. NO.	COUNTRY	REGION	POLICY NAME	DESCRIPTION
1	-	Global	ISO/TC 184/SC 4	This standard establishes a framework for the use of industrial data. The scope includes standardization of the content, meaning, structure, representation, and quality management of the information required to define an engineered product and its characteristics at any required level of detail at any part of its lifecycle from conception through disposal, together with the interfaces required to deliver and collect the information necessary to support any business or technical process or service related to that engineered product during its life-cycle.

2	-	Global	ISO/TC 184	This standard establishes a framework for automation systems and integration. It relates to the standardization in the field of automation systems and their integration for design, sourcing, manufacturing, production and delivery, support, maintenance, and disposal of products and their associated services. Areas of standardization include information systems, automation and control systems, and integration technologies.
3	-	Global	ISO 23247-1	This standard focuses on automation systems and integration. It also provides an overview and general principles to establish a digital twin.
4	-	Global	IEEE 1451.4	IEEE 1451.4 defines a physical connection (Mixed-Mode Interface, or MMI) that is alternately used for Transducer Electronic Data Sheet (TEDS) data and analog signals on either 2, 3, or 4 wires. This standard can be used for a wide variety of sensors and actuators.
5	-	Global	IEEE 1451	This standard establishes the definition and specific requirements for smart transducer interfaces in cyber-physical systems (CPS), industrial internet of things (IIoT), internet of things (IoT), and internet of everything (IoET). In addition, it describes the sensor requirements for smart grids.
6	-	Global	ISO/IEC 27001	ISO/IEC 27001 is an international standard for information security management. In 2005, the International Organization for Standardization and the International Electrotechnical Commission jointly produced the standard, which was then amended in 2013.
7	-	Global	ISO 19650	The ISO 19650 standard is an international standard for applying building information modeling (BIM) to manage information throughout the life cycle of a built asset. It is closely related to the current UK 1192 standards and contains all of the same principles and high-level requirements as the UK BIM Framework.
8	Europe	EU countries	TiSAX	TiSAX is a European automobile industry-standard information security assessment (ISA) catalog based on major components of information security such data protection and third-party connections. TiSAX was developed by the Association of the German Automotive Industry (VDA) in partnership with an association of European automotive manufacturers, called the European Network Exchange (ENX)
9	Europe	UK	BS 1192	It covers anyone involved in the preparation and use of construction data during project design, construction, operation, and decommissioning. It is applicable to construction and infrastructure projects. According to the British Standards Institute (BSI), it could also be valuable for software developers.

10	US	North America	NISTIR 8356 (Draft)	This provides a detailed definition of digital twins, the motivation and vision for their use, common low-level operations, usage scenarios, and example use cases.
11	Europe	Global	ETSI GS ISI 008	This standard describes information security indicators (ISI) and provides a detailed description of an overall organization-wide security information and event management (SIEM) approach. It presents a series of 9 ISI 00x specifications for event detection and interactions between all parts.
12	China	China	T/CCSA 320-2021	This standard specifies the functional and performance requirements for mobile communication terminals to support augmented reality through Video See Through (VST), including technical architecture, business processes, hardware requirements, functional requirements, performance requirements, and power consumption requirements. This standard applies to the design, development, production, and testing of mobile communication terminals that support augmented reality functions, such as mobile phones and tablets.
13	China	China	T/CCSA 310.6-2021	This standard specifies the process specifications of the big data visualization system in the field of public safety, including preprocessing, visualization display, traceability search, process scheduling, and system security. This standard is suitable for guiding the design, development, operation, and maintenance of big data visualization systems in the field of public safety.
14	US	Global	IEC 61850	This standard defines communication protocols for intelligent electronic devices at electrical substations.
15	US	Global	ANSI/ISA95	This ISA standard provides consistent terminology for suppliers and manufacturers to provide consistent information and operations models.
16	Europe	Global	ISO/IEC 27002:2022	This policy gives standards for organizational information security, cybersecurity, and privacy protection practices, including selecting, implementing, and managing control risks.
17	US	Global	ISO 19136:2007 GML	It is a standard for exchanging location-based information, addressing geographic data requirements for smart grid applications.

Source: Secondary Research and Regulatory Body Websites

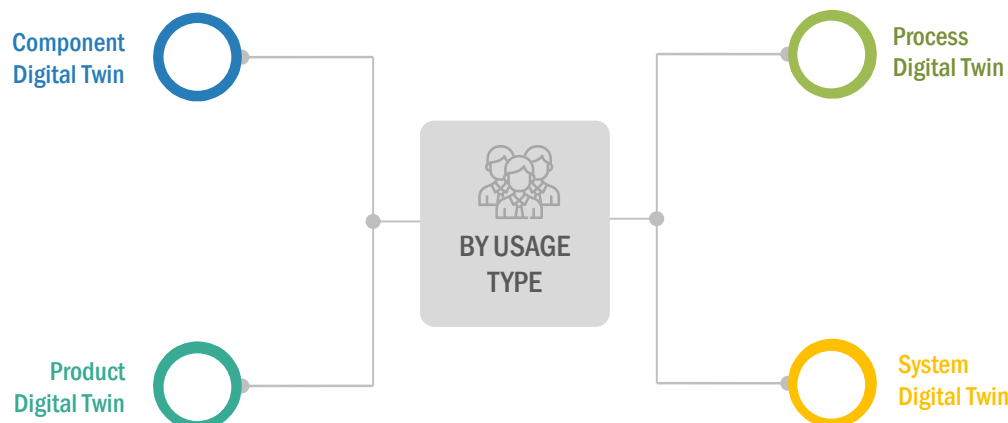
6 DIGITAL TWIN MARKET, BY USAGE TYPE

6.1 INTRODUCTION

A digital twin is a virtual replica of a physical product or process that is used to better understand and anticipate the performance characteristics of the actual process taking place in the real-world to provide real-time feedback and insight into decision-making processes. Digital twins can demonstrate the influence of design changes, environmental conditions, and a plethora of other variables by combining simulation, data analysis, and machine learning skills. Depending on the extent of product magnification, many types of digital twins exist. The types of digital twins depend upon the field of application. Different digital twins frequently coexist within a system or process. The digital twin market, by usage type, has been segmented into three categories: component, product, process, and system. Each type is used for various purposes, such as designing, simulation, performance analysis, controlling, and real-time performance monitoring of any of the abovementioned types throughout the lifecycle.

Several companies are using digital twins to detect and isolate faults, perform diagnostics and troubleshooting, determine the ideal maintenance schedule, optimize asset operation, and generate insights to improve the designing and performance of the next generation of the products, processes, and systems. Digital twin has been adopted for various products, processes, and systems in different verticals, such as aerospace, automotive & transportation, healthcare, infrastructure, oil & gas, energy & utilities, agriculture, retail, telecommunications, and others (semiconductors, water & wastewater, chemicals, and food & beverages).

FIGURE 36 DIGITAL TWIN MARKET, BY USAGE TYPE



Source: Secondary Research, Expert Interviews, Company Websites, and MarketsandMarkets Analysis

The growth of the digital twin market can be attributed to the increasing use of digital twin for systems in various industries. For example, in the aerospace sector, National Aeronautics and Space Administration (NASA) uses digital twins to monitor its space stations and spacecraft to ensure better safety. In the automotive & transportation sector, a start-up, SWIM AI, uses digital twins for the traffic management system in California. Likewise, Tesla uses digital twins in the manufacturing of its cars. COVID-19 has affected every aspect of the business for several industries such as automotive & transportation, aerospace, oil & gas, and infrastructure, and the impact is visible on the usage type market for digital twin. The market growth was stagnated due to the pandemic; however, COVID-19 also increased the level of awareness about digital twin among industrial end users and the market started recovering in 2021. New installations of products/processes/systems will continue to grow over the next four to five years. It is expected that end users will extensively implement system/process digital twins to minimize future losses.

6.2 COMPONENT DIGITAL TWIN

The digital twin of a component/part digitally replicates the complete component based on various parameters. The example of a component digital twin can be a bearing on a rotating piece of a motor, rotor, engine, the heart or lung, wind turbine rotor, and control valve. Component/part twins are the simplest example of a functioning component in a digital twin. Component twins are similar to identical twins; however, they are for components that are slightly less important.

A component twin is linked to a larger system. For example, the bearing of an energy-producing facility. During operation, this bearing can have a digital twin that explains its state, such as estimated mean time between failures (MTBF) or mean time to failure (MTTF). This information can be generated (predicted or modeled) from current data (such as vibration sensor data or sound), as well as data collected during the design or construction process (such as what gear tooth shape was designed or what molding cutter has been used for building it).

6.3 PRODUCT DIGITAL TWIN

The digital twin of a product digitally replicates the complete product based on various parameters. The product twin is typically a sub-component that has a significant impact on the performance to whichever process or system it belongs. Product/asset twins provide visibility at the equipment level. When two or more components work together, they form an asset, and it is also referred as a resource. Product/asset twins allow to investigate how those components interact, generating a large amount of performance data that can be analyzed and transformed into meaningful insights.

A product twin is essentially a collection of part twins with their interactions reflected. From the standpoint of a software product, the product twin is usually the same; hence, part twins are accessed by drilling down from a product twin. The example of a product digital twin can be the generator of an energy production plant. The digital twin of this generator includes many bearings component twins, among other things.

The product twin is typically a sub-component that has a significant impact on the performance to whichever process or system it belongs. Product digital twins are mainly used for predictive maintenance or monitoring. It provides a virtual-physical connection that lets designers, operators, and others dealing with the product to analyze how it will perform under various conditions and make adjustments in the virtual world to ensure that the next physical product will perform exactly as planned in the field. Product digital twins help one to navigate any level of complexity to make the best possible decisions and necessary control measures. All of this eliminates the need for multiple prototypes, reduces total development time, improves the quality of the final manufactured product, and enables faster iterations in response to customer feedback.

6.4 PROCESS DIGITAL TWIN

Process digital twin can be a set of activities or operations, such as a manufacturing process, a surgical process, farming process, and chemical process. A process digital twin can be informed by a set of product twins but focuses more on the process itself rather than the equipment.

A process digital twin can help validate how well any process will work on the field before anything actually goes into production or working conditions. These types of digital twins are mainly used for inventory management and predictive maintenance. By simulating the process using a digital twin and analyzing why things are happening using the digital thread, companies can create a production methodology that is efficient for a variety of conditions and according to the desired requirements and outcomes. A digital twin of a manufacturing process can assist assess how effectively it will perform on the shop floor before anything goes into production. Companies can develop a manufacturing approach that stays efficient under a variety of scenarios by recreating the process using a digital twin.

The production can be optimized even further by creating process digital twins of all the manufacturing equipment involved in performing that specific process. By using the data from process digital twins, businesses can prevent costly downtime to equipment and even predict when preventative maintenance will be necessary. This constant stream of accurate information enables manufacturing operations that are faster, more efficient, and reliable.

6.5 SYSTEM DIGITAL TWIN

A system digital twin is a collection of products and processes that together perform as a system- or network-wide function, such as an oil and gas refinery/plant, production line in a factory, assembly line, traffic network, wind farm, or hospital. A system twin provides visibility into a set of interdependent equipment. A system twin is simply the next step up from a product twin. They are most frequently implemented in the same software product as product or component twins, but they are just a perspective of the entire system. For instance, in an energy production plant, a system twin (depending on its definition) might reflect the past and current condition of a power plant, the entire power plant, or even a partition of a power grid, and anticipate future state. System twins provide insight into asset interactions and may recommend performance improvements.

A system digital twin generates a large amount of data regarding their (different assets within a system) utilization and effectiveness. This data is analyzed and processed for the betterment of the systems and improving the system performance by reducing the faults and taking preventive measures. It also provides actionable insight for informed decision-making. By leveraging system digital twins, companies can:

- Create new business opportunities
- Gain insights to improve virtual models
- Capture, aggregate, and analyze operational data
- Improve product and production system efficiency
- Optimize business
- Design and develop new products

7 DIGITAL TWIN MARKET, BY ENTERPRISE

KEY FINDINGS

- The global digital twin market for large enterprises was estimated at USD 3.8 billion in 2021 and is expected to reach USD 59.3 billion by 2027, at a CAGR of 59.2% during the forecast period.
- Digital twin solutions are still in the nascent stage of development and implementation, and large enterprises have an abundance of resources to incorporate the latest digital twin technologies and solutions. Hence, large enterprises are expected to dominate the market
- The digital twin market for small & medium enterprises was estimated at USD 0.7 billion in 2021 and is expected to reach USD 14.1 billion by 2027, at a higher CAGR of 67.6% during the forecast period.
- Digital twin is gaining popularity among small & medium enterprises. In comparison to larger businesses, SMEs require less capital for digital twin implementation and receive a higher return on investment.
- The digital twin market for large enterprises in North America was estimated at USD 1.3 billion in 2021; it is expected to reach USD 21.4 billion by 2027, growing at the highest CAGR of 60.1% during the forecast period.

7.1 INTRODUCTION

The digital twin market, on the basis of enterprise size, has been categorized into small & medium-sized enterprises (SMEs) and large enterprises. This chapter categorizes the end users (demand side) of digital twin products/solutions on the basis of their enterprise size. It is to be noted that this chapter does not indicate or represent the digital twin providers (supply side) on the basis of their enterprise size.

These organizations are categorized according to the number of employees and global standards. The study of the market according to enterprise size is important since it provides a clear picture of the adoption scenario of digital twin technology, solutions, and services across enterprises of various sizes.

According to the current trends, all types of enterprises, either large or SMEs, are witnessing technological disruptions in the digital age that are impacting engineering, manufacturing, operations, and service legislation. Intelligent and linked goods are expected to deliver continuous and real-time engineering data, while deeper insights into the generated data will enable cognitive contextualization.

As digital twins become more capable, they are being applied to a growing number of uses in SMEs and large enterprises, from designing and testing products and processes to monitoring day-to-day operations and conducting maintenance. Digital twin has emerged as a critical technology for use in the early stages of product development. It can enable enterprises to forecast system performance in the future and be used to create a digital thread with data from connected smart devices. Digital twins reduce the amount of time engineering teams spend iterating a product, lowering the overall product life cycle expenses. On the other hand, the complimentary/related technologies such as artificial intelligence (AI), machine learning (ML), internet of things (IoT), industrial IoT, extended reality (augmented reality, virtual reality, and mixed reality), and blockchain-as-a-service (BaaS) enable faster adoption of digital twin solutions. Furthermore, over decentralized networks, BaaS for IoT can provide better solutions with benefits such as scalability, security, interoperability, cost-effectiveness, and reduced operational complexity. As a result, blockchain may be a better candidate for IoT digital twin networks than centralized networking systems.

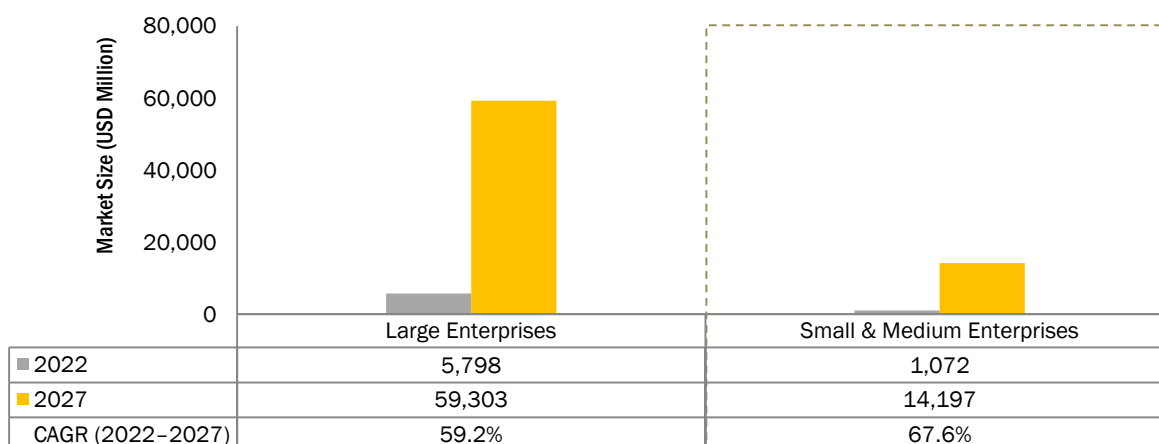
Digital twin solutions implemented across the supply chain collect data in real time over the network in the supply chain. The processes/products/equipment across supply chain can be linked using an IoT network, which ensures the authenticity of the information/products/services and builds trust, transparency, and security throughout the organization as well as among the customers and clients.

FIGURE 37 DIGITAL TWIN MARKET, BY ENTERPRISE



Source: Secondary Research, Press Releases, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

FIGURE 38 DIGITAL TWIN MARKET FOR SMALL & MEDIUM ENTERPRISES TO GROW AT HIGHER CAGR DURING FORECAST PERIOD



Source: Secondary Research, Press Releases, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 17 DIGITAL TWIN MARKET, BY ENTERPRISE, 2018–2021 (USD MILLION)

Enterprise	2018	2019	2020	2021
Large Enterprises	2,548	2,955	3,132	3,831
Small & Medium Enterprises	370	457	516	669
Total	2,918	3,412	3,648	4,500

Source: Secondary Research, Press Releases, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 18 DIGITAL TWIN MARKET, BY ENTERPRISE, 2022–2027 (USD MILLION)

Enterprise	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Large Enterprises	5,798	8,997	14,343	22,970	36,872	59,303	59.2%
Small & Medium Enterprises	1,072	1,758	2,956	4,983	8,407	14,197	67.6%
Total	6,870	10,754	17,299	27,953	45,280	73,500	60.6%

Source: Secondary Research, Press Releases, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

The large enterprises segment held the larger share (~85%) of the digital twin market in 2021 and is expected to dominate the market throughout the forecast period. Digital twin solutions are still in the nascent stage of development and implementation, and large enterprises have an abundance of resources to incorporate the latest digital twin technologies and solutions. Hence, large enterprises are expected to dominate the market.

However, the SMEs segment is expected to grow in the next four to five years as the reachability of digital twin expands. Many startups are offering digital twin products/solutions that are cost-efficient to some extent and, therefore, a perfect fit for SMEs. For instance, Pratiti Technologies (India) offers cost-efficient digital twin solutions for the renewable energy industry. QiO Technologies (UK) offers PARCs and Digital Health Record algorithms to businesses to measure and benchmark their asset performance, and continuously rank next best actions. The firm provides multi-platform compatibility as well as the ability to install on any cloud or IoT platform. Interoperability with legacy IT and OT systems is a key feature of the solutions. Similarly, Kalam Innovations LLP (India), a digital twin provider, offers solutions for SMEs. In developing countries, there are several machines from the Industry 3.0 era. Kalam Innovations LLP, for example, provides digital twin solutions for SMEs using IoT and other sensors. Instead of purchasing highly complex equipment and machinery with incorporated sensors and Industry 4.0 technologies, SMEs can better afford to set up a digital twin. Hence, the SMEs segment is expected to grow at a higher CAGR during the forecast period

7.2 LARGE ENTERPRISES

7.2.1 LARGE ENTERPRISES USE DIGITAL TWIN FOR TRACKING ASSETS THROUGHOUT SUPPLY CHAIN

Organizations with 1,000 or more employees are categorized as large enterprises. The adoption of digital twin products/solutions is high among large enterprises due to the rapid deployment of blockchain technology in the healthcare, energy & utilities, oil & gas, and automotive & transportation industries. The number of businesses that use digital twin solutions integrated with the internet of things (IoT), artificial intelligence (AI), and cloud is growing manifold. The combination of AI, IoT, and cloud, together with the optimization and real-time monitoring offered by digital twin technology, creates the perfect combination for managing large databases. Several stakeholders in the retail supply chain industry have started using digital twin to improve transparency and efficiency in their business operations, as well as to improve the speed of delivery. Some of the major larger enterprises experimenting with the use cases of digital twin technology are Volkswagen (Germany), Polaris (US), Bell Flight (US), Faurecia (France), Doosan (South Korea), Brookfield Properties (Australia), Bye Aerospace (US), TBI (Netherlands), and DYWIDAG (Luxembourg).

Large enterprises use digital twin for asset tracking & management, product traceability, and product monitoring throughout the supply chain. For instance, pharmaceutical companies can monitor production, as well as logistics for COVID-19 testing kits and vaccines, using a digital twin solution. The testing kits and vaccines are temperature-sensitive and should be closely monitored because unnoticed deviations could affect their quality and performance. A digital twin solution provides global and real-time tracking and monitoring of such shipments and ensures the precision of this real-time data by actively monitoring parameters such as temperature and location. The system can be expanded throughout the supply chain of the pharmaceuticals industry to maintain transparency. Several alliances also aim at unlocking the value of the combined usage of digital twin with enterprise resource planning (ERP), product lifecycle management (PLM), computer-aided design (CAD) and 3D CAD, manufacturing process management (MPM), manufacturing operations management (MOM), model-based system engineering (MBSE) extended reality– augmented reality (AR)/virtual reality (VR)/mixed reality (MR), AI, cloud, blockchain, and IoT.

The demand for digital twin products/solutions among large enterprises is expected to be propelled by the need to revolutionize traditional business systems and enhance customer experience. The direct application of digital twin in organizations includes product design & development, predictive maintenance, inventory management, performance monitoring, and business optimization applications within the organization. Aerospace and automotive OEMs, for instance, can incorporate digital twin to design critical parts and components and test their efficiency and performance before actually designing the parts and components in real, thereby saving a lot of money in the actual trial. This also enables in reducing industrial wastes as less resources and materials will be consumed while designing and testing parts and components on digital twin platform.

TABLE 19 LARGE ENTERPRISES: DIGITAL TWIN MARKET, BY REGION, 2018–2021 (USD MILLION)

Region	2018	2019	2020	2021
North America	883	1,025	1,091	1,339
Europe	917	1,061	1,124	1,373
Asia Pacific	557	649	687	840
Rest of the World	190	220	231	280
Total	2,548	2,955	3,132	3,831

Note: Rest of the World includes South America, Africa, and the Middle East.

Source: Secondary Research, Press Releases, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 20 LARGE ENTERPRISES: DIGITAL TWIN MARKET, BY REGION, 2022–2027 (USD MILLION)

Region	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
North America	2,035	3,173	5,085	8,185	13,208	21,387	60.1%
Europe	2,077	3,224	5,142	8,242	13,244	21,298	59.3%
Asia Pacific	1,268	1,963	3,121	4,983	7,973	12,756	58.7%
Rest of the World	417	637	996	1,561	2,448	3,862	56.0%
Total	5,798	8,997	14,343	22,970	36,872	59,303	59.2%

Note: Rest of the World includes South America, Africa, and the Middle East.

Source: Secondary Research, Press Releases, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

The North America region is projected to be the largest digital twin market for large enterprises during the forecast period. It is further expected to grow at the highest CAGR from 2022 to 2027. North America is one the early adopters of digital twin technology and most of the digital twin providers are located in the North American region, including Microsoft (US), IBM (US), Amazon (ASW) (US), and General Electric (US). National Aeronautics and Space Administration (NASA) was an early adopter of digital twin in North America. The region is concentrated with aerospace companies such as Bell Flight (US), Bye Aerospace (US), Lockheed Martin (US), and Boeing (US) who constantly invest in digital twin technology and carry out huge R&D with the technology to optimize their production and reduce the downtime and operational cost. Industries such as healthcare, aerospace, and financial institutions generate a vast amount of data. Analyzing this data on a digital twin with IoT accessibility benefits them with secure communication channels and helps them achieve data provenance. Hence, large enterprises with enormous data are expected to adopt digital twin in the next four to five years.

7.3 SMALL & MEDIUM-SIZED ENTERPRISES

7.3.1 SMES ADOPT DIGITAL TWIN TO GAIN COMPETITIVE ADVANTAGE OVER ESTABLISHED PLAYERS

SMEs face intense competition from large enterprises, and to gain a competitive edge, these enterprises are adopting digital twin solutions, as the technology provides optimization, efficiency, flexibility, and scalability, along with predictive maintenance, which would enable a quick response, timely decisions, and enhanced business productivity.

Digital twins in organizations do not function as a standalone solution; rather, they are part of a comprehensive set of solutions that include IoT, AI, ML, blockchain, and cloud/edge computing. Subscription-based digital twin services allow quick and easy integration of the technology in a medium-sized enterprise by providing ready-to-go solutions rather than building a solution from scratch. Microsoft (US), Amazon Web Services (US), and Robert Bosch (Germany) are a few digital twin providers that offer digital twin solutions to their clients on a subscription basis. However, the cost of operations is high due to the requirement of several intermediaries for trading across regions, which limits the adoption of digital twin solutions by SMEs with limited resources. In addition, SMEs are not well-versed with the benefits of digital twin technology and are slow in adopting such solutions. However, the COVID-19 pandemic, to some extent, created awareness among SMEs regarding the benefits of implementing a digital twin solution for their businesses.

SMEs are the backbone of an economy. However, these organizations face challenges in scaling operations and attracting investors. Moreover, since the technology is considerably new, SMEs also lack technical knowledge. The big players in the digital twin market, along with digital twin startups, are encouraging SMEs to adopt digital twin by offering budget-friendly solutions. Advantages such as reliability, scalability, improved inventory management, quick access to real-time data, performance monitoring, and improved business optimization are also fostering adoption among SMEs.

TABLE 21 SMALL & MEDIUM ENTERPRISES: DIGITAL TWIN MARKET, BY REGION, 2018–2021 (USD MILLION)

Region	2018	2019	2020	2021
North America	202	245	272	349
Europe	93	112	124	157
Asia Pacific	58	77	92	125
Rest of the World	18	23	28	37
Total	370	457	516	669

Note: Rest of the World includes South America, Africa, and the Middle East.

Source: Secondary Research, Press Releases, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 22 **SMALL & MEDIUM ENTERPRISES: DIGITAL TWIN MARKET, BY REGION, 2022–2027 (USD MILLION)**

Region	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
North America	554	900	1,501	2,514	4,216	7,091	66.5%
Europe	247	399	661	1,098	1,827	3,040	65.1%
Asia Pacific	209	355	616	1,069	1,849	3,189	72.5%
Rest of the World	62	104	178	303	515	877	69.9%
Total	1,072	1,758	2,956	4,983	8,407	14,197	67.6%

Note: Rest of the World includes South America, Africa, and the Middle East.

Source: Secondary Research, Press Releases, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

The North America region is expected to dominate the digital twin market for SMEs during the forecast period. Apart from being an early adopter of digital twin technology, the North American region offers a technologically advanced environment for SMEs to grow. The emergence of digital twin startups and generous amount of funding propel the digital twin market in the US. QiO Technologies (US), Datumix (US), Sight Machine (US), Allvision Io (US), and VEERUM (Canada) are the leading and emerging startups in North America.

Another major reason for the rapid adoption of digital twin among SMEs is that SMEs get a better return on investment in digital twins. For instance, a larger enterprise might invest USD 100 million in a digital twin project and receive a return of USD 2 million over time. This is accurate since large businesses have more operations and machinery to cover with digital twin, necessitating a higher capital investment. An SME, on the other hand, might invest USD 10 million on digital twin implementation and see a 10% return on the investment over the same time period. Because SMEs have fewer machines and operations to cover than large corporations, the number of communications (pings or messages) that are exchanged between the device and the digital twin cloud server will be lower as well. This means that, in comparison to larger businesses, SMEs require less capital for digital twin implementation and have a higher return on investment. Hence, the digital twin market for SMEs is expected to grow at a higher CAGR during the forecast period.

8 DIGITAL TWIN MARKET, BY APPLICATION

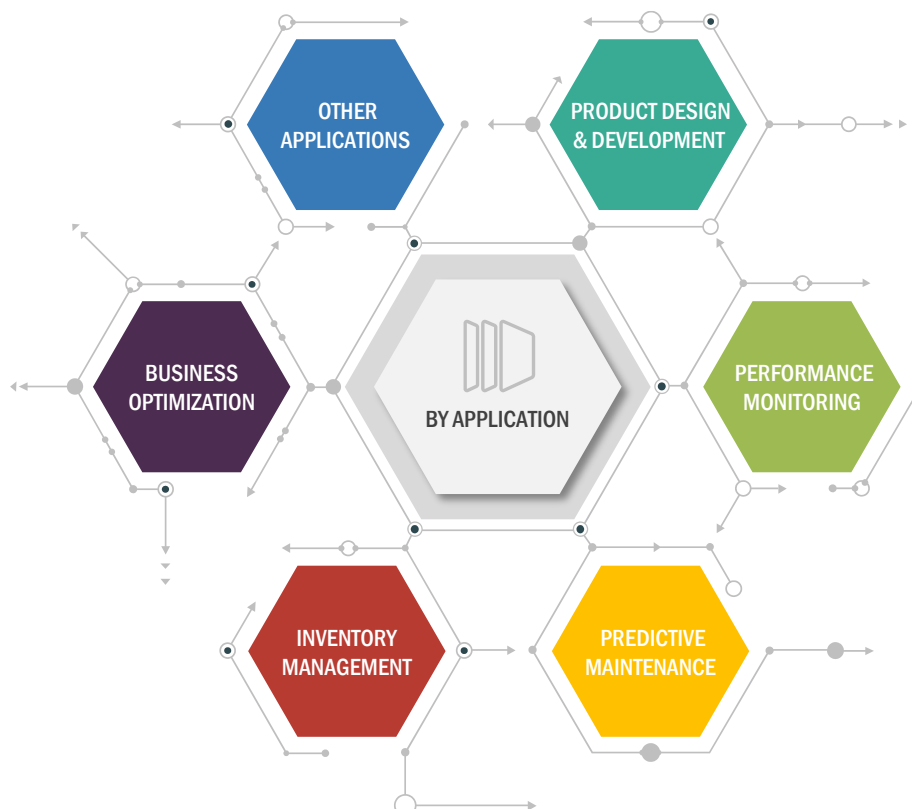
KEY FINDINGS

- The global digital twin market was estimated at USD 4.5 billion in 2021 and is expected to reach USD 73.5 billion by 2027, at a CAGR of 60.6% during the forecast period.
- The growing application of digital twin in the automotive & transportation industry acts as a major driver for the digital twin market. The advent of IoT sensors, electric vehicles, automated guided vehicles (AGV), and connected vehicles enabled the collection of better-quality of data that can be analyzed and interpreted through digital twins, resulting in the overall performance optimization.
- The digital twin market for predictive maintenance applications was estimated at USD 1.8 billion in 2021 and is expected to reach USD 30.8 billion by 2027, at a CAGR of 61.9% during the forecast period.
- Digital twin is gaining popularity for predictive maintenance applications in the automotive & transportation, energy & utilities, infrastructure, and healthcare industries as they optimize the maintenance cycle and strike a balance between corrective and preventative maintenance, thereby resulting in cost savings.
- The digital twin market for business optimization applications was estimated at USD 1.3 billion in 2021; it is expected to reach USD 19.6 billion by 2027, growing at the highest CAGR of 62.0% during the forecast period.

8.1 INTRODUCTION

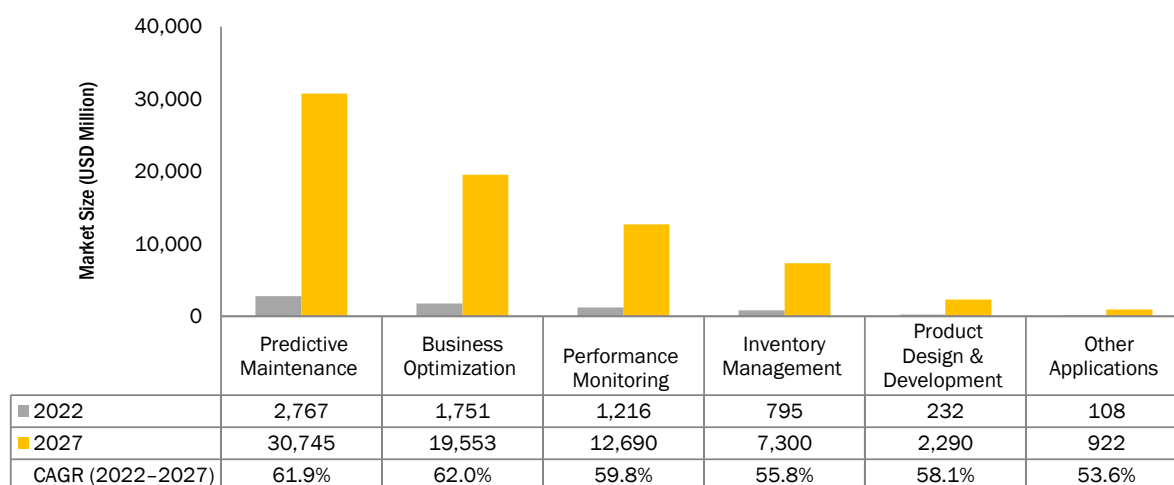
The digital twin market, by application, has been segmented into six categories, namely, predictive maintenance, business optimization, performance monitoring, inventory management, product design & development, and others. End users were initially using digital twins mainly for performance monitoring. However, according to present trends, it is expected that industry players will use digital twin solutions for predictive maintenance and business optimization.

FIGURE 39 DIGITAL TWIN MARKET, BY APPLICATION



Source: Secondary Research, Press Releases, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

FIGURE 40 PREDICTIVE MAINTENANCE SEGMENT TO ACCOUNT FOR LARGEST SIZE OF DIGITAL TWIN MARKET DURING FORECAST PERIOD



Note: Other applications include environmental analysis, customized services, instore planning & security implementation, and energy management

Source: Secondary Research, Press Releases, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 23 DIGITAL TWIN MARKET, BY APPLICATION, 2018–2021 (USD MILLION)

Application	2018	2019	2020	2021
Predictive Maintenance	1,175	1,361	1,459	1,797
Business Optimization	695	849	915	1,137
Performance Monitoring	518	597	641	801
Inventory Management	370	425	444	535
Product Design & Development	107	122	128	155
Other Applications	54	58	62	74
Total	2,918	3,412	3,648	4,500

Note: Other applications include environmental analysis, customized services, instore planning & security implementation, and energy management

Source: Secondary Research, Press Releases, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 24 DIGITAL TWIN MARKET, BY APPLICATION, 2022–2027 (USD MILLION)

Application	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Predictive Maintenance	2,767	4,368	7,082	11,530	18,814	30,745	61.9%
Business Optimization	1,751	2,766	4,488	7,315	11,950	19,553	62.0%
Performance Monitoring	1,216	1,894	3,032	4,874	7,856	12,690	59.8%
Inventory Management	795	1,209	1,887	2,956	4,638	7,300	55.8%
Product Design & Development	232	356	563	895	1,429	2,290	58.1%
Other Application	108	161	248	382	593	922	53.6%
Total	6,870	10,754	17,299	27,953	45,280	73,500	60.6%

Note: Other applications include environmental analysis, customized services, instore planning & security implementation, and energy management

Source: Secondary Research, Press Releases, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

The predictive maintenance application is expected to account for the largest size of the digital twin market throughout the forecast period. It is further expected to grow at the second-highest CAGR during the forecast period. The fact that every organization needs their key equipment to operate at peak efficiency and utilization to realize their return on capital expenditures is the fundamental cause for the emergence of predictive maintenance applications for digital twin. With this technology, companies are focusing on developing a predictive maintenance model that optimizes the maintenance cycle and strikes a balance between corrective and preventative maintenance.

8.2 PRODUCT DESIGN & DEVELOPMENT

8.2.1 DIGITAL TWIN HELPS IN PRODUCT DESIGN & DEVELOPMENT BY MONITORING KEY PERFORMANCE INDICATORS IN REAL TIME

A digital twin of a physical object or an asset combines all available information and analytical models of the object throughout its life cycle. The technology is increasingly being used to describe a means of connecting information about the physical product and its behavior in the real world with 3D digital representation, which is commonly employed during engineering and in other areas of the business. For instance, Dassault Systems (France) introduced the virtual twin concept that enables design and engineering teams to visualize and analyze the products or systems virtually and also provides insights similar to physical behaviors, including stress and vibration as well as behaviors associated with the software and control systems.

It provides end users with insights into their factory performance by monitoring the key performance indicators in real time across all systems in the facility and providing rapid access to vital business data. At the same time, it provides traceability, improves transparency, and tracks issues during operations.

TABLE 25 **PRODUCT DESIGN & DEVELOPMENT: DIGITAL TWIN MARKET, BY INDUSTRY, 2018–2021 (USD MILLION)**

Industry	2018	2019	2020	2021
Automotive & Transportation	9.2	11.5	13.3	17.5
Energy & Utilities	9.4	10.5	10.7	12.6
Infrastructure	11.7	13.2	13.8	16.1
Healthcare	6.6	8.3	9.5	12.3
Aerospace	50.4	57.6	58.0	68.3
Oil & Gas	6.1	6.9	7.0	8.3
Telecommunications	0.0	0.0	0.4	0.6
Retail	0.0	0.0	0.1	0.2
Other Industries	13.7	13.6	14.8	19.3
Total	107.2	121.6	127.6	155.3

Note: Other industries include semiconductors, chemicals, paper & pulp, and food & beverages. The values have been represented in one decimal point to exhibit the difference between the minute values.

Source: Secondary Research, Press Releases, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 26 **PRODUCT DESIGN & DEVELOPMENT: DIGITAL TWIN MARKET, BY INDUSTRY, 2022–2027 (USD MILLION)**

Industry	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Automotive & Transportation	29.4	50.4	88.3	154.9	271.4	475.1	74.5%
Energy & Utilities	18.3	27.1	41.3	62.9	95.7	145.2	51.4%
Infrastructure	24.4	37.7	59.7	95.0	151.1	240.5	58.1%
Healthcare	20.7	35.5	62.5	109.9	193.2	339.2	75.0%
Aerospace	99.8	149.2	229.0	352.8	544.2	840.0	53.1%
Oil & Gas	12.1	18.0	27.6	42.4	65.2	100.3	52.7%
Telecommunications	1.0	1.7	3.0	5.1	8.6	14.7	69.6%
Retail	0.3	0.5	0.9	1.5	2.6	4.4	68.4%
Other Industries	26.3	36.3	51.0	71.0	97.4	131.0	37.9%
Total	232.2	356.5	563.3	895.4	1,429.4	2,290.4	58.1%

Note: Other industries include semiconductors, chemicals, paper & pulp, and food & beverages. The values have been represented in one decimal point to exhibit the difference between the minute values.

Source: Secondary Research, Press Releases, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

The aerospace segment is expected to account for the largest size of the digital twin market for product design & development application during the forecast period. It is further expected to grow at significant CAGR during 2022 to 2027. Digital twin is one of the emerging technologies used in product design and development along with artificial intelligence (AI), machine learning (ML), internet of things (IoT), and cloud computing. The capital required for real part design and development in the aerospace industry is enormous. Even the design of prototypes is costly. Digital twins are used by aerospace companies in R&D to improve the engineering of new parts by allowing them to model their performance in a wide range of

scenario. Aerospace businesses utilize digital twin to develop one or more critical systems, such as the airframe, propulsion and energy storage systems, avionics, and thermal protection system.

8.3 PERFORMANCE MONITORING

8.3.1 PERFORMANCE MONITORING OPTIMIZES OPERATIONAL DOWNTIME, TRIGGERS PRE-EMPTIVE MAINTENANCE, AND MITIGATES COSTLY FAILURES

For monitoring the health and performance of any asset/equipment/machinery, digital twin creates a simulated replica throughout the life cycle of that product. These applications are subsequently used with machine learning algorithms to optimize operational downtime, trigger pre-emptive maintenance, and mitigate costly failures. In a manufacturing organization, product manufacturing, production, asset life cycle management, and health monitoring play a vital role. With the increasing innovation in manufacturing organizations, rising demand for production efficiency and competitive pressure, there would be a rise in demand for digital twins for optimizing processes.

The performance monitoring applications of digital twin include the assessment of production process and the final product itself. The digital twin platform receives data from the product and the manufacturing facilities on a regular basis, resulting in new insights. Performance monitoring improves the dependability and availability of a system. As a successful technique for improving system performance, digital twins facilitate monitoring, modeling, and control of an asset, process, or network. For example, GE (US) provides SmartSignal and Asset Performance Management (APM) solutions that generate digital twins from operational/fleet data of components (pumps or compressors), essential assets (turbines), or asset systems (an entire power station). The solutions assure product quality by facilitating real-time monitoring of asset and process performance, allowing customers to improvise and react to modification while minimizing supply chain impact.

TABLE 27 PERFORMANCE MONITORING: DIGITAL TWIN MARKET, 2018–2021 (USD MILLION)

Industry	2018	2019	2020	2021
Automotive & Transportation	106.4	121.6	128.5	156.6
Energy & Utilities	77.6	91.7	99.1	123.5
Infrastructure	23.1	26.9	29.2	35.3
Healthcare	73.6	88.9	98.1	123.1
Aerospace	77.3	90.8	94.0	113.8
Oil & Gas	54.5	61.4	63.3	75.2
Telecommunications	-	-	0.4	1.4
Agriculture	-	7.9	8.0	9.6
Retail	-	-	0.2	0.3
Other Industries	105.1	107.3	120.3	162.2
Total	517.7	596.5	641.0	800.9

Source: Secondary Research, Press Releases, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 28 PERFORMANCE MONITORING: DIGITAL TWIN MARKET, BY INDUSTRY, 2022–2027 (USD MILLION)

Industry	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Automotive & Transportation	243.6	388.2	635.0	1,042.4	1,713.8	2,820.9	63.2%
Energy & Utilities	191.1	303.2	494.2	809.1	1,327.7	2,183.2	62.8%
Infrastructure	55.1	88.4	145.5	240.6	398.5	661.4	64.4%
Healthcare	200.7	334.7	572.1	980.4	1,681.0	2,883.2	70.4%
Aerospace	170.9	263.3	416.5	661.5	1,052.8	1,678.3	57.9%
Oil & Gas	110.2	165.6	255.2	394.9	611.9	949.3	53.8%
Telecommunications	2.2	3.5	5.8	9.7	16.1	26.8	65.1%
Agriculture	14.2	21.4	33.3	51.9	81.0	126.4	54.9%
Retail	0.5	0.8	1.4	2.3	3.9	6.6	69.2%
Other Industries	227.8	325.4	472.8	681.5	969.3	1,353.4	42.8%
Total	1,216.4	1,894.4	3,031.7	4,874.3	7,856.2	12,689.6	59.8%

Source: Secondary Research, Press Releases, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

The automotive & transportation industry is expected to account for the largest size of the digital twin market for performance monitoring application during the forecast period. A digital twin is anticipated to have all of the data from a product's lifecycle. Using a digital twin at the development stage helps to bridge the data gap and improve simulation test performance for automotives. The first step in observing trends in machine behavior is to collect real-time data from the vehicle sensors and overlay it on the digital version of the vehicle. The advent of IoT sensors and electric vehicles, automated guided vehicles (AGV), and connected vehicles enabled the collection of better quality of data that can be analyzed and interpreted through digital twins, resulting in the overall performance optimization and the avoidance of unplanned downtime or failure. Whereas, in the case of transportation, the digital twin of the physical object or process (for e.g., a traffic signal controller) allows for two-way real-time data interchange between the digital twin and the real-world physical object or process. This data is used for real-time monitoring of the traffic and utilized for adaptive traffic control. In a similar way, digital twin model of routes around a manufacturing center or shipyard is created by using LiDAR and cloud-data point is created to manage the transportation.

8.4 PREDICTIVE MAINTENANCE

8.4.1 DIGITAL TWIN PREDICTS FAILURES BASED ON REAL-TIME DATA COLLECTION

Predictive maintenance offered by digital twin increases content productivity, thereby enabling improved ongoing operations, maintenance, and service. This virtual copy of a product enables better communication and collaboration among organizations or manufacturers for product management, manufacturing, supply chain management, and service offering. For example, a digital twin of a wind turbine with intelligent data processing capability enables predictive maintenance of its behavior and condition. Also, the virtual models combine mathematical models to describe the physics of the turbine operation, with sensor data collected and processed from real assets during real-world operations. This would benefit the wind farm operators to predict structural failure and plan maintenance activities with greater accuracy, in advance. As a result, operators can reduce the maintenance costs and operational downtime during the life cycle of wind turbines.

Digital twin can help in prediction of failures on the basis of real-time data collection and enable just-in-time repair of components, if possible. This method not only replaces components on the verge of failure but also increases component lifetime by decreasing unscheduled maintenance and labor expenses. Businesses can save money and gain a competitive advantage by taking this approach. Digital twin predictive maintenance model gets empowered by physics-based models, statistical models, and machine learning/AI models to help describe, predict, and prescribe the behavior (current and future) of the asset, system, or process. The first step includes the loading of timeseries data in the digital twin server. The next stage is to develop an ML model based on the data obtained, which predicts failure based on fundamental parameters or the performance of external interfacing systems.

TABLE 29 PREDICTIVE MAINTENANCE: DIGITAL TWIN MARKET, BY INDUSTRY, 2018–2021 (USD MILLION)

Industry	2018	2019	2020	2021
Automotive & Transportation	211.2	243.1	258.9	318.1
Energy & Utilities	235.6	274.2	291.9	358.5
Infrastructure	187.2	218.6	238.1	289.3
Healthcare	202.9	239.4	258.3	316.9
Aerospace	56.8	68.0	71.8	88.7
Oil & Gas	200.4	229.0	239.3	288.4
Telecommunications	-	-	0.5	2.0
Agriculture	-	5.9	6.1	7.3
Retail	-	-	0.2	0.2
Other Industries	80.5	83.1	94.0	128.0
Total	1,174.5	1,361.3	1,459.1	1,797.3

Note: The values have been represented in one decimal point to exhibit the difference between the minute values.

Source: Secondary Research, Press Releases, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 30 PREDICTIVE MAINTENANCE: DIGITAL TWIN MARKET, BY INDUSTRY, 2022–2027 (USD MILLION)

Industry	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Automotive & Transportation	498.9	801.6	1,322.2	2,189.0	3,630.0	6,027.9	64.6%
Energy & Utilities	546.7	854.8	1,373.3	2,216.3	3,585.4	5,812.3	60.4%
Infrastructure	454.4	731.5	1,209.6	2,008.2	3,340.9	5,567.4	65.1%
Healthcare	505.3	824.5	1,379.8	2,315.6	3,889.2	6,536.2	66.9%
Aerospace	136.5	215.3	348.5	566.0	920.5	1,498.5	61.5%
Oil & Gas	429.8	656.6	1,030.0	1,622.2	2,559.4	4,036.3	56.5%
Telecommunications	3.1	4.9	7.9	13.0	21.2	34.9	62.7%
Agriculture	10.9	16.6	26.0	40.7	63.9	100.3	55.9%
Retail	0.4	0.6	1.1	1.8	3.0	5.0	67.3%
Other Industries	181.4	261.5	383.4	557.5	799.9	1,126.4	44.1%
Total	2,767.4	4,367.9	7,081.7	11,530.3	18,813.5	30,745.1	61.9%

Note: The values have been represented in one decimal point to exhibit the difference between the minute values.

Source: Secondary Research, Press Releases, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

The predictive maintenance digital twin market for healthcare industry is expected to grow at the highest CAGR during the forecast period. In healthcare, digital twins are used to create a patient's digital twin that captures continuous data from the individual concerning various vitals, medical conditions, drug and therapy responses, and the surrounding ecosystem. Each patient's historical and real-time data aids the ML algorithm in predicting future health issues. The model alerts the patient for prescriptions, dietary habit modifications, and medical consultation based on the patient's lifestyle, daily food habits, and blood sugar data. Digital twin uses AI-powered models to generate more tailored and better treatment plans based on a significant amount of rich data from multiple internet of medical things (IoMT) devices. Using insights from each patient's unique digital twin, the proper therapy may be determined, the success of a given procedure can be predicted, and chronic diseases can be managed.

8.5 INVENTORY MANAGEMENT

8.5.1 DIGITAL TWINS AID END USERS IN INVENTORY OPTIMIZATION ACROSS ENTIRE NETWORK

A digital twin represents a real-world system in virtual space and can provide many benefits. Digital twin solutions can be used in supply chains or across entire systems, where they help to cut down costs and deliver value. The digital visualization provided by a warehouse's digital twin can give timely and accurate visibility to actual inventory levels, and companies can make predictions or decisions about deliveries or stocking. In a fully integrated shipping system, IoT sensors can be placed in individual containers to track location and monitor damage or contamination for handling conditions and predict the most efficient routes. Digital twin for inventory management is mainly useful in process industries.

A company creates digital twin models to redefine its operations by modelling numerous scenarios, including manufacturing, inventory, and product distribution data. To avoid stock-outs and reduce total production and warehousing costs, a supply chain digital twin can input data from demand forecasting processes. As a result, a digital twin can aid inventory optimization in a particular warehouse as well as across the entire network. It handles both the single-echelon and the multi-echelon challenges. The single-

echelon deals with optimizing inventory in a single warehouse whereas multi-echelon is associated with optimizing inventory across many warehouses or the entire network. It accomplishes so by factoring in demand estimates. A digital twin also helps to enhance replenishment plans and adjust inventory levels. It does so in response to demand in order to avoid stockouts and save money. Digital twin has an impact on logistical infrastructure design, operation, and optimization. Digital twins for infrastructure provide perfect blueprints and layouts for new infrastructures. For instance, SenSat (US), a digital twin startup, offers cloud-based digital twin solutions that accurately simulate real-world scenarios in real time. Their method uses visual overviews to analyze a system. At the same time, it interacts with a variety of data sources. Warehouse simulators can help businesses make the most of their space, labor, and material transportation.

TABLE 31 INVENTORY MANAGEMENT: DIGITAL TWIN MARKET, BY INDUSTRY, 2018–2021 (USD MILLION)

Industry	2018	2019	2020	2021
Automotive & Transportation	53.22	59.75	62.03	74.22
Energy & Utilities	60.02	69.31	73.18	89.13
Infrastructure	43.68	49.43	52.18	61.39
Healthcare	49.93	57.19	59.83	71.09
Aerospace	67.58	76.64	76.56	89.39
Oil & Gas	43.76	49.71	51.62	61.81
Telecommunications	-	-	0.51	1.18
Agriculture	-	-	0.12	0.15
Retail	-	11.37	11.06	12.90
Other Industries	51.83	51.63	56.45	74.15
Total	370.02	425.04	443.54	535.42

Note: The values have been represented in two decimal points to exhibit the difference between the minute values.

Source: Secondary Research, Press Releases, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 32 INVENTORY MANAGEMENT: DIGITAL TWIN MARKET, BY INDUSTRY, 2022–2027 (USD MILLION)

Industry	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Automotive & Transportation	113.24	176.83	283.17	454.64	730.23	1,187.76	60.0%
Energy & Utilities	134.77	208.92	332.72	532.19	853.16	1,352.91	58.6%
Infrastructure	93.34	145.36	232.34	372.65	598.45	961.97	59.4%
Healthcare	109.99	173.93	281.74	457.05	740.99	1,213.30	61.6%
Aerospace	129.38	191.72	291.42	444.20	677.45	1,033.24	51.5%
Oil & Gas	91.27	138.10	214.57	334.68	522.90	821.91	55.2%
Telecommunications	1.75	2.64	4.09	6.36	9.91	15.44	54.6%
Agriculture	0.23	0.37	0.60	0.97	1.58	2.61	62.4%
Retail	19.31	29.63	46.73	74.00	117.45	186.87	57.5%
Other Industries	101.44	141.04	199.38	279.48	386.34	523.91	38.9%
Total	794.72	1,208.54	1,886.76	2,956.23	4,638.48	7,299.92	55.8%

Note: The values have been represented in two decimal points to exhibit the difference between the minute values.

Source: Secondary Research, Press Releases, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

The automotive & transportation, aerospace, healthcare, and energy & utilities industries are expected to account for significant shares of the digital twin market for inventory management application during the forecast period. Whether it is water, power, telecommunications, logistics, aerospace, healthcare, mining, or oil & gas, inventory management is being challenged by increased expectations for more effective asset control to optimize the network. The advancements in digital twin technology are transforming the energy and utility industries. The multiple levels of a digital twin enable new insights to be gained that were previously unavailable. The inventory optimization equation relies on a variety of elements, including inventory item standardization, criticality, visibility to previous usage (at the highest level of a corporation possible), and known future work orders to determine which parts are required to maintain high service levels. Some businesses are already adopting digital twins to assist with complex capital improvements and inventory maintenance activities on a significant level. Their application, on the other hand, will become more sophisticated.

8.6 BUSINESS OPTIMIZATION

8.6.1 BUSINESS OPTIMIZATION APPLICATION TO REGISTER HIGHEST GROWTH IN DIGITAL TWIN MARKET DURING FORECAST PERIOD

Business optimization is the analysis and implementation of the overall data gathered through digital twins of systems, processes, or products to increase business efficiency. The virtual models created by digital twin are used to optimize the characteristic parameters for the production of an asset or process. The optimization of internal processes includes operability, manufacturability, sustainability, model implementation, and simulations applied during the entire life cycle of the product or asset. The ongoing modification in designs and models is possible with the help of a digital twin as it allows the data to be easily interconnected with the details of the modified design. Designers and engineers can use digital twin virtualizations to derive insights about the performance of existing equipment/assets and processes over time, use the information to influence and validate new prototypes, and bring better products to market faster at less cost. For example, Tesla manufactures each car by using a digital twin solution, owing to

which the company has a large volume of data and related output, which helps the company to decide the future steps.

The constant flow of data aids in obtaining the most accurate analysis and insights about the asset, hence optimizing the company outcome. Product-based companies and users generate profit from digital twins in a variety of ways. They can help with design optimization, cost reduction, and reduction of time to market, as well as speed up a company's responsiveness to changing client needs.

TABLE 33 BUSINESS OPTIMIZATION: DIGITAL TWIN MARKET, BY INDUSTRY, 2018–2021 (USD MILLION)

Industry	2018	2019	2020	2021
Automotive & Transportation	115.9	137.1	149.9	189.1
Energy & Utilities	62.5	75.0	82.3	104.0
Infrastructure	161.0	186.6	201.9	243.5
Healthcare	55.5	66.2	72.2	89.4
Aerospace	149.3	176.3	183.5	223.4
Oil & Gas	65.3	76.8	82.5	102.1
Telecommunications	-	34.1	34.1	39.0
Agriculture	-	8.9	9.1	10.8
Retail	-	-	0.3	0.7
Other Industries	85.8	88.1	99.3	134.6
Total	695.4	849.2	915.1	1,136.8

Note: The values have been represented in one decimal point to exhibit the difference between the minute values.

Source: Secondary Research, Press Releases, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 34 BUSINESS OPTIMIZATION: DIGITAL TWIN MARKET, BY INDUSTRY, 2022–2027 (USD MILLION)

Industry	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Automotive & Transportation	304.4	501.6	848.2	1,439.1	2,444.8	4,142.3	68.6%
Energy & Utilities	163.2	262.4	433.3	718.5	1,193.5	2,003.2	65.1%
Infrastructure	379.9	607.3	997.3	1,644.4	2,717.1	4,497.2	63.9%
Healthcare	143.7	236.3	398.4	673.7	1,140.2	1,917.8	67.9%
Aerospace	336.9	520.7	826.5	1,317.3	2,103.5	3,364.3	58.4%
Oil & Gas	156.7	246.4	397.5	643.3	1,042.3	1,694.0	61.0%
Telecommunications	59.4	92.8	148.9	240.1	388.0	628.4	60.3%
Agriculture	15.9	23.8	36.7	56.6	87.6	135.4	53.5%
Retail	1.1	1.8	3.0	4.9	8.0	13.2	63.2%
Other Industries	190.0	272.7	398.3	576.9	824.6	1,157.0	43.5%
Total	1,751.2	2,765.9	4,488.1	7,314.9	11,949.6	19,552.9	62.0%

Note: The values have been represented in one decimal point to exhibit the difference between the minute values.

Source: Secondary Research, Press Releases, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

The business optimization digital twin market for automotive & transportation industry is expected to grow at the highest CAGR during the forecast period. Digital twin enables automotive OEMs to optimize expenses and customer value at the same time. Engineers can precisely analyze novel design concepts or vehicle features in a virtual area and test them against various environmental conditions, saving time and money. For instance, Volkswagen (Germany), a global automotive manufacturer has invested in digital twin technology to test modern designed vehicle battery model's electrical and thermal properties. The real-time data allows automotive development teams to communicate more effectively, minimize data analysis time, and streamline vehicle production.

In the infrastructure industry, digital twin plays a vital role in optimization of the entire operations of modern and smart buildings. Cities and huge university campuses benefit from real-time digital twins because they provide a robust platform for modelling, understanding, and analyzing interconnections between buildings, electrical systems, and other public infrastructure. Digital twins collect data from a variety of sources and intelligently combine it with other data sets, capturing context and relationships. For instance, in September 2021, the Brooklyn Navy Yard in the US hired Cityzenith (US), a digital twin software developer, to create digital twins for a group of buildings, systems, and infrastructure at the Navy Yard. The digital twins will be used to automate real-time energy monitoring, onsite renewable energy generation, and carbon offset purchases at the Brooklyn Navy Yard.

8.7 OTHER APPLICATIONS

The others applications segment includes environmental analysis, customized services, instore planning, security implementation, and energy management. These applications will help industry players to provide more personalized and secure services to customers.

The use of digital twin technology and integrated performance analysis tools is already having a significant impact on reducing the built environment's carbon footprint, and these numbers are steadily increasing as more and more building professionals recognize the social, environmental, and economic benefits of implementing a digital twin solution. For instance, Integrated Environmental Solutions (IES), a UK-based software simulation tools and consulting service provider, offers environmental digital twin technology—the Intelligent Communities Lifecycle (ICL). The ICL Digital Twin makes real-time performance optimization of built environment assets a reality. ICL combines physics-based simulation with a three-dimensional model, real-time operational data, machine learning, and artificial intelligence. According to IES, its ICL digital twin can save 12% of energy for an existing net-zero energy building in the tropics. The same technology was utilized to develop a Virtual Campus Energy Model to help the Dundalk Institute of Technology (Ireland) to achieve its goal of becoming a net-zero energy campus, offering analysis and knowledge of the dynamics in their distributed energy network. The solution focuses on delivering both environmental analysis of buildings and providing energy management options.

In terms of secure services, digital twin can be used to strengthen the security of an organization. For instance, Haruspex (Italy), a computer and network security provider, offers complete cyberthreat assessment of clients' system through a digital twin. Haruspex's predictive cyber security solution can generate a virtual duplicate of clients' IT/OT system and model a wide range of attack strategies and techniques automatically. Hundreds of thousands of simulations are conducted on Haruspex's platform to identify potential attack vectors that a potential intruder (hacker, worm, or other) might use against the clients' real system.

TABLE 35 OTHER APPLICATIONS: DIGITAL TWIN MARKET, BY INDUSTRY 2018–2021 (USD MILLION)

Industry	2018	2019	2020	2021
Automotive & Transportation	6.02	7.02	7.58	9.43
Energy & Utilities	7.24	8.11	8.29	9.76
Infrastructure	5.14	5.73	5.97	6.91
Healthcare	5.44	6.34	6.76	8.20
Aerospace	7.14	8.35	8.62	10.41
Oil & Gas	9.33	10.22	10.20	11.72
Telecommunications	-	-	0.58	0.81
Agriculture	-	-	0.20	0.23
Retail	-	-	0.22	0.27
Other Industries	13.22	12.63	13.19	16.49
Total	53.52	58.41	61.61	74.23

Note: The values have been represented in two decimal points to exhibit the difference between the minute values.

Source: Secondary Research, Press Releases, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 36 OTHER APPLICATIONS: DIGITAL TWIN MARKET, BY INDUSTRY 2022–2027 (USD MILLION)

Industry	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Automotive & Transportation	14.99	24.40	40.76	68.34	114.78	193.01	66.7%
Energy & Utilities	14.24	21.25	32.49	49.74	76.07	116.13	52.1%
Infrastructure	10.34	15.82	24.82	39.00	61.28	96.20	56.2%
Healthcare	12.91	20.81	34.38	56.96	94.42	156.56	64.7%
Aerospace	15.55	23.82	37.46	59.17	93.63	148.42	57.0%
Oil & Gas	16.61	24.06	35.68	52.95	78.41	115.76	47.5%
Telecommunications	1.26	2.01	3.29	5.40	8.91	14.70	63.5%
Agriculture	0.34	0.51	0.77	1.18	1.80	2.76	51.9%
Retail	0.41	0.64	1.03	1.67	2.72	4.41	60.9%
Other Industries	21.35	27.92	36.87	47.85	60.58	74.22	28.3%
Total	108.00	161.23	247.55	382.26	592.59	922.16	53.6%

Note: Other industries include semiconductors, chemicals, paper & pulp, and food & beverages. The values have been represented in two decimal points to exhibit the difference between the minute values.

Source: Secondary Research, Press Releases, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

The automotive & transportation industry is expected to account for the largest size of the digital twin market for other applications during the forecast period. Automobiles represent a wide range of complex, interconnected systems, and digital twins are widely employed in vehicle design to improve vehicle performance and increase production efficiency. The automotive & transportation industry requires digital twins for various other applications, including customization services, instore planning, security implementation, and energy management. The infrastructure and energy & utilities industries use digital twins extensively for energy management applications.

9 DIGITAL TWIN MARKET, BY INDUSTRY

KEY FINDINGS

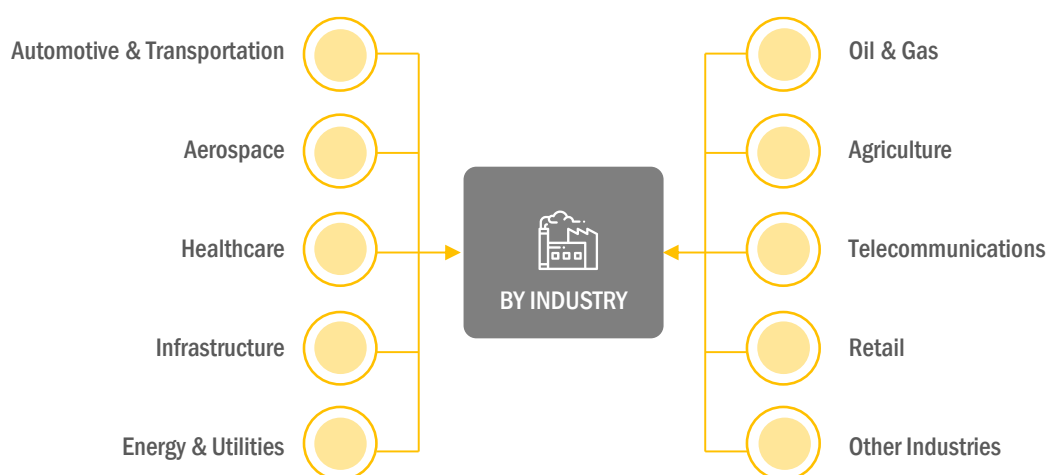
- The automotive & transportation segment was estimated at USD 0.8 billion in 2021 and is expected to reach USD 14.8 billion by 2027, at a CAGR of 65.3% during the forecast period.
- The growth of the automotive & transportation segment can be attributed to the increasing usage of digital twins for designing, simulation, and maintenance, repair, and overhaul (MRO) services.
- The healthcare segment was estimated at USD 0.6 billion in 2021 and is expected to reach USD 13.0 billion by 2027, at the highest CAGR of 67.4% during the forecast period.
- The rapid growth of the healthcare segment is due to the increasing value of patient safety and advancements in patient care systems with the integration of technologies, such as internet of things (IoT), artificial intelligence (AI), and extended reality (XR).
- The infrastructure segment was estimated at USD 0.7 billion in 2021; it is expected to reach USD 12.0 billion by 2027, growing at the highest CAGR of 63.9% during the forecast period.

9.1 INTRODUCTION

By industry, the digital twin market has been segmented into aerospace, automotive & transportation, infrastructure, healthcare, energy & utilities, oil & gas, agriculture, telecommunications, retail, and others. Others includes industries such as semiconductors, chemicals, paper & pulp, and food & beverages.

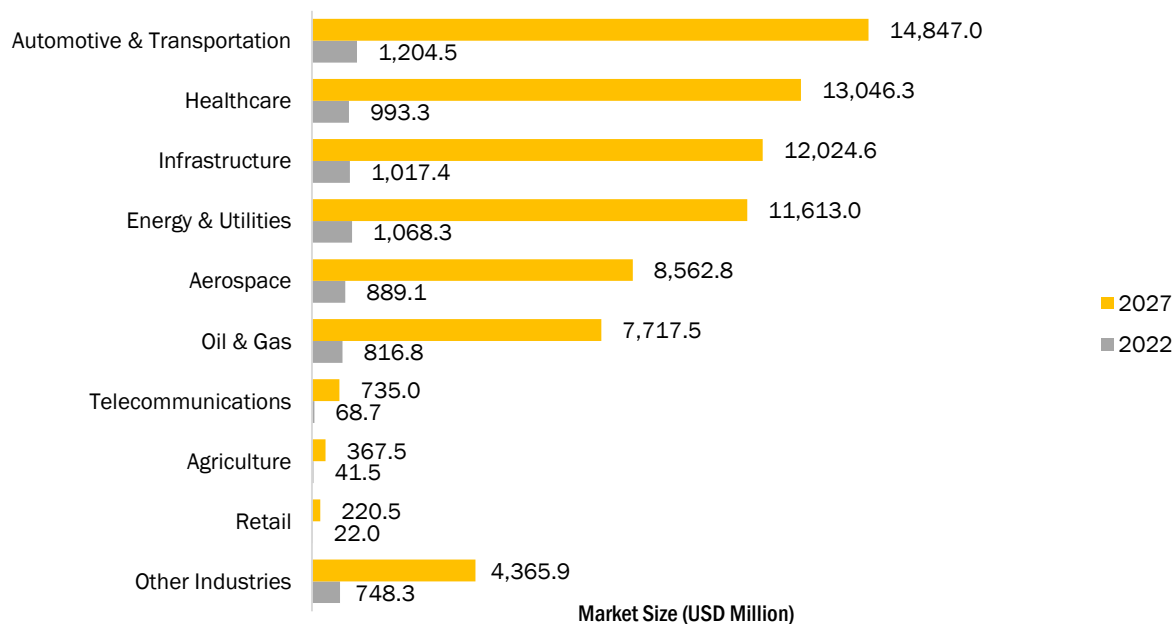
Product-, process-, and system-type digital twins are widely used for designing, monitoring, controlling, and predictive maintenance in several industries. Increasing adoption of emerging technologies such as the Internet of Things (IoT), artificial intelligence (AI), machine learning (ML), and cloud computing are the major enablers of digital twin technology. Increasing global demand to maximize uptime and minimize equipment maintenance costs over their life cycle has led to the implementation of these technologies across various industries.

FIGURE 41 DIGITAL TWIN MARKET, BY INDUSTRY



Source: Secondary Research, Press Releases, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

FIGURE 42 HEALTHCARE SEGMENT EXPECTED TO GROW AT HIGHEST CAGR IN DIGITAL TWIN MARKET DURING FORECAST PERIOD



Note: Other industries include semiconductors, chemicals, paper & pulp, and food & beverages. The values have been represented in one decimal point to exhibit the difference between the minute values.

Source: Secondary Research, Press Releases, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 37 DIGITAL TWIN MARKET, BY INDUSTRY, 2018–2021 (USD MILLION)

Industry	2018	2019	2020	2021
Automotive & Transportation	502	580	620	765
Energy & Utilities	452	529	565	697
Infrastructure	432	500	541	652
Healthcare	394	466	505	621
Aerospace	409	478	492	594
Oil & Gas	379	434	454	547
Telecommunication	-	34	36	45
Agriculture	-	23	24	28
Retail	-	11	12	15
Other Industries	350	356	398	535
Total	2,918	3,412	3,648	4,500

Note: Other industries include semiconductors, chemicals, paper & pulp, and food & beverages.

Source: Secondary Research, Press Releases, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 38 DIGITAL TWIN MARKET, BY INDUSTRY, 2022–2027 (USD MILLION)

Industry	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Automotive & Transportation	1,205	1,943	3,218	5,348	8,905	14,847	65.3%
Energy & Utilities	1,068	1,678	2,707	4,389	7,132	11,613	61.2%
Infrastructure	1,017	1,626	2,669	4,400	7,267	12,025	63.9%
Healthcare	993	1,626	2,729	4,594	7,739	13,046	67.4%
Aerospace	889	1,364	2,149	3,401	5,392	8,563	57.3%
Oil & Gas	817	1,249	1,961	3,090	4,880	7,718	56.7%
Telecommunication	69	108	173	280	453	735	60.6%
Agriculture	42	63	97	151	236	368	54.7%
Retail	22	34	54	86	138	221	58.5%
Other Industries	748	1,065	1,542	2,214	3,138	4,366	42.3%
Total	6,870	10,754	17,299	27,953	45,280	73,500	60.6%

Note: Other industries include semiconductors, chemicals, paper & pulp, and food & beverages.

Source: Secondary Research, Press Releases, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

The automotive & transportation industry is expected to account for the largest share of digital twin market during forecast period. It is further expected to grow at significant CAGR from 2022 to 2027. The growth can be attributed to the increasing usage of digital twins for designing, simulation, MRO (maintenance, repair, and overhaul), production, and after service. The Automotive & transportation industry is the early adopter of technologies, and there is a rising demand for connected things in this sector, which consequently is expected to boost the growth of the market for digital twin.

The digital twin market for healthcare industry is expected to grow at highest CAGR during the forecast period. The rapid growth of the healthcare sector is due to the increasing value of patient safety and advancements in the patient care systems with the integration of technologies, such as internet of things (IoT), artificial intelligence (AI), and extended reality (XR). In addition, the use of digital twins is also growing for monitoring, diagnostics, and prognostics, for example, digital twins of organs such as the heart, lung, and brain are developed that help surgeons and doctors to test various parameters on the twin and then implementing it in the real-life situation. Siemens (Germany) has been developing a digital twin of the heart to help with pharmacological therapy and cardiac catheterization simulations. Dassault Systèmes (France) is also working on the Living Heart Project to create a virtual twin of the human heart. Further, in January 2022, Dassault Systèmes (France) showcased the virtual twin of human brain at CES 2022 (Las Vegas, US) with the help of augmented reality (AR)/virtual reality (VR). Similar projects have been conducted for the lungs, knees, eyes, and other systems by the company. Such developments in the field of healthcare is expected to drive the digital twin market in the foreseeable future.

9.2 AEROSPACE

9.2.1 AEROSPACE INDUSTRY LEVERAGES DIGITAL TWIN FOR PRODUCT DESIGNING & DEVELOPMENT, AND MAINTENANCE PURPOSES

With increasing advancements in digital technologies, rapid change has been witnessed in the operations of the aerospace industry. The industry is leveraging digital technologies for various processes such as manufacturing, designing, maintenance, and development of products. The manufacturing operations in the aerospace industry are quite complex. Hence, manufacturers primarily emphasize on controlling component supplies, improving product quality, shortening the production cycle, and employing efficient manufacturing processes.

Manufacturers are increasingly adopting advanced solutions for manufacturing and designing aircraft engines, machines, and other parts. In the aerospace industry, digital twins help minimize the cost of production, optimize processing, and reduce production time. Digital twins combined with augmented and virtual reality help the companies in better involvement of customers at the earlier stage in the design process, thereby minimizing the need for expensive late-stage changes in the design and saving time.

Governments of various countries worldwide are initiating new projects to transform infrastructure and security systems for developing aircraft, satellites, and other space technologies. The use of digital twins in the aerospace industry facilitates collaborative and synchronized program management to be applied across the product life cycle and value chain. The implementation of digital twin technology in the aerospace industry was initiated by National Aeronautics and Space Administration (NASA) (US) to monitor and operate complex systems in remote and critically hazardous conditions. The technology has the ability to record real-time telemetry data and process it quickly to represent it into model-based simulations, which is valuable in remote, demanding, or dangerous industrial processes. Businesses in the aerospace industry can benefit from digital twins in a variety of ways, including prolonging the life of parts and machines and leveraging data to better the next iteration. Aerospace businesses can employ digital twins in research and development to improve the engineering of new parts by simulating their performance under a wide range of situations. A digital twin can predict how an engine would perform in adverse conditions; or what will the airline's performance be like after 300+ flights. This information may now be gathered and used to enhance parts, forecast when maintenance is required, and increase the life of aircraft.

TABLE 39 AEROSPACE: DIGITAL TWIN MARKET, BY APPLICATION, 2018–2021 (USD MILLION)

Application	2018	2019	2020	2021
Product Design & Development	50.45	57.63	58.03	68.32
Performance Monitoring	77.34	90.77	93.95	113.76
Predictive Maintenance	56.76	68.01	71.82	88.68
Inventory Management	67.58	76.64	76.56	89.39
Business Optimization	149.31	176.27	183.49	223.44
Other Applications	7.14	8.35	8.62	10.41
Total	408.58	477.68	492.48	594.00

Note: Other applications include environmental analysis, customized services, instore planning & security implementation, and energy management. The values have been represented in two decimal points to exhibit the difference between the minute values.

Source: Secondary Research, Press Releases, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 40 AEROSPACE: DIGITAL TWIN MARKET, BY APPLICATION, 2022–2027 (USD MILLION)

Application	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Product Design & Development	99.75	149.19	229.04	352.82	544.16	840.01	53.1%
Performance Monitoring	170.94	263.28	416.47	661.52	1,052.82	1,678.30	57.9%
Predictive Maintenance	136.54	215.32	348.51	566.02	920.50	1,498.48	61.5%
Inventory Management	129.38	191.72	291.42	444.20	677.45	1,033.24	51.5%
Business Optimization	336.92	520.70	826.51	1,317.27	2,103.50	3,364.30	58.4%
Other Applications	15.55	23.82	37.46	59.17	93.63	148.42	57.0%
Total	889.09	1,364.03	2,149.40	3,401.00	5,392.07	8,562.75	57.3%

Note: Other applications include environmental analysis, customized services, instore planning & security implementation, and energy management. The values have been represented in two decimal points to exhibit the difference between the minute values.

Source: Secondary Research, Press Releases, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 41 AEROSPACE: DIGITAL TWIN MARKET, BY REGION, 2018–2021 (USD MILLION)

Region	2018	2019	2020	2021
North America	155.3	182.6	189.4	229.9
Europe	130.7	153.3	158.6	191.9
Asia Pacific	94.0	109.3	112.2	134.6
Rest of the World	28.6	32.4	32.3	37.6
Total	408.6	477.7	492.5	594.0

Note: Rest of the World includes South America, Africa, and the Middle East. The values have been represented in one decimal point to exhibit the difference between the minute values.

Source: Secondary Research, Press Releases, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 42 AEROSPACE: DIGITAL TWIN MARKET, BY REGION, 2022–2027 (USD MILLION)

Region	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
North America	346.2	534.2	846.9	1,347.9	2,149.6	3,433.7	58.2%
Europe	288.1	443.3	700.7	1,112.1	1,768.6	2,800.0	57.6%
Asia Pacific	200.5	306.1	480.0	755.8	1,192.2	1,900.9	56.8%
Rest of the World	54.3	80.3	121.8	185.2	281.6	428.1	51.1%
Total	889.1	1,364.0	2,149.4	3,401.0	5,392.1	8,562.8	57.3%

Note: Rest of the World includes South America, Africa, and the Middle East. The values have been represented in one decimal point to exhibit the difference between the minute values.

Source: Secondary Research, Press Releases, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

The predictive maintenance application is expected to grow at the highest CAGR during the forecast period. It is further expected to account for the second-largest market share from 2022 to 2027. In the aerospace business, digital twins facilitate predictive analysis and fault diagnosis of jet engines. Digital twins are used in the aerospace sector to reduce unplanned downtime for engines and other systems. Not only do businesses get early notice and predictions using this technology but also get a plan of action based on

simulated scenarios that take weather conditions, asset performance, and a variety of other elements into consideration.

The digital twin market for aerospace is expected to grow at the highest CAGR in North America during the forecast period. North American digital twin providers such as General Electric (US), Microsoft (US), PTC (US), and ANSYS (US) are continuously investing in R&D work for aerospace digital twin and offer their solutions across the world. The North American region has one of the busiest flight routes in the world. Digital twin technology facilitates the optimization, predictive maintenance, and planning of flights. This factor further propels the growth of the aerospace segment in the region.

9.3 AUTOMOTIVE & TRANSPORTATION

9.3.1 AUTOMOTIVE & TRANSPORTATION INDUSTRY USES DIGITAL TWIN FOR PRODUCT DESIGN & DEVELOPMENT

The automotive industry is one of the early adopters of digital twin technology. The industry involves various systems and processes at several stages of the complete life cycle of automobile products and services. In order to optimize and enhance these systems and processes, the operators in this industry are adopting digital twins at a significant rate. Automotive product development is complex and time-consuming. Typically, it takes about 4–6 years for a new car model to be launched right from the design to the launch phase. Moreover, many new vehicles in the market are equipped with hundreds of sensors for constant monitoring of critical systems. Effective design is the first and most important phase of the complete product life cycle, as it can impact the long-term sustainability of an automobile manufacturer. Even a small oversight in the product design phase can erode the company's brand value and profitability. For instance, Mercedes Benz launched its A-Class in early 2000 and incurred a product development cost of USD 1.5 billion. After its launch, the vehicle failed a Moose test, resulting in the recall of 2,500 new cars. Subsequently, Mercedes added stability control and redesigned the car's suspension to address the problem. The cost of implementing the change was about USD 250 million. However, in this scenario, if a digital twin is built at the earlier stage, i.e., the designing phase, the company can save a lot of time as well as money.

A digital twin is capable of analyzing performance data collected over time and under different conditions during the manufacturing phase of vehicle design. For example, a digital twin enables visualization of a race car engine to identify the necessity of maintenance of the components that can burn out or get damaged. Even in the after-sales stage, a digital twin is used for collecting feedback. Digital twins help ensure the safety of vehicles by monitoring the systems or parts that need replacement and alerting the relevant teams regarding the necessary change. Also, the manufacturer can analyze the data & performance through predictive analysis to help make the driving experience smarter and safer. For instance, Computer Science Corp. (US) partnered with Microsoft Corporation (US) to build machine learning solutions on an industrial level and used the Microsoft Cortana Intelligence Suite to run the digital twin for continuously simulating new innovative ideas of creating hybrid cars. Similarly, Cummins (US), a motor vehicle manufacturing company, is using ANSYS's (US) SPDM platform and physics-based digital twins for research and development of advanced diesel engines. On the other hand, Faurecia (France), a subsidiary of the Group Forvia, one of the largest automotive suppliers in the world, is using Dassault Systèmes' 3DEXPERIENCE digital twin platform to optimize its automated guided vehicle (AGV) inbound logistics in its factories.

TABLE 43 AUTOMOTIVE & TRANSPORTATION: DIGITAL TWIN MARKET, BY APPLICATION, 2018–2021 (USD MILLION)

Application	2018	2019	2020	2021
Product Design & Development	9.20	11.51	13.25	17.51
Performance Monitoring	106.44	121.57	128.46	156.59
Predictive Maintenance	211.20	243.10	258.90	318.11
Inventory Management	53.22	59.75	62.03	74.22
Business Optimization	115.88	137.07	149.94	189.14
Other Applications	6.02	7.02	7.58	9.43
Total	501.96	580.04	620.16	765.00

Note: The values have been represented in two decimal points to exhibit the difference between the minute values.

Source: Secondary Research, Press Releases, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 44 AUTOMOTIVE & TRANSPORTATION: DIGITAL TWIN MARKET, BY APPLICATION, 2022–2027 (USD MILLION)

Application	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Product Design & Development	29.40	50.37	88.31	154.91	271.44	475.10	74.5%
Performance Monitoring	243.61	388.20	634.99	1,042.40	1,713.76	2,820.93	63.2%
Predictive Maintenance	498.91	801.59	1,322.17	2,188.99	3,630.03	6,027.88	64.6%
Inventory Management	113.24	176.83	283.17	454.64	730.23	1,187.76	60.0%
Business Optimization	304.38	501.59	848.22	1,439.14	2,444.78	4,142.31	68.6%
Other Applications	14.99	24.40	40.76	68.34	114.78	193.01	66.7%
Total	1,204.54	1,942.98	3,217.62	5,348.42	8,905.02	14,847.00	65.3%

Note: The values have been represented in two decimal points to exhibit the difference between the minute values.

Source: Secondary Research, Press Releases, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 45 AUTOMOTIVE & TRANSPORTATION: DIGITAL TWIN MARKET, BY REGION, 2018–2021 (USD MILLION)

Region	2018	2019	2020	2021
North America	191	223	240	299
Europe	171	196	207	254
Asia Pacific	105	122	130	161
Rest of the World	35	40	42	51
Total	502	580	620	765

Source: Secondary Research, Press Releases, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 46 **AUTOMOTIVE & TRANSPORTATION: DIGITAL TWIN MARKET, BY REGION, 2022–2027 (USD MILLION)**

Region	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
North America	477	779	1,305	2,195	3,698	6,236	67.2%
Europe	394	628	1,026	1,682	2,763	4,543	63.0%
Asia Pacific	254	412	686	1,145	1,915	3,207	66.0%
Rest of the World	79	124	201	326	529	861	61.4%
Total	1,205	1,943	3,218	5,348	8,905	14,847	65.3%

Source: Secondary Research, Press Releases, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

The product design & development application segment of the automotive & transportation digital twin market is expected to grow at the highest CAGR during the forecast period. In the automotive sector, digital twins are utilized to generate a virtual replica of connected cars. Automotive companies use the technology to design the ideal automotive product even before production starts. They replicate and assess the manufacturing process as well as any issues that may arise once the car is on the road. It also aids automotive OEMs in providing consumers with truly personalized/customized service. Automotive & transportation is one of the leading industries to adopt digitization and digital twin solutions in North America. Currently, Tesla (US) manufactures each car with the help of digital twins. As a result of the rapid implementation of digital twin by North American automotive OEMs, the digital twin market for the automotive & transport industry is expected to be dominated by the North America region during the forecast period.

9.4 INFRASTRUCTURE

9.4.1 INITIATION OF SMART CITY PROJECTS TO PROPEL DIGITAL TWIN MARKET GROWTH

The digitalization trends in the infrastructure sector includes edge computing/cloud computing, internet of things (IoT), and other connected devices such as sensors, detectors, and smart home devices, which have generated huge opportunities for the future development of cities, buildings, and home infrastructure. Countries such as India, France, Austria, the Netherlands, Germany, and many more across the world have undertaken concrete steps for the development of smart cities. Digital twin is highly adopted in smart city projects as it provides a high-level abstraction on the behavior of the real systems, allowing to know the real-time status, environmental effects, transfer of information, and, most importantly, the simulation scenarios. Such inputs and outputs help in analyzing and controlling various parameters. For instance, Dassault Systems (France) is working on a project in Singapore to create a digital twin called “Virtual Singapore.” The virtual twin would help urban planners and policymakers visualize the data obtained from various sensor networks and intelligent systems deployed and to be deployed in the future and use them to make informed decisions.

Digital twin services are also utilized for digital representation of educational institutes and large university campuses for energy management, security, digital library management, and other purposes. This unified image can be used to merge data streams from different building components and create new services. These can then be used to create new business models and provide real assistance in day-to-day operations. In addition, the consolidating trend of Construction 4.0, which refers to the fourth industrial revolution in the construction industry and focuses on digitization, will increase the demand for digital twin solutions. With the rise in the adoption of Construction 4.0 principles, there will be a massive transformation in the infrastructure as well as the construction equipment used or controlled

autonomously. The aforesaid is expected to lead to the development of digital twins of the assets, products, processes, and systems in the near future.

One of the important technologies enabling the creation of digital twins for buildings is building information modeling (BIM) systems with consistent and structured data management tools. With the help of these technologies, the dynamic performance of buildings can be examined by building simulation tools under various boundary conditions. However, the advancements in the technologies contributing to digitalization of homes such as wireless connectivity, edge computing, analytics, and machine learning are propelling the adoption of digital twins in this sector.

In the construction sector, a digital twin can be used for capturing the spatial and temporal effects of infrastructures and sites to optimize urban sustainability. For instance, “Virtual Singapore,” a part of the Smart Nation Singapore initiative of the government of Singapore, is the world’s first digital twin of an existing city-state, providing an effective way to the population of Singapore to engage in the digital economy. The digital twins provide clients with an entirely new perspective on the buildings/infrastructure they own or run. Historical data is combined with extensive data acquired in real time to provide a detailed image of the facility that is updated and enhanced on a regular basis. This technology offers an entirely new degree of knowledge about a building and how it is operated, resulting in a comprehensive data set that can be used to carry out maintenance activities based on asset predictive analysis.

TABLE 47 INFRASTRUCTURE: DIGITAL TWIN MARKET, BY APPLICATION, 2018–2021 (USD MILLION)

Application	2018	2019	2020	2021
Product Design & Development	11.7	13.2	13.8	16.1
Performance Monitoring	23.1	26.9	29.2	35.3
Predictive Maintenance	187.2	218.6	238.1	289.3
Inventory Management	43.7	49.4	52.2	61.4
Business Optimization	161.0	186.6	201.9	243.5
Other Applications	5.1	5.7	6.0	6.9
Total	431.9	500.4	541.1	652.5

Note: The values have been represented in one decimal point to exhibit the difference between the minute values.

Source: Secondary Research, Press Releases, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 48 INFRASTRUCTURE: DIGITAL TWIN MARKET, BY APPLICATION, 2022–2027 (USD MILLION)

Application	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Product Design & Development	24.4	37.7	59.7	95.0	151.1	240.5	58.1%
Performance Monitoring	55.1	88.4	145.5	240.6	398.5	661.4	64.4%
Predictive Maintenance	454.4	731.5	1,209.6	2,008.2	3,340.9	5,567.4	65.1%
Inventory Management	93.3	145.4	232.3	372.6	598.5	962.0	59.4%
Business Optimization	379.9	607.3	997.3	1,644.4	2,717.1	4,497.2	63.9%
Other Applications	10.3	15.8	24.8	39.0	61.3	96.2	56.2%
Total	1,017.4	1,626.1	2,669.2	4,399.9	7,267.4	12,024.6	63.9%

Note: The values have been represented in one decimal point to exhibit the difference between the minute values.

Source: Secondary Research, Press Releases, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 49 INFRASTRUCTURE: DIGITAL TWIN MARKET, BY REGION, 2018–2021 (USD MILLION)

Region	2018	2019	2020	2021
North America	164	191	207	251
Europe	147	169	182	218
Asia Pacific	91	106	115	140
Rest of the World	30	35	37	44
Total	432	500	541	652

Source: Secondary Research, Press Releases, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 50 INFRASTRUCTURE: DIGITAL TWIN MARKET, BY REGION, 2022–2027 (USD MILLION)

Region	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
North America	394	634	1,046	1,735	2,883	4,800	64.9%
Europe	336	532	865	1,413	2,311	3,764	62.1%
Asia Pacific	220	355	588	977	1,628	2,702	65.1%
Rest of the World	67	106	170	275	446	759	62.3%
Total	1,017	1,626	2,669	4,400	7,267	12,025	63.9%

Note: Rest of the World includes South America, Africa, and the Middle East.

Source: Secondary Research, Press Releases, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

The predictive maintenance application is expected to dominate the digital twin market for infrastructure industry during the forecast period. With digital twin, building maintenance businesses are able to accurately forecast when and for how long certain components of a ventilation system will be in operation, if they behave as predicted, and whether or not it will be essential to replace a filter, for example. This allows for optimal technician scheduling as well as automatic ordering of essential replacement components a few weeks ahead of time.

The digital twin market for infrastructure is expected to grow at the highest CAGR in Asia Pacific during the forecast period. The advent of smart city projects and smart transportation initiatives are expected to boost the demand for digital twin solutions in infrastructure. For instance, through the smart city initiatives, the Government of Orissa, India, is upgrading its road infrastructure in Bhubaneswar and implementing advanced traffic management systems. The Land Transport Authority (LTA), Singapore, along with Intelligent Transportation Society Singapore (ITSS), has developed a master plan, Smart Mobility 2030, to enhance the travelers' experience and the country's existing transportation systems. All these initiatives are expected to drive the digital twin market for infrastructure during the forecast period.

9.5 HEALTHCARE

9.5.1 GROWING TREND OF DIGITALIZATION IN HEALTHCARE TO BOOST ADOPTION OF DIGITAL TWIN TECHNOLOGY

Digital transformation in the healthcare industry is expected to boost the adoption of digital twins. The innovations and technological advancements in the healthcare industry have eased the simulation of the medical devices and monitoring and diagnosis of patients. The use of digital twins in the healthcare industry will reduce the complexities associated with handling and operating medical devices, such as MRI and X-rays instruments, and CT scanners. The use of virtual care appointments, wearable wellness apps, and smartphones as diagnostic tools is likely to increase in the coming years. The IoT-enabled medical devices approved by the Food and Drug Administration (FDA) are already in use in hospitals to track and monitor patient status, in homes to manage long-term conditions and medications, and in clinics to support basic treatment processes. Adopting digital twins of medical systems and processes makes the approaches to care easy, and these new experiences lead to a rise in patient expectations and enable doctors to provide continuous medical care to patients.

Healthcare is one of the most promising industries for digital twins. Digital twins are used in applications/phases such as monitoring, diagnostics, and prognostics. In January 2022, Dassault Systèmes (France) showcased a digital twin of the human brain at the CES event 2022 in Las Vegas, US. The company had earlier developed the digital twin of the human heart called the Living Heart, which took 24 months to build. This virtual heart has been generated by combining 208,561 tetrahedrons containing electrical and muscular properties. Additionally, the doctors at the Bioengineering Institute in Auckland, New Zealand, have created a virtual lung with 300 million alveoli that function similar to flesh and blood. Similarly, Q Bio (US), a digital twin startup, offers the innovative Gemini Digital Twin platform for full-body scan. Q Bio claims that it can capture a full-body scan in 15 minutes without utilizing radiation or breath holds, and that it can do so using new computational physics models that are more precise than traditional MRI for many diagnoses. These innovations would help to monitor the organs and provide relevant treatment in case of any abnormalities.

In addition, the medical device industry has been dynamically adopting digital simulation tools for product life cycle management across the ideation, realization, and utilization phases of a device life cycle. A digital twin offers differentiated capabilities at each phase of the life cycle. Digital twins serve as models for simulation and, thus, enable users to proactively and effectively predict, optimize, and realize the engineering-based performance and quality of both product manufacturing operations and designs. This technology efficiently eliminates the errors in medical devices through a data-driven navigational process, which significantly reduces the issues associated with the quality of medical devices.

A digital twin would help virtualize the hospital system and medical devices to create a safe environment and test the impact of potential changes on system performance. Apart from operations, digital twins can also help improve the quality of health services delivered to patients. For instance, Philips (Netherlands) conducted simulations in the development of a portable oxygen generator for patients with breathing problems. By using digital prototypes, the company was able to perform tests that would have taken months and several iterations if it had to build physical prototypes.

A digital twin of a human body could be developed in an organ-by-organ manner, using analytics software and simulation tools. This will create a highly personalized diagnostics model capable of predicting upcoming illnesses, recommending lifestyle or dietary changes, and even calculating the age at the death of a human being. This, in turn, can help find solutions to extend life through healthier living or early intervention treatments. However, capturing the complex behavior and responses of the highly advanced organs, such as the brain and neural network, will be complex and critical while creating a perfect human digital twin. Meanwhile, current advancements are already helping life sciences companies to optimize their outcomes. Digital human organs are being increasingly prototyped and are likely to become the norm

in the near future. For example, a surgeon can use a digital twin for digital visualization of the heart before operating it. The industry is using digital twin solutions to provide personalized services and for performance monitoring to tackle the pandemic.

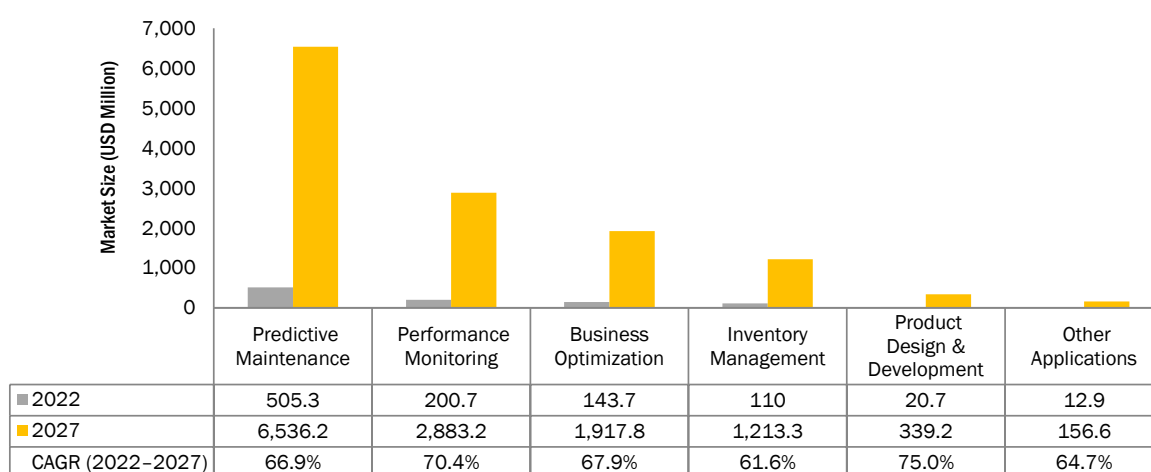
TABLE 51 HEALTHCARE: DIGITAL TWIN MARKET, BY APPLICATION, 2018–2021 (USD MILLION)

Application	2018	2019	2020	2021
Product Design & Development	6.6	8.3	9.5	12.3
Performance Monitoring	73.6	88.9	98.1	123.1
Predictive Maintenance	202.9	239.4	258.3	316.9
Inventory Management	49.9	57.2	59.8	71.1
Business Optimization	55.5	66.2	72.2	89.4
Other Applications	5.4	6.3	6.8	8.2
Total	394.0	466.3	504.6	621.0

Note: Other applications include environmental analysis, customized services, instore planning & security implementation, and energy management. The values have been represented in one decimal point to exhibit the difference between the minute values.

Source: Secondary Research, Press Releases, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

FIGURE 43 PREDICTIVE MAINTENANCE APPLICATION TO ACCOUNT FOR LARGEST SHARE OF DIGITAL TWIN MARKET FOR HEALTHCARE DURING FORECAST PERIOD



Note: The values have been represented in one decimal point to exhibit the difference between the minute values.

Source: Secondary Research, Press Releases, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 52 HEALTHCARE: DIGITAL TWIN MARKET, BY APPLICATION, 2022–2027 (USD MILLION)

Application	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Product Design & Development	20.7	35.5	62.5	109.9	193.2	339.2	75.0%
Performance Monitoring	200.7	334.7	572.1	980.4	1,681.0	2,883.2	70.4%
Predictive Maintenance	505.3	824.5	1,379.8	2,315.6	3,889.2	6,536.2	66.9%
Inventory Management	110.0	173.9	281.7	457.0	741.0	1,213.3	61.6%
Business Optimization	143.7	236.3	398.4	673.7	1,140.2	1,917.8	67.9%
Other Applications	12.9	20.8	34.4	57.0	94.4	156.6	64.7%
Total	993.3	1,625.7	2,728.9	4,593.7	7,739.1	13,046.3	67.4%

Note: The values have been represented in one decimal point to exhibit the difference between the minute values.

Source: Secondary Research, Press Releases, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 53 HEALTHCARE: DIGITAL TWIN MARKET, BY REGION, 2018–2021 (USD MILLION)

Region	2018	2019	2020	2021
North America	138	163	177	218
Europe	158	187	203	251
Asia Pacific	75	88	95	117
Rest of the World	24	27	29	35
Total	394	466	505	621

Source: Secondary Research, Press Releases, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 54 HEALTHCARE: DIGITAL TWIN MARKET, BY REGION, 2022–2027 (USD MILLION)

Region	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
North America	351	576	971	1,641	2,775	4,697	68.0%
Europe	405	671	1,138	1,935	3,294	5,610	69.1%
Asia Pacific	186	304	509	854	1,435	2,416	67.0%
Rest of the World	51	75	111	164	235	324	44.7%
Total	993	1,626	2,729	4,594	7,739	13,046	67.4%

Source: Secondary Research, Press Releases, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

The Europe region is expected to dominate the digital twin market for healthcare during the forecast period. According to the European Federation of Pharmaceutical Industries and Associations (EFPIA), Europe is one of the leading regions for the development of pharmaceuticals and healthcare R&D. The continuous R&D initiatives by European digital twin providers for monitoring human organs is expected to drive the market. For instance, Siemens (Germany) and Dassault Systèmes (France) have been working on virtual representation of the human heart so that diseases can be predicted earlier, and damages can be mitigated in heart patients. Dassault Systèmes (France) has also developed a digital twin of the human brain focused on preventing dementia. The Living Brain Project of the company can be applied to treat traumatic brain injury, neuromodulation, neurodegenerative disease progression, and drug delivery

through cerebrospinal fluid. Hence, the predictive maintenance application is expected to account for the largest share of the digital twin market for the healthcare industry. Such initiatives will further drive the market over the next four to five years.

9.6 ENERGY & UTILITIES

9.6.1 DIGITAL TWIN HELPS UTILITIES TO IMPROVE PRODUCTIVITY AND EMERGENCY RESPONSE

The demand for energy is increasing and is estimated to grow significantly in the next five years. The emerging economies such as Africa, the Middle East, and South-East Asia are the major contributors to the rising demand due to the increasing population, GDP growth, and enhanced access to electricity. Moreover, there is a shift toward the use of renewable energy sources.

In wind farms, digital twins are being used to display wind patterns, electrical output, and wear and tear on equipment. They help improve the efficiency of operations. For instance, GE (US) has initiated the Digital Wind Farm project built on the Predix platform. This software platform creates digital twins of a windmill, which comprise a digital infrastructure for the wind farm, enabling operators to collect, visualize, and analyze unit- and site-level data. Through the constant collection of this data, component information, service reports, and performance data of similar wind turbine models can be obtained from the predictive model for the digital twin. These predictive models serve as the basis of the new design and simulation applications, allowing wind farm operators to optimize maintenance strategy, improve reliability and availability, and increase annual energy production.

Also, the adoption of digital twin technology has created new ways to improve productivity and optimize all critical parts of production lines in the power industry. For instance, GE (US) developed a software and hardware system that creates a virtual digital twin of an entire industrial power plant complex. This Digital Power Plant is powered by GE's Predix industrial internet operating system, which is a collection of physics-based methods and digital technologies that are used to model the present state of every asset in a digital power plant. As the global economy and environment demand real-time control and precision, this technology enables monitoring and managing every aspect of the power generation ecosystem to generate electricity efficiently and safely. In addition, GE (US) also built a digital twin that generated 100 million hours of virtual power plant data and is demonstrated to Aluminium Bahrain B.S.C. (Alba) (Bahrain), which was planning to build a single-site aluminum smelter that would generate a large amount of ~800 megawatts electricity. The twin of this gas turbine represented how the turbines would react to different situations in Bahrain.

In August 2021, ENGIE (France), one of the world's largest providers of energy efficiency services, approached ANSYS for simulation technologies to assist ENGIE in making the transition to carbon-free energy. ENGIE Lab CRIGEN is developing an ultra-fast and high-fidelity platform to deliver the quality of 3D computational fluid dynamics (CFD) results in real-time, reducing companies' costs, environmental impact and time to market. Similarly, Doosan (South Korea), a global leader in the energy & utilities industry, recognized that establishing a digital twin of an existing wind farm would be the most cost-efficient and effective approach to combine all of the numerous data sources required to understand its wind farm operations. A wind farm digital twin integrates real-time Internet of Things (IoT) data, as well as other wind farm and meteorological data, with physics-based and machine learning models to accurately anticipate output for each turbine in the farm.

Water utility companies employ digital twins to assure a constant supply of water and to be better prepared for emergencies. They can acquire an accurate assessment of how the current water system functions, discover issues before they occur, and model what-if situations using digital copies. Water utilities build virtual models of water systems based on data captured by sensors and actuators in the actual system. They also leverage data from water industry information systems, including computer maintenance

management systems (CMMS), geographic information systems (GIS), and supervisory control and data acquisition (SCADA).

TABLE 55 ENERGY & UTILITIES: DIGITAL TWIN MARKET, BY APPLICATION, 2018–2021 (USD MILLION)

Application	2018	2019	2020	2021
Product Design & Development	9.4	10.5	10.7	12.6
Performance Monitoring	77.6	91.7	99.1	123.5
Predictive Maintenance	235.6	274.2	291.9	358.5
Inventory Management	60.0	69.3	73.2	89.1
Business Optimization	62.5	75.0	82.3	104.0
Other Applications	7.2	8.1	8.3	9.8
Total	452.4	528.9	565.4	697.5

Note: The values have been represented in one decimal point to exhibit the difference between the minute values.

Source: Secondary Research, Press Releases, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 56 ENERGY & UTILITIES: DIGITAL TWIN MARKET, BY APPLICATION, 2022–2027 (USD MILLION)

Application	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Product Design & Development	18.3	27.1	41.3	62.9	95.7	145.2	51.4%
Performance Monitoring	191.1	303.2	494.2	809.1	1,327.7	2,183.2	62.8%
Predictive Maintenance	546.7	854.8	1,373.3	2,216.3	3,585.4	5,812.3	60.4%
Inventory Management	134.8	208.9	332.7	532.2	853.2	1,352.9	58.6%
Business Optimization	163.2	262.4	433.3	718.5	1,193.5	2,003.2	65.1%
Other Applications	14.2	21.3	32.5	49.7	76.1	116.1	52.1%
Total	1,068.3	1,677.7	2,707.3	4,388.7	7,131.6	11,613.0	61.2%

Note: The values have been represented in one decimal point to exhibit the difference between the minute values.

Source: Secondary Research, Press Releases, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 57 ENERGY & UTILITIES: DIGITAL TWIN MARKET, BY REGION, 2018–2021 (USD MILLION)

Region	2018	2019	2020	2021
North America	167	197	211	262
Europe	154	179	191	235
Asia Pacific	100	117	125	155
Rest of the World	32	36	38	45
Total	452	529	565	697

Source: Secondary Research, Press Releases, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 58 ENERGY & UTILITIES: DIGITAL TWIN MARKET, BY REGION, 2022–2027 (USD MILLION)

Region	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
North America	403	637	1,032	1,682	2,746	4,529	62.2%
Europe	358	560	899	1,450	2,345	3,800	60.4%
Asia Pacific	238	375	606	984	1,602	2,578	61.0%
Rest of the World	69	107	170	273	439	706	59.4%
Total	1,068	1,678	2,707	4,389	7,132	11,613	61.2%

Source: Secondary Research, Press Releases, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

The predictive maintenance segment is expected to grow at the highest CAGR in the digital twin market for energy & utilities industry during the forecast period. Utility companies can use digital twins to swiftly predict the possible leaks and prevent water loss. They can evaluate different techniques of water system operations using virtual simulations to improve emergency response, increase water supply dependability, and save energy. Similarly, energy companies can use digital twins to improve product performance throughout the operation and provide insight into predictive analytics and asset performance management decisions, regardless of the environment.

9.7 OIL & GAS

9.7.1 OIL & GAS INDUSTRY USES DIGITAL TWIN FOR PREDICTIVE MAINTENANCE

In the oil & gas industry, a digital twin is used to predict the behavior of machines, systems, factories, and processes, which is necessary for future productivity and infrastructure growth. Digital transformation in the oil & gas industry plays a crucial role in managing, optimizing, and analyzing big data related to plant operations. IoT analytic platforms are useful in providing real-time analytics that directly drives process improvements and production optimization. The twin continuously detects and creates new relevant data to optimize business outcomes. Every smart machine within every system gets connected, after which the data is sent to communication site links and eventually to the cloud or control centers for consolidation at the application level. As the digital twin depends on IoT for creating a virtual model, the oil & gas industry has substantial opportunities to enhance the efficiencies and workflows at a lower cost.

Moreover, these virtual models of processes and assets can be used with knowledge management systems to train new workers. With the help of smart manufacturing and digital twin, enterprises (Oil & gas companies) will be able to respond quickly to customer demands, minimize energy consumption, and improve material usage. Organizations will be able to meet new requirements faster, resulting from variations in market demand, feedstock, and price. It will also help organizations improve innovation, productivity, competitiveness, and sustainability.

TABLE 59 OIL & GAS: DIGITAL TWIN MARKET, BY APPLICATION, 2018–2021 (USD MILLION)

Application	2018	2019	2020	2021
Product Design & Development	6.14	6.88	7.03	8.28
Performance Monitoring	54.46	61.42	63.30	75.21
Predictive Maintenance	200.36	229.04	239.33	288.39
Inventory Management	43.76	49.71	51.62	61.81
Business Optimization	65.34	76.82	82.50	102.10
Other Applications	9.33	10.22	10.20	11.72
Total	379.39	434.08	453.97	547.50

Note: The values have been represented in two decimal points to exhibit the difference between the minute values.

Source: Secondary Research, Press Releases, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 60 OIL & GAS: DIGITAL TWIN MARKET, BY APPLICATION, 2022–2027 (USD MILLION)

Application	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Product Design & Development	12.07	18.01	27.57	42.37	65.17	100.33	52.7%
Performance Monitoring	110.24	165.55	255.23	394.92	611.94	949.25	53.8%
Predictive Maintenance	429.85	656.61	1,030.01	1,622.19	2,559.43	4,036.25	56.5%
Inventory Management	91.27	138.10	214.57	334.68	522.90	821.91	55.2%
Business Optimization	156.74	246.40	397.48	643.30	1,042.30	1,693.99	61.0%
Other Applications	16.61	24.06	35.68	52.95	78.41	115.76	47.5%
Total	816.77	1,248.72	1,960.56	3,090.40	4,880.15	7,717.50	56.7%

Note: The values have been represented in two decimal points to exhibit the difference between the minute values.

Source: Secondary Research, Press Releases, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 61 OIL & GAS: DIGITAL TWIN MARKET, BY REGION, 2018–2021 (USD MILLION)

Region	2018	2019	2020	2021
North America	137	155	160	191
Europe	133	149	152	179
Asia Pacific	76	89	95	116
Rest of the World	34	42	47	61
Total	379	434	454	547

Note: Rest of the World includes South America, Africa, and the Middle East.

Source: Secondary Research, Press Releases, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 62 OIL & GAS: DIGITAL TWIN MARKET, BY REGION, 2022–2027 (USD MILLION)

Region	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
North America	283	430	671	1,050	1,646	2,585	55.6%
Europe	259	383	581	884	1,344	2,045	51.2%
Asia Pacific	176	274	436	697	1,116	1,790	59.0%
Rest of the World	98	162	273	460	773	1,297	67.5%
Total	817	1,249	1,961	3,090	4,880	7,718	56.7%

Note: Rest of the World includes South America, Africa, and the Middle East.

Source: Secondary Research, Press Releases, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

The predictive maintenance application is expected to hold the largest share of the digital twin market for oil & gas industry during the forecast period. Oil & gas companies use extremely complex and expensive equipment, such as boilers, pumps, generators, and piping networks, which consume a lot of power and generate a large amount of heat. The oil and gas business can profit from predictive digital twins in a way that no other industry can. Oil and gas engineers can more precisely predict real-world behaviors of equipment or systems with the help of simulation where data is lacking. Predictive engineering analytics can cover the data gap in cases when data is unavailable or insufficient, resulting in a predictive digital twin that simulates the asset's or system's real-world behavior.

9.8 AGRICULTURE

9.8.1 FOCUS ON REAL-TIME MONITORING OF CROPS AND AGRICULTURAL MACHINES TO BOOST DIGITAL TWIN ADOPTION

The use of digital twins in the agriculture industry helps in remote monitoring, crop and soil health monitoring, and livestock and farm management. Digital twin in farm management considerably removes the constraints regarding place, time, and human observation. The digital twin of a farm or field helps farmers and other stakeholders to act and take precautions immediately in case of deviations and emergencies. For example, Dattabot (Indonesia), a big data analytics company, leveraged the Predix platform offered by GE (US) as its development backbone and developed HARA solutions, which comprise features such as data acquisition, powerful analytics, and precision agriculture insights. This solution is designed to transform the agricultural practices in Indonesia in the modern digital industrial age.

Amazon Web Services (AWS) (US) is among the few digital twin providers in the market offering digital twin solutions for the agriculture industry. The company offers a predictive digital twin solution for the agriculture industry. The predictions from the digital twin models can be exhibited in context-relevant visualizations (e.g., 2D dashboard/3D/AR/VR). For instance, in the agriculture industry, AWS offers digital twin support to Descartes Labs (Mexico), a customer of AWS, that uses satellite imagery to analyze crop health at a continental size. It also analyzes pricing, supply and demand, and other market data to forecast commodity markets. End customers utilize these estimates to determine the optimum hedging or trading strategies, which can result in millions of dollars in savings or trading profits.

TABLE 63 AGRICULTURE: DIGITAL TWIN MARKET, BY APPLICATION, 2018–2021 (USD MILLION)

Application	2018	2019	2020	2021
Performance Monitoring	-	7.91	8.01	9.58
Predictive Maintenance	-	5.93	6.09	7.33
Inventory Management	-	-	0.12	0.15
Business Optimization	-	8.90	9.14	10.83
Other Applications	-	-	0.20	0.23
Total	-	22.75	23.56	28.13

Note: The values have been represented in two decimal points to exhibit the difference between the minute values.

Source: Secondary Research, Press Releases, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 64 AGRICULTURE: DIGITAL TWIN MARKET, BY APPLICATION, 2022–2027 (USD MILLION)

Application	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Performance Monitoring	14.16	21.44	33.31	51.91	80.99	126.42	54.9%
Predictive Maintenance	10.90	16.61	25.96	40.71	63.89	100.33	55.9%
Inventory Management	0.23	0.37	0.60	0.97	1.58	2.61	62.4%
Business Optimization	15.87	23.82	36.67	56.64	87.56	135.39	53.5%
Other Applications	0.34	0.51	0.77	1.18	1.80	2.76	51.9%
Total	41.51	62.73	97.31	151.41	235.83	367.50	54.7%

Note: The values have been represented in two decimal points to exhibit the difference between the minute values.

Source: Secondary Research, Press Releases, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 65 AGRICULTURE: DIGITAL TWIN MARKET, BY REGION, 2018–2021 (USD MILLION)

Region	2018	2019	2020	2021
North America	-	4.55	4.70	5.59
Europe	-	6.82	7.02	8.33
Asia Pacific	-	9.10	9.51	11.46
Rest of the World	-	2.27	2.33	2.74
Total	-	22.75	23.56	28.13

Note: The values have been represented in two decimal points to exhibit the difference between the minute values.

Source: Secondary Research, Press Releases, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 66 AGRICULTURE: DIGITAL TWIN MARKET, BY REGION, 2022–2027 (USD MILLION)

Region	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
North America	8.26	12.49	19.40	30.22	47.12	73.50	54.8%
Europe	12.20	18.29	28.13	43.42	67.06	103.64	53.4%
Asia Pacific	17.06	25.99	40.65	63.77	100.13	157.29	55.9%
Rest of the World	3.99	5.96	9.12	14.01	21.52	33.08	52.6%
Total	41.51	62.73	97.31	151.41	235.83	367.50	54.7%

Note: The values have been represented in two decimal points to exhibit the difference between the minute values.

Source: Secondary Research, Press Releases, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

The performance monitoring application is expected to dominate the digital twin market for agriculture industry during the forecast period. The monitoring of the health of crops is a vital in the agriculture industry and digital twin plays a prominent role in continuous and real-time monitoring of crops and their health over time. Digital twins not only monitor crops but also monitor the health cattle in agricultural farms, machineries, equipment, and water levels.

9.9 TELECOMMUNICATIONS

9.9.1 5G COMMERCIALIZATION WOULD FUEL DEMAND FOR DIGITAL TWIN IN TELECOMMUNICATIONS

Product/process/system digital twins can help communication service providers to optimize their network solutions, processes, and customers services, and how they impact one another. Digital twins can be used for network design, tower management, and field service management. Once 5G gets commercialized across the globe, the demand for digital twin solutions is expected to increase at a significant rate.

In the telecommunications industry, a digital twin can monitor and manage complicated telecom networks while simultaneously gathering data from social media. A digital twin can monitor and augment these complex systems in real time as operators add layers of technology, such as connected devices and additional frequency bands. For instance, Nokia (Finland) unveiled the Nokia 5G Digital Design concept, which aims to emulate 5G use cases. The platform can use digital twin technology to instantly monitor and assess the impact of 5G deployment while also delivering automated recommendations with the integration of machine learning algorithms. Nokia also collaborated with digital twin provider Robert Bosch (Germany) for studying the use of 5G for industrial automation and control systems in manufacturing. As IoT implementations provide access to large data and broad digital ecosystems, high-fidelity digital twins are becoming easier to build and maintain. In the future, digital twin technology will be one of the most important software tools for transforming product development.

TABLE 67 TELECOMMUNICATIONS: DIGITAL TWIN MARKET, BY APPLICATION, 2018–2021 (USD MILLION)

Application	2018	2019	2020	2021
Product Design & Development	-	-	0.36	0.64
Performance Monitoring	-	-	0.40	1.39
Predictive Maintenance	-	-	0.47	1.98
Inventory Management	-	-	0.51	1.18
Business Optimization	-	34.12	34.15	39.00
Other Applications	-	-	0.58	0.81
Total	-	34.12	36.48	45.00

Note: The values have been represented in two decimal points to exhibit the difference between the minute values.

Source: Secondary Research, Press Releases, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 68 TELECOMMUNICATIONS: DIGITAL TWIN MARKET, BY APPLICATION, 2022–2027 (USD MILLION)

Application	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Product Design & Development	1.05	1.74	2.97	5.06	8.62	14.70	69.6%
Performance Monitoring	2.18	3.52	5.83	9.68	16.10	26.83	65.1%
Predictive Maintenance	3.06	4.86	7.91	12.95	21.24	34.91	62.7%
Inventory Management	1.75	2.64	4.09	6.36	9.91	15.44	54.6%
Business Optimization	59.40	92.78	148.91	240.08	388.02	628.43	60.3%
Other Applications	1.26	2.01	3.29	5.40	8.91	14.70	63.5%
Total	68.70	107.54	172.99	279.53	452.80	735.00	60.6%

Note: The values have been represented in two decimal points to exhibit the difference between the minute values.

Source: Secondary Research, Press Releases, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 69 TELECOMMUNICATIONS: DIGITAL TWIN MARKET, BY REGION, 2018–2021 (USD MILLION)

Region	2018	2019	2020	2021
North America	-	14.0	15.1	18.7
Europe	-	10.9	11.7	14.4
Asia Pacific	-	6.5	6.9	8.4
Rest of the World	-	2.7	2.9	3.4
Total	-	34.1	36.5	45.0

Note: The values have been represented in one decimal point to exhibit the difference between the minute values.

Source: Secondary Research, Press Releases, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 70 TELECOMMUNICATIONS: DIGITAL TWIN MARKET, BY REGION, 2022–2027 (USD MILLION)

Region	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
North America	28.8	45.5	73.8	120.4	196.6	321.9	62.0%
Europe	22.1	34.6	55.7	90.2	146.4	238.9	61.0%
Asia Pacific	12.8	20.0	32.1	51.8	83.6	135.2	60.1%
Rest of the World	5.0	7.4	11.3	17.2	26.1	39.0	50.9%
Total	68.7	107.5	173.0	279.5	452.8	735.0	60.6%

Note: The values have been represented in one decimal point to exhibit the difference between the minute values.

Source: Secondary Research, Press Releases, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

Digital twins can help forecast breakdown points during periods of extraordinarily heavy network demand, such as after natural catastrophes, by allowing modelling of existing network architecture. This twin is particularly valuable because it can account for everything, from weather patterns to the position of street furniture, pedestrian walkways, and road markings, including their potential impact on signal strength and overall network health. 5G-based digital twins enable machine operators to monitor and control many aspects of a production process without having to access individual machine data, decreasing burden and offering a more accurate view of the present operational condition. As a result, it enhances the overall business optimization.

9.10 RETAIL

9.10.1 DIGITAL TWIN ENABLES TRACKING OF PRODUCT ACROSS SUPPLY CHAIN FOR INVENTORY MANAGEMENT

Digitalization in a retail store does not need personal data and helps optimize the customer experience and in data-driven marketing that focuses on the individual. In a retail store, operators need to know the object identity, which enables a marketing function to send marketing and sales messages directly to the digital twin. For example, a customer with its registered objects in an application will get an alert whenever a message from a marketing function arrives in one of its digital twins, and the customer or operator can decide to disable the alert or take access away for the third parties connected to it. At present, object marketing is one of the powerful new marketing tools for maintaining the balance between vendors and customers. In addition, object owners can connect with each other to share their product experience by sending messages among each digital twins created in smartphones. The customers could disable this community feature as per their needs.

The use of digital twin in consumer goods offers optimized distribution of goods to the customers through shipment tracking and monitoring, efficient delivery route design, security, and loss prevention. Additionally, digital twin can facilitate preventive maintenance, a prediction regarding the loss of raw materials, intelligent consumption of energy, and tracking of assets.

Dassault Systèmes (France), a 3D modelling and digital twin provider, collaborated with Store Electronic Systems (France), a retail technology business, to create digital twins of retail establishments. A 3D model of a store's architecture is populated with data from electronic shelf labels to create a detailed virtual depiction of the store and its contents in the system. The model is updated in real time to reflect the quantity of items stored on the shelves due to links to the store's inventory and point-of-sale systems. Dassault Systèmes claims that its approach might be used for a variety of applications, including guiding shoppers to the products on their shopping list. The approach can also help to optimize merchandise layouts and prevent stock-outs.

TABLE 71 RETAIL: DIGITAL TWIN MARKET, BY APPLICATION, 2018–2021 (USD MILLION)

Application	2018	2019	2020	2021
Product Design & Development	-	-	0.13	0.20
Performance Monitoring	-	-	0.18	0.29
Predictive Maintenance	-	-	0.16	0.23
Inventory Management	-	11.37	11.06	12.90
Business Optimization	-	-	0.26	0.74
Other Applications	-	-	0.22	0.27
Total	-	11.37	12.01	14.63

Note: The values have been represented in two decimal points to exhibit the difference between the minute values.

Source: Secondary Research, Press Releases, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 72 RETAIL: DIGITAL TWIN MARKET, BY APPLICATION, 2022–2027 (USD MILLION)

Application	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Product Design & Development	0.33	0.54	0.91	1.54	2.61	4.41	68.4%
Performance Monitoring	0.48	0.79	1.35	2.30	3.90	6.62	69.2%
Predictive Maintenance	0.38	0.62	1.05	1.78	3.01	4.96	67.3%
Inventory Management	19.31	29.63	46.73	74.00	117.45	186.87	57.5%
Business Optimization	1.14	1.82	2.98	4.89	8.04	13.23	63.2%
Other Applications	0.41	0.64	1.03	1.67	2.72	4.41	60.9%
Total	22.04	34.06	54.06	86.19	137.73	220.50	58.5%

Note: The values have been represented in two decimal points to exhibit the difference between the minute values.

Source: Secondary Research, Press Releases, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 73 RETAIL: DIGITAL TWIN MARKET, BY REGION, 2018–2021 (USD MILLION)

Region	2018	2019	2020	2021
North America	-	4.89	5.19	6.36
Europe	-	3.64	3.82	4.63
Asia Pacific	-	2.27	2.41	2.94
Rest of the World	-	0.57	0.59	0.69
Total	-	11.4	12.0	14.6

Note: Rest of the World includes South America, Africa, and the Middle East. The values have been represented in two decimal points to exhibit the difference between the minute values.

Source: Secondary Research, Press Releases, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 74 RETAIL: DIGITAL TWIN MARKET, BY REGION, 2022–2027 (USD MILLION)

Region	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
North America	9.64	14.98	23.92	38.35	61.63	99.23	59.4%
Europe	6.93	10.64	16.79	26.61	42.26	67.47	57.6%
Asia Pacific	4.45	6.90	10.98	17.56	28.15	44.98	58.8%
Rest of the World	1.02	1.53	2.37	3.66	5.68	8.82	54.0%
Total	22.0	34.1	54.1	86.2	137.7	220.5	58.5%

Note: Rest of the World includes South America, Africa, and the Middle East. The values have been represented in two decimal points to exhibit the difference between the minute values.

Source: Secondary Research, Press Releases, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

The inventory management application is expected to account for the largest share of the digital twin market for retail industry during the forecast period. The retail industry is focusing on the construction of digital twins that follow product flow across supply chains. This can be done by developing systems that may extract significant insights from aggregated data produced by vast numbers of very simple models. This enables to manage the inventory by keeping a real-time track record of products across the supply chain. Meanwhile, firms are building increasingly complex customer behavior models for retail and e-commerce applications. Many organizations are already using data on past purchases, web surfing history, and social media activity to target advertising and promotions to specific customers and circumstances. These models may evolve into consumer digital twins over time, which are complicated, multi-attribute behavioral models that aim to forecast future actions and influence purchase decisions proactively.

9.11 OTHER INDUSTRIES

Other industries include the semiconductors, chemicals, paper & pulp, and food & beverages industries.

The semiconductors industry is at the forefront of adopting digital twin technology. The technology is specifically adopted at the initial stage, i.e., designing and simulation phase, in the semiconductors industry. In February 2022, ANSYS (US), a provider of digital twins, announced that it will assist LG Electronics (South Korea) on strengthening its sustainability and digital transformation efforts. To help LG achieve its goal, ANSYS will offer its CAE and advanced simulation solutions, including artificial intelligence/machine learning (AI/ML), high-performance computing, and digital twins. Similarly, Siemens signed a partnership agreement with Bridg (Florida) for creating a digital twin in the semiconductor manufacturing process. Siemens PLM provided an in-kind software grant valued at more than USD 30 million for the Bridg wafer fab in NeoCity. This digital twin program established the digital duplicates for model-based systems that lay the foundation for a semiconductor design for manufacturing. The twining helped Bridg develop its commercialization infrastructure, including capabilities for proof of concept, custom development, and pilot production.

The chemicals industry has been seeking opportunities to implement digital technologies, which would help attain operational excellence or offer new products in combination with digital components. The adoption of digital twins in this industry would help to digitally monitor processing and enable the direct route for the production of chemicals. However, the timing and scale of impact cannot be predicted, as these developments in the chemicals industry could fundamentally change the rules of engagement for chemical production. Siemens AG (Germany) released the STAR-CCM+ software with new features to help product development organizations to enhance and accelerate software ability. This software is capable of digitally simulating and predicting product performance in the real world by creating digital twins. STAR-CCM+ v12.02 is an updated version that includes multi-component gas/liquid species and solid ion models used to analyze electrochemical reactions occurring in solid oxide fuel cells (SOFCs). STAR-CCM+

enables users to create a digital twin to improve the efficiency of SOFCs, which are increasingly being used to generate efficient and clean electricity.

Food and beverages production consists of complex processes, such as clean-in-place, traceability, capturing mislabeled products, and sterilization. Along with these, the industry needs to be compliant with government regulations for food safety, nutrition and health, and sustainability. Smart manufacturing and digital twin offer food & beverage businesses efficient connectivity, data collection capabilities, and speed. They can simplify food safety controls by collecting operational data throughout a production environment. For instance, Bühler (Switzerland), a major developer of equipment (optical sorters and others) for processing meals and producing materials, is gaining more data and enhancing operations by co-innovating digital twins to drive digital transformation with Schneider Electric. Bühler has integrated Schneider Electric's (France) augmented reality application as part of its digital transformation. It used these augmented reality systems in a test to give customized digital information to machine users based on their needs. Similarly, Siemens (Germany) demonstrated the utilization of a digital twin in the process and discrete workflows in the food & beverages industry. The company offers a range of software solutions that provide digital twin capability across the entire value chain of the dairy industry. Moreover, the dairy industry is affected not only by factors such as a high cost sensitivity and mass production but also the growing diversity of available flavors and innovative packaging concepts, which can be overcome using digital twin technology.

Digital twins have become a vital tool for paper manufacturers and machinery businesses that upgrade their operations in order to meet the rising demand for domestic paper products. The industry faces various operational challenges such as unscheduled downtime, paper breaks, lack of stability, and a scarcity of raw materials. In the industry, it is essential for a manufacturer to monitor and manage workflows and optimize transportation and material storage. This can be achieved through the implementation of digital twin. Digital twin technology is adopted by other industries for various processes such as for enhancement of the product development process, optimization of processes, and improvement of performance of the systems to prevent faults or breakdowns.

TABLE 75 OTHER APPLICATIONS: DIGITAL TWIN MARKET, BY APPLICATION, 2018–2021 (USD MILLION)

Application	2018	2019	2020	2021
Product Design & Development	13.75	13.62	14.81	19.34
Performance Monitoring	105.06	107.31	120.33	162.21
Predictive Maintenance	80.55	83.07	94.05	127.98
Inventory Management	51.83	51.63	56.45	74.15
Business Optimization	85.80	88.10	99.31	134.58
Other Applications	13.22	12.63	13.19	16.49
Total	350.21	356.36	398.14	534.75

Note: Other applications include environmental analysis, customized services, instore planning & security implementation, and energy management. The values have been represented in two decimal points to exhibit the difference between the minute values.

Source: Secondary Research, Press Releases, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 76 OTHER APPLICATIONS: DIGITAL TWIN MARKET, BY APPLICATION, 2022–2027 (USD MILLION)

Application	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Product Design & Development	26.30	36.33	51.00	70.98	97.37	130.98	37.9%
Performance Monitoring	227.83	325.40	472.77	681.49	969.34	1,353.43	42.8%
Predictive Maintenance	181.43	261.50	383.35	557.49	799.88	1,126.40	44.1%
Inventory Management	101.44	141.04	199.38	279.48	386.34	523.91	38.9%
Business Optimization	189.99	272.74	398.25	576.93	824.63	1,156.96	43.5%
Other Applications	21.35	27.92	36.87	47.85	60.58	74.22	28.3%
Total	748.33	1,064.93	1,541.63	2,214.22	3,138.14	4,365.90	42.3%

Note: Other applications include environmental analysis, customized services, instore planning & security implementation, and energy management. The values have been represented in two decimal points to exhibit the difference between the minute values.

Source: Secondary Research, Press Releases, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 77 OTHER APPLICATIONS: DIGITAL TWIN MARKET, BY REGION, 2018–2021 (USD MILLION)

Region	2018	2019	2020	2021
North America	133.08	135.81	152.18	204.99
Europe	117.32	118.39	131.16	174.69
Asia Pacific	75.29	77.29	87.10	118.00
Rest of the World	24.51	24.87	27.69	37.08
Total	350.21	356.36	398.14	534.75

Note: Rest of the World includes South America, Africa, and the Middle East. The values have been represented in two decimal points to exhibit the difference between the minute values.

Source: Secondary Research, Press Releases, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 78 OTHER APPLICATIONS: DIGITAL TWIN MARKET, BY REGION, 2022–2027 (USD MILLION)

Region	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
North America	287.69	410.59	596.10	858.62	1,220.39	1,702.70	42.7%
Europe	242.75	343.03	493.07	703.14	989.39	1,366.53	41.3%
Asia Pacific	166.55	239.02	348.92	505.33	722.12	1,012.89	43.5%
Rest of the World	51.34	72.30	103.55	147.12	206.25	283.78	40.8%
Total	748.33	1,064.93	1,541.63	2,214.22	3,138.14	4,365.90	42.3%

Note: Rest of the World includes South America, Africa, and the Middle East. The values have been represented in two decimal points to exhibit the difference between the minute values.

Source: Secondary Research, Press Releases, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

10 REGIONAL ANALYSIS

KEY FINDINGS

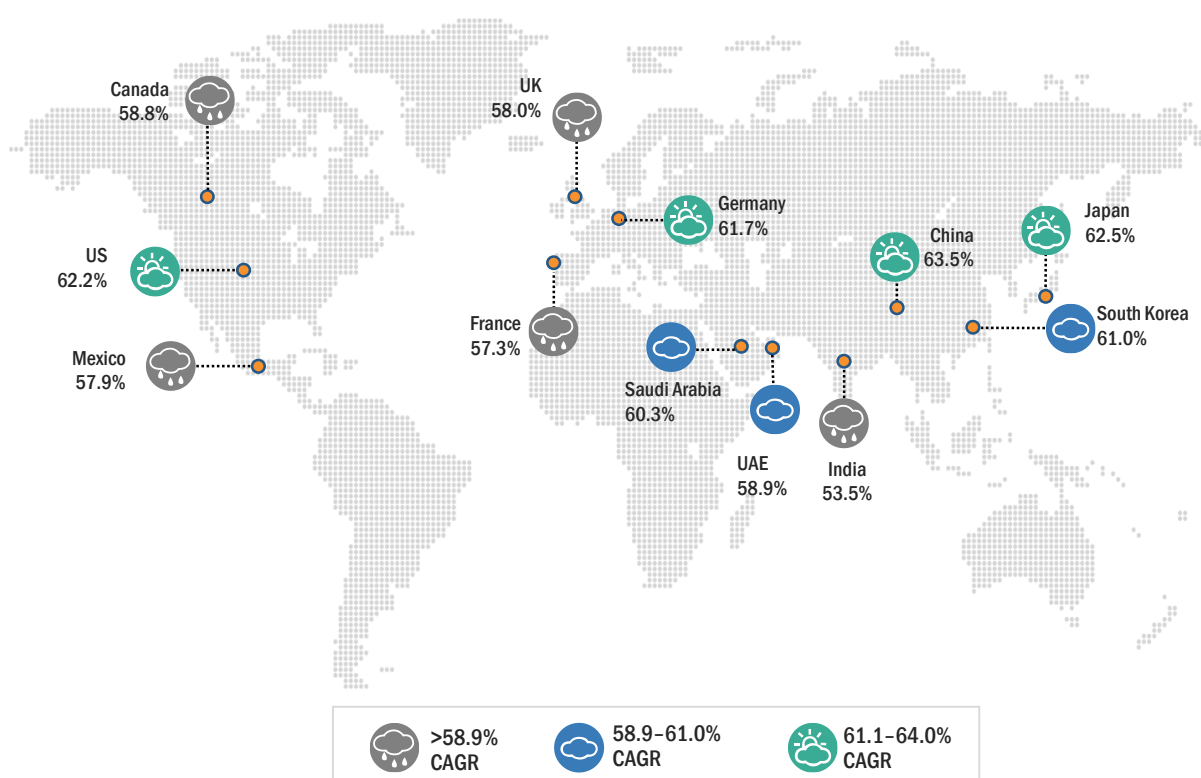
- The digital twin market in North America was estimated at USD 1.7 billion in 2021. It is expected to reach USD 28.5 billion by 2027, growing at the highest CAGR of 61.5% from 2022 to 2027.
- The market in North America is mainly driven by the rising emphasis on digital twin in manufacturing industries to reduce cost and improve supply chain, and development (R&D) investments by manufacturers to drive innovation in their manufacturing processes.
- The US is expected to account for the largest share of the digital twin market in North America during the forecast period. The market in the US is expected to grow at a CAGR of 62.2% during the forecast period to reach USD 23.9 billion by 2027.
- Increasing initiatives to upgrade aging infrastructure and surging adoption in automotive & transportation are expected to be the key drivers for the rapid growth of the digital twin market in the US.
- The digital twin market in Europe was estimated at USD 3.2 billion in 2021. It is expected to grow at the highest CAGR of 59.9% to reach USD 24.3 billion by 2027.

10.1 INTRODUCTION

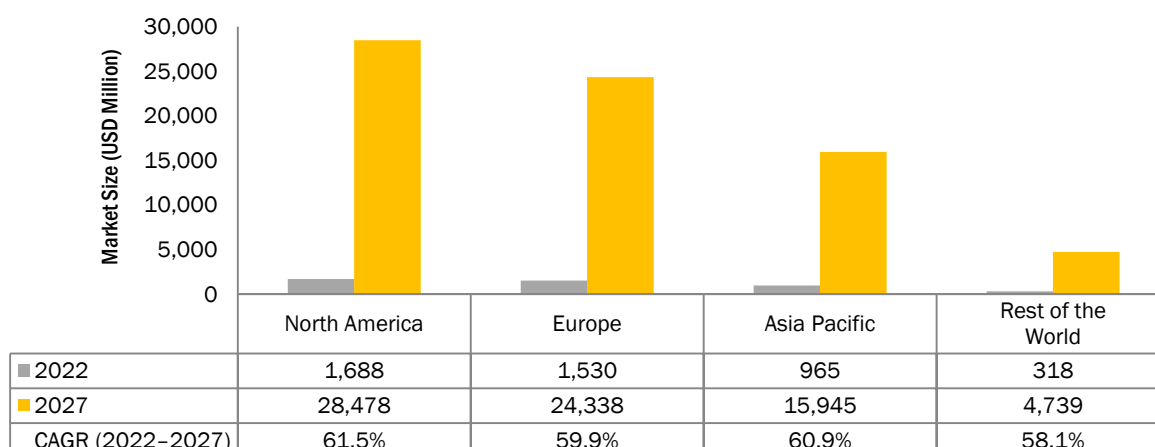
The digital twin market based on region has been segmented into four regions—North America, Europe, Asia Pacific, and the Rest of the World (RoW). The RoW segment includes the Middle East, Africa, and South America. North America has emerged as a prominent region in terms of the development and adoption of digital twins across various industries such as aerospace, infrastructure, energy & utilities, and other industries. The region is expected to witness high adoption of digital twin products/solutions in the next four to five years.

North America dominated the global digital twin market by capturing a share of nearly ~37.5% in 2021, followed by Europe with a share of 34.0%. The North America market is projected to grow at the highest CAGR during the forecast period due to the increasing adoption of digital twin in aerospace, energy & utilities, automotive & transportation, retail, telecommunications, and other industries. The market growth in the region is also expected to be driven by the growing development in related technologies in the region such as 5G networking, internet of things (IoT), extended reality, artificial intelligence (AI), and machine learning (ML). In North America, the US is expected to be the largest and fastest-growing market for digital twin during the forecast period.

FIGURE 44 DIGITAL TWIN MARKET IN CHINA TO REGISTER HIGHEST CAGR FROM 2022 TO 2027



Source: Secondary Research, Press Releases, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

FIGURE 45 NORTH AMERICA TO DOMINATE DIGITAL TWIN MARKET DURING FORECAST PERIOD


Note: Rest of the World includes the Middle East, Africa, and South America.

TABLE 79 DIGITAL TWIN MARKET, BY REGION, 2018–2021 (USD MILLION)

Region	2018	2019	2020	2021
North America	1,085	1,270	1,363	1,688
Europe	1,010	1,173	1,247	1,530
Asia Pacific	616	726	779	965
Rest of the World	208	243	259	318
Total	2,918	3,412	3,648	4,500

Source: Secondary Research, Press Releases, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 80 DIGITAL TWIN MARKET, BY REGION, 2022–2027 (USD MILLION)

Region	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
North America	2,589	4,073	6,586	10,698	17,424	28,478	61.5%
Europe	2,325	3,623	5,803	9,339	15,071	24,338	59.9%
Asia Pacific	1,477	2,318	3,737	6,051	9,822	15,945	60.9%
Rest of the World	479	741	1,174	1,864	2,963	4,739	58.1%
Total	6,870	10,754	17,299	27,953	45,280	73,500	60.6%

Source: Secondary Research, Press Releases, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

The North America region accounted for the largest share of the digital twin market in 2021. It is further expected to dominate the market throughout the forecast period while growing at the highest CAGR from 2022 to 2027. The major factors driving the growth of the digital twin market in North America are the rising demand for smart sensors due to increasing automation in industries; growing adoption of technologies such as IoT, and Industrial IoT (IIoT); and the increasing need to optimize productivity and reduce operational and maintenance costs. North America is one of the early adopters of digital twin technology. The region leads in terms of advanced automation technology and the use of digital twin and related/complementary technologies. All these factors contribute to the growth of the market in North

America. Improving business outlook in the US and the government's steps to reform tax codes, significant package announcements for manufacturing and infrastructure companies, and the presence of major digital twin technology providers are boosting the adoption of digital twin in the region. Factors such as growing focus on product design & development; business optimization; increasing stringent government regulations for the workplace and personal safety (considering human-centric approach for digital twin); and rising awareness of the need to control and assure output quality in aerospace, energy & utilities, automotive & transportation, and food & beverages industries are driving the demand for real-time performance monitoring and predictive maintenance applications of digital twin in North America.

10.2 NORTH AMERICA

The market in North America has been further segmented into the US, Canada, and Mexico. North America is the leading region in the digital twin market owing to the increasing adoption of the technology in the aerospace, automotive & transportation, and energy & utilities industries. In recent years, the development of various software and deployment of automation solutions in the manufacturing industries have improved production lines and downstream operations. These developments have positively impacted the growth of the digital twin market in North America.

Increased R&D in the field of the IoT and IIoT and surging demand for efficient and cost-effective technologies for product manufacturing are also supporting the regional market growth. Further, the adoption of automation and enabling technologies in the above-mentioned industries is rising owing to the need to gain real-time data and visibility into production activities. Key factors driving growth in this region are rising emphasis on digital twin in manufacturing industries to reduce cost and improve supply chain, and R&D investments by manufacturers to drive innovation in their manufacturing processes.

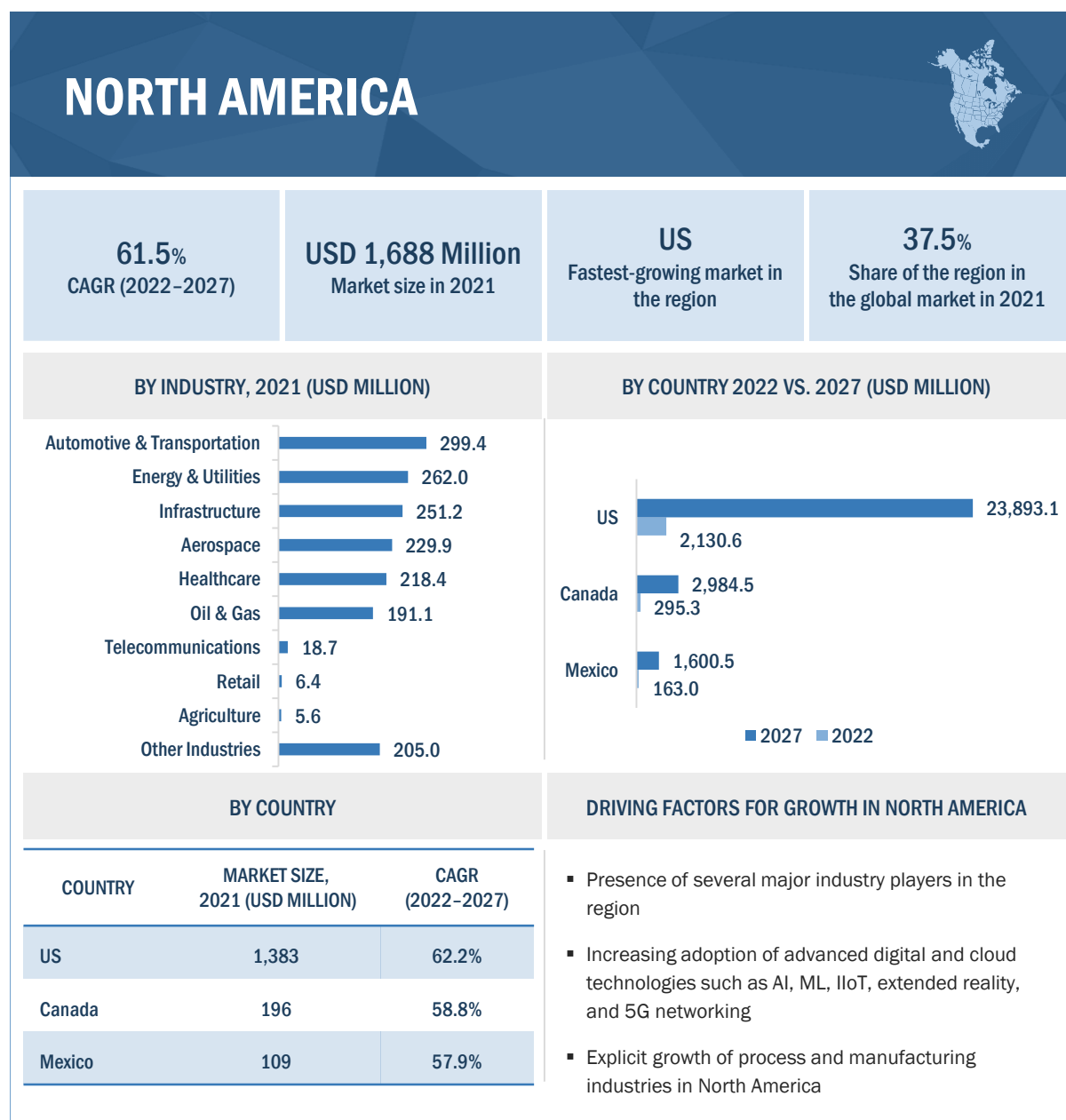
North America is a major hub for technological innovations and an early adopter of digital twin and related technologies. The presence of major digital twin providers in the region such as General Electric (US), Bentley Systems (US), IBM (US), Emerson (US), Microsoft (US), ANSYS (US), Amazon Web Services (US), Altair (US), and Oracle (US) also raises the demand and awareness regarding digital twin solutions. These vendors are investing significantly in the digital twin market. For instance, in August 2021, General Electric (US) introduced upgrades to its on-premises analytics software, Proficy CSense. The software utilizes AI and machine learning technologies, along with process digital twins, to identify problems, root causes, predict future performance, and automate actions.

North America has an established ecosystem for digital twin practices and the presence of large automotive & transportation, aerospace, chemical, energy & utilities, and food & beverage companies in the US. These industries are replacing legacy systems with advanced solutions to improve performance efficiency and reduce overall operational costs, resulting in the growth of the technology market in this region. Intensifying pressure to reduce costs, growing competition, and the modernization of factories are the major factors driving the growth of the digital twin market. R&D activities at both academic and industry levels are broadening the scope of IoT applications in different industries, especially in the US. This is boosting the market for digital twin in this region. The increased R&D in the field of the IoT and IIoT and increasing demand for efficient and cost-effective technologies to improve asset performance and optimize business operation are also supporting the regional market growth. Rapid digitalization across various verticals, the increasing adoption of smart devices, and technological advancements are fueling the growth of industrial sensors, digital twins, and artificial intelligence (AI) in manufacturing markets in North America. The growth of the region is also attributed to the increasing number of SMEs. For SMEs, it is easier to incorporate new technologies and digitalize themselves. Hence, the increasing number of SMEs in the region is making it more susceptible to industry automation and digitization.

Furthermore, the electrical transmission and distribution infrastructure in North America is also unprepared for modern threats and natural hazards. An electrical digital twin enables more flexible and reliable operations without being affected by these threats and hazards. Moreover, the rising number of projects related to the integration of renewables, smart cities, smart communities, smart grids, refurbished

energy storage, and electric vehicle (EV) charging infrastructures is the key reason for the growing adoption of advanced digital technologies such as AI, AR, IIoT, and electrical digital twins.

FIGURE 46 NORTH AMERICA: DIGITAL TWIN MARKET SNAPSHOT



Source: Secondary Research, Press Releases, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 81 NORTH AMERICA: DIGITAL TWIN MARKET, BY COUNTRY, 2018-2021 (USD MILLION)

Country	2018	2019	2020	2021
US	879	1,033	1,113	1,383
Canada	132	152	160	196
Mexico	74	85	90	109
Total	1,085	1,270	1,363	1,688

Source: Secondary Research, Press Releases, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 82 NORTH AMERICA: DIGITAL TWIN MARKET, BY COUNTRY, 2022–2027 (USD MILLION)

Country	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
US	2,131	3,365	5,462	8,907	14,563	23,893	62.2%
Canada	295	457	727	1,162	1,861	2,985	58.8%
Mexico	163	251	396	629	1,000	1,600	57.9%
Total	2,589	4,073	6,586	10,698	17,424	28,478	61.5%

Source: Secondary Research, Press Releases, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 83 NORTH AMERICA: DIGITAL TWIN MARKET, BY INDUSTRY, 2018–2021 (USD MILLION)

Industry	2018	2019	2020	2021
Automotive & Transportation	190.7	222.6	240.3	299.4
Energy & Utilities	167.4	196.7	211.3	262.0
Infrastructure	164.1	191.0	207.4	251.2
Healthcare	137.9	163.5	177.2	218.4
Aerospace	155.3	182.6	189.4	229.9
Oil & Gas	136.6	154.7	160.1	191.1
Telecommunications	-	14.0	15.1	18.7
Agriculture	-	4.5	4.7	5.6
Retail	-	4.9	5.2	6.4
Other Industries	133.1	135.8	152.2	205.0
Total	1,085.1	1,270.3	1,363.0	1,687.6

Note: Other industries include semiconductors, chemicals, paper & pulp, and food & beverages. The values have been represented in one decimal point to exhibit the difference between the minute values.

Source: Secondary Research, Press Releases, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 84 NORTH AMERICA: DIGITAL TWIN MARKET, BY INDUSTRY, 2022–2027 (USD MILLION)

Industry	2022	2023	2024	2025	2026	2027	CAGR (2020-2026)
Automotive & Transportation	477.1	778.9	1,305.3	2,195.2	3,697.6	6,235.7	67.2%
Energy & Utilities	403.3	636.6	1,032.4	1,681.8	2,746.4	4,529.1	62.2%
Infrastructure	394.1	633.6	1,046.3	1,735.0	2,882.7	4,800.2	64.9%
Healthcare	350.7	576.3	971.1	1,641.0	2,775.4	4,696.7	68.0%
Aerospace	346.2	534.2	846.9	1,347.9	2,149.6	3,433.7	58.2%
Oil & Gas	283.1	430.0	670.5	1,049.7	1,646.2	2,585.4	55.6%
Telecommunications	28.8	45.5	73.8	120.4	196.6	321.9	62.0%
Agriculture	8.3	12.5	19.4	30.2	47.1	73.5	54.8%
Retail	9.6	15.0	23.9	38.4	61.6	99.2	59.4%
Other Industries	287.7	410.6	596.1	858.6	1,220.4	1,702.7	42.7%
Total	2,589.0	4,073.2	6,585.7	10,698.3	17,423.8	28,478.1	61.5%

Note: Other industries include semiconductor, chemical, paper & pulp, and food & beverages. The values have been represented in one decimal point to exhibit the difference between the minute values.

Source: Secondary Research, Press Releases, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 85 NORTH AMERICA: DIGITAL TWIN MARKET, BY ENTERPRISE, 2018–2021 (USD MILLION)

Enterprise	2018	2019	2020	2021
Large Enterprises	883	1,025	1,091	1,339
Small & Medium Enterprises	202	245	272	349
Total	1,085	1,270	1,363	1,688

Source: Secondary Research, Press Releases, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 86 NORTH AMERICA: DIGITAL TWIN MARKET, BY ENTERPRISE, 2022–2027 (USD MILLION)

Enterprise	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Large Enterprises	2,035	3,173	5,085	8,185	13,208	21,387	60.1%
Small & Medium Enterprises	554	900	1,501	2,514	4,216	7,091	66.5%
Total	2,589	4,073	6,586	10,698	17,424	28,478	61.5%

Source: Secondary Research, Press Releases, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

In 2021, the US had the largest size of the digital twin market in North America. The market in the US is further expected to grow at the highest CAGR and dominate the North American region throughout the forecast period. The US leads the adoption of digital twin in most industries, including aerospace, automotive & transportation, infrastructure, energy & utilities, telecommunications, retail, and other industries. The country has been one of the early adopters of digital twin technology. US-based manufacturers invested in new technologies and disruptive tools and are already seeing the returns on the investments made. The adoption of digital twin is helping manufacturers in the US to increase productivity, improve industrial efficiency with better quality control, and mitigate industrial risks.

10.2.1 US

10.2.1.1 Increasing demand from automotive and transportation industries drives digital twin market

The growth of the market in the US is driven by the presence of multiple manufacturers, software developers, and technology providers, who play a vital role in the development of digital twins. The major players in the digital twin market in the US include IBM, Microsoft, Amazon Web Services, General Electric, ANSYS, Oracle, and PTC. These players have started offering products and solutions that help create digital twins, which form a core part of the digital transformation of various industries. For instance, GE (US) has built digital twins of critical jet engine components to predict the business outcomes related to the remaining life of the components. The twins of gas turbines are used to deliver the data regarding the desired electrical power output at the lowest possible fuel consumption, which, along with the twins of wind turbines, optimize the process of electricity production from wind farms. These digital twins provide real-time updated and customized information that enables GE businesses and customers to make timely decisions and interventions for continued profitability and maximized performance at a low cost.

The manufacturers and organizations in the country are focusing on the digital transformation of their design operations, especially in terms of 3D designs. Digital twin is one of the digital approaches that utilizes software simulation tools for creating a 3D model of an asset. For example, the Digital Manufacturing and Design Innovation Institute (DMDII) has the capabilities, innovative spirit, and collaborative expertise to transform the American manufacturing sector by offering assistance in the digital revolution. DMDII has received funds from the US Federal Government for R&D, which aims to demonstrate tools offered by Siemens that represent the digital twin of interactive kiosks for the processes ranging from product design to production.

According to the BP Statistical Review of World Energy 2021, the US renewable energy sources, including biomass, geothermal, hydropower, wind, and solar PV systems, accounted for approximately 17.5% of the total global electricity generation in 2020 at both utility-scale and distributed systems. The share of renewables increased by 14.1% in 2020 as compared to 2019, owing to reduced cost and policy support. The US is the most mature market in terms of smart grid deployment, advanced IT infrastructure, presence of several enterprises, and availability of technical expertise.

According to the US Department of Energy (DOE), nearly 72% of the power infrastructure in the US is more than 25 years old, ~60.0% of the distribution poles are 30–50 years old, and ~70.0% of transformers are 25 years old, most of which are approaching the end of their operational life. Aging power generation and distribution infrastructure, increasing need to deliver clean and reliable energy, and a high focus on renewable energy generation are among a few major factors leading to increased investments in power infrastructure. To accommodate this growing share of renewables and distributed energy resources (DERs), utility operators are planning to invest in the development of digital twin to enhance operational sensing of their assets and better control power generation and grid components.

The automotive & transportation industry is a major adopter of digital twins in the US. The country is one of the early adopters of electric vehicles. In addition, in the automotive & transportation industry, Tesla Motors is using a digital twin solution for every car it manufactures for better performance and business improvement. Tesla (US), General Motors (US), Rivian, (US), and Lucid Motors (US) are the major automotive manufacturers driving the EV market not only in North America but also in the global market. Tesla alone accounted for the highest share in the global plug-in electric vehicle market in 2021. The overall global plug-in electric vehicle market share enjoyed by Tesla (US) and General Motors in 2021 was more than 21 %. The introduction of attractive, new options by practically all OEMs, as well as improved availability of the Tesla Model 3 and Y, boosted sales in North America. Thus, the strong EV market is further driving the digital market in the country.

10.2.2 CANADA

10.2.2.1 Growing adoption of renewables and advanced technologies to boost demand for digital twin

Canada is expected to be the second-largest country-level market in North America during the forecast period. Emerging digital technologies such as mobile apps, digital platforms, and cloud architectures have increased productivity and innovation in enterprises in the private and public sectors. From the past few years, the Information and Communications Technology Council (ICTC) has been conducting R&D for estimating the contributions made by various emerging technologies to the Canadian economy. These technological innovations are projected to propel the growth of the Canadian digital twin market.

According to Global Affairs Canada (GAC), the country has the lowest cost structures and lowest business tax burden among advanced economies, with an overall cost advantage of 8.2% over the US and 4.5% over Japan in the automotive manufacturing sector. This further attracts various companies to Canada, thereby boosting the digital twin market for the automotive industry in the country. In knowledge-based areas such as R&D services, which include product testing services, development, and testing, Canadian companies typically enjoy 27.8% lower costs relative to their US-based counterparts. Such cost advantages are significant for foreign investors in several industries (e.g., as aerospace, automotive & transportation), where product development and testing make up for considerable expenses during the development phase of a product. As a result, Canada offers lucrative opportunities for the adoption of digital twin for product design & development applications in many industries.

Further, Air Canada and WestJet are two major airlines in Canada in terms of traffic and capacity growth. These airlines are expected to contribute to the growth of the digital twin market in Canada for maintenance, repair, and operations (MRO). Additionally, the country is an innovation hub with regard to breakthrough technologies in the energy industry. According to the Canadian Electricity Association, increasing investments in the deployment of digital twin would enable utilities to address outage management, reduce maintenance costs, and improve operations and reliability. Canadian utilities are expected to benefit from the digitalization strategy as they have a larger network of assets compared to the number of customers because of lower population density compared to countries such as the US, the UK, and China. The country is witnessing growth in the adoption of advanced digital technologies such as IIoT, AI, ML, and digital twins. For instance, the smart meter penetration in this region is about 51%, and the development in network communication has also ramped up the deployment of digital twin technology in Canada. Ontario has registered the highest number of smart meter deployments in the country in recent years and is estimated to witness the highest adoption of smart grid and electrical digital twin technologies. In addition to electric grid modernization, Canada is emphasizing on increasing its renewable energy capacity, which would further support the market growth.

10.2.3 MEXICO

10.2.3.1 Government initiatives for developing automotive and energy & utilities industries to support digital twin market

Mexico is an emerging economy with moderate growth prospects for the digital twin market. It is a rapidly evolving country in North America. The innovations in and developments of digital platforms are creating new opportunities for the production, distribution, and manufacturing of assets in the electronics & electrical and machine manufacturing industries. As a result of these developments, the digital twin market in Mexico is expected to gain significant traction in the near future.

According to International Renewable Energy Agency (IRENA), Mexico plans to generate 46.0% of electricity from renewable sources by 2030. The country is rich in renewable energy sources, including solar, geothermal, wind, hydropower, and biomass. It is one of the attractive markets for investments owing to its significant potential for renewable energy sources. With the rising electricity demand, additional investments in electric power infrastructures, and increasing renewable energy capacity additions, the

opportunities for the implementation of electrical digital twins are increasing in the country. The adoption of digital twin technologies will enable the utility and power transmission & distribution operators in the country to monitor the integration of renewable energy resources with the grid. In addition, the digital twin systems of assets can also create new revenue streams for solar and wind operators and accelerate the decarbonization of its electrical systems.

According to a report by the World Trade Organization (WTO), Mexico's total export value is dominated by consumer car automobiles, automatic-data processing machines, motor vehicle parts, and motor vehicles for goods transport. According to another report by the WTO, Mexico's move toward automation and digitalization will continue in the coming years, driven mostly by the automotive industry, thereby creating significant demand for digital twin in the industry. Factors such as proximity to the US and government measures to augment manufacturing activities are expected to fuel the growth of the Mexican digital twin market.

10.3 EUROPE

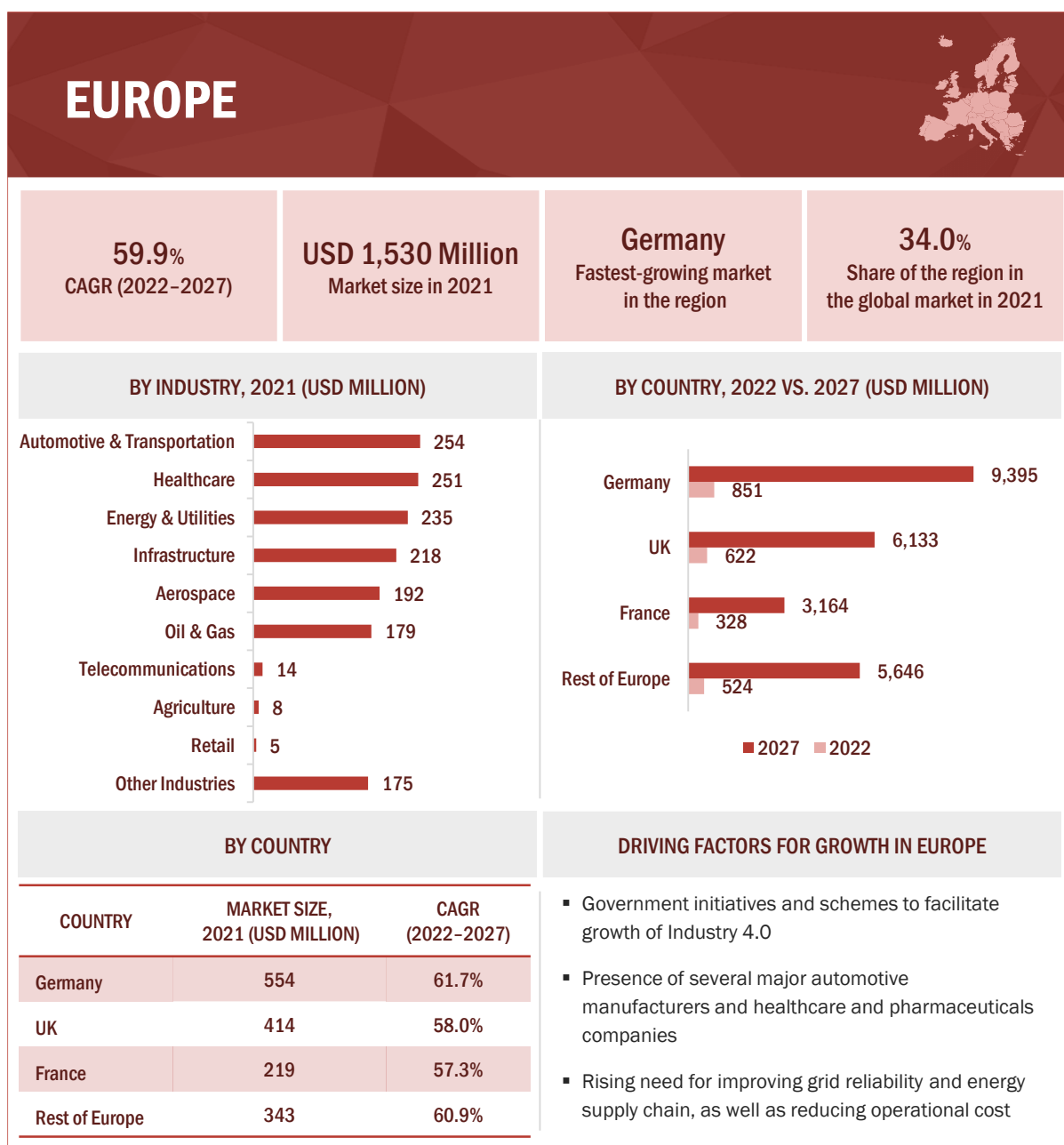
The digital twin market in Europe comprises Germany, the UK, France, and Rest of Europe. This region accounted for a share of 34% of the digital twin market in 2021. According to the European Commission, the European Union (EU) is the largest economy and trading block in the world, and EU members account for 16% of world imports and exports. Growth in the automotive & transportation, healthcare, energy & utilities, and infrastructure industries in Europe will lead to the growth of the digital twin market in the region.

Germany, being a global automotive manufacturing hub, is a key market for digital twin in Europe. According to Organisation Internationale des Constructeurs d'Automobiles (OICA), in 2020, the total number of all types of vehicles registered in Europe was 16.7 million and was about 9.2 million in the second half of 2021. The presence of major digital twin providers such as Siemens (Germany), Robert Bosch (Germany), Dassault Systèmes (France), Schneider (France), DNV (Norway), ANDRITZ (Austria), Software AG (Germany), and Johnson Controls (Ireland) drives the digital twin market in the region.

The adoption of digital twins in Europe is high at present due to the increasing digitalization of end-user industries. These next-generation factories in the region have been adopting digital twins in manufacturing and maintenance activities. Algoryx Simulation AB (Sweden) is working within a European consortium that addresses the virtual commissioning of smart factories. The consortium is receiving grants from EU for the past few years to develop standards and tools to enable the implementation of virtual twins of factories. During the designing phase of the factories, virtual twins are used to design, plan, and optimize the real factory operations. After the factory is built, the virtual twin is used for virtual commissioning, and eventually, when the factory is involved in the full production, digital twin is used to monitor, optimize, and upgrade the factory.

In tandem with the European Institute of Innovation & Technology (EIT) Health Germany, European countries are using digital twins to improve medication prescribing. For instance, ExactCure is a digital health firm based in France that is tackling the problem of preventing drug side effects. ExactCure developed a digital twin, a digital clone of the patient based on information such as age, gender, kidney status, genotype, smoking status, or any other unique aspect that has been shown to have an impact on a given prescription. This digital twin can then be utilized to simulate drug effects and interactions in patients' bodies depending on their unique traits. Further, the developments made by European companies in the area of healthcare digital twin are also supporting the growth of the market in the region.

The European Union makes efforts and promotes R&D programs to aid its manufacturing sector to fully benefit from emerging digital technologies. Many companies in Europe have implemented digital twin to improve their industrial functioning, while several companies are making efforts to implement these smart technology solutions. Hence, smart manufacturing boom in the coming years will create great opportunities for digital twin providers in Europe.

FIGURE 47 EUROPE: DIGITAL TWIN MARKET SNAPSHOT

Note: Rest of Europe includes the Netherlands, Norway, Spain, Italy, Sweden, Belgium, and Scandinavian countries.

Source: Secondary Literature, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 87 EUROPE: DIGITAL TWIN MARKET, BY COUNTRY, 2018–2021 (USD MILLION)

Country	2018	2019	2020	2021
Germany	353	415	447	554
UK	283	325	342	414
France	151	173	182	219
Rest of Europe	222	260	278	343
Total	1,010	1,173	1,247	1,530

Note: Rest of Europe includes the Netherlands, Norway, Spain, Italy, Sweden, Belgium, and Scandinavian countries.

Source: Secondary Research, Press Releases, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 88 EUROPE: DIGITAL TWIN MARKET, BY COUNTRY, 2022–2027 (USD MILLION)

Country	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Germany	851	1,340	2,170	3,530	5,757	9,395	61.7%
UK	622	958	1,516	2,412	3,845	6,133	58.0%
France	328	503	793	1,256	1,993	3,164	57.3%
Rest of Europe	524	821	1,323	2,142	3,476	5,646	60.9%
Total	2,325	3,623	5,803	9,339	15,071	24,338	59.9%

Note: Rest of Europe includes the Netherlands, Norway, Spain, Italy, Sweden, Belgium, and Scandinavian countries.

Source: Secondary Research, Press Releases, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 89 EUROPE: DIGITAL TWIN MARKET, BY INDUSTRY, 2018–2021 (USD MILLION)

Industry	2018	2019	2020	2021
Automotive & Transportation	170.7	195.6	207.4	253.7
Energy & Utilities	153.8	179.2	191.0	234.8
Infrastructure	146.9	169.0	181.6	217.5
Healthcare	157.6	187.1	203.1	250.7
Aerospace	130.7	153.3	158.6	191.9
Oil & Gas	132.8	148.7	152.1	179.4
Telecommunications	-	10.9	11.7	14.4
Agriculture	-	6.8	7.0	8.3
Retail	-	3.6	3.8	4.6
Other Industries	117.3	118.4	131.2	174.7
Total	1,009.8	1,172.8	1,247.5	1,530.1

Note: The values have been represented in one decimal point to exhibit the difference between the minute values.

Source: Secondary Research, Press Releases, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 90 EUROPE: DIGITAL TWIN MARKET, BY INDUSTRY, 2022–2027 (USD MILLION)

Industry	2022	2023	2024	2025	2026	2027	CAGR (2020-2026)
Automotive & Transportation	394.4	627.8	1,025.9	1,682.4	2,763.0	4,543.2	63.0%
Energy & Utilities	358.0	559.5	898.6	1,449.8	2,344.7	3,799.8	60.4%
Infrastructure	336.0	532.1	865.3	1,412.8	2,311.4	3,763.7	62.1%
Healthcare	405.4	670.6	1,137.6	1,935.1	3,293.9	5,609.9	69.1%
Aerospace	288.1	443.3	700.7	1,112.1	1,768.6	2,800.0	57.6%
Oil & Gas	259.1	383.1	581.0	883.5	1,344.2	2,045.1	51.2%
Telecommunications	22.1	34.6	55.7	90.2	146.4	238.9	61.0%
Agriculture	12.2	18.3	28.1	43.4	67.1	103.6	53.4%
Retail	6.9	10.6	16.8	26.6	42.3	67.5	57.6%
Other Industries	242.8	343.0	493.1	703.1	989.4	1,366.5	41.3%
Total	2,324.8	3,623.0	5,802.8	9,339.2	15,071.1	24,338.2	59.9%

Note: The values have been represented in one decimal point to exhibit the difference between the minute values.

Source: Secondary Research, Press Releases, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 91 EUROPE: DIGITAL TWIN MARKET, BY ENTERPRISE, 2018–2021 (USD MILLION)

Enterprise	2018	2019	2020	2021
Large Enterprises	917	1,061	1,124	1,373
Small & Medium Enterprises	93	112	124	157
Total	1,010	1,173	1,247	1,530

Source: Secondary Research, Press Releases, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 92 EUROPE: DIGITAL TWIN MARKET, BY ENTERPRISE, 2022–2027 (USD MILLION)

Enterprise	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Large Enterprises	2,077	3,224	5,142	8,242	13,244	21,298	59.3%
Small & Medium Enterprises	247	399	661	1,098	1,827	3,040	65.1%
Total	2,325	3,623	5,803	9,339	15,071	24,338	59.9%

Source: Secondary Research, Press Releases, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

Europe is moving toward renewable energy resources by bringing in solutions such as solar panels and electric cars. Germany is among the world leaders in the automotive and healthcare industries, which are key end-user industries for digital twin. Therefore, the healthcare industry is expected to dominate the digital twin market during the forecast period. Further, the developments made by European companies in the area of healthcare digital twin are also favoring the growth of the market in the region. Industries like automotive & transportation, and energy & utilities are also likely to exhibit significant adoption of the technology during the forecast period.

Germany accounted for the largest share of the digital twin market in Europe in 2021. It is further projected to grow at the highest CAGR between 2022 and 2027. According to the International Trade Administration, Germany has the largest economy in Europe, with a very strong automobile, electronics, and healthcare sectors that are major adopters of digital twin products/solutions.

The digital twin market in Europe for SMEs is expected to grow at a higher CAGR during the forecast period. The adoption rate of digital twin among SMEs is expected to be higher as compared to large enterprises. A startup based in Italy, Nemo's Garden, is using Siemens (Germany) digital twin technologies to demonstrate how the distribution of advanced automation technology as a service is extending potential for new underwater agriculture. Nemo's Garden is using Siemens' cloud-based Xcelerator as a Service product and is focusing on sustainable underwater agricultural cultivation. A sub-aqua biosphere—an underwater greenhouse designed to use the ocean's temperature stability, evaporative water generation, CO₂ absorption, abundance of oxygen, and inherent pest protection to create an environment for the underwater cultivation of herbs, fruits, and vegetables—is Nemo's Garden's key innovation. As of April 2022, Nemo's Garden had already demonstrated the practicality of its concept, and now they are planning to use the Xcelerator portfolio to develop their prototype into a global offering.

10.3.1 GERMANY

10.3.1.1 Rising production of EVs to support digital twin market expansion

Industry 4.0 is an ambitious project undertaken by the German government to digitalize the manufacturing industry in order to increase resource efficiency and industrial output. Several regulations in Germany have propelled the pre-installation simulation of wind turbines operating in a wind farm grid and mandated the use of digital twins in the wind farm certification process across the country. These initiatives are expected to have a positive impact on the growth of the digital twin market in this country. Siemens (Germany), Robert Bosch (Germany), SAP (Germany), and Software AG (Germany) are some of the major players in the digital twin market.

According to the European Federation of Pharmaceutical Industries and Associations (EFPIA) report 2021, Germany is among the world leaders in the healthcare industry, which plays an important role in driving Germany's digital twin market. Siemens (Germany) is continuously investing in developing digital twins for human body organs such as heart, brain, and lungs.

Companies are attracted to Germany because it is Europe's largest economy and most populous nation. According to the Central Intelligence Agency (CIA), Germany is the world's fifth-largest economy in terms of purchasing power parity (PPP) and producing automobiles, precision engineering products, electronic and communications equipment, chemicals, medical, and pharmaceutical and healthcare products. With increasing automation and initiatives for digitization, these sectors act as great enablers for the adoption of digital twin technology in Germany.

With dynamics in the automotive industry shifting towards electronic vehicles (EVs), automotive manufacturing processes have become complicated. To keep up with the recent trends in the manufacturing sector, the German government laid out systematic plans to enable manufacturers to replace their legacy systems with cutting-edge Industry 4.0 technologies. As a result, a significant number of smart factories in Germany are equipped with technologies such as artificial intelligence, blockchain, industrial robots, and digital twin solutions.

Additionally, the German federal government is promoting and supporting digital transformation and has undertaken initiatives to incorporate IIoT processes and encourage cloud adoption to increase digitization and productivity of the power sector. Several regulations in Germany have fueled the pre-installation simulation of wind turbines operating in a wind farm grid and mandated the use of digital twins in the wind

farm certification process across the country. These initiatives are expected to have a positive impact on the growth of the digital twin market in this country.

10.3.2 FRANCE

10.3.2.1 Growing investments in vehicle manufacturing to boost demand for digital twins

France is focusing on R&D to enhance its technological capabilities and has introduced new solutions to modernize the industrial sector. Its response toward IoT adoption has been positive. The government has an ambitious plan to revive the industries based in France with cloud computing and IoT. Technologies such as IoT and cloud computing play a vital role in the development of digital twins; hence, the rise in the adoption of these technologies is expected to drive the growth of the digital twin market in France.

Aerospace, automotive, and pharmaceuticals are the most important industries in France. The country is one of the early adopters of electric or hybrid vehicles. Renault (France) is investing billions in electric vehicle production facilities to address the increasing demand for such vehicles. The introduction of new facilities, coupled with government incentives for electric vehicles, is expected to boost the digital twin market in France. The French automotive industry is continuously making efforts to increase competitiveness by reducing costs and increasing productivity through automation. The French government planning to have one million electric and hybrid vehicles on the road by 2022; subsequent investments in vehicle manufacturing are expected to boost the market for digital twin for designing & development of auto parts, predictive maintenance, and other applications.

The country has set ambitious green energy targets for renewable energy adoption and development of EV charging infrastructures to achieve carbon neutrality by 2050. According to International Renewable Energy Agency (IRENA) 2021, France installed a solar energy capacity of 929 MW and wind capacity additions of 955 MW in 2020. The share of renewable energy in the country's energy mix is growing, representing 25.3% of France's gross final energy consumption for 2022. With the increasing adoption of renewable energy sources, there has been a significant strain on the country's grid operators to provide reliable power and accommodate fluctuations in power supply and bi-directional flow of power.

10.3.3 UK

10.3.3.1 Steady development of renewable energy infrastructure and adoption of 5G and IoT to drive market growth

According to the Central Intelligence Agency (CIA), the UK is the third-largest economy in Europe, with its manufacturing industry accounting for ~10% of its economic output. The UK government, in its budget for 2020, announced that the corporation tax main rate (for all profits except ring fence profits) would be 19% for the fiscal years beginning April 1, 2020, and 2021, thus making the environment for industries more favorable.

The UK has played an integral role in the evolution of industrial operations. The IoT is considered as the prime driver for the next industrial revolution. The UK government has appointed the Ordnance Survey (UK), the national mapping agency for the UK, to generate a smart map comprising a digital twin that would determine the best locations for the radio antennae needed to establish a 5G network. Thus, the growing adoption of new technologies with digitization is expected to boost the economic growth of the UK, and consequently drive the growth of the digital twin market in the near future.

In June 2021, Samsung (South Korea) announced its entry in Europe with a Vodafone (UK) 5G network deal. Vodafone, a British telecommunication company, chose Samsung Electronics to supply its 5G network equipment in the UK under a new agreement. With the advent of 5G connectivity, the market for digital twins is expected to not only flourish in the UK but also across the entire European region. AI is required for a fully operational and efficient 5G network. The use of 5G networks allows for the integration

of machine learning and AI at the network edge. 5G allows multiple IoT devices to connect at the same time, generating a massive amount of data that must be processed using machine learning and AI.

The phasing out of coal and increasing investments in wind and solar power generation are expected to increase direct investments in the upgrade of transmission and distribution networks in the UK. According to the UK Department for Business, Energy, and Industrial Strategy, the UK government plans on implementing digital energy strategies to reduce capital expenditure and operational cost of power projects, optimize power sector operations, and asset performance management. Power utilities in the UK apply advanced digital technologies, such as IoT, AI, blockchain, and digital twin, to enable high interoperability of data, flexible processes, and asset performance management.

10.3.4 REST OF EUROPE

Rest of Europe includes the Netherlands, Norway, Spain, Italy, Sweden, Belgium, and Scandinavian countries. The region accounts for a significant share of the European digital twin market and is expected to grow at a significant CAGR during the forecast period.

Spain is one of the developed countries in Europe, and the automotive industry is one of the key contributors to the overall economy of Spain. Most of the country's automotive products are exported (to ~130 countries). Many major automobile manufacturing companies are based in Spain. All these manufacturers focus on leveraging industrial automation and communication technologies/products to ensure greater production efficiency. Among all the European countries, the industrial 3D printing market in Spain is expected to grow at the fastest rate during the forecast period. Government support and investments from private companies are the major factors driving the digital twin market in Spain.

Italy has a diversified industrial economy, with developed infrastructure and a high per capita gross domestic product (GDP). Italy is also well known for its automotive manufacturing sector. The country's largest industrial enterprise and the sole mass manufacturer of passenger cars, Fiat Automobiles S.p.A (Italy), is a well-known brand. Fiat Group leads Italy's automotive industry, accounting for ~90% of domestic car production. For the automotive industry, smart factories are essential for increasing efficiency and productivity. Italy has a huge industrial base with advanced technological, architectural, and design capabilities. Companies based in Italy are continuously focusing on collaborations to develop highly efficient technologies.

Further, the presence of digital twin startup companies in other parts of Europe also drives the digital twin market in the region. FEops (Belgium), Artemis Technologies (Ireland), and MEIK LABS Spa (Chile) are few such startups.

10.4 ASIA PACIFIC

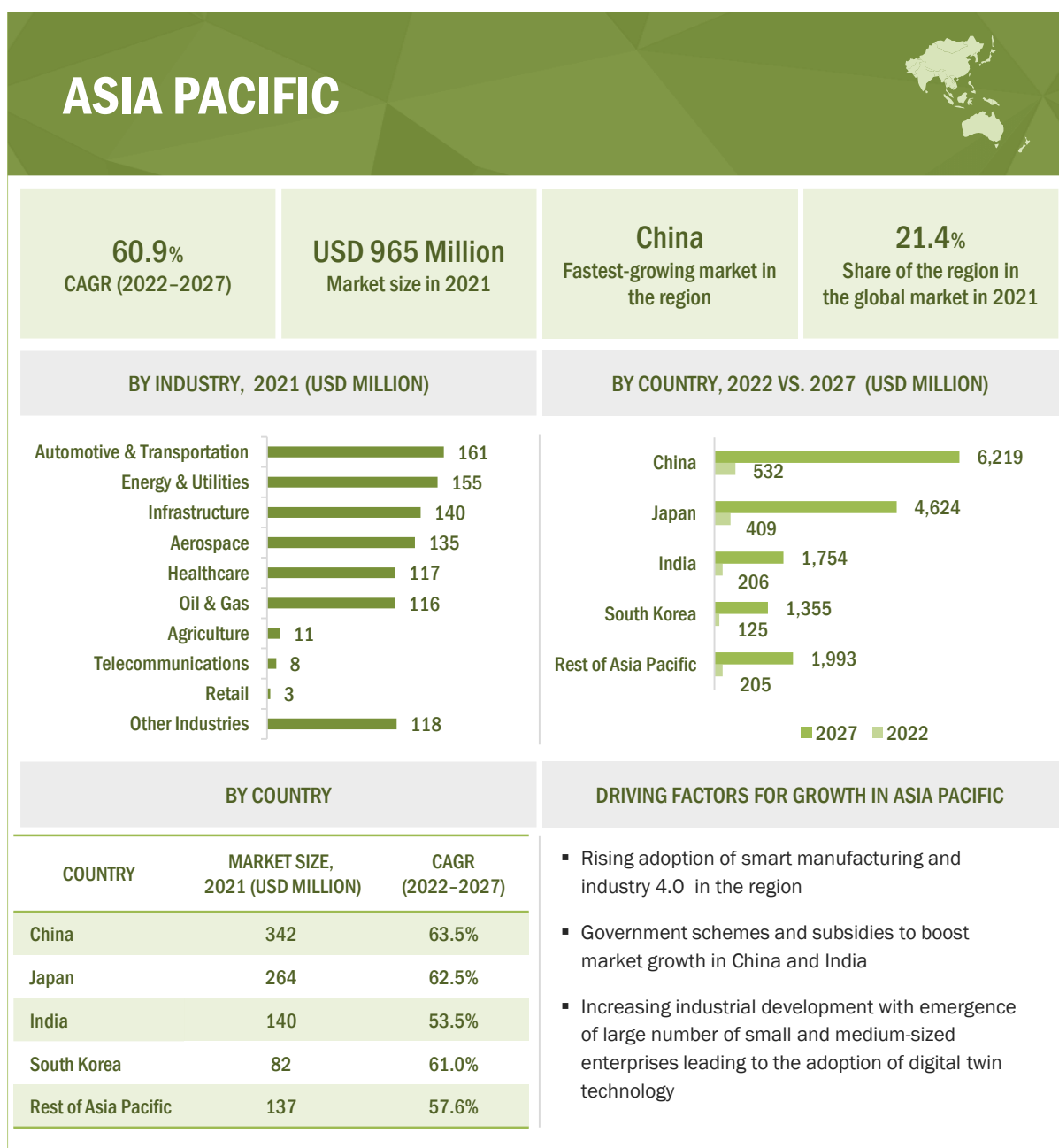
The digital twin market in Asia Pacific is further segmented into China, India, Japan, South Korea, and Rest of Asia Pacific. Rest of Asia Pacific includes Taiwan, Australia, New Zealand, Singapore, Malaysia, and the Philippines. Asia Pacific accounted for 21.4% of the digital twin market in 2021. The region is a major manufacturing hub in the world and expected to provide ample growth opportunities to the digital twin market.

Countries like China, India, and Japan are not just the manufacturing hubs of several end-user industries but are also prime consumers for the end products and services. Asia Pacific is the largest consumer of electronics/electrical devices, automotive, and telecommunications to mention a few. The region hosts a large number of small and medium-sized enterprises (SMEs). The need for huge investments to successfully implement industrial control systems security projects restricts SMEs from adopting a strong security practice within their organizations. However, factors such as the increasing complexity of businesses and the unregulated nature of the internet are expected to encourage SMEs to adopt digital twin solutions extensively. Further, many Asian countries are highly populated and the geriatric population is becoming a concern for the future. The aging population, especially of China, is likely to generate high

demand for medical care in the foreseeable future, thereby boosting the demand for digital twins in the healthcare industry.

The commercialization of IIoT applications is in the introductory stages in the developing economies of Asia Pacific. However, countries such as China, Japan, and India have undertaken numerous initiatives to encourage the implementation of IIoT in their respective domestic industries. Dense population, growing per capita income, and large-scale industrialization and urbanization are a few of the major factors driving the growth of IIoT in Asia Pacific. Due to increasing maintenance costs, the manufacturing and energy & utilities industries have been genuinely considering the adoption of advanced technologies, including digital twin for applications such as predictive maintenance, to optimize maintenance processes and reduce operational costs.

Asia Pacific is a manufacturing hub; hence, motor vehicle suppliers such as Volkswagen (Germany), Toyota (Japan), Renault-Nissan (Netherlands), and Daimler (Germany), and local manufacturers such as Tata Motors (India) and Hyundai Motors (South Korea) are investing in the development of smart factories in the region. Most of the key manufacturers from various industries have their manufacturing plants in Asia Pacific due to the low labor costs and the availability of skilled labor in the region. The increasing automation in process and discrete industries is one of the major factors that is expected to drive the growth of the digital twin market in Asia Pacific.

FIGURE 48 ASIA PACIFIC: DIGITAL TWIN MARKET SNAPSHOT

Note: Rest of Asia Pacific includes Taiwan, Australia, New Zealand, Singapore, Malaysia, and the Philippines.

Source: Secondary Literature, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 93 ASIA PACIFIC: DIGITAL TWIN MARKET, BY COUNTRY, 2018–2021 (USD MILLION)

Country	2018	2019	2020	2021
China	210	251	273	342
Japan	164	195	211	264
India	100	114	118	140
South Korea	50	60	65	82
Rest of Asia Pacific	91	106	112	137
Total	616	726	779	965

Note: Rest of Asia Pacific includes Taiwan, Australia, New Zealand, Singapore, Malaysia, and the Philippines.

Source: Secondary Research, Press Releases, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 94 ASIA PACIFIC: DIGITAL TWIN MARKET, BY COUNTRY, 2022–2027 (USD MILLION)

Country	2022	2023	2024	2025	2026	2027	CAGR (2022-2027)
China	532	849	1,391	2,288	3,772	6,219	63.5%
Japan	409	647	1,054	1,723	2,822	4,624	62.5%
India	206	309	477	737	1,138	1,754	53.5%
South Korea	125	196	317	514	834	1,355	61.0%
Rest of Asia Pacific	205	316	498	790	1,255	1,993	57.6%
Total	1,477	2,318	3,737	6,051	9,822	15,945	60.9%

Note: Rest of Asia Pacific includes Taiwan, Australia, New Zealand, Singapore, Malaysia, and the Philippines.

Source: Secondary Research, Press Releases, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 95 ASIA PACIFIC: DIGITAL TWIN MARKET, BY INDUSTRY, 2018-2021 (USD MILLION)

Industry	2018	2019	2020	2021
Automotive & Transportation	105.4	121.9	130.4	160.9
Energy & Utilities	99.5	116.8	125.4	155.3
Infrastructure	90.7	105.8	115.2	139.9
Healthcare	74.9	88.3	95.2	116.7
Aerospace	94.0	109.3	112.2	134.6
Oil & Gas	75.9	88.6	94.6	116.4
Telecommunications	-	6.5	6.9	8.4
Agriculture	-	9.1	9.5	11.5
Retail	-	2.3	2.4	2.9
Other Industries	75.3	77.3	87.1	118.0
Total	615.6	725.9	778.9	964.7

Note: The values have been represented in one decimal point to exhibit the difference between the minute values.

Source: Secondary Research, Press Releases, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 96 ASIA PACIFIC: DIGITAL TWIN MARKET, BY INDUSTRY, 2022–2027 (USD MILLION)

Industry	2022	2023	2024	2025	2026	2027	CAGR (2020-2026)
Automotive & Transportation	254.5	412.3	685.9	1,145.2	1,915.1	3,207.0	66.0%
Energy & Utilities	238.3	374.9	606.0	984.0	1,601.8	2,578.1	61.0%
Infrastructure	220.0	354.8	587.5	977.0	1,627.7	2,701.9	65.1%
Healthcare	186.2	303.9	508.8	854.1	1,434.9	2,416.2	67.0%
Aerospace	200.5	306.1	480.0	755.8	1,192.2	1,900.9	56.8%
Oil & Gas	176.3	273.6	435.9	697.1	1,116.5	1,790.5	59.0%
Telecommunications	12.8	20.0	32.1	51.8	83.6	135.2	60.1%
Agriculture	17.1	26.0	40.7	63.8	100.1	157.3	55.9%
Retail	4.4	6.9	11.0	17.6	28.1	45.0	58.8%
Other Industries	166.5	239.0	348.9	505.3	722.1	1,012.9	43.5%
Total	1,476.8	2,317.6	3,736.8	6,051.5	9,822.2	15,944.9	60.9%

Note: The values have been represented in one decimal point to exhibit the difference between the minute values.

Source: Secondary Research, Press Releases, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 97 ASIA PACIFIC: DIGITAL TWIN MARKET, BY ENTERPRISE, 2018–2021 (USD MILLION)

Enterprise	2018	2019	2020	2021
Large Enterprises	557	649	687	840
Small & Medium Enterprises	58	77	92	125
Total	616	726	779	965

Source: Secondary Research, Press Releases, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 98 ASIA PACIFIC: DIGITAL TWIN MARKET, BY ENTERPRISE, 2022–2027 (USD MILLION)

Enterprise	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Large Enterprises	1,268	1,963	3,121	4,983	7,973	12,756	58.7%
Small & Medium Enterprises	209	355	616	1,069	1,849	3,189	72.5%
Total	1,477	2,318	3,737	6,051	9,822	15,945	60.9%

Source: Secondary Research, Press Releases, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

China accounted for the largest share of the digital twin market in Asia Pacific in 2021. It is further expected to dominate the market throughout the forecast period. China has a high potential for creating digital transformation in the manufacturing sector. The dense population, growing per capita income, large-scale industrialization and urbanization, and increasing adoption of IoT are the factors driving the growth of the digital twin market in the country.

The digital twin market for SMEs in Asia Pacific is expected to grow at a higher CAGR during the forecast period as the reachability of digital twin expands. The Asia Pacific region has the highest number of small & medium enterprises that drive the backbone of economies, especially in developing countries. The rapid

industrial development in the region encourages SMEs to invest in digital twin solutions. Further, the return on investment is high for SMEs as compared to large enterprises for adopting digital twin. In addition, many startups are offering digital twin products/solutions that are cost-efficient to some extent and, therefore, a perfect fit for SMEs. Hence, the digital twin market for SMEs in Asia Pacific region is expected to grow at a higher CAGR during the forecast period.

10.4.1 CHINA

10.4.1.1 Rapid adoption of Industry 4.0 principles to propel market growth in China

China is a key country-level market in the Asia Pacific region. According to the International Trade Centre, China is the world's largest exporter and importer of manufactured goods. The growing population has resulted in the growth of several industries, such as electronics, automotive & transportation, telecommunications, energy & utilities, and other industries. Thus, China is the largest market for digital twins that are mainly utilized in the electronics, automotive, and telecommunications end-user industries.

The country is one of the major manufacturing hubs in the world. The Chinese government is actively adopting new and improved manufacturing strategies to develop its manufacturing sector, and reform and strengthen various industries. IIoT is increasingly being adopted in China owing to the heavy financial and strategic involvement of the government in R&D of IoT-based solutions. The Government of China is committed to developing the nation's IIoT capabilities. As China is a major hub for semiconductor fabrication and manufacturing industries, its involvement in the implementation and development of IIoT is crucial for the growth of these industries. The government is actively adopting new and improved strategies to develop its manufacturing sector and reform and strengthen other industries. Growing adoption of automation technologies and government support to spur industrialization are expected to drive the growth of the digital twin market.

The government is working toward implementing the Made in China 2025 program, a 10-year government-sponsored program aimed at putting China on equal footing with Western industrial countries. The plan is focused on automation and the development of cyber-physical systems that will boost lean production and cut dependency on an outdated cheap-labor model. The continuously increasing labor costs in the country is prompting companies, such as Foxconn Technology Company Limited (Taiwan), to work on developing industrial robots, targeting about ~50% automation in Chinese factories by 2022. Government support encourages end users to adopt digital twin solutions to enhance operational efficiency.

10.4.2 JAPAN

10.4.2.1 Heightened focus on incorporation of advanced technologies in manufacturing to fuel market growth

Being the first country to be industrialized in Asia Pacific, Japan has a competitive advantage over other countries in the region. Japan's major exports include automobiles, consumer electronics, semiconductors, and iron and steel, among others. Lean manufacturing or just-in-time manufacturing is a manufacturing trend that emerged in Japan and is popular with major manufacturing companies in the country as well as globally. As quality has always been the priority for Japan-based manufacturers, their products have a high value in the global market. The country has now launched the Industrial Value Chain Initiative to explore Industry 4.0 options across the manufacturing sector. Members of this initiative include major businesses in the electric, IT, machinery, and automobile sectors, including Mitsubishi Electric, Fujitsu, Nissan Motor, and Panasonic.

Many companies in Japan are stepping up the adoption of IoT to achieve digital innovation. Recently, Toshiba Corporation (Japan) released SPINEX, a new IoT architecture that combines the latest technologies of the Toshiba Group with the experience and expertise of a wide range of industries and worksites, including the energy, social infrastructure, and manufacturing fields. SPINEX is a collection of industrial IoT

solutions that integrates technologies, products, and services to transform customer business models. According to the Japanese government's panel on future investment, post the outbreak of COVID-19, there is a need for the shift of the manufacturing base of high value-added products to the country and other goods across Southeast Asia. This shift will lead to an upsurge in the demand for digital twins in new facilities during the forecast period.

Further, the government has made several policy reforms to accelerate the decarbonization of the energy sector and reduce dependency on fossil fuel-fired power generation. In July 2021, the government released a draft of the revised Strategic Energy Plan, which proposed a target of a 36–38% share of renewables and a 20–22% share of nuclear power in the country's energy mix by 2030. Such large-scale investments in renewable energy and strengthening policy support in the coming years are likely to boost the adoption of digital technologies such as digital twins to improve operational flexibility.

10.4.3 INDIA

10.4.3.1 Rising government initiatives for smart manufacturing to boost digital twin market in India

The Government of India is encouraging foreign players to establish their manufacturing plants in India to encourage the development of the Indian manufacturing industry with the help of state-of-the-art manufacturing technologies offered by the players operating globally. Thus, this initiative by the government is encouraging foreign players to invest in the manufacturing industry in India, which would increase the adoption of digital twin technology in new plants, leading to the growth of the digital twin market in this region.

The National Manufacturing Policy, an initiative by the Indian government, aims to increase the share of manufacturing in GDP to 20% by 2025, which will lead to an increase in manufacturing activities in the automation sector and fuel the requirement for power. Such initiatives and developments in India are expected to increase the implementation of digital twins for power supplies in the near future.

In the recent past, the government has come up with various initiatives supporting the growth of the manufacturing industry. The government intends to increase this contribution further by adopting advanced industrial automation technologies and developing related infrastructure. The “Make in India” initiative is designed to help manufacturing industries grow and compete globally. The “Digital India” initiative is expected to increase internet penetration in the country, with companies such as Google (US) and Facebook (US) lending their technical expertise toward the expansion of internet connectivity. Another initiative spearheaded by the Ministry of Heavy Industries is “Samarth Udyog Bharat 4.0,” with the aim of accelerating the adoption of smart manufacturing solutions through knowledge sharing, increasing awareness, and contextual solution creation in India. The role of the government to promote the adoption of advanced technologies in the country is increasing industrial automation in India, which is consequently projected to boost the growth of the smart manufacturing market. This, in turn, is expected to drive the market for digital twin in India.

India's power system is undergoing a transformation towards cleaner electricity. The country has set an ambitious national-level renewables target of 175 GW by 2022 and a national renewables ambition of increasing its capacity to 450 GW by 2030. However, according to the International Energy Agency, as of 2020, conventional generation still dominated India's capacity mix, with coal and lignite accounting for 205 GW (55%) and gas representing 25 GW (7%) of the total capacity. To support the integration, the country plans to develop Green Energy Corridors that support the massive renewable energy capacity addition targets. The rising penetration of renewable energy, both at utility scale and in decentralized mode, would necessitate the implementation of digital technologies such as digital twins for better forecasting and asset management in India. Also, digital twin technologies can help overcome the poor grid connectivity and transmission losses in the country by improving the quality and supply of power. For instance, in November 2019, General Electric was awarded a contract by the Jammu & Kashmir Power

Development Department (JKPDD) to provide supervisory control and data acquisition (SCADA) and advanced distribution management solution (ADMS) for the distribution operations in the cities of Jammu and Srinagar.

10.4.4 SOUTH KOREA

10.4.4.1 Increasing government investments in advanced manufacturing practices to spur market growth

South Korea is one of the rapidly growing economies in Asia Pacific. The country has emerged as an important nation for the advancement of cellular technologies. The Government Offices of South Korea are encouraging R&D on technologies associated with IoT. The manufacturing industry in South Korea also plays a key role in its economic development. South Korea is home to various large-scale manufacturing companies operating in the displays, transport, and logistics segments. The facilities of these companies are spread across large areas and have a huge workforce. SynusTech, a South Korea-based automated material handling system (AMHS) provider, and KAIST, a research university, have developed an advanced MES/MCS/xCS solution with AI and digital twin technologies. This solution will help the industry players to enhance their performance of the IoT delivery and operation by more than 30% as compared to that of the existing solutions.

South Korea is one of the largest producers of motor vehicles in the world but had lost market share to other countries over the past few years. Kia Motors (South Korea) has moved its production to other countries. However, there may be new opportunities for automotive electronics to be manufactured through joint ventures between automotive electronics manufacturers, such as Samsung, LG Electronics, and Hyundai Motors. South Korea is the third-largest manufacturer of electronic integrated circuits. Technological developments in manufacturing are expected to boost the market for digital twin in South Korea.

The South Korean government is focusing on the development of the manufacturing sector in the country. It has introduced various initiatives and programs to encourage advanced manufacturing practices and investment in the manufacturing sector. For instance, in September 2020, amidst the COVID-19 pandemic, the Government of South Korea declared a USD 4.5-billion investment plan for the artificial intelligence industry in the country. This plan was approved by the Presidential Committee on the Fourth Industrial Revolution. The funds will be invested over five years in large-scale AI initiatives, specifically for defense, science, and public safety. The government is highly focusing on the development of the manufacturing sector. Increasing international trade is one of the driving factors for the growth of the South Korean economy. The digital revolution in the industrial manufacturing sector in South Korea has encouraged the adoption of advanced manufacturing solutions across various industries.

10.4.5 REST OF ASIA PACIFIC

In this report, Rest of Asia Pacific comprises Taiwan, Australia, New Zealand, Singapore, Malaysia, and the Philippines. This market segment is expected to witness significant growth in the coming years, owing to the increase in the use of digital twin in various applications in the region. For instance, Dassault Systems (France) is working on a project in Singapore to create a digital twin called “Virtual Singapore,” which is likely to be launched by the end of the year. The virtual twin would help urban planners and policymakers visualize the data obtained from various sensor networks and intelligent systems deployed and to be deployed in the future and use them to make informed decisions.

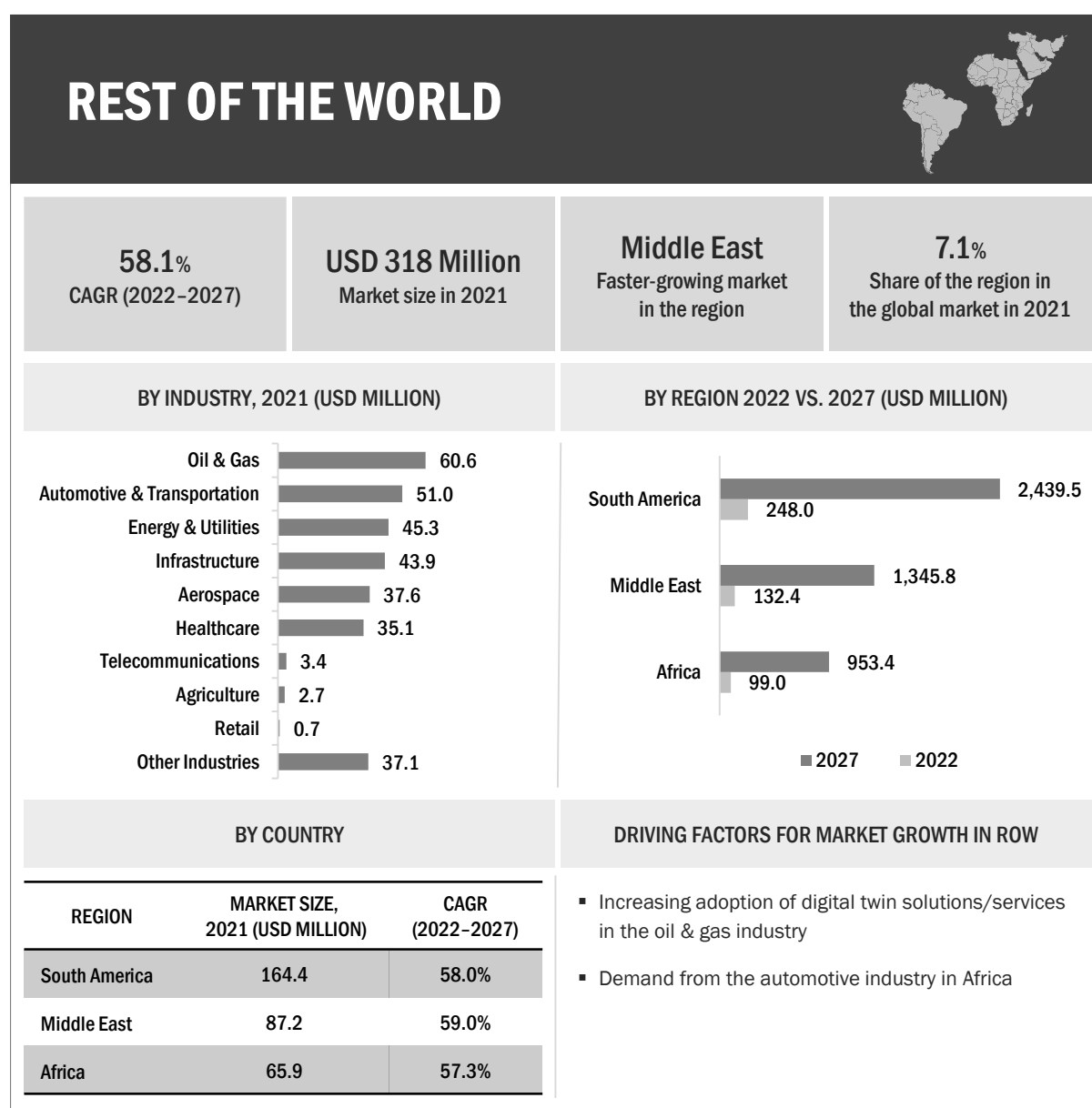
Infrastructure developments in the manufacturing industry in other countries of the region are expected to offer growth opportunities for the market players. In Taiwan and Singapore, government initiatives and support are encouraging the adoption of Industry 4.0 principles. Many companies in China are shifting their manufacturing units to Vietnam due to disputes with the Chinese government. This has led to the growth of the manufacturing sector in Vietnam. To further boost the development of the manufacturing sector, the

government is encouraging manufacturers to adopt state-of-the-art Industry 4.0 technologies in their manufacturing plants. Other governments in the region are also implementing various schemes for the deployment of AI, blockchain, and IoT technologies in manufacturing units.

10.5 REST OF THE WORLD

In the digital twin market, the Rest of the World has been segmented into South America, the Middle East, and Africa. The Rest of the World held a relatively smaller share of the overall market as compared to other regions. The oil & gas industry plays a major role in the Rest of the World digital twin market. The globalization of supply chain networks and the rising adoption of on-cloud solutions are driving the growth of the smart manufacturing market in the region. As the Middle East and Africa have huge reserves of oil and gas, this industry offers a major scope for the digital twin market. Growing competition in the global market is forcing companies in the process industry in the region to adopt innovative cost reduction techniques to increase productivity. Digital twin and related technologies provide an advantage in analyzing real-time data that helps keep production costs low.

FIGURE 49 REST OF THE WORLD: DIGITAL TWIN MARKET SNAPSHOT



Source: Secondary Literature, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 99 REST OF THE WORLD: DIGITAL TWIN MARKET, BY REGION, 2018–2021 (USD MILLION)

Region	2018	2019	2020	2021
South America	108.0	126.1	134.0	164.4
Middle East	56.1	66.0	70.6	87.2
Africa	43.8	50.9	53.9	65.9
Total	207.9	243.0	258.6	317.6

Note: The values have been represented in one decimal point to exhibit the difference between the minute values.

Source: Secondary Research, Press Releases, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 100 REST OF THE WORLD: DIGITAL TWIN MARKET, BY REGION, 2022–2027 (USD MILLION)

Region	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
South America	248.0	382.8	606.0	961.7	1,526.7	2,439.5	58.0%
Africa	132.4	205.8	327.8	523.7	836.8	1,345.8	59.0%
Middle East	99.0	152.2	239.9	379.1	599.2	953.4	57.3%
Total	479.4	740.7	1,173.7	1,864.5	2,962.8	4,738.8	58.1%

Note: The values have been represented in one decimal point to exhibit the difference between the minute values.

Source: Secondary Research, Press Releases, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 101 REST OF THE WORLD: DIGITAL TWIN MARKET, BY INDUSTRY, 2018–2021 (USD MILLION)

Industry	2018	2019	2020	2021
Automotive & Transportation	35.14	39.96	42.03	51.00
Energy & Utilities	31.66	36.14	37.70	45.34
Infrastructure	30.23	34.59	36.92	43.93
Healthcare	23.64	27.45	29.13	35.15
Aerospace	28.60	32.38	32.28	37.62
Oil & Gas	34.15	42.06	47.11	60.59
Telecommunications	-	2.73	2.85	3.43
Agriculture	-	2.27	2.33	2.74
Retail	-	0.57	0.59	0.69
Other Industries	24.51	24.87	27.69	37.08
Total	207.94	243.01	258.63	317.58

Note: The values have been represented in two decimals point to exhibit the difference between the minute values.

Source: Secondary Research, Press Releases, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 102 REST OF THE WORLD: DIGITAL TWIN MARKET, BY INDUSTRY, 2022–2027 (USD MILLION)

Industry	2022	2023	2024	2025	2026	2027	CAGR (2020-2026)
Automotive & Transportation	78.56	123.92	200.56	325.66	529.35	861.1	61.4%
Energy & Utilities	68.69	106.70	170.29	272.98	438.59	706.1	59.4%
Infrastructure	67.28	105.57	170.08	275.04	445.53	758.8	62.3%
Healthcare	51.01	74.95	111.48	163.53	234.88	323.5	44.7%
Aerospace	54.33	80.33	121.80	185.17	281.59	428.1	51.1%
Oil & Gas	98.19	162.06	273.17	460.13	773.23	1,296.5	67.5%
Telecommunications	4.98	7.40	11.27	17.17	26.13	39.0	50.9%
Agriculture	3.99	5.96	9.12	14.01	21.52	33.1	52.6%
Retail	1.02	1.53	2.37	3.66	5.68	8.8	54.0%
Other Industries	51.34	72.30	103.55	147.12	206.25	283.8	40.8%
Total	479.41	740.71	1,173.68	1,864.46	2,962.75	4,738.8	58.1%

Note: The values have been represented in two decimals point to exhibit the difference between the minute values.

Source: Secondary Research, Press Releases, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 103 REST OF THE WORLD: DIGITAL TWIN MARKET, BY ENTERPRISE, 2018–2021 (USD MILLION)

Enterprise	2018	2019	2020	2021
Large Enterprises	190	220	231	280
Small & Medium Enterprises	18	23	28	37
Total	208	243	259	318

Source: Secondary Research, Press Releases, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 104 REST OF THE WORLD: DIGITAL TWIN MARKET, BY ENTERPRISE, 2022–2027 (USD MILLION)

Enterprise	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Large Enterprises	417	637	996	1,561	2,448	3,862	56.0%
Small & Medium Enterprises	62	104	178	303	515	877	69.9%
Total	479	741	1,174	1,864	2,963	4,739	58.1%

Source: Secondary Research, Press Releases, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

South America is expected to dominate the digital twin market in the Rest of the World throughout the forecast period. The Middles East, on the other hand, is likely to be the fastest-growing market for digital twin in RoW. The market in the Middle East is driven by the high-tech investment in infrastructure and the presence of an established oil & gas sector.

10.5.1 SOUTH AMERICA

10.5.1.1 Surging adoption of IIoT in mining and other industries to drive digital twin market

South America includes Argentina, Bolivia, Brazil, Venezuela, Colombia, and Chile. South America has become a key region for mining and exploration investments. Chile, Peru, and Colombia are key mining markets in South America. Chile has been one of South America's fastest-growing economies in recent decades. In 2018, manufacturing production improved in the country due to the growth of the food and chemicals industries. Also, the country witnessed significant progress in wholesale trade and commercial services. MEIK LABS Spa (Chile) and DigitalTwin Technology (Germany) are companies that offer digital twin for mining industry in the region.

South America is home to large industries that can commercially take advantage of IIoT. The Ministry of Science, Technology, Innovation, and Communication (MCTIC) (Brazil) and the Brazilian Development Bank (BNDES) announced a set of initiatives to develop and accelerate the adoption of IoT in Brazil. The Government of Brazil has opened low-interest-rate financial lines for Brazil-based companies worth USD 1.4 billion through BNDES designated for smart industry projects for the next 4 to 5 years. Companies such as Siemens (Germany), Microsoft (US), and SAP (Germany) have their presence in Brazil and are contributing toward the growth of digital twin solutions.

The major industries in South America include oil & gas and petroleum, which contribute majorly to the country's GDP. The use of smart manufacturing in South American countries compared with the developed countries is low. Hence, companies in the region have started adopting emerging technologies in cooperation with larger companies from other regions without further delay. A large number of new startups and businesses are emerging in the countries of South America. This is creating a considerable ground for industrialization. The major focus of the revolutionized companies will be to increase productivity and improve production efficiency.

10.5.2 MIDDLE EAST

The Middle East includes Saudi Arabia, the United Arab Emirates (UAE), and Rest of the Middle East. The region has vast reserves of oil and gas; the industry offers a major scope for the growth of the smart manufacturing market in the Rest of the World. The manufacturing sector in Saudi Arabia has shown noteworthy growth and has the potential to become a technology-driven industry at par with global levels due to the increasing implementation of IIoT-related software and services. One of the factors that can boost the adoption of digital twin and related technologies in this region is increasing partnerships among industry vendors, process licensors, equipment experts, and consultants to provide innovative solutions to industrial problems.

TABLE 105 MIDDLE EAST: DIGITAL TWIN MARKET, BY COUNTRY, 2018–2021 (USD MILLION)

Country	2018	2019	2020	2021
Saudi Arabia	23.0	27.3	29.4	36.7
UAE	18.0	21.1	22.6	27.9
Rest of the Middle East	15.2	17.6	18.6	22.7
Total	56.1	66.0	70.6	87.2

Note: Rest of the Middle East includes Kuwait, Oman, and Qatar. The values have been represented in one decimal point to exhibit the difference between the minute values.

Source: Annual Reports, Press Releases, SEC Filings, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 106 MIDDLE EAST: DIGITAL TWIN MARKET, BY COUNTRY, 2022–2027 (USD MILLION)

Country	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Saudi Arabia	56.1	87.9	141.1	227.3	366.0	593.2	60.3%
UAE	42.3	65.7	104.7	167.2	267.0	429.3	58.9%
Rest of the Middle East	34.0	52.1	82.0	129.3	203.8	323.3	56.9%
Total	132.4	205.8	327.8	523.7	836.8	1,345.8	59.0%

Note: Rest of the Middle East includes Kuwait, Oman, and Qatar. The values have been represented in one decimal point to exhibit the difference between the minute values.

Source: Annual Reports, Press Releases, SEC Filings, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

10.5.2.1 Saudi Arabia

10.5.2.1.1 Implementation of advanced technologies in oil & gas and other industries to augment market growth

Saudi Arabia is one of the rapidly growing economies in the Middle East. The country has a significant wealth of oil reserves. Companies involved in the production of crude oil in the country are implementing digital twin. The technology would help companies in accelerating the daily crude oil production while ensuring efficient and safe operations. Digital twin would also help companies in designing of new assets and facilities. Thus, the implementation of digital twin technology in crude oil production plants to monitor several processes is estimated to generate huge opportunities for future developments in the oil & gas industry in the country.

The demand for energy is increasing rapidly, which is forcing the energy industry to utilize the available resources efficiently and reduce asset downtime while enhancing its performance. Digital twin plays a vital role in reducing downtime and increasing productivity. Companies in Saudi Arabia are focusing on developing digital and advanced solutions for the energy and utilities industry. Moreover, the country has placed itself at the forefront of renewable energy supply by developing the largest solar PV plant. The plant requires continuous monitoring to avoid wastage of solar energy. The implementation of digital twin would help solar plants in monitoring their performance and schedule timely maintenance. Thus, the implementation of digital twin technology in the energy and utilities industry is estimated to create new ways to improve productivity and optimize all critical parts of production lines in the power industry in the country.

The innovations and technological advancements in the healthcare industry have eased the simulation of medical devices and monitoring and diagnosis of patients. The use of digital twins in the healthcare industry helps in eliminating errors in medical devices through data-driven navigation processes. This reduces issues associated with handling and operating medical devices such as CT scanners, X-ray systems, and MRI instruments. The Kingdom of Saudi Arabia (KSA) is expected to be one of the fastest-growing health markets in the Gulf Cooperation Council. The country is focusing on the expansion of the healthcare industry. For instance, Saudi Arabia has signed a memorandum of understanding with Pfizer Inc. and AstraZeneca plc for the production of vaccines in the country. As the production of vaccines is a complicated process, the implementation of digital twin technology will help in managing the production process. The KSA's Vision 2030 program for the healthcare industry is also focusing on the adoption of digital transformation systems. Thus, the adoption of digital twin technology in the healthcare industry is likely to create opportunities for the growth of the market.

Further, the use of digital twins in the water & wastewater industry helps in simulating operational scenarios in real time to check treatment plant capacity and the water output quality in the real world. Digital twins allow utility workers to visualize the asset, check status, and perform simulations and analysis. This allows utility workers to understand the past and present performance of their water systems, which helps them in predicting the forecast performance, identifying failures before it happens, and simulating the impact of potential changes in the virtual world. Thus, many water utilities are implementing digital twin technology in water treatment plants to improve the design of new facilities, support staff training, and optimize ongoing operations. For instance, ACCIONA, a Spanish company, has used digital twin in the construction of the largest desalination plant in Saudi Arabia. The technology helps the company in testing and commissioning of the plant remotely in order to analyze and optimize productivity.

10.5.2.2 UAE

10.5.2.2.1 Government initiatives to promote smart manufacturing to increase demand for digital twin in UAE

The United Arab Emirates (UAE) is one of the largest economies in the Middle East and contributes significantly to the digital twin market growth. The manufacturing companies in the country are investing substantially to advance digitization and technology integration. They are planning to elevate their performance in the coming years by responding positively to emerging technologies. At the same time, compared with the developed economies, this performance would be somewhat lower. The government is taking special initiatives to develop programs to promote smart manufacturing. One of such initiatives include UAE AI Strategy 2031, UAE AI Strategy 2031 aims at positioning UAE as a major hub for AI by 2031. This initiative involves investing in the latest AI techniques and employing modern technologies to develop life in the UAE.

10.5.2.3 Rest of the Middle East

Rest of the Middle East includes Kuwait, Oman, and Qatar. These countries currently account for a small share in the digital twin market. The government of rest of middle east countries are focusing on initiatives to leverage the benefits of Industry 4.0 and related technologies including digital twin. Some of these initiatives include Qatar National Vision 2030, and New Kuwait Vision 2035. Qatar National Vision 2030 is an initiative by the Qatar government to make the country an advanced society that provides its people a high standard of living. This initiative will ensure the economic growth and social development of the country.

The Middle East countries are rich in oil and other related products, the extraction and refining of which constitutes a major industry. IoT and AI, and blockchain technologies offer several opportunities for oil & gas companies in this region to integrate their platforms globally, automate their processes, improve efficiency, reduce dependency on humans, and provide enhanced safety. The growing initiatives to automate the oil & gas vertical, and the introduction of smart cities has increased AI and IoT adoption in the region, especially in the Middle East. Industries such as petrochemicals and heavy machinery are expected to witness a drastic change, which will, in turn, drive the growth of the Industry 4.0 market.

10.5.3 AFRICA

10.5.3.1 Demand from energy & utilities industry to underpin market growth in Africa

The economic and political scenario is improving in many areas of Africa, and the governments are showing a keen interest in improving the industrial sector of the region. Over the past few years, many consumer electronics industry players have been focusing on Africa to tap the large consumer base. Although the disposable income of Africa is lower than the rest of the regions, Africa offers lucrative opportunities for telecommunications, mobile electronics, and white consumer electronics goods. Increasing count of mobile subscribers is the factor driving the telecommunications infrastructure industry in Africa. As a result, telecom and power supply devices are expected to witness positive growth in the next 5 years. These developments in Africa are expected to drive the growth of the digital twin market during the forecast period.

According to the Ministry of Mines and Energy of Brazil, the country's installed power generation capacity is projected to reach 212.5 GW by 2024. The ministry also stated that an investment of USD 531.0 billion is required by 2024 to modernize the power infrastructure in the country and is expected to add 15 GW of centralized power generation capacity by 2025. The government plans to invest USD 62 billion in utilized-scale power generation and USD 10 billion in distributed generation by 2029. Furthermore, total investments in the power transmission sector are projected to reach USD 22 billion by 2029, with USD 15 billion allocated to the expansion and upgrade of transmission lines and USD 7 billion in substations. In addition, according to the Brazilian Energy Planning Agency's (EPE) Energy Expansion Plan (PDE) for 2019–2029, the country plans on increasing the renewables capacity and is targeting a share of 48% in the power generation mix by 2029. The integration of renewables and management of distributed energy resources are expected to provide ample opportunity to technology vendors providing IoT, AI, AR, and digital twin solutions. The government is investing in biomass power generation, solar power generation, thermal power generation, and offshore wind power generation to mitigate the risk of power shortages during periods of severe drought. It has set a target to deploy 3.5 GW and 7 GW of solar PV power plants by 2023 and 2024, respectively. Thus, increased government investments to upgrade the power infrastructure of Brazil is expected to fuel the growth of the digital twin market in the country.

In addition, the increase in aircraft deliveries and the high demand for MRO services in this region are expected to be the key factors driving the growth of the digital MRO market in Africa. Leading airlines in this region are adopting MRO for fleet maintenance, which, in turn, is projected to drive the growth of the digital twin market in Africa.

11 COMPETITIVE LANDSCAPE

11.1 OVERVIEW

The competitive landscape section includes the market share and ranking analysis of the key players in the digital twin market. It describes the key growth strategies adopted by the major market players from January 2020 to June 2022 to expand their global presence and increase their market share. Several factors, such as annual reports, revenues, digital twin products/solutions and complementary technologies, key developments, and geographic presence of the companies are analyzed, along with the industries served by the companies, among other factors, to arrive at the ranking of the leading companies in the market.

Product launches, deals (mergers, acquisitions, partnerships, cooperation, alliances, collaborations, agreements, contracts, and investments), and others (expansions) were the major growth strategies adopted by the players in the digital twin market. These strategies enabled the players to efficiently cater to the growing demand for digital twin from the end users and expand their global footprint by offering their products in all the major regions.

The inorganic growth strategy of deals was adopted by a majority of players in the digital twin market. However, product launches and other strategies were also adopted by many players in the market from 2020 to 2022. The developments related to financial announcements and the expansion of management teams were not considered while analyzing the recent developments.

11.2 KEY PLAYER STRATEGIES/RIGHT TO WIN

Different companies are deploying different strategies to win in this market.

TABLE 107 OVERVIEW OF STRATEGIES DEPLOYED BY DIGITAL TWIN COMPANIES

KEY PLAYER	PRODUCT TYPE	PRODUCT LAUNCH	ORGANIC/INORGANIC DEVELOPMENTS (JANUARY 2020–MAY 2022)	REGIONAL PRESENCE
General Electric (US)	Digital twin solution	GE launched a solution that uses artificial intelligence (AI) to build a machine learning (ML) digital twin model of a gas turbine to continuously identify ideal flame temperatures and fuel splits to minimize emissions and acoustics.	Both inorganic and organic growth strategies—product launches, acquisitions, agreements	Major presence in North America; significant presence in Asia Pacific, Europe, and the Rest of the World
Microsoft (US)	Digital twin solution	Microsoft announced the general availability of Azure Digital Twins. By integrating assets such as internet of things devices and current business systems, Azure Digital Twins allows customers to model settings such as buildings, factories, farms, energy networks, railways, stadiums, and cities.	Both inorganic and organic growth strategies—product launches, partnerships, and collaborations	Major presence in North America; strong presence Asia Pacific, Europe, and the Rest of the World

Siemens (Germany)	Digital twin solution	At LogiMAT 2022 (Germany) event, Siemens demonstrated how a digital twin may assist satisfy emerging intralogistics demands with the Dynamic Gapper application.	Both inorganic and organic growth strategies—product launches, acquisitions, partnerships, cooperation, and collaborations	Major presence in Europe; strong presence in North America, Asia Pacific, and the Rest of the World
Amazon (Amazon Web Service) (US)	Digital twin solution	AWS launched the new version of its Vertex visualization software platform for digital twin. It provides end-user enterprises with real-time PLM-aligned data.	Both inorganic and organic growth strategies—includes product launches and partnerships	Major presence in the North America; strong presence Asia Pacific, Europe, and the Rest of the World
Dassault Systèmes (France)	Digital twin solution	The company showcased its anticipated digital twin solution for healthcare. Virtual twin experiences open up new possibilities for life sciences and healthcare by allowing researchers, medical, surgical, and other health-related disciplines to understand, model, search, test, and treat the human body and organs.	Both inorganic and organic growth strategies—product launches, acquisitions, contracts, and partnerships	Major presence in the Americas; strong presence in Asia Pacific and Europe

Source: Annual Reports, Press Releases, SEC Filings, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

11.2.1 PRODUCT/SOLUTION PORTFOLIO

The top companies offer digital twin product/solution for both large enterprises and small & medium enterprises. These companies have invested in R&D to offer high-quality digital twin platforms, software solutions, and excellent services across the world. With the advent of 5G technology, growing deployment of artificial intelligence (AI) and machine learning (ML) algorithms, and increasing focus on digital twins for predictive maintenance, digital twin providers are now focusing on providing specialized digital twin for the healthcare industry. Digital twin in healthcare enables to better understand disease development in patients over time and understand the response to new medications, therapies, or surgical interventions. For instance, Siemens Healthineers is developing a digital twin of the heart to help with pharmacological therapy and cardiac catheterization simulations. Similarly, Dassault Systèmes' Living Brain project is guiding epilepsy therapy and tracking the evolution of neurodegenerative illnesses. Similar projects have been conducted for the lungs, knees, eyes, and other human organs by the company.

11.2.2 REGIONAL FOCUS

All companies are well-diversified globally, although it has been observed that a few of these companies are located in North America and Europe as the regions are the early adopters of digital twin technology. However, these companies have a well-established network of sales and services all over the world, thereby providing excellent reach to their clients. General Electric (US), Microsoft (US), Amazon Web Services (US), and Dassault Systèmes (France) have major presence in North America and significant presence in Asia Pacific, Europe, and the Rest of the World. Siemens (Germany) has major presence in Europe, and strong presence in North America, Asia Pacific, and the Rest of the World.

11.2.3 INDUSTRY FOOTPRINT

These companies offer digital twins products/solutions for different industries, namely, automotive & transportation, aerospace, energy & utilities, infrastructure, healthcare, oil & gas, agriculture, telecommunications, retail, and others (semiconductors, chemicals, paper & pulp, and food & beverages). Companies provide customized software to ensure that the clients achieve the desired outputs for business optimization and other applications. In case of digital twin, the entire solution—starting from building blocks, sensors, layout of design (both software and hardware), type of platform and products/solutions are fully customized for the client.

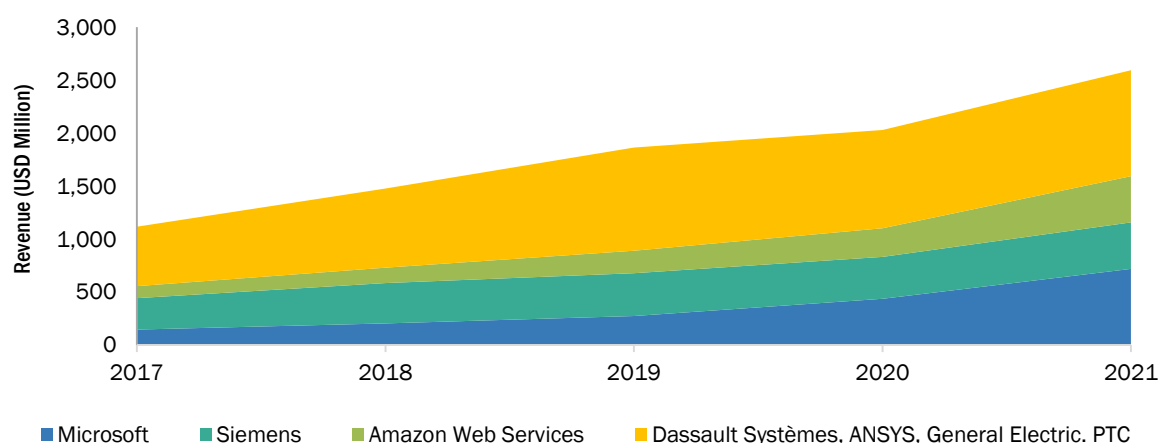
11.2.4 ORGANIC/INORGANIC PLAY

The top companies have primarily leveraged organic and inorganic growth strategies to expand their product portfolio and geographic reach. Some companies have focused more on product launches, while others have focused more on expanding their operations through mergers, acquisitions, partnerships, cooperation, alliances, collaborations, agreements, contracts, investments, and others (expansions). For instance, General Electric (US), Siemens (Germany), and Dassault Systèmes (France) engaged in a few acquisitions, agreements, collaborations, and partnerships from January 2020 to June 2022, while General Electric (US), and Siemens (Germany) collectively launched five or more new digital twin products/solutions from January 2020 to June 2022.

11.3 TOP COMPANY REVENUE ANALYSIS

The digital twin market is a competitive one due to the strong presence of a few well-established brands/market players in the global digital twin market. Players operating in this market are developing and offering digital twin for various industries with specific customizations. Primary interviews were conducted with key opinion leaders, such as CEOs, directors, and industry experts and other executives, to validate revenues and market shares. Historical analysis of revenues helped us understand revenue patterns of the parent market over the past five years. This information further helped anticipate the future growth of key players in the digital twin market.

FIGURE 50 DIGITAL TWIN MARKET: REVENUE ANALYSIS OF TOP COMPANIES, 2017–2021



Note: For each of the above-mentioned company, primary interviews were conducted with key opinion leaders, such as CEOs, directors, and industry experts, and other executives to validate revenues and market shares.

Source: Annual Reports, Press Releases, SEC Filings, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

From the figure provided above, it can be inferred that the business revenue for most of the major players stagnated in FY 2020 due to the COVID-19 pandemic during the period; short-term demand fluctuations amid an uncertain market outlook; and medium-term lockdown in the major markets (infrastructure, automotive & transportation, aerospace, and oil & gas) during February–June 2020. However, the market saw a growth in 2021; the business revenue for most of the major players has increased many folds in the last one year, and it is expected to increase further on account of the increasing demand for digital twin solutions in various industries, as businesses became more aware about the technology. The outbreak of the COVID-19 pandemic increased the awareness about digital twins and acted as catalyst for the market as more business approach for digitization. The increased adoption of digital twin in the healthcare and manufacturing industries has played a major role in boosting the growth of digital twin and related technologies. Similarly, increasing R&D in digital twin and complementary/related technologies (AI, ML, internet of things (IoT), industrial IoT, and others) acts as a major driver for the digital twin market.

11.4 MARKET SHARE ANALYSIS, 2021

The digital twin market witnessed exponential growth in 2021 as compared to 2020. Since 2021, the market for digital twin has seen significant growth, which continued throughout 2021. The increasing automation and Industry 4.0 developments escalated the demand for digital twin solutions in the healthcare, automotive & transportation, infrastructure, telecommunications, and other industries. The demand from the healthcare industry is driven by growing use cases of digital twins. Through the digital twin solutions, the efficacy and interactions of medications in the body of a patient based on their personal characteristics such as age, gender, kidney status, and others are monitored. With the rapid spread of the coronavirus, healthcare and pharmaceuticals industries are at the forefront of combating COVID-19 with digital twin applications. Further, the technology is used widely for predictive maintenance across various industries. Digital twin-based predictive maintenance uses real-time sensor records about the health and working conditions of a product, process, or system/unit and analyzes it against historical data about its failure modes and their criticality. This section provides a market share analysis of the top companies offering digital twin products/solutions.

TABLE 108 DEGREE OF COMPETITION, DIGITAL TWIN MARKET (2021)

DEGREE OF COMPETITION	COMPETITIVE
Microsoft	13–15 %
Siemens	12–13 %
Amazon (Amazon Web Services)	10–11 %
Dassault Systèmes, ANSYS, General Electric, PTC	15–21 %
Others	40–50 %

DEGREE OF COMPETITION IS DEFINED AS BELOW:

- Fragmented: When the top five players have a total market share < 25%
- Competitive: When the top five players have a total market share of 25–50%
- Consolidated: When the top five players have a total market share > 50%

Note: In the table above, companies have been ranked based on their market presence; client/customer base; revenue generated through digital twin products/solutions; and key products/solutions launches and developments, partnerships, and acquisitions. To analyze the ranking of the top market players, sales of only digital twin products/solutions have been considered. The sales of complementary/related technologies were not considered.

Source: Annual Reports, Press Releases, SEC Filings, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

The global digital twin market is competitive in nature; the top players, namely, General Electric (US), Microsoft (US), Siemens (Germany), Amazon Web Services (US), ANSYS (US), Dassault Systèmes (France), and PTC (US) collectively accounted for about 50–60% market share in 2021. Other players in the market contributed to a market share of about 40–50%. Factors such as the market presence of companies, their client/customer base, total revenue generated by them from the sales of digital twin products/solutions, and key developments undertaken by them have been considered for ranking the top players in the digital twin market. This segment studies the growth strategies such as product launches, mergers, acquisitions, partnerships, cooperation, alliances, collaborations, agreements, contracts, investments, and expansions adopted by market players from January 2020 to June 2022 to expand their global presence and increase their shares in the digital twin market. The table above gives a detailed ranking and share of the key market players.

MICROSOFT

Microsoft offers significantly diversified software products and is involved in the development, licensing, and providing support for the same. The company provides cloud-based services. Microsoft offers Azure Digital Twins allows developers to create next-generation IoT applications that model entire environments in real time. Azure Digital Twins can assist predict the future of any connected environment by tracking its history.

Microsoft focuses on bringing these digital twins to life with a live execution environment that tracks twin evolution. In addition, powerful query APIs also provide meaningful insights into the behavior of simulated settings, and it connects with Azure data analytics that helps in creation of a powerful digital twin to optimize the businesses of the clients. Microsoft is developing new digital tools, which will enable its cloud-connected machines to predict the exact point at which an equipment needs maintenance, averting possible breakdowns. Many industrial manufacturers are leveraging the capability of HoloLens to develop a virtual service business, which allows senior technicians to guide other technicians remotely across the globe. This gives the company a competitive advantage in the digital twin market. The company also spent about 12% to 13% of its revenue every year in the last three years for the development of innovative solutions. It has more than 10,000 patents in its digital solution for Azure portfolio.

SIEMENS

Siemens is a global technology provider that specializes in manufacturing, infrastructure, transportation, and healthcare. Siemens operates through the following reportable business segments: Digital Industries, Smart Infrastructure, Siemens Healthineers, Mobility, Portfolio Companies, Siemens Financial Services, and Reconciliation to Consolidated Financial Statements.

Siemens works on making digital twins more accessible to consumers and reducing the time it takes to develop and adapt them. To this end, the company devised the Digital Twin Operations (DT Ops) strategy. With DT Ops, the digital twin is approached in a modular manner. It contains best practices and tools for managing the lifecycles of digital twins by combining modelling and simulation of digital twins (DT) with IT operations (Ops). The company continues to invest in R&D for Product Digital Twins (using digital twins for efficient design of new products), Production Digital Twins (using digital twins in manufacturing & production planning), and Performance Digital Twins (using digital twins capture, analyze, and act on operational data). Siemens has extensive experience in development of key components of digital twin such as IoT, artificial intelligence (AI), machine learning (ML), as well as other high-end digital software and related technologies that work in tandem with digital twins. Siemens has also invested in the creation of digital twin for additive manufacturing factories.

AMAZON (AMAZON WEB SERVICE)

Amazon divides its operations into these segments: Online Stores, Third-party Seller Services, Subscription Services, Advertising Services, Physical Stores, and Amazon Web Services (AWS). Amazon Web Services Inc (AWS) is a subsidiary of Amazon that provides cloud computing services. The company offers digital twin offerings through AWS. It has created a four-level digital twin index: 1–Descriptive, 2– Informative, 3–Predictive, and 4–Living Digital Twin.

Amazon is strongly committed to R&D for its AWS services, including digital twin. AWS constantly focuses on the development of digital twin tools, software solutions, and platforms. The company also places high emphasis on digital twin complementary technologies such as machine learning solutions, internet of things, and cloud computing services. The company's revenue grew at a quick rate in 2020—30% year over year (YoY) in 2020 on a USD 35 billion yearly revenue base in 2019. The revenue continued to grow at a rate of 37% YoY in 2021. The company's strength lies in its strong clientele base and long-term customer relationships. For instance, AWS offers its IoT TwinMaker to INVISTA (US) for improving productivity and efficiency in field operations. INVISTA is a subsidiary of Koch Industries, which serves the nylon and specialty materials industries. Similarly, Carrier (US), a leading provider of healthy, safe, sustainable, and intelligent building and cold chain solutions, also uses IoT TwinMaker for monitoring operational data for smart buildings. John Holland, one of Australia's leading integrated infrastructure, rail, multi-modal transport, and building companies, also uses AWS IoT TwinMaker for environmental impact monitoring in construction.

11.5 COMPANY EVALUATION QUADRANT, 2021

In this section, we have mapped 15 players operating in the digital twin market. These players have been ranked based on two major factors—business strategy excellence and strength of product portfolio. The detailed rating has been based on market share/rank and product footprint of the companies belonging to the digital twin ecosystem. The market share/rank section describes the approaches used by key players to expand their presence in the market. The product footprint section describes the number of digital twin products/solutions launched by market players. Factors such as the focus on product innovations and product offerings are rated based on products and services offered by each market player.

11.5.1 STAR

Star players generally receive high scores in most of the evaluation criteria. They have an established product portfolio with a very strong market presence. Star companies provide mature and reputable products and services. They offer digital twin products/solutions for various industries, including automotive & transportation, aerospace, energy & utilities, infrastructure, healthcare, oil & gas, agriculture, telecommunications, retail, and others (semiconductor, chemical, paper & pulp, and food & beverages); and operate across almost all the regions worldwide. Star companies primarily focus on their product portfolio, brand equity, and tie-ups with clients, integrators, and partners to stay ahead of the curve. For instance, Microsoft (Japan) offers various types of digital twin solutions that allow users to create knowledge graphs based on digital models of entire environments; buildings, industries, farms, energy networks, railways, stadiums, and more—even entire cities—could be among these environments. Microsoft delivers its products/solutions in nearly 250 countries with the help of partners. General Electric (US), Siemens (Germany), Amazon Web Services (US), ANSYS (US), Dassault Systèmes (France), and PTC (US) also fall in this category.

11.5.2 EMERGING LEADER

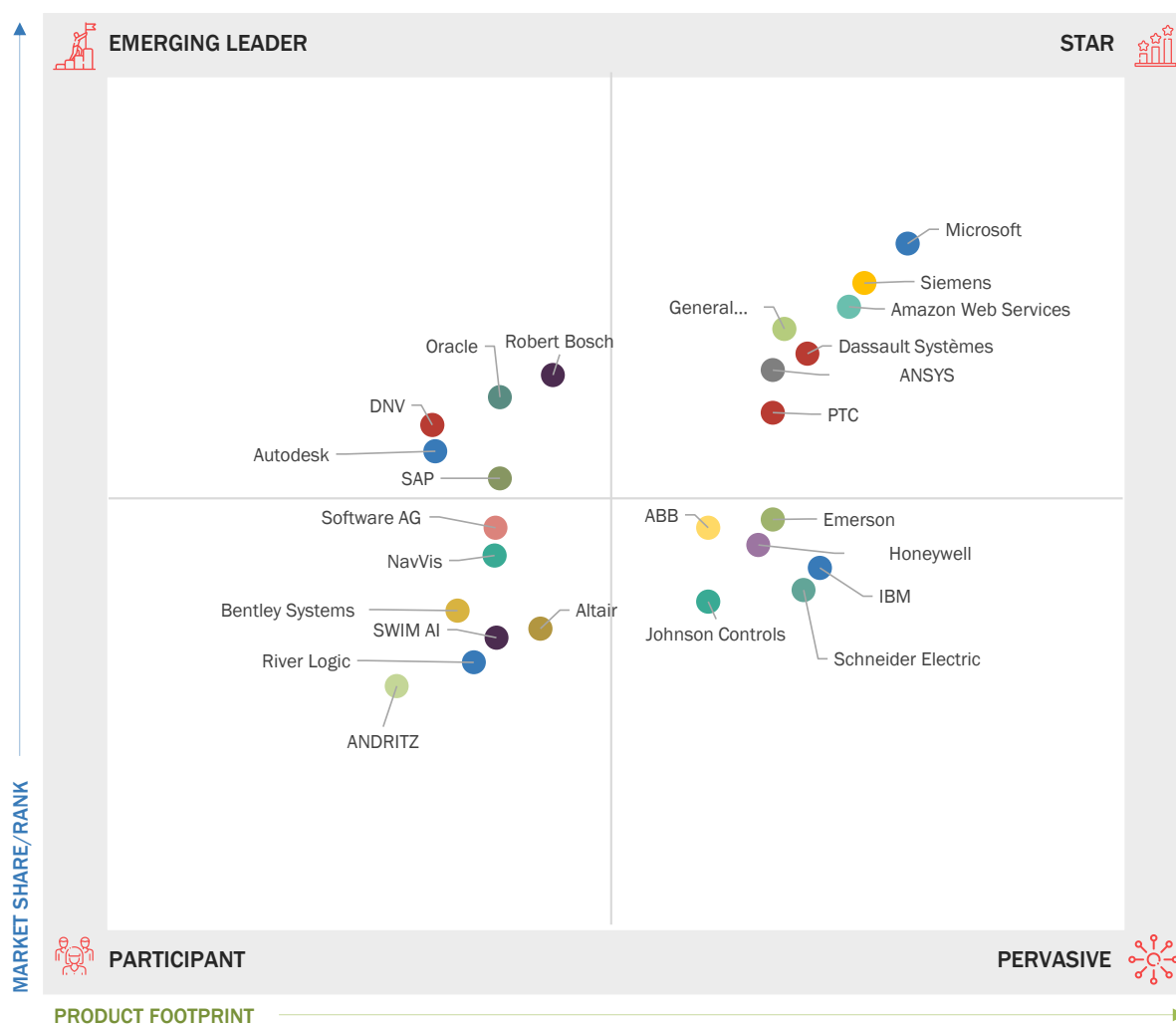
Companies categorized under this are those who have demonstrated substantial product innovations compared to their competitors. They have an enhanced product portfolio. However, with their unique portfolio and strategies, they have the potential to become market leaders in 3–4 years. These companies have clearly differentiated their products from the rest. Innovation is offered in terms of product offerings that are unique to the market and are not generally offered by most companies. Robert Bosh (Germany) offers digital buildings twin specifically designed for developing the digital replicas of buildings. In terms of building management, digital twin opens more options for the clients. A digital building twin will allow for the creation of a virtual representation of a physical building, complete with all of the building's systems, gadgets, sensors, and actuators. Other companies included in this category are Oracle (US), DNV (Norway), Autodesk (US), and SAP (Germany).

11.5.3 PERVASIVE

Pervasive players are well-established companies with effective business strategies. They offer innovative solutions, services, and hardware portfolios and have an extensive network of channel partners and resellers to increase the deployment of their solutions across a multitude of end-user markets. They generally focus on a specific type of technology related to the product. Like stars, these companies also have good tie-ups with clients, integrators, and partners. Emerson (US), ABB (Switzerland), Honeywell (US), IBM (US), Johnson Controls (Ireland), and Schneider Electric (France) are a few of the pervasive players in the digital twin market.

11.5.4 PARTICIPANT

Participants are companies that offer niche products and have started gaining recognition in the market. Unlike other established players, they lack effective business strategies. These players are mostly new entrants in the market and require more time to gain significant traction in the market. For instance, Software AG offers IoT digital twin solutions for designing, managing, and visualizing digital twins of complex assets, asset hierarchies, devices as well as manufacturing production process. With Cumulocity IoT device management from Software AG, users can streamline onboarding and manage all of their devices from a single platform. A few of the other participants in the digital twin market are NavVis (Germany), Bentley Systems (US), Altair (US), SWIM AI (US), River Logic (US), and ANDRITZ (Austria).

FIGURE 51 DIGITAL TWIN MARKET: COMPANY EVALUATION QUADRANT, 2021

Note: For market rank, factors such as product features and functionality, product innovation, branding, quality and reliability, vision alignment, effectiveness of growth strategies, productivity, and cost efficiencies have been considered.

Source: Annual Reports, Press Releases, SEC Filings, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

11.6 STARTUP/SME EVALUATION QUADRANT, 2021

For most of the common industries, the market is dominated by established market players such as Microsoft (US), General Electric (US), Siemens (Germany), Amazon Web Services (US), ANSYS (US), Dassault Systèmes (France), and PTC (US). However, there are several SMEs that offer different types of innovative and customized digital twin products/solutions. The digital twin market is competitive, and there is a huge segment of potential clients who are still working on Industry 3.0 principles and are in need of adopting digital twin to optimize their business operations. This serves as a lucrative market for startup players to acquire such clients and deliver innovative projects for them. Digital twin startups have the potential to cater to the local and small businesses in several regions. The startups in the digital twin market are responsible for creating digital twin for niche applications in agriculture, telecommunications, and retail.

11.6.1 PROGRESSIVE COMPANIES

Sight Machine (US), Sensat (UK), Cosmo Tech (France), Artemis Technologies (Ireland), and FEops (Belgium) are progressive companies in the digital twin market. These players offer a strong digital twin solution portfolio. These vendors have been marking their presence in the market by offering solutions as per client's requirements and adopting growth strategies to achieve their target.

11.6.2 RESPONSIVE COMPANIES

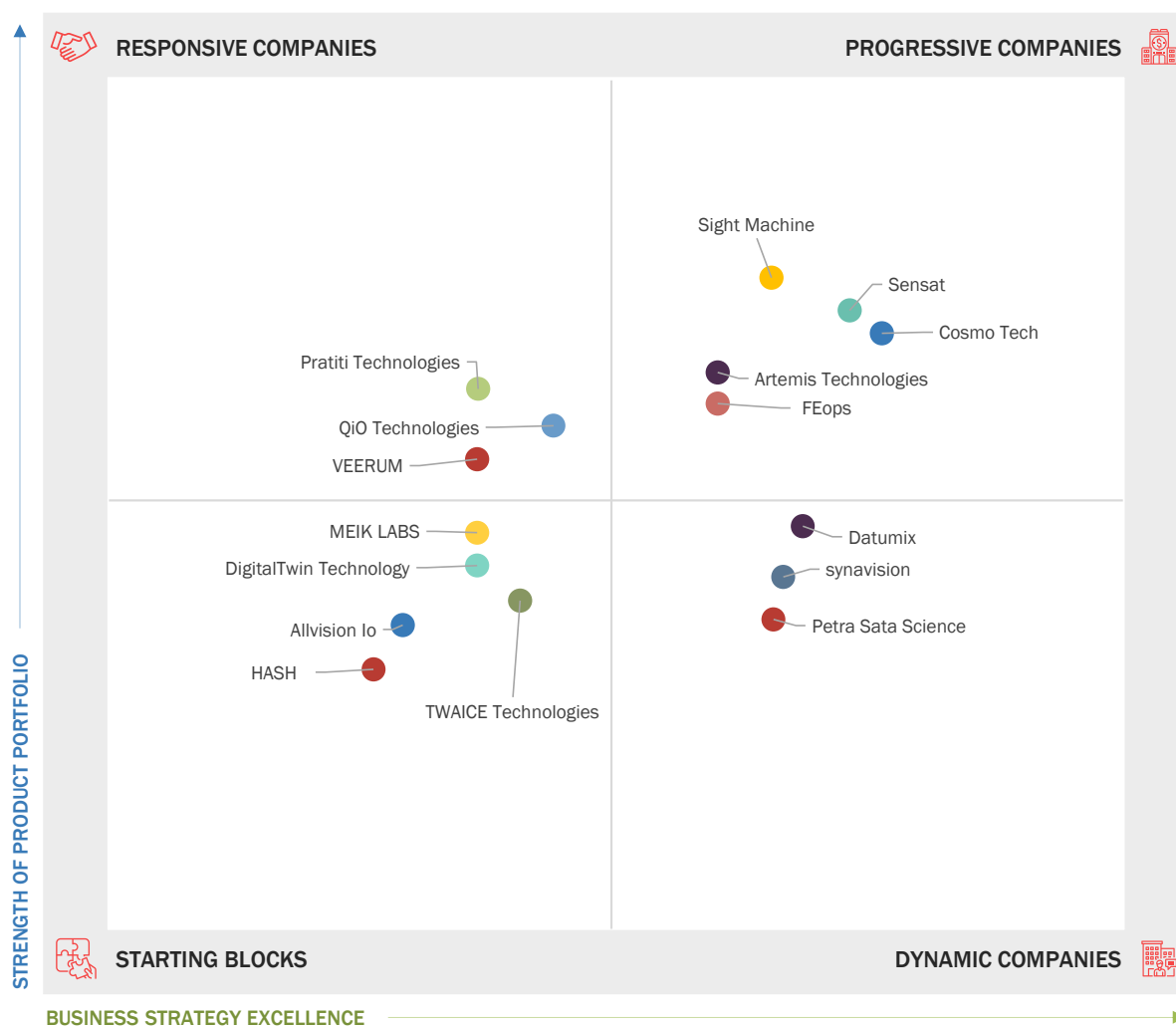
Pratiti Technologies (India), QiO Technologies (UK), and VEERUM (Canada) have been recognized as responsive companies in the digital twin market. They possess innovative solutions to cater to future demands. These companies are focused on improving their product portfolios and have robust potential to build strong business strategies to expand their businesses and stay at par with progressive companies. Responsive companies have been at the forefront of responding to the emerging market trends such as AI, ML, and IoT technology.

11.6.3 DYNAMIC COMPANIES

Dynamic companies offer innovative solutions portfolios and have an extensive network of channel partners and resellers to increase the deployment of their products/services across a multitude of end-user markets. Datumix (US), synavision (Germany) and Petra Sata Science (Australia) are the dynamic companies in this market.

11.6.4 STARTING BLOCKS

MEIK LABS (US), DigitalTwin Technology (Germany), TWAICE Technologies (Germany), Allvision Io (US), and HASH (UK) have been recognized as the starting blocks in the digital twin market. Most of the starting blocks have limited capabilities for boosting their sales across regions.

FIGURE 52 DIGITAL TWIN MARKET, STARTUPS/SME EVALUATION QUADRANT, 2021

Source: Press Releases, Interviews with Experts, and MarketsandMarkets Analysis

11.7 DIGITAL TWIN MARKET: COMPANY FOOTPRINT (25 COMPANIES)

TABLE 109 COMPANY FOOTPRINT

COMPANY	INDUSTRY FOOTPRINT	ENTERPRISE FOOTPRINT	REGION FOOTPRINT	COMPANY FOOTPRINT
General Electric	2.80	4.00	4.00	3.52
IBM	2.80	4.00	3.20	3.24
Microsoft	3.20	4.00	4.00	3.68
Siemens	3.20	4.00	4.00	3.68
PTC	2.80	4.00	4.00	3.52
ANSYS	3.20	4.00	4.00	3.68
SAP	2.67	4.00	4.00	3.47
Oracle	3.20	4.00	4.00	3.68
Robert Bosch	2.40	4.00	4.00	3.36
Dassault Systèmes	2.80	4.00	4.00	3.52
SWIM AI	1.60	4.00	1.33	2.11
Emerson	1.20	4.00	1.33	1.95
ABB	0.80	4.00	4.00	2.72
Honeywell	1.60	4.00	4.00	3.04
Schneider Electric	2.80	4.00	4.00	3.52
NavVis	1.20	4.00	4.00	2.88
DNV	1.20	4.00	4.00	2.88
Autodesk	1.20	4.00	4.00	2.88
ANDRITZ	0.40	4.00	4.00	2.56
Software AG	1.20	4.00	4.00	2.88
Bentley Systems	0.80	4.00	4.00	2.72
River Logic	1.20	4.00	1.33	1.95
Altair	2.00	4.00	4.00	3.20
Johnson Controls	0.80	4.00	4.00	2.72
Amazon Web Services	3.20	4.00	4.00	3.68

Source: Annual Reports, Press Releases, SEC Filings, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 110 COMPANY INDUSTRY FOOTPRINT (25 COMPANIES)

COMPANY	AEROSPACE	AUTOMOTIVE & TRANSPORTATION	INFRASTRUCTURE	HEALTHCARE	ENERGY & UTILITIES	OIL & GAS	AGRICULTURE	TELECOMMUNICATION	RETAIL	OTHERS	TOTAL SCORE
General Electric	Y	Y	Y	Y	Y	Y	N	N	N	Y	2.80
IBM	Y	Y	Y	Y	Y	Y	N	N	N	Y	2.80
Microsoft	Y	Y	Y	Y	Y	Y	N	N	Y	Y	3.20
Siemens	Y	Y	Y	Y	Y	Y	N	N	Y	Y	3.20
PTC	Y	Y	N	Y	Y	Y	N	N	Y	Y	2.80
ANSYS	Y	Y	N	Y	Y	Y	N	Y	Y	Y	3.20
SAP	Y	Y		Y	Y	N	N	N	Y	Y	2.67
Oracle	Y	Y	Y	Y	Y	Y	N	N	Y	Y	3.20
Robert Bosch	Y	Y	Y	N	Y	N	N	N	Y	Y	2.40
Dassault Systèmes	Y	Y	Y	Y	Y	N	N	N	Y	Y	2.80
SWIM AI	N	N	Y	N	Y	N	N	Y	N	Y	1.60
Emerson	N	N	N	N	Y	Y	N	N	N	Y	1.20
ABB	N	N	N	N	Y	N	N	N	N	Y	0.80
Honeywell	N	Y	N	Y	N	Y	N	N	N	Y	1.60
Schneider Electric	N	Y	Y	Y	Y	Y	N	N	Y	Y	2.80
NavVis	N	Y	N	N	N	Y	N	N	N	Y	1.20
DNV	N	N	N	N	Y	Y	N	N	N	Y	1.20
Autodesk	N	N	Y	Y	N	N	N	N	N	Y	1.20
ANDRITZ	N	N	N	N	N	N	N	N	N	Y	0.40
Software AG	N	Y	N	N	Y	N	N	N	N	Y	1.20
Bentley Systems	N	N	N	N	Y	N	N	N	N	Y	0.80
River Logic	N	N	N	N	Y	Y	N	N	N	Y	1.20
Altair	Y	Y	N	Y	Y	N	N	N	N	Y	2.00
Johnson Controls	N	N	Y	N	N	N	N	N	N	Y	0.80
Amazon Web Services	N	Y	Y	Y	Y	Y	Y	N	Y	Y	3.20

Source: Annual Reports, Press Releases, SEC Filings, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 111 COMPANY ENTERPRISE FOOTPRINT (25 COMPANIES)

COMPANY	LARGE ENTERPRISE	SMES	TOTAL SCORE
General Electric	Y	Y	4.00
IBM	Y	Y	4.00
Microsoft	Y	Y	4.00
Siemens	Y	Y	4.00
PTC	Y	Y	4.00
ANSYS	Y	Y	4.00
SAP	Y	Y	4.00
Oracle	Y	Y	4.00
Robert Bosch	Y	Y	4.00
Dassault Systèmes	Y	Y	4.00
SWIM AI	Y	Y	4.00
Emerson	Y	Y	4.00
ABB	Y	Y	4.00
Honeywell	Y	Y	4.00
Schneider Electric	Y	Y	4.00
NavVis	Y	Y	4.00
DNV	Y	Y	4.00
Autodesk	Y	Y	4.00
ANDRITZ	Y	Y	4.00
Software AG	Y	Y	4.00
Bentley Systems	Y	Y	4.00
River Logic	Y	Y	4.00
Altair	Y	Y	4.00
Johnson Controls	Y	Y	4.00
Amazon Web Services	Y	Y	4.00

Source: Annual Reports, Press Releases, SEC Filings, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 112 COMPANY REGION FOOTPRINT (25 COMPANIES)

COMPANY	NORTH AMERICA	EUROPE	ASIA PACIFIC	SOUTH AMERICA	MIDDLE EAST	AFRICA	TOTAL SCORE
General Electric	Y	Y	Y	Y	Y	Y	4.00
IBM	Y	Y	Y	Y	Y	Y	4.00
Microsoft	Y	Y	Y	Y	Y	Y	4.00
Siemens	Y	Y	Y	Y	Y	Y	4.00
PTC	Y	Y	Y	Y	Y	Y	4.00
ANSYS	Y	Y	Y	Y	Y	Y	4.00
SAP	Y	Y	Y	Y	Y	Y	4.00
Oracle	Y	Y	Y	Y	Y	Y	4.00
Robert Bosch	Y	Y	Y	Y	Y	Y	4.00
Dassault Systèmes	Y	Y	Y	Y	Y	Y	4.00
SWIM AI	Y	N	N	N	Y	N	1.33
Emerson	Y	Y	Y	Y	Y	Y	4.00
ABB	Y	Y	Y	Y	Y	Y	4.00
Honeywell	Y	Y	Y	Y	Y	Y	4.00
Schneider Electric	Y	Y	Y	Y	Y	Y	4.00
NavVis	Y	Y	Y	Y	Y	Y	4.00
DNV	Y	Y	Y	Y	Y	Y	4.00
Autodesk	Y	Y	Y	Y	Y	Y	4.00
ANDRITZ	Y	Y	Y	Y	Y	Y	4.00
Software AG	Y	Y	Y	Y	Y	Y	4.00
Bentley Systems	Y	Y	Y	Y	Y	Y	4.00
River Logic	Y	Y	N	N	N	N	1.33
Altair	Y	Y	Y	Y	Y	Y	4.00
Johnson Controls	Y	Y	Y	Y	Y	Y	4.00
Amazon Web Services	Y	Y	Y	Y	Y	Y	4.00

Source: Annual Reports, Press Releases, SEC Filings, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

11.8 COMPETITIVE BENCHMARKING

TABLE 113 DIGITAL TWIN MARKET: DETAILED LIST OF KEY STARTUP/SMES

COMPANY NAME	CATEGORY	OWNERSHIP STATUS	HQ LOCATION	YEAR FOUNDED	EMPLOYEES	FINANCING STATUS	LATEST FUNDING ROUND	TOTAL FUNDING
Pratiti Technologies	Digital Twin Software/ Solution Provider	Private	Pune, Maharashtra, India	2015	150	-	-	-
QiO Technologies	Digital Twin Software/ Solution Provider	Private	Farnborough, Hampshire, UK	2015	11–50	-	-	-
Petra Sata Science	Digital Twin Software/ Solution Provider	Private	Toowong, Queensland, Australia	2015	11–50	-	-	-
Datumix	Digital Twin Software/ Solution Provider	Private	San Francisco, California, US	2017	-	-	-	-
synavision GmbH	Digital Twin Software/ Solution Provider	Public	Bielefeld, Germany	2010	11–50	-	-	-
Sight Machine	Digital Twin Software/ Solution Provider	Private	San Francisco, California, US	2011	50–70	-	Series C April 2019	USD 29.4 Million
Sensat	Digital Twin Software/ Solution Provider	Private	London, England	2017	11–50	-	Series A October 2019	USD 10.0 Million
Simplify GmbH	Digital Twin Software/ Solution Provider	Private	Berlin, Germany	2013	10–20	-	-	-
TWAICE Technologies GmbH	Digital Twin Software/ Solution Provider	Private	Munich, Bavaria, Germany	2018	-	-	Series B over 6 rounds, April 2022	USD 74.4 Million
MEIK LABS Spa	Digital Twin Software/Solutio n Provider	Private	Ñuñoa, Santiago, Metropolitan a, Chile	2018	-	-	-	-
Allvision Io	Digital Twin Software/ Solution Provider	Private	Pittsburgh, Pennsylvania , US	2018	-	-	January 2019	USD 3.2 Million
DigitalTwin Technology GmbH	Digital Twin Software/ Solution Provider	Private	Cologne, North Rhine- Westphalia,	2018	-	-	-	-

Germany

HASH	Digital Twin Software/ Solution Provider	Private	London, England	2019	-		
Artemis Technologies	Digital Twin Software/ Solution Provider	Private	Belfast, Lisburn, Northern Ireland	2015	10–20		
VEERUM	Digital Twin Software/ Solution Provider	Private	Calgary, Alberta, Canada	2014	11–50	Series A February 2021	USD 7.4 Million
Cosmo Tech	Digital Twin Software/ Solution Provider	Private	Lyon, Rhone, France	2010	50–100	Series B over 3 rounds, September 2018	USD 32.9 Million
FEops	Digital Twin Software/ Solution Provider						
Q Bio	Digital Twin Software/ Solution Provider						

TABLE 114 DIGITAL TWIN MARKET: COMPETITIVE BENCHMARKING OF KEY STARTUPS/SMES

COMPANY NAME	ENTERPRISE SIZE		INDUSTRY										GEOGRAPHY					
	LARGE ENTERPRISE	SMES	AEROSPACE	AUTOMOTIVE & TRANSPORTATION	INFRASTRUCTURE	HEALTHCARE	ENERGY & UTILITIES	OIL & GAS	AGRICULTURE	TELECOMMUNICATION	RETAIL	OTHERS	NORTH AMERICA	EUROPE	ASIA PACIFIC	SOUTH AMERICA	MIDDLE EAST	AFRICA
Pratiti Technologies	N	Y				Y	Y					Y	Y	Y	Y			
QiO Technologies	Y	Y				Y	Y			Y		Y	Y	Y	Y			
Petra Sata Science	Y											Y	Y		Y			
Datumix	Y	Y		Y								Y	Y					
synavision	Y	Y			Y								Y	Y	Y			
Sight Machine	Y	Y		Y								Y	Y					
Sensat	Y	Y		Y	Y		Y							Y				
TWAICE Technologies	Y	Y										Y	Y	Y				
MEIK LABS Spa	Y	Y			Y							Y	Y	Y				
Allvision Io	Y			Y									Y					
DigitalTwin Technology	Y	Y			Y							Y		Y				
HASH		Y										Y		Y				
Artemis Technologies	Y											Y		Y				
VEERUM	Y	Y					Y	Y				Y	Y					
Cosmo Tech	Y	Y	Y	Y	Y		Y				Y			Y				
FEops	Y					Y								Y				
Q Bio	Y					Y								Y				
Pratiti Technologies		Y				Y	Y					Y	Y	Y	Y			

11.9 COMPETITIVE SCENARIO AND TRENDS

The Digital twin market is competitive, owing to the presence of the key players. The key strategies adopted by these players to enhance their digital twin portfolio, increase their market share, and expand their presence in the market include product launches, mergers, acquisitions, partnerships, cooperation, alliances, collaborations, agreements, contracts, investments, and expansions. With the adoption of these strategies as well as political and economic factors, the Digital twin market has witnessed a substantial change from January 2020 to June 2022.

11.9.1 PRODUCT LAUNCHES

The product launch strategy accounted for ~38% of the overall growth strategies adopted by the market players. Various companies have adopted this strategy to strengthen their product portfolio and cater to the changing needs of customers. General Electric (US), Oracle (US), ANSYS (US), PTC (US), Siemens (Germany), Microsoft (US), Dassault Systèmes (France), Amazon Web Services (US), and IBM (US) are a few of the players that adopted this strategy.

TABLE 115 PRODUCT LAUNCHES, JANUARY 2022–MAY 2022

MONTH & YEAR	COMPANY	PRODUCT TYPE	PRODUCT NAME	PRODUCT DESCRIPTION
May 2022	Siemens (Germany)	Digital Twin solution	Unnamed digital twin solution for intralogistics	At LogiMAT 2022 (Germany) event, Siemens demonstrated how a digital twin may assist satisfy emerging intralogistics demands with the Dynamic Gapper application. Operators of logistics facilities will also be benefited from this digital twin solution as well, because commissioning times can be cut by up to 40% with digital twin support.
May 2022	Oracle (US)	NFT support for blockchain based digital twin	Oracle Blockchain Platform	Oracle announced native non-fungible tokens (NFT) support on its market-leading enterprise blockchain – Oracle Blockchain Platform. B2C businesses can utilize NFTs to represent consumer-accessible digital twins or direct digital assets and sell them through an e-commerce website based on NFT marketplaces. These marketplaces are supported by the Oracle Blockchain Platform ledger and smart contracts created by the Blockchain App Builder.
April 2022	General Electric (US)	AI/ML technology for digital twin	AI/ML tool	GE Research (US) and GE Renewable Energy (France), subsidiaries of GE, collaborated and developed a cutting-edge artificial intelligence (AI)/machine learning (ML) technology that has the potential to save the worldwide wind industry billions of dollars in logistical expenses over the next decade. GE's AI/ML tool uses a digital twin of the wind turbine logistics process to accurately predict and streamline logistics costs. Based on the current industry growth forecasts, AI/ML might enable a 10% decrease in logistics costs, representing a global cost savings to the wind sector of up to USD 2.6 billion annually by 2030.

February 2022	ANSYS (US)	Solution upgrade	ANSYS Twin Builder 2022 R1	By merging physical and virtual sensors, ANSYS Twin Builder 2022 R1 provides predictive analytics with unrivalled accuracy. New features enable faster deployment of user's digital twin, easier workflows, and web application for online interaction with the user model.
January 2022	General Electric (US)	AI- and ML-based software solution for building digital twin	Autonomous Tuning	GE Digital announced the release of the Autonomous Tuning software that ensures gas turbines operate with optimal combustion, resulting in lower emissions and fuel consumption. Autonomous Tuning uses artificial intelligence (AI) to build a machine learning (ML) digital twin model of a gas turbine to continuously identify ideal flame temperatures and fuel splits to minimize emissions and acoustics. Changes in ambient temperature, gas fuel characteristics, and degradation are detected by the on-premises software, which sends real-time automatic modifications to the controls every two seconds.

Source: Company Websites, Company Presentations, and Press Releases

11.9.2 DEALS

Deals accounted for ~61% of all the strategies adopted by the market players. Deals include mergers, acquisitions, joint ventures, and investments. Companies adopted these strategies to extend their geographic reach and expand their product portfolios. The major players that adopted these strategies are General Electric (US), Oracle (US), ANSYS (US), PTC (US), Siemens (Germany), Microsoft (US), Dassault Systèmes (France), Amazon Web Services (US), SAP (Germany), Robert Bosch (Germany), and IBM (US).

TABLE 116 DEALS, JANUARY 2022–MAY 2022

MONTH & YEAR	DEAL TYPE	COMPANY 1	COMPANY 2	DESCRIPTION	DEAL SIZE
May 2022	Partnership	Microsoft (US)	Willow (Australia)	Willow and Microsoft have a multi-faceted strategic partnership that includes product design, product development, and co-marketing. Both firms previously worked together on the creation of Azure Digital Twins, participated in the private preview in 2018, and were named as a launch partner for public availability in 2020. Microsoft launched a global case study in late 2020 highlighting WillowTwin on Brookfield Properties' (US) flagship – One Manhattan West building.	NA

March 2022	Partnership	Microsoft (US)	Newcrest (Australia)	Microsoft announced a strategic partnership with Newcrest. The mining business of Newcrest would adopt Azure as its preferred cloud provider globally, as well as work on digital twins and a sustainability data model. Both organizations are working together on projects, including the use of digital twins to improve operational performance and developing a high-impact sustainability data model.	NA
February 2022	Collaboration	ANSYS (US)	Amazon Web Services (US)	ANSYS and Amazon Web Services announced a strategic collaboration to transform cloud-based engineering simulations. The partnership will allow ANSYS products to be deployed on AWS, making simulation workloads more user-friendly while also providing scalability and flexibility with easy access to software and storage solutions from any web browser. ANSYS can integrate AWS' comprehensive framework, which includes products such as computation, storage, internet of things (IoT), digital twin, analytics, and machine learning.	NA
February 2022	Acquisition	Robert Bosch (Germany)	Hoerbuerger (Germany)	Bosch Building Technologies (division of Bosch) officially completed the acquisition of Hoerbuerger (Germany). Hoerbuerger has a broad range of products and services centered on building automation and energy management. Digitalization, networking, and artificial intelligence are expanding Bosch's capacity to transform commercial buildings into smart living and working spaces that boost people's security, safety, and comfort while also ensuring higher sustainability. The acquisition of Hoerbuerger is aligned with the Bosch initiative for digital building twin.	NA
January 2022	Alliance	PTC (US)	Schaeffler (Germany)	Schaeffler will use PTC technologies to create an integrated, end-to-end IT landscape as part of a new strategic alliance. PTC's efforts to further develop and evolve market-specific standard solutions to better satisfy the demands of automotive and industrial companies would benefit from Schaeffler's input. The new partnership will expand on Schaeffler's strong twenty-year relationship with PTC, which began with Schaeffler's adoption of PTC's CAD and PLM products. These tools are critical for the introduction and development of Schaeffler's digital twin initiatives as well	NA

as the visualization of 3D models (such as those utilized in augmented reality projects).

Source: Company Websites, Company Presentations, and Press Releases

11.9.3 OTHERS

Other strategies (expansions) accounted for only ~1% of the overall growth strategies adopted by the market players. Oracle (US) was the only player that adopted this growth strategy.

TABLE 117 EXPANSIONS, APRIL 2022

MONTH & YEAR	COMPANY NAME	DEVELOPMENT TYPE	DESCRIPTION
April 2022	Oracle (US)	Expansion	Oracle inaugurated the Oracle Industry Innovation Lab in Deerfield, Illinois, US. The Oracle Industry Innovation Lab is a unique facility for the company's customers to incubate 5G ecosystems and develop cross-industry partnerships. Within the lab, Oracle Communications technology is providing secure and reliable livestreaming of large amounts of data to enable applications and devices for public safety, digital twin, and remote operations experiences.

Source: Company Websites, Company Presentations, and Press Releases

12 COMPANY PROFILES

12.1 KEY PLAYERS

12.1.1 GENERAL ELECTRIC

12.1.1.1 Business overview

General Electric (GE) is a technology and financial services company involved in the development and manufacturing of products that are used in the generation, transmission, distribution, control, and utilization of electricity. The company operates through the Aviation, Healthcare, Power, Renewable Energy, and Corporate segments. GE conducts its business in the areas of additive manufacturing, aviation, capital, digital, healthcare, power, renewable energy and research, and licensing.

The Corporate segment comprises the earnings of the Global Research Center and GE Digital business, as well as the remaining GE Capital operations, including the run-off Insurance business. The company offers its Digital Twin Creation service via the GE Research division located in Niskayuna, New York, US. GE Research is the research and development division of General Electric. Through its Digital Twin Creation, GE builds complex physical assets that are crucial to critical infrastructure, such as energy generation and distribution, air and train transportation, and healthcare. The company builds digital twins for applications in several industries, including aerospace, power & utilities, and automotive & transportation.

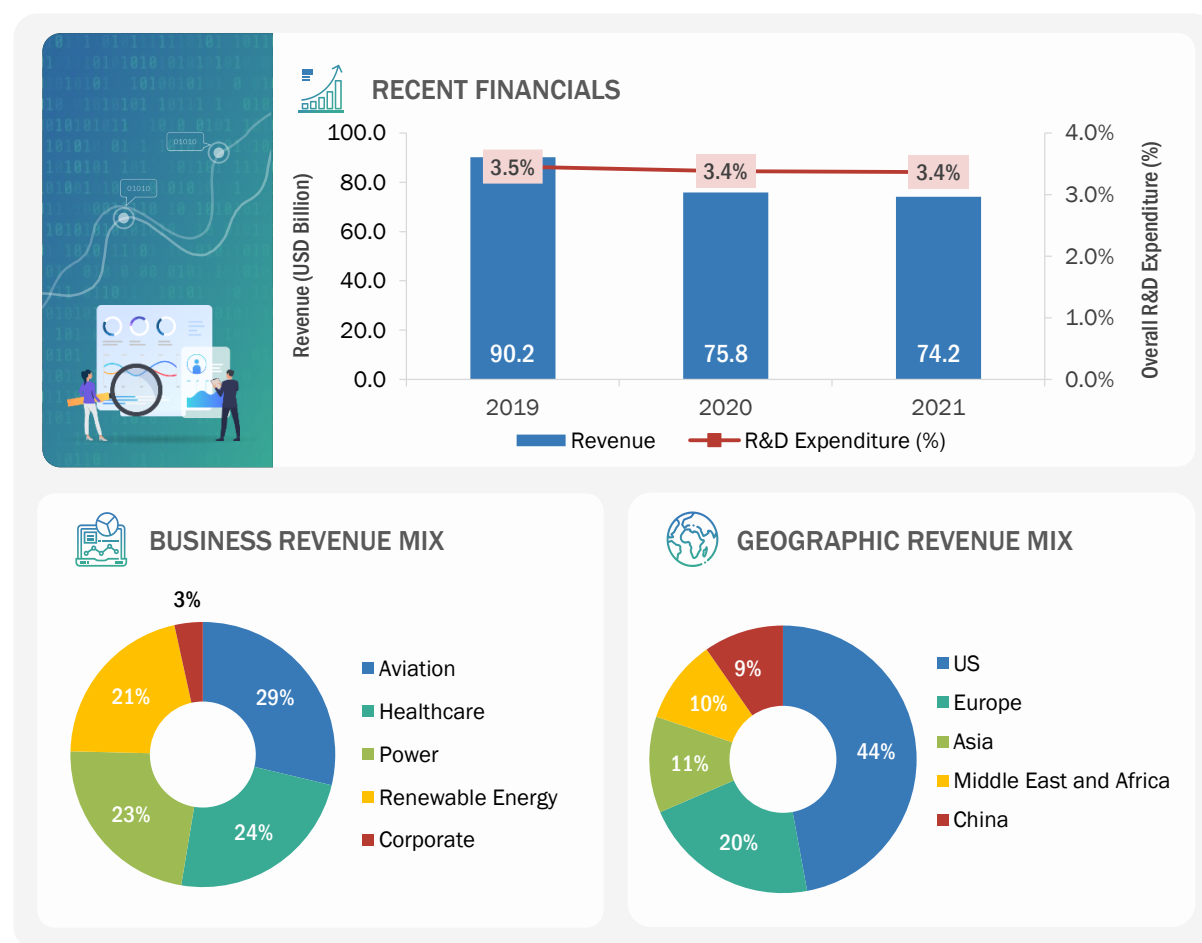
The business model of GE focuses on the development of the Industrial Internet of Things (IIoT) ecosystem. The company has started developing industrial software solutions based on big data under its Intelligent Platforms division. The company is one of the founding members of the Industrial Internet Consortium, which is a non-profit organization. In 2016, GE Digital, a US-based subsidiary of General Electric, made its Predix platform operational for the first time. The Predix platform is a digital software solution for various industries such as telecommunications, power, manufacturing, oil & gas, and mining. GE's digital twin software uses machine learning and advanced analytics to monitor vast amounts of industrial data in real time, including HMI/SCADA systems, time series, alarms, events, and weather and ambient temperature readings. GE Digital, the US-based subsidiary of General Electric, provides digital twin platforms and digital twin software for analytics orchestration and execution of digital twin technologies. GE provides more dimensionality to increase the performance of the asset or system modeled using GE's digital twin. The GE digital twin incorporates the fundamental physics of GE systems, including steady-state and dynamic behavior, as well as how to effectively provide the desired outcome. The GE Digital Twin footprint extends into the embedded control layer. As of December 2021, the company claimed the completion of 50,000 units of single Predix Edge customer deployment across the globe, with almost 400 billion data points processed per year.

The organization serves consumers and communities in over 175 countries and is dedicated to addressing the world's most serious sustainability concerns in energy, health, and aviation. The company provides its product and services in the Americas, Asia Pacific, Europe, Africa, and the Middle East.

TABLE 118 GENERAL ELECTRIC: BUSINESS OVERVIEW

Founding Year	1892
Headquarters – Country	US
Headquarters – City	Boston
Ownership Type	Public (NYSE: GE)
Employee Count (as of December 2021)	168,000

Source: Company Website

FIGURE 53 GENERAL ELECTRIC: COMPANY SNAPSHOT

Note: In the Recent Financials block, the R&D expenditure has been calculated as the percentage of the total revenue. The financial year of the company ends on December 31. The Corporate segment includes Global Research Center and Digital business.

We Source: Company Website and Annual Reports

12.1.1.2 Products/solutions offered

PRODUCT TYPE	PRODUCT/ SOLUTION	DESCRIPTION	INDUSTRY
Digital Twin Solution	Digital Twin Creation	Through the Digital Twin Creation solution, the GE Research team builds custom digital twins for clients. Customer's desired end results serve as the foundation of a digital twin. Based on the data obtained from the design requirements, environmental and operating settings, GE works backwards from the desired outcome.	Aerospace, healthcare, power & utilities, automotive & transportation, and others
Digital twin solution	- Predix Platform – - Predix Essentials - Predix Edge - Predix Platform security - Predix Cloud	Predix Platform is the IoT edge-to-cloud foundation shared by GE Digital applications such as asset performance management (APM) and operations performance management (OPM). It offers secure and scalable edge-to-cloud data connectivity, analytics processing, and a variety of additional underlying services that support client's digital applications and IoT activities. Predix enables the flexible orchestration and execution of GE digital twin analytics, as well as customer-developed analytics, at the edge and in the cloud. Predix streamlines the process of building dynamic digital models of real assets and systems. This enables clients to design and operate their own digital twin apps, as well as gather knowledge and insights about performance and future operations projections. Predix Platform features information/operational technology (IT/OT) connectivity, data processing fabric, analytics execution, edge-to-cloud security, cloud scalability, and a few additional digital application features such as data visualization, dashboarding, alerting, case management, and many more features that support the workflows.	Aerospace, healthcare, power & utilities, automotive & transportation, and others
Digital twin solution	SmartSignal	SmartSignal aids in the detection, diagnosis, forecasting, and prevention of equipment failures. SmartSignal's early detection and time-to-action forecasting are powered by unique digital twin analytics, resulting in improved reliability with maximum operation and maintenance (O&M) efficiency.	Aerospace, healthcare, power & utilities, automotive & transportation, and others

Source: Company Website

12.1.1.3 Recent developments

12.1.1.3.1 Product launches

MONTH & YEAR	PRODUCT TYPE	PRODUCT NAME	PRODUCT DESCRIPTION
April 2022	AI/ML technology for digital twin	AI/ML tool	GE Research (US) and GE Renewable Energy (France), subsidiaries of GE, collaborated and developed a cutting-edge artificial intelligence (AI)/machine learning (ML) technology that has the potential to save the worldwide wind industry billions of dollars in logistical expenses over the next decade. GE's AI/ML tool uses a digital twin of the wind turbine logistics process to accurately predict and streamline logistics costs. Based on the current industry growth forecasts, AI/ML might enable a 10% decrease in logistics costs, representing a global cost saving to the wind sector of up to USD 2.6 billion annually by 2030.
January 2022	AI- and ML-based software solution for building digital twin	Autonomous Tuning	GE Digital announced the release of the Autonomous Tuning software that ensures gas turbines operate with optimal combustion, resulting in lower emissions and fuel consumption. Autonomous Tuning uses artificial intelligence (AI) to build a machine learning (ML) digital twin model of a gas turbine to continuously identify ideal flame temperatures and fuel splits to minimize emissions and acoustics. Changes in ambient temperature, gas fuel characteristics, and degradation are detected by the on-premises software, which sends real-time automatic modifications to the controls every two seconds.
January 2022	Data management software solution	Proficy Historian 2022	The GE Proficy Historian 2022 is a powerful industrial time-series and alarms & events (A&E) data collection software for on-premises and cloud-based storage and analysis. It collects industrial data at a very high speed, normalizes data, and stores data securely; distributes data; and enables for quick retrieval and analysis. The decentralized data management can reduce maintenance costs and downtime for large utilities and industrial companies with distributed assets. It also improves data security and enables enterprise-wide industrial data management.
August 2021	Analytics software	Proficy CSense	GE Digital announced upgrades to its leading on-premises analytics software, Proficy CSense. The software uses AI and machine learning technologies to create process digital twins, which allows engineers to merge data from different sources to quickly identify problems, determine root causes, predict future performance, and automate actions.
June 2021	Software suite	Proficy Operations Analytics Software	Proficy Operations Analytics was added to GE Digital's industry-leading Proficy family of software solutions. This solution increases efficiency with cloud-based, enterprise-wide predictive analytics. Self-provisioning processes automatically provide context to digital twins, allowing for automated analytics on all process and event data. Thirty pre-built industrial predictive applications can be simply assigned to digital twins without requiring any data science or application development.

April 2021	Software release (new version)	SmartSignal	GE Digital released a new version of its SmartSignal predictive maintenance software, which assists industrial firms in proactively predicting, diagnosing, forecasting, and preventing equipment faults. SmartSignal can be purchased separately or as part of the GE Digital APM software package. New features include unique time-to-action analytics to forecast when a malfunction may occur; improved visualization and analysis tools; and continued expansion of the digital twin analytics and content catalog, which now covers over 300 industrial equipment classes from multiple manufacturers.
June 2020	Software suite	Advanced Distribution Management Solutions (ADMS)	GE Digital updated its Advanced Distribution Management Solutions (ADMS) software suite for secure and reliable management of the distribution grid and predictive analytics for autonomous operations. This enhancement will help in the import of low-voltage networks from a geographic information system (GIS) into an existing ADMS system, which creates a network digital twin of the entire network, allowing teams from the control room to the field to view how their decisions could impact the wider grid.
June 2020	Software suite	Asset Performance Management (APM)	GE Digital launched the latest version of its Asset Performance Management (APM) software suite to allow companies in the asset and process-intensive industries, such as oil & gas, chemicals, and power, to bring together data from all equipment across the plant and quantify risk reduction.
December 2019	Data management software solution	Historian 8.0	GE Digital added significant enhancements to its Proficy Historian offering with the launch of Historian 8.0. The updates include tag mapping, advanced trend analysis, and the ability to define an asset model, which allows users to put data in context with business needs.
April 2019	Cloud solution	Manufacturing Data Cloud (MDC)	GE Digital launched the Predix Manufacturing Data Cloud (MDC), a cloud solution built to consolidate and transform manufacturing data across plants for enterprise cloud storage and analysis. Predix MDC will allow manufacturers with operational analysis in the cloud offer flexibility in deployment, help reduce the size of on-premises systems and make them run more efficiently.

Source: Company Website and Press Releases

12.1.1.3.2 Deals

MONTH & YEAR	DEAL TYPE	COMPANY 1	COMPANY 2	DESCRIPTION	DEAL SIZE
December 2021	Acquisition	General Electric (US)	Opus One (Canada)	In 2021, GE Digital (a subsidiary of GE) acquired Opus One Solutions Energy Corporation (Opus One), a Canada-based software company that helps electric utilities optimize energy planning, operations, and market management. The software solution developed by GE is opening the way for an autonomous grid that can detect problems and discover the most cost-efficient solutions, which is a critical step in hastening	NA

				the energy transition. GE Digital and Opus One's combined technology portfolio will enable widespread and quick adoption of renewables and distributed energy resources (DERs) at all points on the grid.
June 2019	Agreement	General Electric (US)	Wabtec Corporation (US)	GE Digital (a subsidiary of GE) and Wabtec Corporation (US), a leading supplier of locomotive products, signed a multi-year agreement to continue the commercialization of applications such as Predix Operations Performance Management (OPM), and Predix Asset Performance Management (APM) solutions, with a primary focus on the mining domain.

Source: Company Website

12.1.1.4 MnM view

12.1.1.4.1 Key strengths/right to win

General Electric's strength is its global business network, which enables it to offer high-quality digital twin platforms, software solutions, and excellent services in locations all over the world. By leveraging its network, General Electric can anticipate and elicit a wide range of customers' needs. GE offers the Digital Twin Framework (GENIX) platform that helps to build digital twins for client's specific business needs. GENIX makes it possible to create digital twins that represent a single asset, an integrated system of assets, or a fleet of assets. Cloud-based digital twins become a single source of truth for all information about an asset, including data about its history and present state, condition, and performance.

General Electric has continuously made innovations in its products and solutions. For instance, the company expanded its digital twin portfolio with the launch of the latest version of its Proficy Historian 2022 offering with updates such as tag mapping and advanced trend analysis. Similarly, GE also updated its SmartSignal software solution with the release of its new version in April 2021. The new version of SmartSignal digital twin solution enables industrial end users to predict, diagnose, forecast, and prevent downtime of critical equipment.

GE offers solutions for the overall digital twin ecosystem that work in tandem and are well interlinked with each other. For instance, GE offers Digital Ghost, which is a real-time, active cyber defense for securing industrial assets and critical infrastructure from both malicious cyber-attacks and naturally occurring faults. Critical assets in today's power plants, such as gas turbines, are all governed by physics, which GE fully comprehends and uses to develop digital twins of these equipment. Digital Ghost, on the other hand, continuously monitors the asset's activity using these digital twins, knowledge of the underlying control systems, and very advanced artificial intelligence algorithms. Even though the operator's user interface suggests everything is fine, the Digital Ghost can recognize if the machine is behaving abnormally due to a cyber-attack.

12.1.1.4.2 Strategic choices made

GE has strengthened its position in the market by investing in technology, globalization, partnerships, acquisitions, and agreements. The company has been investing in disruptive innovation, which is likely to improve its industrial productivity in the future. For instance, in January 2022, GE Digital, a subsidiary of General Electric, launched the GE Proficy Historian 2022, a historian software solution that collects industrial time-series and alarms & events (A&E) data to analyze asset and process performance to drive business value. Predix MDC will allow manufacturers with operational analysis to reduce the size of on-

premises systems and make them run more efficiently. Proficy Historian 2022 features a flexible and scalable design—from sensor to enterprise—that makes it ideal for industrial internet deployments. For instance, GE's Proficy Historian can help to create a single industrial data repository across the client's entire plant for improved visibility and insights, resulting in increased plant capacity. A global chemical company used GE's Proficy Historian solution to increase its capacity by 20%. Similarly, GE's Proficy Historian can help large power monitoring and diagnostics centers to save tens of millions of dollars with a reduction in unplanned downtime and IT infrastructure costs.

With the expansion of several new developments in several software suits and Predix platform in the last three years, General Electric aims to capitalize on its strengths in the digital twin business to maintain its position as one of the global leaders in the digital twin market. GE is also focusing on inorganic growth strategies to enhance its digital twin market position. For instance, in December 2021, GE Digital acquired Opus One Solutions Energy (US) to accelerate the adoption of renewables and distributed energy resources (DERs) at all points on the grid. The acquisition will bring together a skilled team of Opus One engineers and management dedicated to grid innovation, as well as a century of continuous GE innovation. This combined effort will create an interoperable portfolio that will assist utilities in forecasting, managing, optimizing, and trading renewable energy. Moreover, General Electric is making investments in Egypt to launch new renewable projects. The company should focus on expanding its presence in the Middle East & Africa, as the logistics and mining industries are witnessing healthy growth in these regions. Digital twins are used in these industries to reduce design iterations, optimize storage locations, compress schedules, and reduce production costs.

12.1.1.4.3 Weaknesses and competitive threats

The company experienced a decline of 2.2% in revenue in FY 2021 over FY 2020, owing to the impact of rising inflationary costs of transportation and commodities on various industries. GE has operations in nearly every country, serves clients in over 175 countries, and the company generated ~56% of its revenue outside the US in 2021. As a result of dense global network, GE's sales are affected by local and regional economic environments and policies in the US and other markets that the company serves, including interest rates, interest rates, monetary policy, inflation, economic growth, recession, commodity prices, currency volatility, currency controls or other limitations on the ability to expatriate cash, sovereign debt levels and actual or anticipated defaults on sovereign debt in the Middle East & Africa. The company can focus on pursuing strategic alliances to expand its partner network in the region. Short-term demand fluctuations amid an uncertain market outlook may hamper the revenue generation of the company. In addition, the effect of above-mentioned risks continues to be heightened by the impact of the COVID-19 pandemic in 2021. As a result, the growth of the digital twin market also stagnated to some extent in 2020 and continued to remain impacted in 2021–22 with a slower growth rate as compared to predicted growth rate by GE in the absence of COVID-19. GE faces fierce competition to retain existing clients, as new emerging players are performing well with innovative product launches in the field of digital solutions. A few of the major competitors of GE are IBM (US), Microsoft (US), and Robert Bosch (Germany).

12.1.2 MICROSOFT

12.1.2.1 Business overview

Microsoft offers significantly diversified software products and is involved in the development, licensing, and providing support for the same. The company provides cloud-based services. The company operates through three business segments: Intelligent Cloud, More Personal Computing, and Productivity and Business Processes.

Microsoft's Intelligent Cloud segment includes public, private, and hybrid cloud server solutions and services that can empower modern businesses and developers. The following are the main components of this segment: server products and cloud services, including Azure; SQL Server, Windows Server, Visual Studio, System Center, and related client access licenses (CALs); and GitHub. In addition, Enterprise Services, which include Premier Support and Microsoft Consulting, are also a part of the Intelligent Cloud segment. Azure Digital Twins allows developers to create next-generation IoT applications that model entire environments in real time. Azure Digital Twins can assist predict the future of any connected environment by tracking its history.

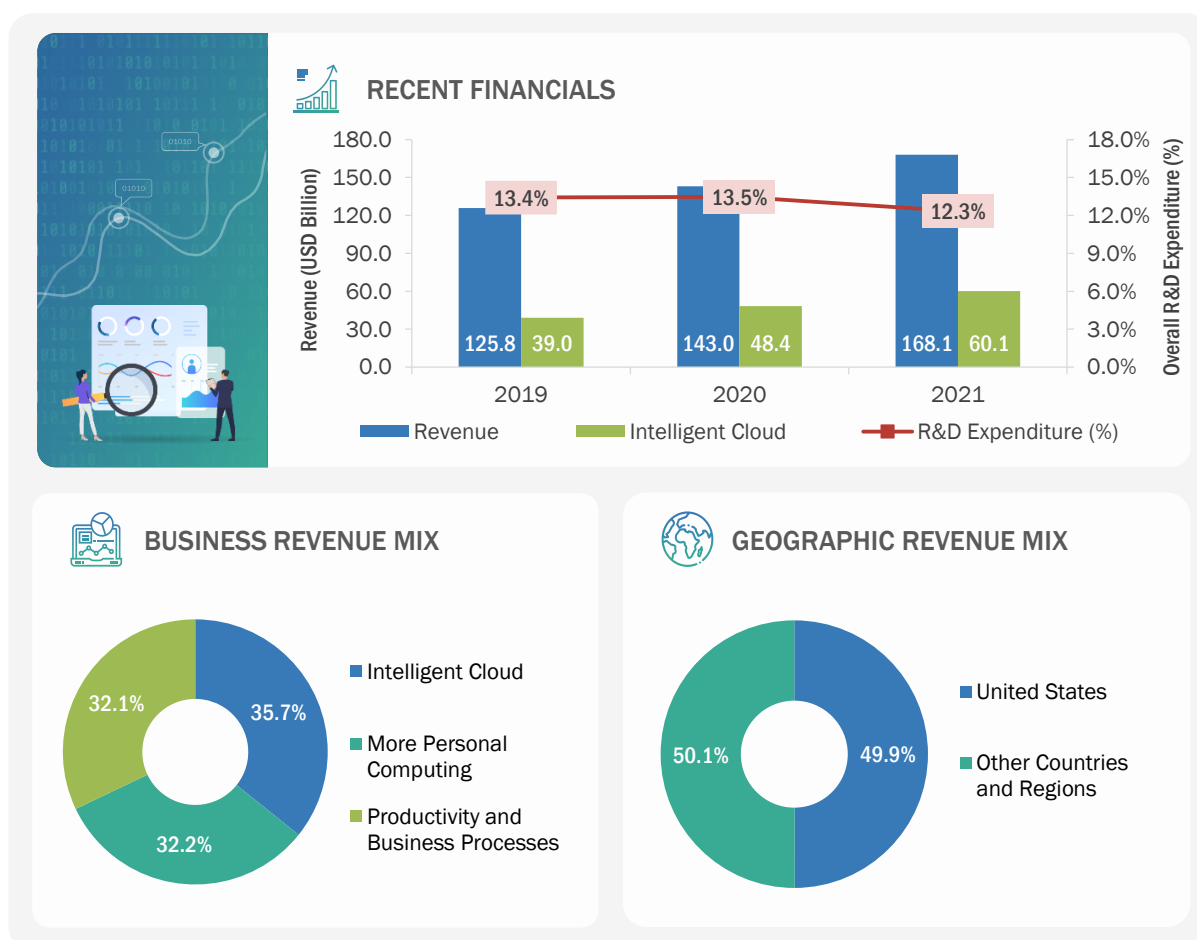
Microsoft offers a wide range of products and solutions for IoT, artificial intelligence (AI), AR, and big data analytics. The AI business of the company is involved in offering products such as Cortana Analytics, Project Active Directory Application Mode (ADAM), Microsoft Pix, Microsoft Azure, and Project Malmo. The company's cloud-based computing services include Bing, Microsoft Azure, Microsoft Dynamics CRM Online, Microsoft Office 365, OneDrive, Skype, Xbox Live, and Yammer. In addition to selling individual products and services, the company offers suites of products and services. The services provided by the company include consulting, product and solution support services, and training and certification of computer system integrators and developers. Azure is a cloud platform offered by Microsoft that targets enterprise markets. Combined Azure with the Predix platform creates digital twins and increases the possibility of cross-domain application areas. The company has been working on the Microsoft HoloLens to create a digital twin and allow manufacturers and their task workers to have relevant data that is entirely immersive to visualize in the context of the real world. With this module, users will be able to connect real-world operating data to digital assets (live or historical IoT data), which allows users to visualize operating conditions, identify anomalies in performance or equipment health /integrity, determine trends and patterns in data over time, and simulate future conditions or operations situations. Microsoft's commercial cloud revenue, which includes Azure, Office 365 Commercial, the commercial portion of LinkedIn, Dynamics 365, and other commercial cloud properties, was USD 69.1 billion, USD 51.7 billion, and USD 38.1 billion in fiscal years 2021, 2020, and 2019, respectively.

Microsoft delivers products/services/solutions in nearly 250 countries with the help of partners. The company caters to customers from Europe, North America, Asia Pacific, South America, the Middle East, and Africa through a global network of its partners.

TABLE 119 MICROSOFT: BUSINESS OVERVIEW

Founding Year	1975
Headquarters Country	US
Headquarters City	Redmond, Washington
Ownership Type	Public (NASDAQ: MSFT)
Employee Count (as of June 2021)	181,000

Source: Company Website

FIGURE 54 MICROSOFT: COMPANY SNAPSHOT

Note: In the Recent Financials section, the R&D expenditure has been calculated as the percentage of the total revenue. The financial year of the company ends on June 30.

Source: Company Website and Annual Reports

12.1.2.2 Products/solutions offered

PRODUCT TYPE	PRODUCT/ SOLUTION	DESCRIPTION	INDUSTRY
Digital twin solution	Azure Digital Twin	Azure Digital Twins is a platform as a service (PaaS) offering that allows users to create knowledge graphs based on digital models of entire environments. Buildings, industries, farms, energy networks, railways, stadiums, and more—even entire cities—could be among these environments. It helps to gain insights that will assist users/clients in developing better products, reducing expenses, and improving customer experiences.	Infrastructure, aerospace, power & utilities, automotive & transportation, and others

Source: Company Website

12.1.2.3 Recent developments

12.1.2.3.1 Product launches

MONTH & YEAR	PRODUCT TYPE	PRODUCT NAME	PRODUCT DESCRIPTION
December 2020	Digital Twin	Azure Digital Twins	Microsoft announced the general availability of Azure Digital Twins. By integrating assets such as internet of things devices and current business systems, Azure Digital Twins allows customers to model settings such as buildings, factories, farms, energy networks, railways, stadiums, and cities. Azure Digital Twins interacts with Azure data, analytics, and AI services to enable customers track the past and forecast the future.

Source: Company Website and Press Releases

12.1.2.3.2 Deals

MONTH & YEAR	DEAL TYPE	COMPANY 1	COMPANY 2	DESCRIPTION	DEAL SIZE
May 2022	Partnership	Microsoft (US)	Willow (Australia)	Willow and Microsoft have a multi-faceted strategic partnership that includes product design, product development, and co-marketing. Both firms previously worked together on the creation of Azure Digital Twins, participated in the private preview in 2018, and were named as a launch partner for public availability in 2020. Microsoft launched a global case study in late 2020 highlighting WillowTwin on Brookfield Properties' (US) flagship – One Manhattan West building.	NA
March 2022	Partnership	Microsoft (US)	Newcrest (Australia)	Microsoft announced a strategic partnership with Newcrest. The mining business of Newcrest would adopt Azure as its preferred cloud provider globally, as well as work on digital twins and a sustainability data model. Both organizations are working together on projects, including the use of digital twins to improve operational performance and developing a high-impact sustainability data model.	NA

November 2021	Partnership	Microsoft (US)	Cosmo Tech (France)	Microsoft partnered with Cosmo Tech, a French start-up specialized in Simulation Digital Twin technology. Cosmo Tech's Simulation Digital Twin complements Microsoft's Azure Digital Twin (ADT) PaaS offering by bringing a 360-degree, dynamic, and simulated vision that delivers the prescriptive and predictive analytics that ADT customers seek. The goal of the collaboration is to add a predictive layer to the Redmond firm's real-time graph models, which will be enabled by Cosmo Tech's Simulation Digital Twins.	NA
May 2021	Partnership	Microsoft (US)	Mars (US)	Microsoft and Mars announced the expansion of their long-standing partnership with a new agreement to help Mars accelerate its digital transformation by leveraging the Microsoft Azure platform to improve operational speed and intelligent manufacturing supply chains. The strategy calls for a cloud-first approach, with AI, IoT, and digital twins used across the board, an emphasis on intelligent supply chains, and hyper-relevant consumer experiences. Mars' global portfolio of confectionery, pet care, pet services, and food businesses are working together to combine transparent and responsible data, AI, and digital technologies.	NA
December 2020	Collaboration	Microsoft (US)	Johnson Control (Ireland)	Microsoft and Johnson Controls, the world's leading provider of smart and sustainable buildings, announced a global partnership to digitally revolutionize how buildings and spaces are designed, built, and operated. Both businesses have launched OpenBlue Digital Twin and Azure Digital Twin integration. As a key partner for Azure Digital Twins, Johnson Control's OpenBlue Digital Twin is a comprehensive platform that will support the entire ecosystem of building and device management technologies with digital cloud technologies.	NA
October 2020	Partnership	Microsoft (US)	ANSYS (US)	ANSYS partnered with Microsoft to deliver public cloud high-performance computing (HPC) and digital twin technologies that will dramatically advance the state-of-the-art in engineering simulation across a variety of industries. Customers may boost productivity, reduce development costs, and shorten time to market by combining ANSYS simulation solutions with Microsoft Azure cloud, HPC, digital twin, and IoT services.	NA

September 2020	Partnership	Microsoft (US)	Telstra (Australia)	Microsoft and Telstra extended their long-standing partnership with intentions to collaboratively develop cloud-based products on Azure while leveraging the Telstra Data Hub. The two companies will use IoT, edge computing, artificial intelligence, and digital twin capability to develop solutions in areas like asset tracking, supply chain management, telematics, and smart spaces.	NA
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Source: Company Website

12.1.2.4 MnM view

12.1.2.4.1 Key strengths/right to win

Microsoft debuted Azure Digital Twins in Orlando, Florida, in October 2018, allowing customers to create digital models and knowledge graphs using AI algorithms. By integrating assets such as IoT (Internet of Things) sensors and current business systems, Azure Digital Twins allows customers to model environments such as factories, farms, energy networks, buildings, stadiums, railways, and cities. It uses models that employ the Digital Twins Definition Language (DTDL) to describe twins in terms of their components, state attributes, instructions, telemetry events, and relationships. With an open modeling language, users can easily model and construct digital representations of networked settings. Model cities, factories, farms, energy networks, railways, stadiums, and even entire cities can be created with the help of digital twin, modeling language, and simulation. Microsoft focuses on bringing these digital twins to life with a live execution environment that tracks twin evolution. In addition, powerful query APIs also provide meaningful insights into the behavior of simulated settings, and it connects with Azure data analytics that helps in creation of a powerful digital twin to optimize the businesses of the clients.

Microsoft is developing new digital tools, which will enable its cloud-connected machines to predict the exact point at which an equipment needs maintenance, averting possible breakdowns. Many industrial manufacturers are leveraging the capability of HoloLens to develop a virtual service business, which allows senior technicians to guide other technicians remotely across the globe. This gives the company a competitive advantage in the digital twin market. The company also spent about 12% to 13% of its revenue every year in the last three years on the development of innovative solutions. It has more than 10,000 patents in its digital solution for Azure portfolio.

12.1.2.4.2 Strategic choices made

The company is focusing on inorganic growth strategies, such as partnerships and collaborations, to maintain its position in the digital twin market. For instance, Microsoft is working with ANSYS (US), a global leader in engineering simulation, to extend its Microsoft Azure digital twins with ANSYS Twin Builder to enable clients to significantly improve operations. The partnership is expected to empower Microsoft's enterprise clients to predict an asset's future performance and reduce downtime expenses more accurately. This will enable users to minimize product maintenance costs and speed up the launch of high-quality products.

Furthermore, Microsoft is also working with a few other companies on the development of digital twin technology. Telstra (Australia), Willow (Australia), Johnson Control (Ireland), Newcrest (Australia) are a few of the companies involved in the development and deployment of digital twin technology. Microsoft is also working with close collaboration with startups. For instance, Microsoft has partnered with Cosmo Tech (France), a French startup, to enrich its digital twins. Cosmo Tech's Simulation Digital Twin platform, which is hosted on Microsoft Azure, gives Microsoft's enterprise customers a unified, comprehensive data model

that is connected to the real world and predicts the evolution of their complex organizations in uncertain environments, allowing them to make better decisions.

12.1.2.4.3 Weaknesses and competitive threats

Microsoft faces high competition from other companies who offer digital twin solutions. A few of its competitors, including General Electric (US), IBM (US), PTC (US), and Siemens (Germany), are rapidly increasing their market share in the digital twin market. The company follows the “Freemium” pricing strategy, allowing users to use basic features for free but pay a premium for additional features. Though a wonderful strategy to get a fast-mover advantage, but such a strategy might have an adverse impact on the company’s revenue in the long run.

12.1.3 SIEMENS

12.1.3.1 Business overview

Siemens is a global technology provider that specializes in manufacturing, infrastructure, transportation, and healthcare. Siemens operates through the following reportable business segments: Digital Industries, Smart Infrastructure, Siemens Healthineers, Mobility, Portfolio Companies, Siemens Financial Services, and Reconciliation to Consolidated Financial Statements. The company focuses on areas such as electrification, automation, and digitalization. Siemens allows its customers to improve their businesses by connecting the physical and digital worlds, enabling them to create actual value for clients.

The company offers cloud-based engineering solutions that connect users with a private cloud. Users can access the plant controller with the help of the totally Integrated Automation (TIA) Portal Cloud Connector, or via MindSphere (cloud solution by Siemens) for additional digital services offered by Siemens. Additionally, PLCSim Advanced with interfaces is used as simulation software such as Plant Simulation and Process Simulation. A Simatic S7-1500 controller can be simulated as a digital twin with PLCSim Advanced to achieve efficient virtual commissioning. The digital twin offerings by Siemens are not confined to one particular business segment of the company. The Smart Infrastructure segment of Siemens continues to promote the digitalization of its switching and control products with built-in intelligence, cloud connectivity, and more remote diagnostics and edge computing capacity. Siemens digital twins of products, building systems, and grids provide customer values from online configuration and parametrization to operation and maintenance planning. Similarly, the Siemens Mobility segment develops efficient vehicle platforms with low lifetime costs and maximum customization options with continuous R&D investments in connectivity, simulation, digital twin, data analytics and AI, additive manufacturing, and software systems and processes.

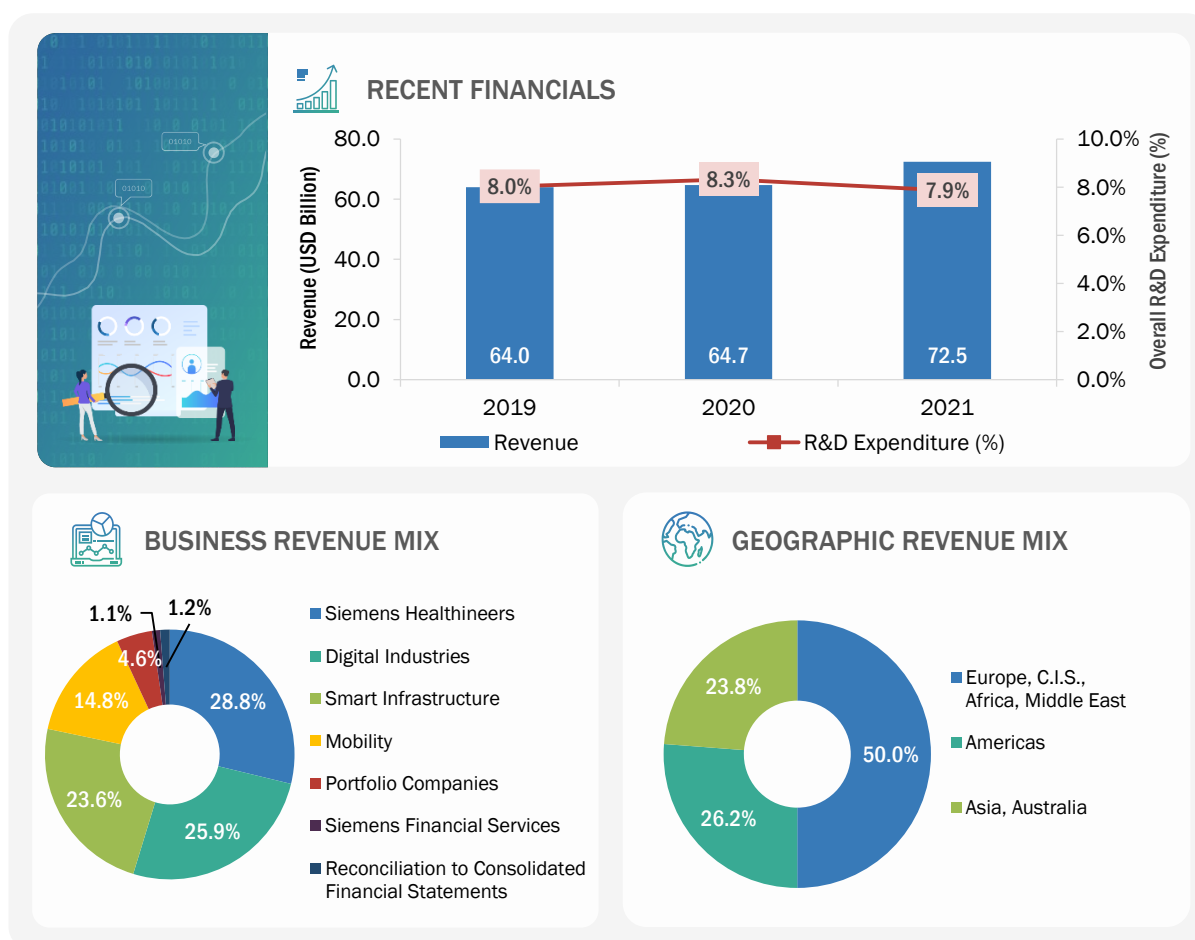
Siemens works on making digital twins more accessible to consumers and reducing the time it takes to develop and adapt them. To this end, the company devised the Digital Twin Operations (DT Ops) strategy. With DT Ops, the digital twin is approached in a modular manner. It contains best practices and tools for managing the lifecycles of digital twins by combining modeling and simulation of digital twins (DT) with IT operations (Ops). Individual components are scalable and reusable, allowing the users to work much faster. Siemens' Digital Enterprise segment helps to solve sustainability challenges by bringing the physical and digital worlds together in a continuous data flow. The comprehensive digital twin is an important aspect of that. It gathers information about products throughout their full lifecycle, from conception to manufacturing and deployment. Siemens cutting-edge solutions enable the industry to better interpret this data and make more efficient use of restricted resources.

Siemens operates in more than 200 countries. The company has a strong global presence, with R&D centers in more than 39 countries that focus on creating innovative solutions. The company has offices in the US, India, Germany, Singapore, China, Japan, South Korea, and Italy.

TABLE 120 SIEMENS: BUSINESS OVERVIEW

Founding Year	1847
Headquarters Country	Germany
Headquarters City	Munich
Ownership Type	Public (ETR: SIE)
Employee Count (as of September 2021)	303,000

Source: Company Website

FIGURE 55 SIEMENS: COMPANY SNAPSHOT

Note: In the Recent Financials block, the R&D expenditure has been calculated as the percentage of the total revenue. The financial year of the company ends on September 30. The currency conversion rates considered for 2019, 2020 and 2021 are EUR 1 = USD 1.09368, EUR 1 = USD 1.17052, and EUR 1 = USD 1.16417 respectively.

Source: Company Website and Annual Reports

12.1.3.2 Products/solutions offered

PRODUCT TYPE	PRODUCT/ SOLUTION	DESCRIPTION	INDUSTRY
Digital twin solution	Simcenter, NX, Teamcenter	Siemens offers digital twin in automation, machining, as well as production, that play an important role in streamlining machine manufacturing engineering processes. This digital twin solution in the automation environment allows process stages to be executed in parallel rather than sequentially. Digital twins also improve processes while the machine is in operation, whether as a digital twin during machining or as a digital twin providing real-time collision protection. Simcenter, NX, Teamcenter, and Capital are a few examples of software solutions offered by Siemens for digital twin.	Aerospace, infrastructure, retail, energy & utilities, healthcare, and others

Source: Company Website

12.1.3.3 Recent developments

12.1.3.3.1 Product launches

MONTH & YEAR	PRODUCT TYPE	PRODUCT NAME	PRODUCT DESCRIPTION
May 2022	Digital Twin solution	Unnamed digital twin solution for intralogistics	At LogiMAT 2022 (Germany) event, Siemens demonstrated how a digital twin may assist satisfy emerging intralogistics demands with the Dynamic Gapper application. Operators of logistics facilities will also be benefited from this digital twin solution as well, because commissioning times can be cut by up to 40% with digital twin support.
September 2021	Product expansion	Xcelerator	Siemens showcased extended hardware and software offering for Sinumerik One at EMO Milano (Italy) event 2021. It included: ▪ New NCU 1740 for higher scalability of Sinumerik One ▪ Sinumerik Run MyRobot /Direct Control V2.0 and Run MyRobot /Direct Handling V2.0 available for Sinumerik One ▪ New functions for the Digital Twin in Create MyVirtual Machine and Run MyVirtual Machine
September 2020	Software upgrade (one complete, digital twin for the process industry)	PlantSight	Siemens' PlantSight May 2020 version included expanded features for digital reliability. The upgraded version of PlantSight allows project data generated during modernization or redesign to be transferred into the context of facility operations. The operator can rearrange the major windows and customize the information to their preferences. As a result, the operator is constantly aware of all pertinent information. The Digital Twin and its assets are offered for facility operations and can be coupled with simulation and real-time data to make the best decisions possible throughout everyday operations.
June 2020	Software solution	Teamcenter X	Siemens launched the Teamcenter X, a new product lifecycle management (PLM) solution that will enable companies to quickly implement, scale, and integrate industry-leading PLM technology across functional disciplines. Teamcenter X is a software as a service (SaaS) offering that enables businesses of all sizes to quickly gain value without the IT resources that on-premises PLM implementations require. Moreover, access to Teamcenter X on the cloud will enable the creation of a complete digital twin, with multi-domain design and bill of materials integration.
June 2020	Software solution	NX Sketch	Siemens (Digital Industries Software segment) announced a new 2D concept capture solution NX Sketch, the industry's first artificial intelligence-powered CAD sketching technology. The Siemens Xcelerator portfolio's NX software is a leading solution for creating the most comprehensive digital twin. This technology allows for substantial freedom in concept design sketching, as well as the ability to work with imported data, allowing for rapid design iteration on historical data and the ability to deal with tens of thousands of curves in a single sketch.

Source: Company Website and Press Releases

12.1.3.3.2 Deals

MONTH & YEAR	DEAL TYPE	COMPANY 1	COMPANY 2	DESCRIPTION	DEAL SIZE
December 2021	Acquisition	Siemens (Germany)	EcoDomus (US)	Siemens signed an agreement to acquire a digital twin software for buildings from EcoDomus, a US-based startup. With this acquisition, Siemens Smart Infrastructure segment expanded its digital building portfolio, including its cloud-based building operations twin software and its flagship building management platform Desigo CC.	NA
November 2021	Partnership	Siemens (Germany)	Hyundai Motor Company and Kia Corporation (South Korea)	Siemens announced a new technology partnership with Hyundai Motor Company and Kia Corporation (Hyundai and Kia) to help them accelerate their digital transformation and create a new mobility future. Hyundai and Kia opted for Siemens' software, which incorporates the world's most comprehensive digital twin.	NA
June 2021	Cooperation	Siemens (Germany)	Threedy (Germany)	Siemens and Threedy cooperated on a visual computing platform for immersive experiences. This collaboration concentrated on unlocking digital model data for drive systems throughout the product lifetime, accessing CAD data over the web, augmented reality, and virtual reality, and combining IIoT with visual computing for design collaboration and troubleshooting. Due to the integration of Threedy's visualization solution, users can have access to visual digital twin information that was previously only available to engineers.	NA
June 2020	Acquisition	Siemens (Germany)	UltraSoC (UK)	Siemens acquired UltraSoC (UK), a provider of instrumentation and analytics solutions with exceptional capabilities in cybersecurity, to integrate UltraSoC's technology into its Xcelerator product portfolio. The combination of Siemens and UltraSoC technology is expected to benefit the entire semiconductor product lifecycle, including structural, electrical, and functional capabilities of system on chips (SoCs). It also supports Siemens's comprehensive digital twin with UltraSoC providing monitoring of the real device.	NA

June 2020	Collaboration	Siemens (Germany)	Acciona (Spain)	Siemens collaborated with Acciona (Spain), a conglomerate dedicated to the development of infrastructure and renewable energy, to create a digital twin for water treatment plants in the Middle East. The digital twin will help analyze, optimize, improve productivity, and reduce operating times.	NA
May 2020	Collaboration	Siemens (Germany)	Atos (France)	Siemens and Atos (France), a global leader in digital transformation, collaborated to create a new digital twin solution for the pharmaceutical industry, with an aim to improve production efficiency and flexibility in the manufacturing of pharmaceuticals.	NA
January 2020	Partnership	Siemens (Germany)	Arm (UK)	Siemens partnered with Arm (UK), a leading semiconductor IP (Intellectual Property) provider, to bring together leading integrated circuit methodologies, processes, and tools to help the automotive supply chain solve design and verification challenges. Arm automotive IP, along with Siemens PAVE360 digital twin environment, will enable automakers to develop and validate differentiated safety-enabled systems, ICs, and software solutions in the context of an entire vehicle.	NA

Source: Company Website

12.1.3.4 MnM view

12.1.3.4.1 Key strengths/right to win

Siemens is one of the leading developers of digital twin software and solutions. The company continues to invest in R&D for Product Digital Twins (using digital twins for efficient design of new products), Production Digital Twins (using digital twins in manufacturing & production planning), and Performance Digital Twins (using digital twins to capture, analyze, and act on operational data). Siemens has extensive experience in development of key components of digital twins such as IoT, artificial intelligence (AI), machine learning (ML), as well as other high-end digital software and related technologies that work in tandem with digital twins. Siemens has also invested in the creation of a digital twin for additive manufacturing factories.

The company majorly focuses on the development of innovative and sustainable solutions. This would increase the business of the company and simultaneously safeguard its competitiveness in the global market. R&D efforts of the Power and Gas division are focused on developing innovative products and solutions for enhancing efficiency, flexibility, and economy in the power generation and oil & gas industries. Siemens is actively contributing to digital transformation by consistently enhancing the Digital Enterprise portfolio in the field of cloud data platforms, which serve as the foundation for new digital business models for data management over the cloud. Siemens offers client-centric specific digital twin solutions that solve mission-critical challenges of clients. For instance, in July 2020, Siemens created the digital twin for HERX's (UK) additive manufacturing plant using end-to-end technologies, allowing HERX to optimize design and speed production processes for personalized bicycle helmets even before production begins. Similarly, a few other global brands such as Sierra Space (US), Daimler (Germany), Bespoke Eyewear (UK), and Bye

Aerospace (US) also rely on Siemens for their digital twin design and implementation. The company has a very strong client base and is a trusted provider of the digital twin in the market.

12.1.3.4.2 Strategic choices made

Siemens adopts both organic and inorganic strategies for growing its business. It has formed numerous acquisitions, strategic partnerships, and collaborations for managing resources and expanding its product portfolio and market reach. For instance, in December 2021, Siemens signed an agreement to acquire digital twin software for buildings from EcoDomus, a US-based startup company. Leveraging the acquired data creation and visualization capabilities, Siemens' digital building software portfolio will help customers by providing increased insights into the operation of their buildings, real-time issue identification and resolution, and better space and energy utilization. Customers will be able to make their facilities more sustainable, comfortable, and safe places to live and work while also streamlining procedures and lowering operating costs. Similarly, in November 2021, Siemens partnered with automotive companies Hyundai Motor Company (South Korea) and Kia Corporation (a subsidiary of Hyundai); both the companies will use Siemens' Xcelerator portfolio's NX software and Teamcenter software solutions for their core design and data management platforms. This will provide Hyundai's teams with a new working environment, paving the way for significant advancements in future automotive development.

The adoption of organic strategies (of software upgrades and new launches) helps the company expand its market share in the digital twin ecosystem. For instance, in May 2022, Siemens (Digital Industries Software segment) announced that its transformation to a software-as-a-service (SaaS)-driven business (for Xcelerator software tools) is business is successful and gaining traction. As of the end of Q1 FY2022, cloud-related annual recurring revenue (ARR) had topped USD 200 million, accounting for 6% of the overall ARR. This growth continued to accelerate in the second quarter as well. The Xcelerator is a collection of software tools aimed at helping businesses get better results faster. In addition, its integrated digital twin and model-based system engineering (MBSE) capabilities make it easier for customers to coordinate suppliers, partners, and industry-specific ecosystems. It also allows them to optimize value chains—from product design, development, and production to post-sales enhancement.

12.1.3.4.3 Weaknesses and competitive threats

The involvement of Siemens in multiple lawsuits in recent years might have an adverse impact on the brand image of the company. Stringent regulations such as European Union's regulation on data security and protection has increased the cost of doing business in European countries. Siemens relies heavily on the European region for the majority of its business; any political turbulence in the region can affect the sales of the company.

12.1.4 AMAZON (AWS)

12.1.4.1 Business overview

Amazon divides its operations into three segments: Online Stores, Third-party Seller Services, Subscription Services, Advertising Services, Physical Stores, and Amazon Web Services (AWS). Amazon Web Services Inc (AWS) is a subsidiary of Amazon that provides cloud computing services. The company offers a comprehensive range of cloud infrastructure services, including compute, storage, databases, analytics, networking, mobile, developer tools, augmented reality and virtual reality, robotics, game tech, machine learning, management tools, content delivery, media services, customer engagement, app streaming, and security, identity, and compliance.

The company offers digital twin offerings through AWS. It has created a four-level digital twin index: 1–Descriptive, 2– Informative, 3–Predictive, and 4–Living Digital Twin.

L1–Descriptive concentrates on the physical system's engineering design and visual depiction (its structure). A 2D engineering diagram (such as a process or piping and instrumentation diagram), a building information model (BIM), or a complicated high-fidelity 3D/AR/VR model can be used for this purpose.

L2–Informative focuses on integrating IoT sensor and maintenance data from the physical system and presenting it to the end user in a contextually appropriate manner, such as a 2D dashboard or a 3D contextual visualization (e.g., its state). It allows the end user to comprehend the current state of the physical system and includes rudimentary analytics that may be used to generate alarms.

L3–Predictive focuses on unmeasured quantity forecasts (e.g., virtual sensors, machine learning-based anomaly detection), as well as employing predictive models to comprehend future states under ongoing operations when future behavior is consistent with past behavior. These models can be scientific first principles-based, data-driven (e.g., utilizing AI/ML), or a combination of the two.

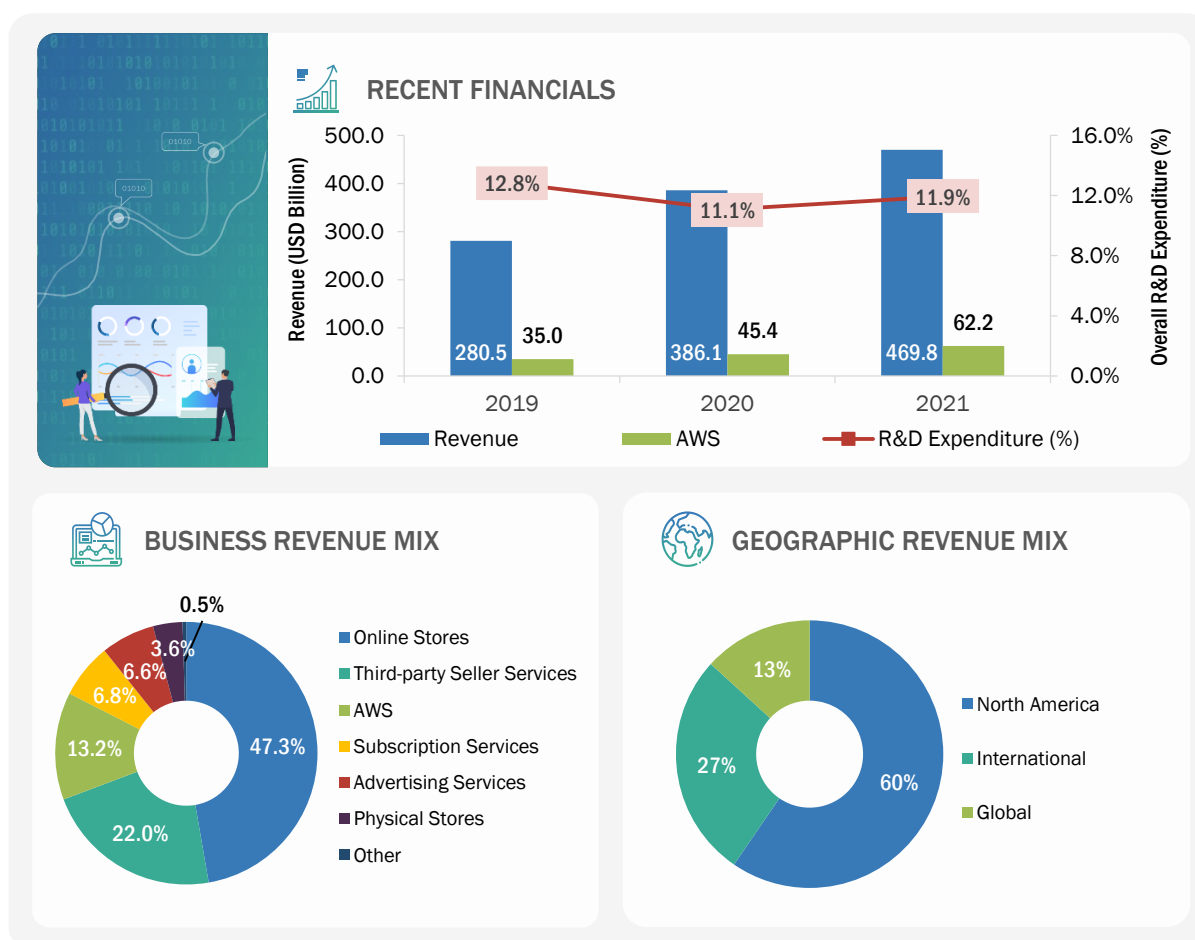
L4–Living Digital Twin focuses on updatable models that can be aggregated to the fleet level if necessary. According to AWS, the L4 level symbolizes the pinnacle of the digital twin journey. The ability for the model to be updated depending on data from the physical entity and environment is the major difference between L3 Predictive and L4 Living.

The company serves many industries, including automotive & transportation, infrastructure, healthcare, oil & gas, energy & utilities, and retail. Amazon offers its digital twin services/solutions through AWS across all the major regions, including Asia Pacific, the Americas, the Middle East, Africa, and Europe. The US, the UK, Australia, China, France, Germany, India, Brazil, Canada, Ireland, Japan, Korea, and Singapore are among a few major countries where AWS caters its services.

TABLE 121 AMAZON: BUSINESS OVERVIEW

Founding Year	1994
Headquarters Country	US
Headquarters City	Seattle, Washington
Ownership Type	Public (NASDAQ: AMZN)
Employee Count (as of December 2021)	1,608,000

Source: Company Website

FIGURE 56 AMAZON: COMPANY SNAPSHOT

Note: In the Recent Financials section, the R&D expenditure has been calculated as the percentage of total revenue. The financial year of the company ends on December 31.

Source: Company Website and Annual Reports

12.1.4.2 Products/solutions offered

PRODUCT TYPE	PRODUCT/SOLUTION	DESCRIPTION	INDUSTRY
Digital twin solution	AWS IoT TwinMaker	AWS IoT TwinMaker is a tool that offers solutions to create digital twins in order to improve building operations, production output, and equipment performance. It allows to generate a comprehensive perspective of operations faster and with less effort by combining existing data from numerous sources, creating virtual representations of any physical location, and combining current 3D models with real-world data.	Automotive & transportation, retail, oil & gas, healthcare, energy & utilities, infrastructure, agriculture, and others
Digital twin solution	Vertex	Vertex is a visualization software platform for digital twin. Vertex Software's cloud-based digital twin technology facilitates team communication by unlocking 3D data and delivering synchronized images to internal and external teams. Customers can use the platform to create 3D visual collaboration solutions such as design collaboration, 3D-enabled product, and full-factory	Automotive & transportation, retail, oil & gas, healthcare, energy & utilities, infrastructure, agriculture, and others

IoT visualization, assembly work instructions, sales configurators, and field service automation, among other things.

Source: Company Website

12.1.4.3 Recent developments

12.1.4.3.1 Product launches

MONTH & YEAR	PRODUCT TYPE	PRODUCT NAME	PRODUCT DESCRIPTION
July 2020	Visualization software platform for digital twin	Vertex	AWS launched the new version of its Vertex visualization software platform for digital twin. It provides end-user enterprises with real-time PLM-aligned data. Through AWS's system of engagement and cloud architecture, users can have flexible, ad-hoc, and linked collaboration capabilities.

Source: Company Website and Press Releases

12.1.4.3.2 Deals

MONTH & YEAR	DEAL TYPE	COMPANY 1	COMPANY 2	DESCRIPTION	DEAL SIZE
November 2021	Partnership	AWS (US)	Siemens (Germany)	Amazon Web Services (AWS) and Siemens expanded their partnership to make cloud-based digital transformation easier. One area of collaboration is digital twin technology, where Siemens and AWS will use AWS IoT TwinMaker to create digital twins that include different data sources to accelerate adoption and democratize new digital twin solutions.	NA

Source: Company Website

12.1.4.4 MnM view

12.1.4.4.1 Key strengths/right to win

Amazon is strongly committed to R&D for its AWS services, including digital twin. AWS constantly focuses on the development of digital twin tools, software solutions, and platforms. The company also places high emphasis on digital twin complementary technologies such as machine learning solutions, internet of things, and cloud computing services. The company's revenue grew at a quick rate in 2020—30% year over year (YoY) in 2020 on a USD 35 billion yearly revenue base in 2019. The revenue continued to grow at a rate of 37% YoY in 2021.

The company's strength lies in its strong clientele base and long-term customer relationships. For instance, AWS offers its IoT TwinMaker to INVISTA (US) for improving productivity and efficiency in field operations. INVISTA is a subsidiary of Koch Industries, which serves the nylon and specialty materials industries. Similarly, Carrier (US), a leading provider of healthy, safe, sustainable, and intelligent building and cold chain solutions, also uses IoT TwinMaker for monitoring operational data for smart buildings. John Holland, one of Australia's leading integrated infrastructure, rail, multi-modal transport, and building companies,

also uses AWS IoT TwinMaker for environmental impact monitoring in construction. The company often collaborates with other players in the market to offer best solutions to its customers. For instance, AWS offers its digital twin services in collaboration with other digital twin providers such as ANSYS (US), NavVis (Germany), and Siemens (Germany).

12.1.4.4.2 Strategic choices made

Amazon is expanding its business by adopting organic, as well as inorganic, strategies such as product launches and partnerships. In November 2021, AWS extended its partnership with Siemens (Germany), another digital twin provider. Through this partnership, Siemens and AWS will be using AWS IoT TwinMaker to offer digital twins solutions for their customers. Customers can use AWS IoT TwinMaker to develop increasingly powerful digital twin solutions. Siemens' Xcelerator portfolio is already integrated with over 60 AWS services, and with the addition of AWS IoT TwinMaker, customers can use AWS IoT TwinMaker to develop increasingly powerful digital twin solutions that are compatible with Siemens' design, simulation, and manufacturing software.

AWS is the preferred cloud provider for General Electric (US), a global digital twin provider. Similarly, AWS has partnered with ANSYS (US) and NavVis (Germany) to offer digital twin solutions in the market. AWS and NavVis partnered to offer solutions that enable their enterprise customers digitize their physical assets, allowing them to become leaner, more efficient, and more competitive. The NavVis Digital Factory Solution gives manufacturing executives unprecedented visibility into their production network's capabilities and status. Whereas the AWS SageMaker is used with NavVis solutions to provide spatial intelligence to a predictive maintenance routine or to enable item recognition during the scanning and development of a digital twin. In addition, AWS also offers supporting technologies for digital twin. For instance, Amazon offers Amazon SageMaker, a cloud machine-learning platform that helps in the development of digital twin for enterprises. SageMaker is a cloud-based platform that allows developers to construct, train, and deploy machine-learning models. Developers can also use SageMaker to install machine learning models on embedded systems and edge devices.

12.1.4.4.3 Weaknesses and competitive threats

The company heavily relies on the North American market; it generates almost 60% of its total revenue from the region. Amazon can face high losses due to global trade conflicts and any political turbulence in North American countries. Further, AWS also faces tough competition from established companies such as General Electric (US), IBM (US), Oracle (US), SAP (Germany), and Siemens (Germany).

12.1.5 DASSAULT SYSTÈMES

12.1.5.1 Business overview

Dassault Systèmes offers software solutions based on the 3DEXPERIENCE platform that enables customers to create innovative new products and services through virtual experiences. In 2018, Dassault Systèmes launched Power'By as part of 3DEXPERIENCE R2018x and introduced the 3DEXPERIENCE Marketplace. The goal of Power'By is to enable the company's customers to immediately benefit from the value of the 3DEXPERIENCE platform without the requirement to migrate legacy data. There are three levels: enabling social collaboration, leveraging hybrid data for product configuration and bill of materials, and fully utilizing the 3DEXPERIENCE platform's capabilities.

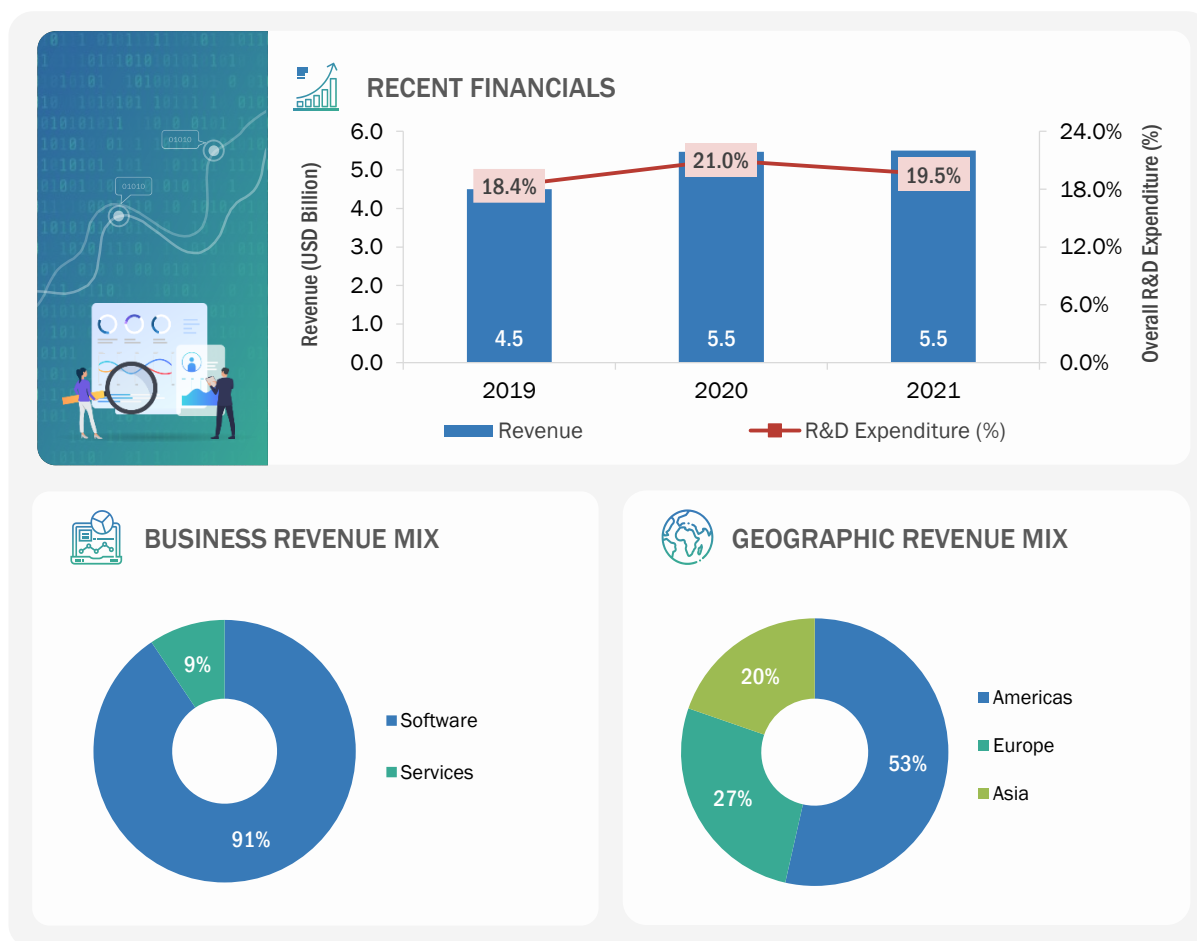
The 3DEXPERIENCE platform offered by the company is the only platform that is both a system of operations for running a business and a business model for transforming business. Many businesses (including retail, healthcare, and automotive) can employ virtual twin technologies supplied by Dassault Systèmes' solutions to build new, visionary models and objects, and use digital twin simulation in the design process to assess performance, dependability, and safety. It also helps clients to manage product creation from early planning to final release, including all developmental milestones, by accelerating cross-team cooperation and managing product creation from early planning to final release.

Dassault Systèmes currently owns over 180 sites worldwide, with more than 300,000 enterprise customers and 25 million users across 11 industries worldwide. The company offers its solutions and services in 135 countries, with more than 150 scientific & research partners, and more than 8,700 technology & marketplace partners worldwide.

TABLE 122 DASSAULT SYSTÈMES: BUSINESS OVERVIEW

Founding Year	1981
Headquarters Country	France
Headquarters City	Vélizy-Villacoublay
Ownership Type	Public (EPA: DSY)
Employee Count (as of 2021)	20,496

Source: Company Website

FIGURE 57 DASSAULT SYSTÈMES: COMPANY SNAPSHOT

Note: In the Recent Financials section, the R&D expenditure has been calculated as the percentage of the total revenue. The financial year of the company ends on December 31. The financial year of the company ends on December 31. The currency conversion rates considered for 2019, 2020 and 2021 are EUR 1 = USD 1.11986, EUR 1 = USD 1.22824, and EUR 1 = USD 1.1324, respectively.

Source: Company Website and Annual Reports

12.1.5.2 Products/solutions offered

PRODUCT TYPE	PRODUCT/SOLUTION	DESCRIPTION	INDUSTRY
Digital twin solution	3DEXPERIENCE digital twin platform	Dassault Systèmes' 3DEXPERIENCE platform offers engineers cloud-based software to design a virtual twin solution. This one cloud-based platform not only assists businesses in resolving issues but also brings the physical and virtual worlds together for unique demands and objectives.	Aerospace, retail, energy & utilities, infrastructure, healthcare, automotive & transportation, others

Source: Company Website

12.1.5.3 Recent developments

12.1.5.3.1 Product launches

MONTH & YEAR	PRODUCT TYPE	PRODUCT NAME	PRODUCT DESCRIPTION
July 2020	Software upgrade	3DEXPERIENCE	Dassault Systèmes announced the enhancement of collaborative data science capabilities of the 3DEXPERIENCE platform. Customers of Dassault Systèmes can leverage the most of their data assets by participating in new virtual twin experiences that translate valuable knowledge discovered in unstructured text data into usable content and insights.
February 2020	Digital twin for healthcare	3DEXPERIENCE Things to Life (under development)	Dassault Systèmes revealed its five-year strategy plan with the launch of 3DEXPERIENCE Things to Life, which focuses on revolutionizing the treatment of patients and assisting them in living a better life by enabling the virtual twin experience of the human body. Virtual twin experiences open up new possibilities for life sciences and healthcare by allowing researchers, medical, surgical, and other health-related disciplines to understand, model, search, test, and treat the human body as precisely, safely, and effectively as other industrial disciplines do with cars, buildings, and airplanes.

Source: Company Website and Press Releases

12.1.5.3.2 Deals

MONTH & YEAR	DEAL TYPE	COMPANY 1	COMPANY 2	DESCRIPTION	DEAL SIZE
February 2020	Partnership	Dassault Systèmes (France)	Cadence (US)	Cadence and Dassault Systèmes formed a strategic alliance aimed at offering electromechanical virtual twin experiences for faster product development in complex electronic systems. This partnership leverages the Cadence Allegro platform and the Dassault Systèmes 3DEXPERIENCE platform to optimize the entire value chain for electromechanical systems modeling, design, simulation, and product lifecycle management.	NA
December 2021	Contract	Dassault Systèmes (France)	Renault Group (France)	Renault Group, an automotive OEM and Dassault Systèmes strengthened their collaboration with a new contract to accelerate the car manufacturer's transformation with the 3DEXPERIENCE platform. Renault Group is using Dassault Systèmes' 3DEXPERIENCE cloud platform to design programmes for new vehicles and mobility services, in a first-of-its-kind deployment for an industrial organization of this size. This enterprise platform will give Renault a new backbone for sharing all product-related data in real time	NA

throughout the product life cycle, as well as managing virtual twins of its many product configurations.

July 2021	Acquisition	Dassault Systèmes (France)	Iterop (France)	Dassault Systèmes acquired Iterop, a business process management business leveraging BPMN 2.0 standards—a neutral, graphical language. Dassault Systèmes and Iterop will improve the 3DEXPERIENCE platform and 3DS Outscale, as well as will extend inclusive innovation through the cloud.	NA
April 2021	Partnership	Dassault Systèmes (France)	Bouygues Construction	Dassault Systèmes and Bouygues Construction expanded their partnership to accelerate Bouygues Construction' digital transformation in the construction industry.. The two companies will concentrate on accelerating research and development of new cloud-based, mobile-enabled approaches using a construction virtual twin.	NA
November 2020	Acquisition	Dassault Systèmes (France)	NuoDB (US)	Dassault Systèmes acquired NuoDB, a leader in cloud-native distributed SQL databases . The acquisition was focused on advancing Dassault Systèmes' 3DEXPERIENCE platform cloud and data science strategy.	NA
July 2020	Acquisition	Dassault Systèmes (France)	PROXEM (France)	Dassault Systèmes acquired PROXEM, a firm specialized in semantics software and services based on artificial intelligence to strengthen the collaborative data science capabilities of the 3DEXPERIENCE platform.	NA

Source: Company Website

12.1.5.4 MnM view

12.1.5.4.1 Key strengths/right to win

The company's core strength lies in its advanced research and development initiatives for a strong foundation of digital twin. The company owns more than 700 families of patents, and about 41% of the total human capital of Dassault Systèmes work in R&D. Further, the company is also investing in unique digital twin solutions for applications in the healthcare industry. In February 2020, Dassault Systèmes unveiled digital twin, which aims to revolutionize healthcare for the treatment of patients and help them live better lives by enabling virtual twin experiences of the human body. As a next step in its strategic vision, in January 2022, at CES 2022 (Las Vegas), the company showcased the virtual twin of the human brain, and a few participants were able to meet their virtual twins and discover the possibilities that virtual twins of the human body bring to the healthcare industry.

12.1.5.4.2 Strategic choices made

The company adopts both organic and inorganic growth strategies to expand its share in the digital twin market. The company collaborated with a few automotive OEMs to enhance its digital twin solutions for the automotive industry. In April 2021, Atos (France), Dassault Systèmes (France), Groupe Renault (France), STMicroelectronics (Switzerland), and Thales (France) joined forces to create the Software République, a new ecosystem for intelligent mobility innovation. The partners seek to develop and market systems and software together using their complementary expertise to create an enhanced and sustainable mobility offer for cities, regions, businesses, and citizens. These new products and services will benefit from artificial intelligence, cybersecurity, networking, embedded electronics, and virtual twin technology. This open innovation ecosystem, which was formed by five automotive and technology industry experts, will welcome new members and foster open cooperation. Additionally, the company also undertook a few acquisitions in 2020 and 2021 to expand its product portfolio and grow its customer base.

12.1.5.4.3 Weaknesses and competitive threats

Dassault Systèmes is expanding its digital twin solutions portfolio, but at present, it is facing tough competition from the well-established players in the market such as General Electric (US), IBM (US), ANSYS (US), Microsoft (US), and Siemens (Germany). The company should also focus on expanding its market presence in the Asia Pacific region (the lowest revenue-generating regional market for the company). The Asia Pacific region is expected to be one of the most lucrative markets in the future and by foraying into the region at this stage, Dassault Systèmes may gain a competitive advantage.

12.1.6 ANSYS

12.1.6.1 Business overview

ANSYS is a major player in engineering simulation software and services. The industry-leading portfolio of engineering simulation software offered by ANSYS includes solutions for creating digital twins to analyze real-time operational data to create awareness and facilitate faster decision-making while maximizing machine efficiency in various industries. The company's simulation tools create digital twins, which optimize the operations of devices or systems. ANSYS' Simplorer system-level modeling tool is used to build digital twins that define the complex interaction between machine components, assemblies, and subsystems accurately. ANSYS SCADE suite is a product line in the ANSYS embedded software family that can be used to control a digital twin and develop a human-machine interface (HMI) using the same control software. ANSYS 18 empowers users to create digital twins with the assistance of partner applications, such as GE Predix and PTC ThingWorx.

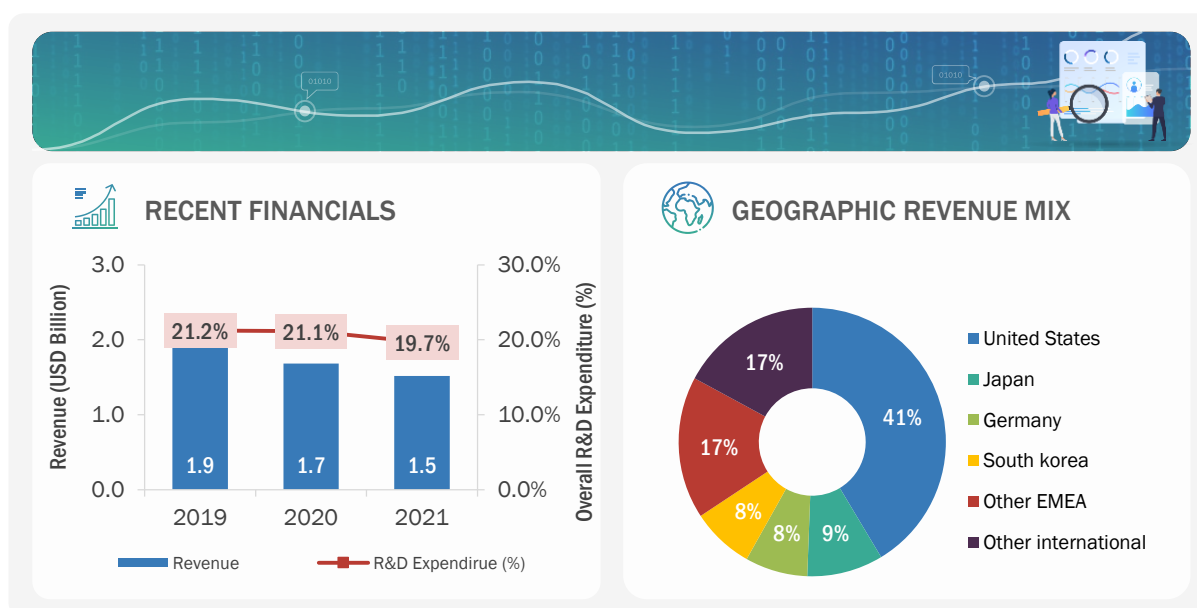
Clients can use ANSYS Twin Builder to create comprehensive virtual prototypes of real-world systems. These can be used to manage the whole product and asset lifecycle. This digital twin simulation paradigm enables clients to boost savings exponentially over time by scheduling maintenance around predictive approaches that improve with real-world testing and response. Engineers may use this data to extract more value from existing assets, avoiding unplanned downtime and lowering operating costs while working at peak efficiency, all while using agnostic Internet of Things (IoT) platforms.

ANSYS has more than 75 strategic sales locations worldwide, with a network of channel partners in more than 40 countries. In addition, the company owns over 250 global partners that provide hardware, connectivity, and software solutions that host or integrate with ANSYS software solutions. ANSYS also collaborates with universities to help the academic institutes in utilizing ANSYS software in teaching and research; ANSYS has over 3,300 university customers in 95 countries.

TABLE 123 ANSYS: BUSINESS OVERVIEW

Founding Year	1970
Headquarters Country	US
Headquarters City	Canonsburg, Pennsylvania
Ownership Type	Public (NASDAQ: ANSS)
Employee Count (as of December 2021)	5,100

Source: Company Website

FIGURE 58 ANSYS: COMPANY SNAPSHOT

Note: In the Recent Financials block, the R&D expenditure has been calculated as the percentage of the total revenue. The financial year of the company ends on December 31.

Source: Company Website and Annual Reports

12.1.6.2 Products/solutions offered

PRODUCT TYPE	PRODUCT/SOLUTION	DESCRIPTION	INDUSTRY
Digital twin solution	ANSYS Twin Builder	ANSYS Twin Builder is an open solution that allows engineers to create simulation-based digital twins—digital representations of assets with real-world or virtual sensor inputs. Users can swiftly create a digital twin, a connected clone of an in-service asset, with ANSYS Twin Builder. This enables better lifetime management and genuine predictive maintenance, which saves money for the user.	Aerospace, automotive & transportation, retail, healthcare, energy & utilities, telecommunication, and others

Source: Company Website

12.1.6.3 Recent developments

12.1.6.3.1 Product launches

MONTH & YEAR	PRODUCT TYPE	PRODUCT NAME	PRODUCT DESCRIPTION
February 2022	Solution upgrade	ANSYS Twin Builder 2022 R1	By merging physical and virtual sensors, ANSYS Twin Builder 2022 R1 provides predictive analytics with unrivaled accuracy. New features enable faster deployment of user's digital twin, easier workflows, and web applications for online interaction with the user model.

Source: Company Website and Press Releases

12.1.6.3.2 Deals

MONTH & YEAR	DEAL TYPE	COMPANY 1	COMPANY 2	DESCRIPTION	DEAL SIZE
February 2022	Collaboration	ANSYS (US)	Amazon Web Services (US)	ANSYS and Amazon Web Services announced a strategic collaboration to transform cloud-based engineering simulations. The partnership will allow ANSYS products to be deployed on AWS, making simulation workloads more user-friendly while also providing scalability and flexibility with easy access to software and storage solutions from any web browser. ANSYS can integrate AWS' comprehensive framework, which includes products such as computation, storage, internet of things (IoT), digital twin, analytics, and machine learning.	NA
September 2021	Collaboration	ANSYS (US)	Rockwell Automation (US)	ANSYS and Rockwell Automation collaborated to bring digital twin connectivity to industrial control systems, allowing users to better design, deploy, and perform industrial operations. By implementing simulation-based digital twins, engineering teams can garner new insights to speed up the innovation and cut costs from the design phase to the production phase.	NA
May 2021	Acquisition	ANSYS (US)	Phoenix Integration (US)	ANSYS acquired Phoenix Integration, Inc., the premier provider of software that enables model-based engineering (MBE) and model-based systems engineering (MBSE). The acquisition will help ANSYS to expand its solution offering, allowing its customers to connect a variety of engineering tools in multi-tool workflows for broad and robust model-based engineering.	NA
October 2020	Acquisition	ANSYS (US)	Analytical Graphics Inc. (US)	ANSYS entered into a definitive agreement to acquire Analytical Graphics, Inc. (AGI), a premier provider of mission-driven simulation, modeling, testing, and analysis software for aerospace, defense, and intelligence applications. The acquisition aimed to broaden ANSYS' solution offering, allowing users to address problems by simulating everything from the chip to a client's complete mission, such as tracking an orbiting satellite and its periodic link to ground stations.	USD 700 million

Source: Company Website

12.1.6.4 MnM view

12.1.6.4.1 Key strengths/right to win

ANSYS has made significant investments in R&D and emphasizes frequent, integrated product releases. In the last five years, the company invested significantly in R&D, which is about 18% to 21% of the total revenue. The company's product development strategy centers for the ongoing development and innovation of new technologies help to increase productivity and provide engineering simulation solutions. These centers will help its customers to integrate products and solutions into enterprise-based product life cycle management (PLM) systems. The company focuses on extending its full product line with new functional modules along with CAD, electronic CAD (ECAD), PLM products, and the development of new products.

An established network of global partners, along with channel partners in more than 40 countries, enables ANSYS to distribute its digital twin solutions and simulation technology to a broader consumer base. Customers can use Ansys Twin Builder to create comprehensive virtual prototypes of real-world systems. These can be used to manage the whole product and asset lifecycle. Further, Ansys Twin Builder is a multi-technology platform that not only allows engineers to create simulation-based digital twins but also allows clients to runtime capabilities for their twin, deploy their twin, as well as validate their twins, along with benefits of advanced asset management and third-party tool integration.

12.1.6.4.2 Strategic choices made

The company focuses on inorganic growth strategies such as collaborations to strengthen its position in the digital twin market. ANSYS is working with Microsoft (US) to enhance ANSYS Twin Builder with Microsoft Azure digital twin to enable clients to significantly improve operations. Through the collaboration, manufacturers that model and connect assets using Azure digital twins can optimize asset production and operations. In May 2020, ANSYS joined Microsoft (US), Dell (US), and Lendlease (Australia) on the Digital Twin Consortium's steering council to help advance the acceptance and deployment of physics-based digital twin technologies across numerous industries. The consortium is a transnational community of forward-thinking technologists who are influencing digital twin creation, usage, and standards.

In the long run, ANSYS is focusing on digital twins and additive manufacturing simulation. The company's strategy of pervasive engineering simulation is aligned with the market growth of digital twins. Further, Microsoft (US), Rockwell Automation (US), PTC (US), Amazon Web Services (US), Synopsys (US), Autodesk (US), and SAP SE (Germany) are the leading strategic partners of ANSYS. These strategic partners help the company to expand its technology in developing and complementary solutions across the whole supply chain of various industries. For example, SAP SE and ANSYS have formed a strategic partnership based on the integration of ANSYS digital twin technology into SAP cloud platforms in order to improve asset predictive maintenance and operational efficiency.

12.1.6.4.3 Weaknesses and competitive threats

Due to the timing, duration, and value of multi-year lease contracts, as well as the company's reliance on high renewal rates for yearly lease and maintenance contracts, the company's revenue has become more volatile. The company undertook multiple acquisitions in the last three years, which resulted in cash outflow. Such continuous cash outflows may impact the overall operations of the company in the future.

12.1.7 IBM

12.1.7.1 Business overview

International Business Machines (IBM) Corporation provides integrated solutions that leverage IT and knowledge of business processes. The company operates its businesses through five segments—Software, Consulting, Infrastructure, Financing, and Others. IBM provides software solutions that are designed to assist clients in becoming more data-driven, as well as automating, securing, and modernizing their environments.

The research team of IBM has been focusing on innovations in AI, analytics, security, cloud computing, quantum computing, data-centric systems, and so on. These technologies are adopted in various industries, such as healthcare, education, and financial services. IBM offers digital twin solutions through its operation unit, Watson Internet of Things. IBM Watson IoT Platform connects IoT devices and aids in streaming a large volume of data to the cloud. This data helps create digital twin representations and derive intelligent insights or relate the data to a high-level context data gathered from a manageable asset or system.

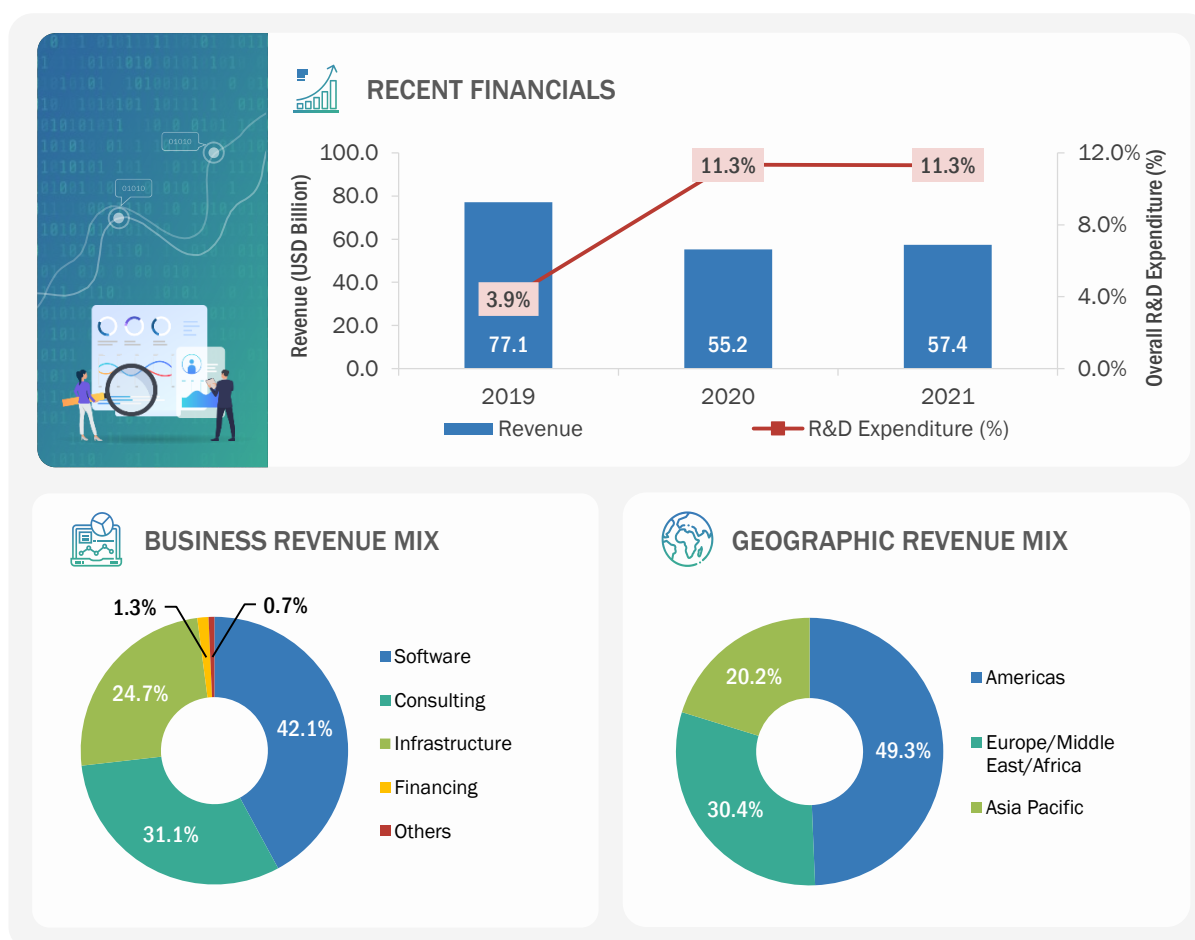
IBM also offers IBM Digital Twin Exchange for asset-intensive industries. Digital twin clients may virtually create, monitor, test, and build products to market products faster, as well as make precise forecasts about their performance over time. The IBM Digital Twin Exchange is a resource for asset-intensive businesses, including organizations using IBM Maximo enterprise asset management, to manage enterprise-wide digital assets. Customers that use digital twins can explore, buy, download, and manage digital twins from IBM. The IBM Digital Twin Exchange supports a wide range of downloadable digital twin types, including bill of materials, 3D CAD files, engineering manuals, building information models, and other downloadable digital twin formats. Even the content creators can list their digital twins on this platform, manage their inventory, and set pricing for their created digital twin.

IBM operates in more than 175 countries and is continuing to shift its business to the higher-value segments of enterprise IT. IBM helps clients design their digital transformation, build open hybrid cloud architectures, orchestrate applications across environments, and optimize their key workflows and business processes. The company's overseas operations are carried out through dedicated sales teams for artificial intelligence, business operations, cloud computing, software development, analytics, and IT infrastructure. The company has a strong network of partners across the world and provides services in the US, Germany, the Netherlands, China, India, Japan, Poland, Australia, and France.

TABLE 124 IBM: BUSINESS OVERVIEW

Founding Year	1911
Headquarters Country	US
Headquarters City	Armonk, New York
Ownership Type	Public (NYSE: IBM)
Employee Count (as of December 2021)	345,000

Source: Company Website

FIGURE 59 IBM: COMPANY SNAPSHOT

Note: In the Recent Financials block, the R&D expenditure has been calculated as the percentage of the total revenue. The financial year of the company ends on December 31. The pie chart numbers are rounded off up to one decimal place, and there could be instances where the total might not add up to 100.

Source: Company Website and Annual Reports

12.1.7.2 Products/solutions offered

PRODUCT TYPE	PRODUCT/ SOLUTION	DESCRIPTION	INDUSTRY
Digital twin solution	IBM Digital Twin Exchange	The Digital Twin Exchange is a marketplace that connects companies wishing to buy digital twins with members of IBM's partner ecosystem, who can offer them fault codes, maintenance plans, 3D models, and other data connected to physical assets. It's the first step toward experimenting with digital twins. Clients can then use their AI and IoT solutions to fully realize the benefits of digital twins. The digital twins, for example, can be used in tandem with IBM Maximo enterprise asset management to forecast asset performance and maintenance based on the data available.	Aerospace, oil & Gas, infrastructure, automotive & transportation, and others

Digital twin solution	Watson Internet of Things:		
	▪ IBM Rational Rhapsody	From production to operations, IBM Digital Twin leverages Watson IoT to connect the real and digital worlds by bringing together numerous digital threads and data streams that may be adapted to the needs of a specific user. Since businesses can access and analyze data throughout the operational phase and then feed the data back into the design phase, the IBM Digital Twin stands out for its analytical components, and cognitive insights produced using Watson.	Aerospace, healthcare, power & utilities, infrastructure, oil & gas, automotive & transportation, and others
	▪ Rational Engineering Lifecycle Manager		
	▪ IBM Rational Lifecycle Integration Adapters		
	▪ IBM Maximo Enterprise Asset Management		
	▪ IBM Maximo Asset Performance Management		

Source: Company Website

12.1.7.3 Recent developments

12.1.7.3.1 Product launches

MONTH & YEAR	PRODUCT TYPE	PRODUCT NAME	PRODUCT DESCRIPTION
February 2020	Digital twin solution	IBM Digital Twin Exchange	IBM launched the IBM Digital Twin Exchange as its platform to offer digital twin solutions. IBM Twin Exchange provides various types of digital twin content, such as bills of materials, maintenance plans, failure modes, parts lists, operational models, 2D/3D files, and other digital twin contents. It allows users to search and download digital twin content according to their requirement.

Source: Company Website and Press Releases

12.1.7.3.2 Deals

MONTH & YEAR	DEAL TYPE	COMPANY 1	COMPANY 2	DESCRIPTION	DEAL SIZE
August 2020	Partnership	IBM (US)	Big Iron Consulting (Australia)	Big Iron Consulting partnered with IBM on the Digital Twin Exchange to share its digital twin resources. Big Iron Consultancy, based in Australia, is a global mining asset management and life cycle costing consulting firm. Through this partnership, Big Iron Consultancy will be able to connect to IBM's 3,000+ Maximo install base by becoming a content provider on the Digital Twin Exchange. As of May 2022, Big Iron had 120 products listed on IBM Digital Twin Exchange, where it has set up an online store.	NA

June 2020	Partnership	IBM (US)	Siemens (Germany)	IBM and Siemens (Germany) partnered to develop a new solution that optimizes the service lifecycle management (SLM) of assets by dynamically connecting real-world maintenance activities and asset performance back to design decisions and field modifications. This solution combines elements of the Xcelerator portfolio from Siemens and IBM Maximo to improve product performance, maintenance, and operations.	NA
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Source: Company Website

12.1.7.4 MnM view

12.1.7.4.1 Key strengths/right to win

With the number of applications (e.g., asset management using augmented reality) of digital twin expanding continuously across various industries, IBM is investing significantly in the R&D of digital twin technology. IBM's Maximo lab services enable many visual and vocal (natural language processing) options for client's workforce. This allows the clients to see their assets in a new light and have immediate access to vital information. Clients can then use an AR helmet with voice/video in the visor to share their findings with others. As a result, this makes 'interacting' as the next step in the evolution of work of digital twin.

Digital twins are also important in the systems engineering lifecycle, and are a core component of IBM Engineering Systems Design, where teams can use model-based systems engineering (MBSE) to speed up product design, development, and innovation in aerospace and defense, an industry where safety is nonnegotiable. IBM offers high-quality digital twins through its IBM Digital Twin Exchange platform that contributes towards the expansion of digital twins across various industries, including aerospace, oil & gas, and automotive & transportation.

12.1.7.4.2 Strategic choices made

IBM is expanding its business by adopting organic and inorganic strategies of product launches and partnerships, respectively. It focuses highly on the development of new products in the field of IoT, as well as on machine learning, to transform businesses, lives, and societies. For instance, in February 2020, IBM launched its Digital Twin Exchange platform, where content creators can list their digital twins on a subscription basis and clients/users can purchase these solutions as per their requirements. Similarly, in November 2019, IBM launched Maximo Asset Monitor, an AI-powered monitoring solution designed to help maintenance and operation leaders understand and improve the performance of their high-value physical assets. The product is an extension of IBM's market-leading IBM Maximo capabilities, which will help unlock essential insights with AI-powered anomaly detection and provide enterprise-wide visibility into critical equipment performance. This would help in faster problem identification and help reduce downtime.

In June 2020, IBM partnered with Siemens (Germany) and announced the availability of a new solution to maximize asset service lifecycle management (SLM) by dynamically integrating real-world maintenance activities and asset performance back to design decisions and field adjustments. Using parts from Siemens Digital Industries Software's Xcelerator range and IBM Maximo, this new solution creates an end-to-end digital thread between equipment makers and their owners/operators. The company intends to maintain its growth trajectory by assisting businesses in creating and managing a closed-loop, end-to-end digital twin that breaks down traditional silos in service innovation and revenue generation.

12.1.7.4.3 Weaknesses and competitive threats

IBM faces high competition from other companies who operate in the same segment and provide digital twin solutions, such as Siemens (Germany), PTC (US), and General Electric (US).

12.1.8 PTC

12.1.8.1 Business overview

PTC is among the leading providers of computer software and services. The company offers a wide range of innovative products such as computer-aided design (CAD) modelling software, PLM, and service life cycle management (SLM) solutions. Various manufacturers use these solutions to create, operate, and service products. The company's offering includes IoT solutions, which enable a secure connection with smart things, management and analysis of data, and quick creation of applications for customers. It has two business units—Software and Professional Services. The offerings of the Software business unit comprise Core product software (CAD and PLM), Growth product software (IoT, AR, Onshape, and Arena), and Focused Solution Group (FSG). Whereas the sales of new licenses and software upgrades are handled by the Professional Services business unit.

It offers PTC ThingWorx, a purpose-built platform to fundamentally transform how the companies use data to connect, analyze, manage, monetize, and experience all “things” in a smart, connected world. The ThingWorx platform is used to generate a dashboard linked to sensors from systems and reality of physical twin into the world of digital twin, allowing development engineers to see the real-world conditions of the system and to conduct simulations and analyses using live data. PTC's diverse range of CAD, PLM, IoT, and AR technologies combine to form a robust digital thread foundation that enables faster data continuity across customers' businesses and improved team collaboration. Enabled by physics and data analytics, PTC's digital twins can provide continuous monitoring and predictive maintenance insights to increase product efficiency, eliminate unplanned downtime, and allow development of service-based models.

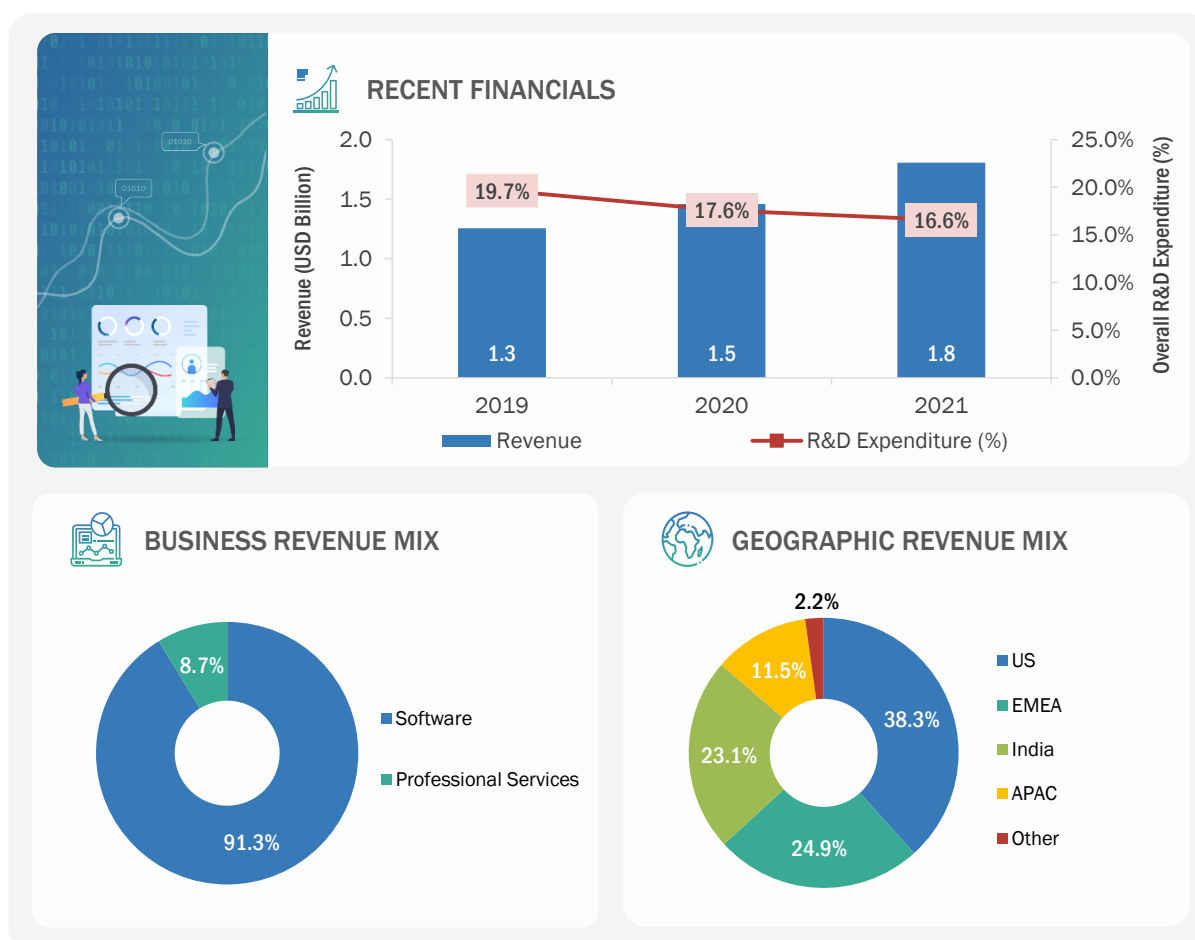
In June 2018, PTC formed a partnership with ANSYS (US) to help customers innovate faster by providing a best-in-class simulation-driven design solution. PTC integrated ANSYS' Discovery Live technology into Creo to develop Creo Simulation Live. PTC and ANSYS have created a digital twin service that extends service optimization solutions by combining simulation and IoT capabilities. The simulation-enhanced digital twin uses ThingWorx IoT data with ANSYS Twin Builder simulation to create a virtual reproduction of a connected asset's life and experience.

PTC serves industries such as aerospace, automotive, electronics and high tech, industrial machinery & equipment, life sciences, oil and gas, and retail & consumer products. PTC's technologies allow industrial organizations to create a closed loop of data that is shared throughout their entire value chain. Polaris (US), Volvo Group (Sweden), and Elekta (UK) are a few of the major clients of PTC's digital twin solutions. The company offers its software and solutions across all the major regions, including Asia Pacific, the Americas, the Middle East, Africa, and Europe. PTC's ThingWorx Kepware Server code base is trusted on more than 75,000 locations across the world.

TABLE 125 PTC: BUSINESS OVERVIEW

Founding Year	1985
Headquarters Country	US
Headquarters City	Boston, Massachusetts
Ownership Type	Public (NASDAQ: PTC)
Employee Count (as of September 2021)	6,709

Source: Company Website

FIGURE 60 PTC: COMPANY SNAPSHOT

Note: In the Recent Financials section, the R&D expenditure has been calculated as the percentage of the total revenue. The financial year of the company ends on September 30.

Source: Company Website and Annual Reports

12.1.8.2 Products/solutions offered

PRODUCT TYPE	PRODUCT/ SOLUTION	DESCRIPTION	INDUSTRY
Digital twin solution	Digital Twin solution– - ThingWorx - Kepware Server - Onshape	PTC offers unique solutions for incorporating advanced simulation models into a digital twin application. PTC's PLM solution manages digital and physical assets. With PLM's Digital Mirrors Physical, the digital twin provides a more advanced replica of the physical counterpart. PLM can choreograph how data streams are processed to provide a real-time digital mirror picture of what's happening in the physical world. It is possible due to PLM's capacity to relate information from numerous sources, including the physical product itself. On the other hand, PTC's SaaS-based CAD software Onshape enables real-time collaboration, agile engineering changes, and seamless PLM integration within digital twins.	Aerospace, automotive & transportation, retail, oil & gas, healthcare, energy & utilities, and others

Source: Company Website

12.1.8.3 Recent developments

12.1.8.3.1 Product launches

MONTH & YEAR	PRODUCT TYPE	PRODUCT NAME	PRODUCT DESCRIPTION
March 2021	AR for digital twin	Vuforia Engine Area Targets	The Vuforia Engine Area Targets is first augmented reality (AR) platform offering in the market to facilitate the creation of immersive AR experiences for spaces up to 300,000 square feet. Area Targets users can construct photorealistic, survey-grade digital twins of areas like as factories, malls, and offices using Matterport and Leica 3D scanners, as well as NavVis' (US) indoor mobile mapping equipment.

Source: Company Website and Press Releases

12.1.8.3.2 Deals

MONTH & YEAR	DEAL TYPE	COMPANY 1	COMPANY 2	DESCRIPTION	DEAL SIZE
January 2022	Alliance	PTC (US)	Schaeffler (Germany)	Schaeffler will use PTC technologies to create an integrated, end-to-end IT landscape as part of a new strategic alliance. PTC's efforts to further develop and evolve market-specific standard solutions to better satisfy the demands of automotive and industrial companies would benefit from Schaeffler's input. The new partnership will expand on Schaeffler's strong twenty-year relationship with PTC, which began with Schaeffler's adoption of PTC's CAD and PLM products. These tools are critical for the introduction and development of Schaeffler's digital twin initiatives as well as the visualization of 3D models (such as those utilized in augmented reality projects).	NA
September 2021	Collaboration	PTC (US)	Volvo Group (Sweden)	PTC and the Volvo Group announced expansion of their collaboration around digital engineering. The Volvo Group decided to integrate its CAD platforms; PTC will serve the primary CAD and PLM solutions for comprehensive vehicle design in the Volvo Group. Volvo Group ensures the validation of right design iterations, systems requirements, and product checklists by comparing as-designed to as-configured data with the help of digital twin.	NA
April 2021	Agreement	PTC (US)	Technion (Israel)	PTC has agreed to create a Research & Development facility on its Haifa campus as part of a long-term strategic agreement with the Technion – Israel Institute of Technology. PTC and Technion will join forces to develop and improve advanced manufacturing learning processes under the	USD 5 Million

				parameters of the agreement. PTC has also set aside a fund for collaborative research in the areas of industrial IoT, augmented reality, simulation, and generative design.	
June 2020	Alliance	PTC (US)	Microsoft (US), Rockwell Automation (US)	PTC extended its alliances with Rockwell Automation (US), a global leader in industrial automation and digital transformation, and Microsoft (US), a leading developer of computer software and hardware, to develop a cloud solution for industrial enterprises that enables real-time production performance monitoring.	NA
January 2020	Partnership	PTC (US)	Howden (UK)	Howden partnered with PTC and demonstrated the immersive experiences for their customers by employing upcoming technologies such as industrial IoT and augmented reality. Howden's mission is to reduce business risk for its customers by enhancing the uptime of deployed equipment using digital twin technology. Uptime– Howden's connected field maintenance service, relies heavily on customer-facing digital twins. This solution provides real-time performance statistics of deployed assets using digital twins and underlying technology (ThingWorx, Vuforia).	NA

Source: Company Website

12.1.8.4 MnM view

12.1.8.4.1 Key strengths/right to win

Recognizing the significance of digital twin and related technologies, PTC is strongly committed to R&D. Since the last 5 years, the company has spent approximately 17% to 20% of its revenue on R&D. PTC constantly benchmarks its software solution and platform development roadmap against the industry to ensure that its solutions specifications are in line with customers' expectations. It focuses on developing augmented reality, IoT, digital twin, CAD, and PLM towards solving industry 4.0's biggest obstacle: connectivity.

PTC is continuously working to improve its digital technology suite in order to enable increasingly sophisticated digital twins. Within digital twins, PTC's SaaS-based CAD software, Onshape, enables real-time collaboration, agile engineering modifications, and seamless PLM connectivity. A real-time digital twin of a physical space that captures not just the virtualization of the area but also the movement of people and goods within it. Harnessing these capabilities PTC offers platform for creating digital twin for its customers/clients. For instance, Polaris (US) relies of PTC solutions for reducing design and validation process time; and offer better customer experience through improved product performance and fit to customer requirements. Digital twins enable "voice of the product" that replaces usage assumptions with facts, accelerating time to market with optimized features and designs for discrete manufacturers through the lens of a product. The PTC solution enables Polaris to drive customer-focused and industry-leading innovations by using a product lifecycle management (PLM) system to develop and coordinate a digital thread while leveraging augmented reality and the digital twin.

12.1.8.4.2 Strategic choices made

PTC is expanding its business by adopting organic as well as inorganic strategies such as product launches, alliances, and agreements, partnerships, and collaborations. PTC is among the major players in the IoT platforms market that continues to expand its customer base. In the last 5 years, PTC has spent more than ~USD 500 million in acquisitions and organic investments in the IoT space, which includes the adoption of Axeda, ThingWorx, and ColdLight. The company is continuously developing and launching innovative products and technologies to maintain its position in the market. For instance, in June 2020, PTC launched the latest version of its ThingWorx Industrial IoT Platform, the ThingWorx 9.0, designed to enhance scalability and simplify solution development across the enterprise value chain. Additionally, PTC has signed strategic alliances with Bosch (Germany), GE (US), Microsoft (US), and Rockwell Automation (US) to leverage their IoT capabilities and understand the demand for IIoT.

In 2018, PTC and ANSYS (US) signed a partnership agreement. Through this partnership, both companies have accelerated the innovation in digital twins and are providing customers a world-class simulation-driven design solution. PTC and ANSYS have created a Digital Twin service that extends Service Optimization solutions by combining simulation and IoT capabilities. The Simulation-Enhanced Digital Twin uses ThingWorx IoT data with ANSYS Twin Builder simulation to create a virtual reproduction of a connected asset's life and experience. Taking advantage of this partnership in concordance with growing opportunities in digital twin market, PTC and ANSYS continue their partnership which will allow PTC to expand its digital twin market in the near future.

12.1.8.4.3 Weaknesses and competitive threats

The company heavily relies on the Indian and the US market with more than 23% of its total revenue generated from India and ~38% from the US. PTC can face high losses due to global trade conflicts and any political turbulence in the countries. It has a consolidated business in India and the US and needs to reduce its high dependence on these two markets and diversify its geographical reach to have better stability in the market. Further, PTC also faces tough competition with a number of companies whose offerings address one or more specific functional areas covered by PTC. PTC faces competition with large established companies such as General Electric (US), IBM (US), Amazon (US), Oracle (US), SAP (Germany), Siemens (Germany), and Software AG (Germany), as well as customers' homegrown solutions.

12.1.9 SAP

12.1.9.1 Business overview

SAP develops enterprise application software for various industries and is one of the major players in terms of business applications and analytics software in the digital commerce market. SAP Asset Intelligence Network offered under the SAP Leonardo offering, which runs on the SAP Cloud Platform, creates digital twins. SAP S/4HANA Cloud Platform, along with SAP IoT Application Enablement, offers microservices that allow other systems to retrieve enhanced digital twin data, including business semantics, hierarchy data, product master data, and customer contract documentation.

A digital twin can be established using a S/4HANA-enabled operational model to address the complete breadth of finance and accounting, supply chain, procurement, master data for clients, and HR. The client's cloud-based SAP S/4HANA operational environment model is enhanced with AI and integrated controls to optimize the overall organization's efficiency and sustainability. Digital manufacturing requires the creation and use of a digital twin based on Industrial IoT (IIoT). Companies that use SAP Intelligent Asset Management can increase product quality, streamline manufacturing processes, and generate new revenue streams. For firms that already use SAP solutions, using a digital twin via SAP IAM is even more beneficial. This is because integration with existing SAP systems, such as enterprise resource planning (ERP) or machine control in the form of manufacturing execution systems (MES), ensures that all necessary processes, from purchasing to warehousing and actual production to after-sales services, mesh seamlessly in the sense of networking from the top floor to shop floor.

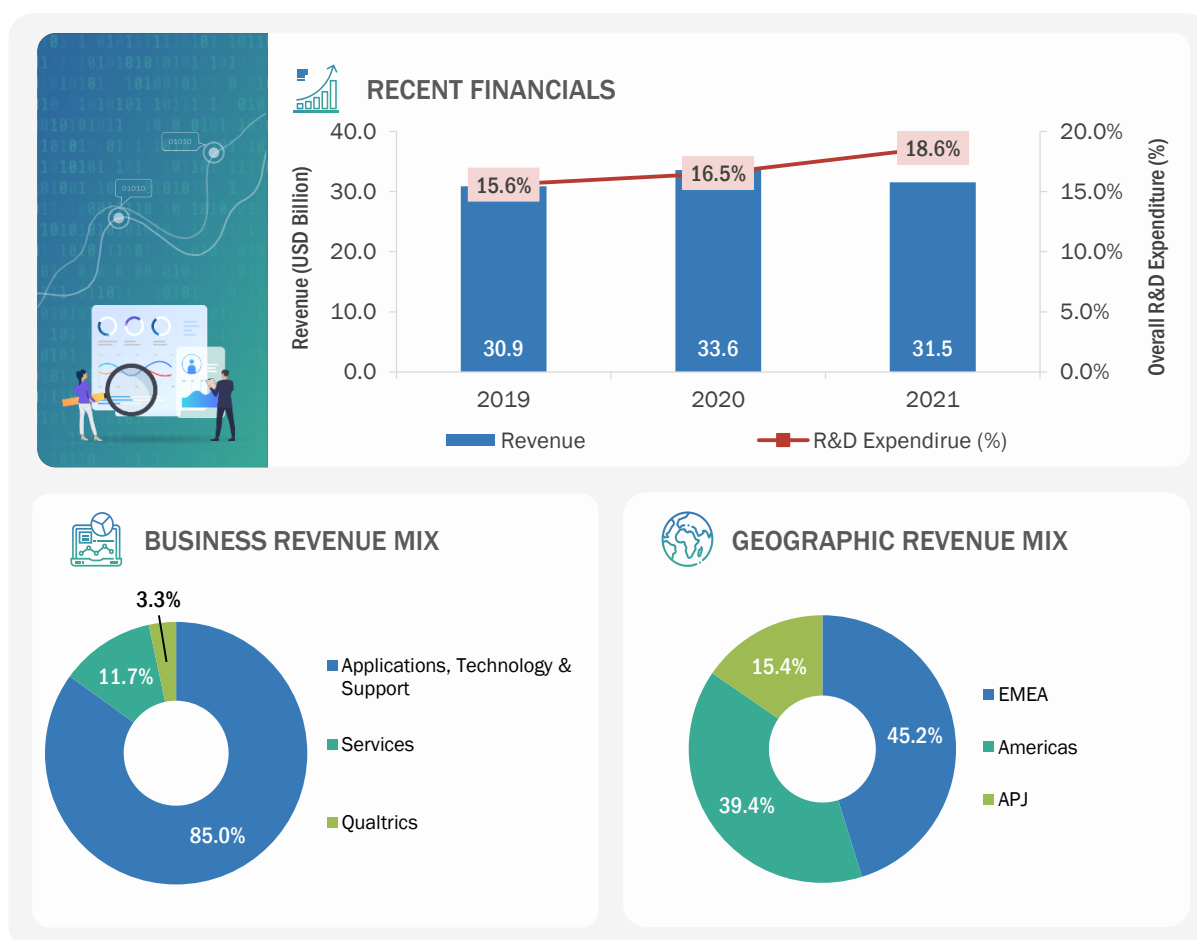
SAP and Flexso (Belgium) together offers the Flexso IoT Connected Digital Twin Accelerator, an all-in-one packaged solution from SAP and Flexso (Belgium). The solution accelerates and streamlines the client's IoT project by combining SAP Business Technology Platform with Flexso's prebuilt project approach and accelerators, code templates, and reusable components. As a result, the pre-configured software that is fitted directly to clients' needs at a fixed fee in record time ensures a quick time-to-value for clients' IoT project.

The company has a diversified product portfolio, with a large customer base and a wide network of partners to distribute products, solutions, and services. SAP offers its solutions in more than 140 countries, with a network of more than 22,700 SAP partner. SAP has a massive client base, ranging from standalone users to major companies and large enterprises. About 99 of the 100 largest companies in the world are SAP customers and ~240 million cloud users leverage SAP solutions.

TABLE 126 SAP: BUSINESS OVERVIEW

Founding Year	1972
Headquarters Country	Germany
Headquarters City	Weinheim
Ownership Type	Public (ETR: SAP)
Employee Count (as of December 2021)	104,364

Source: Company Website

FIGURE 61 SAP: COMPANY SNAPSHOT

Note: In the Recent Financials block, the R&D expenditure has been calculated as the percentage of the total revenue. The financial year of the company ends on December 31. The currency conversion rates considered for 2019, 2020 and 2021 are EUR 1 = USD 1.11986, EUR 1 = USD 1.22824, and EUR 1 = USD 1.1324 respectively.

Source: Company Website and Annual Reports

12.1.9.2 Products/solutions offered

PRODUCT TYPE	PRODUCT/SOLUTION	DESCRIPTION	INDUSTRY
Digital twin solution	Digital twin solution	Digital twins can be created with the help of the company's solutions such as SAP S/4HANA Cloud, SAP Product Lifecycle Management, SAP Digital Manufacturing, and SAP Asset Intelligence Network.	Automotive & transportation, retail, healthcare, energy & utilities, and others
Digital twin solution	My Digital Twin	- SAP offers the My Digital Twins application help for SAP S/4HANA cloud for Projects. Clients can use the My Digital Twins application to perform the following: - Search digital twins - Create, edit, and view digital twins - Import model documents within a digital twin to view the digital representation of a model - Navigate to the digital twin objects of a digital twin using	Automotive & transportation, retail, healthcare, energy & utilities, and others

the component, spatial, and model documents hierarchies

Source: Company Website

12.1.9.3 Recent developments

12.1.9.3.1 Deals

MONTH & YEAR	DEAL TYPE	COMPANY 1	COMPANY 2	DESCRIPTION	DEAL SIZE
May 2021	Partnership	SAP (Germany)	Siemens (Germany)	Siemens and SAP expanded their partnership to use digital twins to integrate plant floor operations, OEM remote condition monitoring, and product development to make collaboration easier across an asset's entire lifecycle.	NA

Source: Company Website

12.1.9.4 MnM view

12.1.9.4.1 Key strengths/right to win

SAP boasts of a strong portfolio of digital twin solutions—SAP S/4HANA Cloud, SAP Product Lifecycle Management, SAP Digital Manufacturing, and SAP Asset Intelligence Network. The existing and established market of these solutions also enables SAP to easily reach majority of the potential markets for digital twin. Further, the positive outlook of the semiconductors industry because of the IoT and cloud computing revolution has created new avenues for the company to expand its offerings. Digital manufacturing requires the creation and use of a digital twin based on industrial internet of things (IIoT). Companies that use SAP Intelligent Asset Management can increase product quality, streamline manufacturing processes, and generate new revenue streams. SAP Intelligent Asset Management (IAM) is the best option for its use, particularly for firms who already use SAP solutions, but not exclusively.

12.1.9.4.2 Strategic choices made

SAP focuses on developing new partnerships to expand its customer base. For instance, in May 2022, the company established a partnership with Siemens (Germany) to use digital twin to integrate plant floor operations. SAP expanded its longstanding partnership with Siemens to co-sell existing digital products focused on production and supply chains. The businesses also seek to develop a new industrial cloud-based service that will use a model-based digital thread to close the loop between IT and OT systems. Siemens plans to offer SAP's Asset Intelligence Network, Asset Strategy and Performance Management application, and SAP Enterprise Portfolio and Project Management package, with an initial focus on discrete manufacturing. Siemens' Teamcenter software service lifecycle management (SLM) suite will be offered by SAP in exchange.

12.1.9.4.3 Weaknesses and competitive threats

Premium pricing policy in an ASP-oriented market can distract the potential customers toward competitors. Few emerging companies are offering innovative digital solutions at reasonable pricing such as aPriori Technologies (US), Cosmo Tech (France), and SWIM AI (US). New technologies developed by competitors could make them market disruptors. Further, the speed of product launches from SAP has slowed down, which may hamper its market share as the market for digital twin is growing rapidly.

12.1.10 ORACLE

12.1.10.1 Business overview

Oracle is among the major providers of products and services for all aspects of corporate information technology (IT) environments, such as applications, platforms, and infrastructure. Oracle operates through three business segments: Cloud and license, Hardware, and Services. Oracle's cloud and license business offers enterprise applications and infrastructure technologies via cloud and on-premise deployment models, such as Oracle's cloud services and license support products; as well as cloud license, and on-premise license offerings.

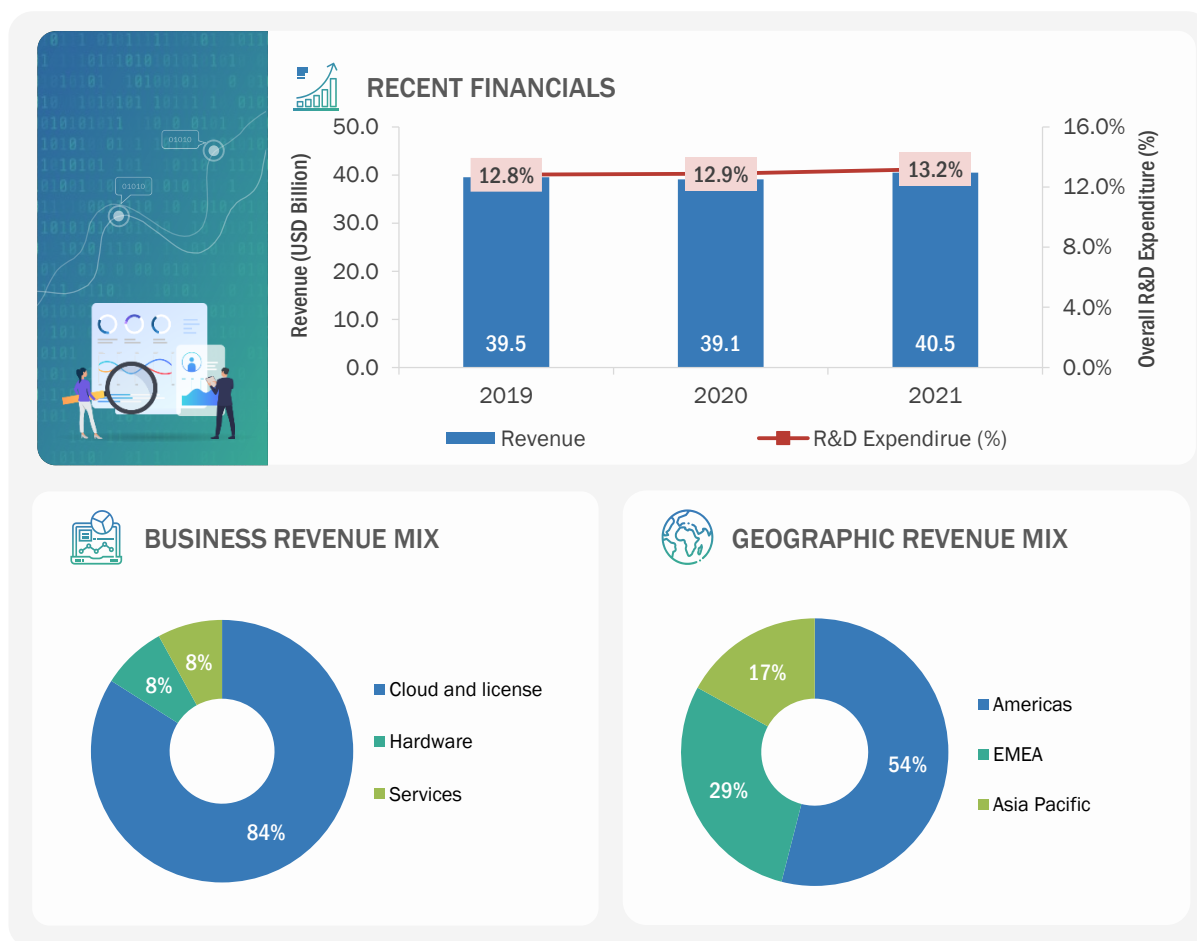
Oracle Cloud Infrastructure improves performance, security, and cost effectiveness. It is built to make it simple for companies to shift workloads from on-premises systems to the cloud, as well as between cloud and on-premises and other clouds. Oracle Cloud apps give clients access to cutting-edge software that help them innovate, expand sustainably, and become more resilient. For instance, the Oracle IoT Asset Monitoring Cloud offers the most comprehensive implementation of digital twin through three ways—virtual twin, predictive twin, and twin projection. Virtual twin is a device virtualization that derives the common data format used for asynchronous browser/server communication, which is observed for determining product performance. Predictive twins created through analytics models are built using a variety of techniques to overcome the complexity of problems that need to be resolved. Twin projection is developed by projecting insights generated by the virtual twin regarding industrial business applications that connect through IoT, which is an integral part of business infrastructure.

The company has its presence in North America, South America, Europe, the Middle East & Africa, and Asia Pacific.

TABLE 127 ORACLE: BUSINESS OVERVIEW

Founding Year	1977
Headquarters Country	US
Headquarters City	Austin, Texas
Ownership Type	Public (NYSE: ORCL)
Employee Count (as of May 2021)	132,000

Source: Company Website

FIGURE 62 ORACLE: COMPANY SNAPSHOT

Note: In the Recent Financials section, the R&D expenditure has been calculated as the percentage of the total revenue. The financial year of the company ends on May 31.

Source: Company Website and Annual Reports

12.1.10.2 Products/solutions offered

PRODUCT TYPE	PRODUCT/SOLUTION	DESCRIPTION	INDUSTRY
Digital twin solution	Oracle IoT Digital Twin	Virtual Twin, Predictive Twin, and Twin Projections are the three pillars of the Oracle IoT digital twin implementation. The IoT digital twin simulator allows users to create simulated devices for their environment without the need to connect and set up hardware. These emulated devices can generate adjustable live data, alarms, and events. The IoT digital twin simulator also provides ready-to-use simulation models that users can use to create simulated devices.	Automotive & transportation, retail, healthcare, oil & gas, infrastructure, energy & utilities, and others

Source: Company Website

12.1.10.3 Recent developments

12.1.10.3.1 Product launches

MONTH & YEAR	PRODUCT TYPE	PRODUCT NAME	PRODUCT DESCRIPTION
April 2022	Digital Twin for real-time asset monitoring	-	Oracle Cloud Infrastructure and Maplesoft (Canada) co-created a digital twin virtual environment for updated, real-time asset monitoring. Clients can now combine operational data at scale with physics-based models to create incredibly adaptable digital twins using Oracle cloud technology and Maplesoft simulation tools. While monitoring production assets, these twins can give a physical meaning to a detected anomaly and point to the required appropriate corrective action instantly.
May 2022	NFT support for blockchain-based digital twin	Oracle Blockchain Platform	Oracle announced native non-fungible tokens (NFT) support on its market-leading enterprise blockchain – Oracle Blockchain Platform. B2C businesses can utilize NFTs to represent consumer-accessible digital twins or direct digital assets and sell them through an e-commerce website based on NFT marketplaces. These marketplaces are supported by the Oracle Blockchain Platform ledger and smart contracts created by the Blockchain App Builder.

Source: Company Website and Press Releases

12.1.10.3.2 Deals

MONTH & YEAR	DEAL TYPE	COMPANY 1	COMPANY 2	DESCRIPTION	DEAL SIZE
May 2021	Partnership	Oracle (US)	Maplesoft (Canada)	The Oracle Cloud Infrastructure and Maplesoft partnered together for the creation of a Digital Twin virtual environment for updated real-time asset monitoring application.	NA

Source: Company Website

12.1.10.3.3 Others

MONTH & YEAR	COMPANY NAME	DEVELOPMENT TYPE	DESCRIPTION
April 2022	Oracle (US)	Expansion	Oracle inaugurated the Oracle Industry Innovation Lab in Deerfield, Illinois, US. The Oracle Industry Innovation Lab is a unique facility for the company's customers to incubate 5G ecosystems and develop cross-industry partnerships. Within the lab, Oracle Communications technology is providing secure and reliable livestreaming of large amounts of data to enable applications and devices for public safety, digital twin, and remote operations experiences.

Source: Company Website

12.1.10.4 MnM view

12.1.10.4.1 Key strengths/right to win

Oracle's strength is its global business expansion and development network, which enables it to offer one of the best cloud services and digital twin solutions in locations all over the world. Oracle also offers unique type of digital twin based on blockchain technology. In May 2022, Oracle announced NFT Support for its Oracle Blockchain Platform. NFTs represent real-world digital twins or native digital assets on the blockchain – with the most popular categories in present times, including unique digital art, iconic sport, or other events videos, and memorabilia linked to teams or sports superstars. Further, NFTs can be used by B2C businesses to represent consumer-accessible digital twins or direct digital assets and make them available on their e-commerce websites.

12.1.10.4.2 Strategic choices made

Oracle strengthens its core competencies that enable sustainable growth. The company has identified critical need of today's business—cloud and IoT—and continues to provide these solutions in the market that offer better reliability with the creation of digital twin. The company is emerging as a strong player with an increasing percentage of total revenues generated through the cloud platform. Oracle PLM Cloud is among the rapidly growing enterprise products, which support the Digital Thread. For instance, the company offers Oracle IoT applications for asset, production, and fleet, and the connected worker solutions are built on IoT cloud. These solutions support the development of a digital twin. In April 2022, in partnership with software provider Maplesoft (Canada), Oracle co-created a digital twin virtual environment for real-time asset monitoring. This digital twin can help clients to detect anomaly in their business and can suggest the corrective measure to the clients immediately.

12.1.10.4.3 Weaknesses and competitive threats

As a market leader in several business, Oracle has had to deal with numerous litigation and issues that have harmed its brand image. Further, in the digital twin market, Oracle needs to consider the fierce competition from prominent players, which often limits the market share of Oracle. Microsoft (US), ANSYS (US), PTC (US), Siemens (Germany), SAP (Germany), and General Electric (US) are few of the major competitors of Oracle in the digital twin market.

12.1.11 ROBERT BOSCH

12.1.11.1 Business overview

Robert Bosch is a global supplier of technology and services and is among the major players in the IoT ecosystem. The company operates through four major business segments—Mobility Solutions, Industrial Technology, Consumer Goods, and Energy and Building Technology. As a leading IoT provider, Bosch offers solutions for smart homes, Industry 4.0, and connected mobility. Bosch offers smart home, Industry 4.0, and linked mobility solutions. It leverages its expertise in sensor technology, software, services, and IoT cloud to offers connected, cross-domain solutions from a single source.

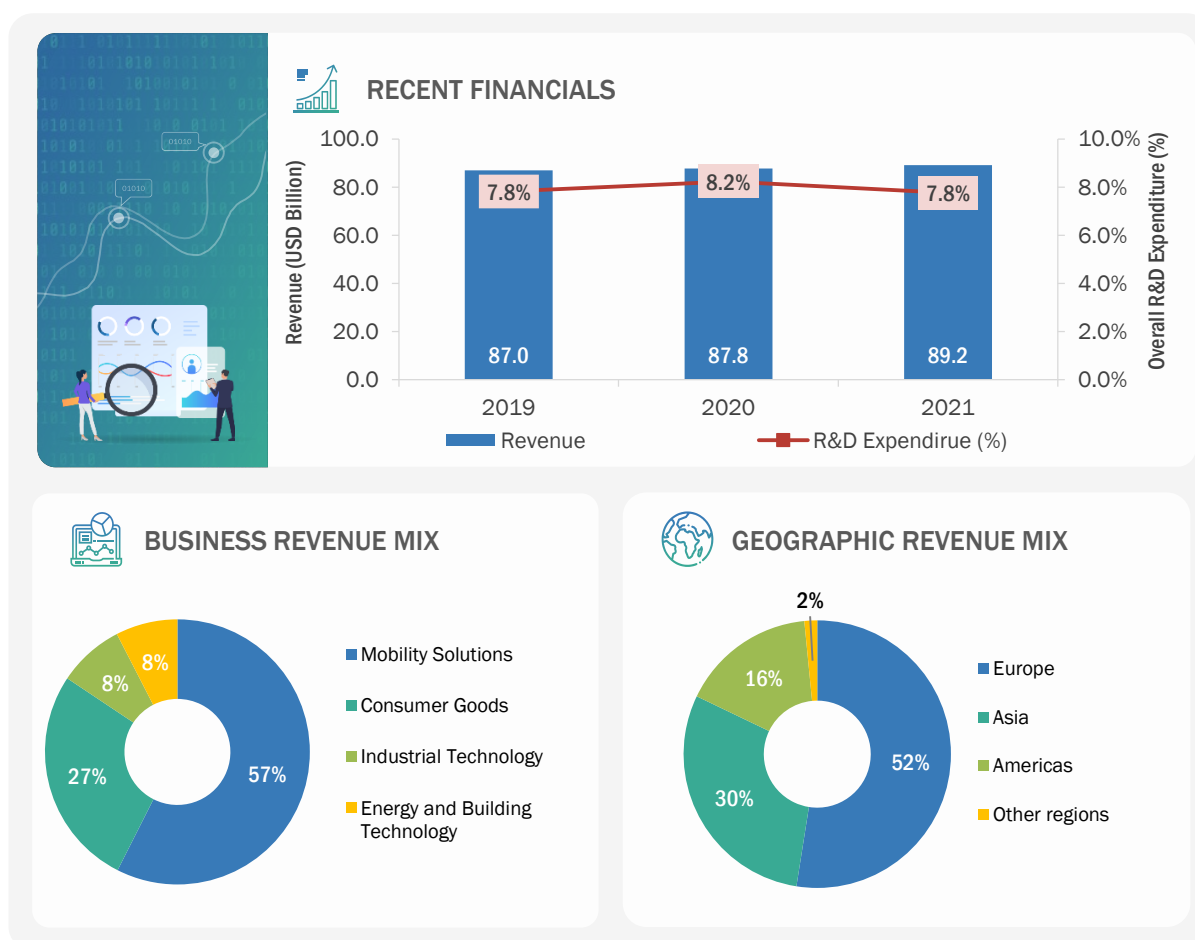
The Bosch IoT cloud comprises three layers—SaaS, PaaS, and IaaS. According to Bosch, an IoT cloud enables the creation of a digital twin of the physical asset based on readouts from machine components as well as additional sensors. The digital twin in the Bosch IoT cloud is used to establish new application areas, including automobile and industrial manufacturing. Bosch also offers digital buildings twin. In terms of building management, the digital twin opens more options for the clients. A digital building twin will allow for the creation of a virtual representation of a physical building, complete with all of the building's systems, gadgets, sensors, and actuators. Bosch has the finest foundation for harnessing this potential and establishing new services for its customers, owing to its many years of experience as a system integrator and expertise in the field of IoT.

Robert Bosch and about 440 subsidiaries and regional firms are a part of the Bosch Group, which operates in more than 60 countries. Bosch's global production, engineering, and sales network, which includes sales and service partners, spans nearly every country worldwide. The company owns a network of more than 60 regional offices and 4,000 associates worldwide. The company offers products, solutions, and services in major regions, including the Americas, Asia Pacific, and Europe. A few of the major regional offices of Robert Bosch are located in the US, Germany, India, the Netherlands, Singapore, and Switzerland. Bosch Rexroth (Germany), BSH Hausgeräte (Germany), Dremel (US), and ETAS (Germany) are few major subsidiaries of the company.

TABLE 128 ROBERT BOSCH: BUSINESS OVERVIEW

Founding Year	1886
Headquarters Country	Germany
Headquarters City	Gerlingen
Ownership Type	Public (BOM: 500530)
Employee Count (as of December 2021)	402,614

Source: Company Website

FIGURE 63 ROBERT BOSCH: COMPANY SNAPSHOT

Note: In the Recent Financials section, the R&D expenditure has been calculated as the percentage of the total revenue. The financial year of the company ends on December 31. The currency conversion rates considered for 2019, 2020 and 2021 are EUR 1 = USD 1.11986, EUR 1 = USD 1.22824, and EUR 1 = USD 1.1324 respectively.

Source: Company Website and Annual Reports

12.1.11.2 Products/solutions offered

PRODUCT TYPE	PRODUCT/SOLUTION	DESCRIPTION	INDUSTRY
Digital twin solution	Digital twin solution	Digital twins of tools, automobiles, sensors, and other web-enabled items use open APIs to depict the virtual status of a client's connected equipment and allow them to interact. To create an even more powerful user experience, both Bosch IoT Device Management with Bosch IoT Edge Agent can be used together.	Aerospace, automotive & transportation, retail, infrastructure, energy & utilities, and others
Digital twin solution	Digital building twin	The digital building twin offered by the company enables a lot of practical building services. For example, a maintenance firm will be able to forecast which component of a ventilation system has previously been in use, when and for how long, whether it will perform as predicted, or whether the filter, for example, needs to be replaced in the future.	Infrastructure, energy & utilities

Source: Company Website

12.1.11.3 Recent developments

12.1.11.3.1 Product launches

MONTH & YEAR	PRODUCT TYPE	PRODUCT NAME	PRODUCT DESCRIPTION
June 2020	Digital twin solution	Digital building twin	Bosch Building Technologies, a division of Bosch Group, developed an energy- and cost-saving solution for its internal operations. The Energy Platform uses Microsoft's Azure digital twin to monitor and analyze energy consumption in real time, helping manage business processes and provide actionable insights to operators.

Source: Company Website and Press Releases

12.1.11.3.2 Deals

MONTH & YEAR	DEAL TYPE	COMPANY 1	COMPANY 2	DESCRIPTION	DEAL SIZE
February 2022	Acquisition	Robert Bosch (Germany)	Hoerburger (Germany)	Bosch Building Technologies (division of Bosch) officially completed the acquisition of Hoerburger (Germany). Hoerburger has a broad range of products and services centered on building automation and energy management. Digitalization, networking, and artificial intelligence are expanding Bosch's capacity to transform commercial buildings into smart living and working spaces that boost people's security, safety, and comfort while also ensuring higher sustainability. The acquisition of Hoerburger is aligned with the Bosch initiative for digital building twin.	NA
October 2021	Collaboration	Robert Bosch (Germany)	Carnegie Mellon University (US)	Bosch and Carnegie Mellon University expanded their collaboration, with an emphasis on how the virtual and physical worlds interact. Bosch and Carnegie Mellon will collaborate on research into industrial spatial computing applications and the utilization of tools like safe and real-time digital twins in manufacturing. This would entail looking into how a future XR-instrumented factory could have digital counterparts for people, machinery, and manufactured items. Bosch Research will also invest USD 3 million in a cooperative project with CMU to investigate real-time digital twins and augmented reality (AR)/virtual reality (VR)/extended reality (XR) in manufacturing contexts.	USD 3 million

Source: Company Website

12.1.11.4 MnM view

12.1.11.4.1 Key strengths/right to win

The company's core strength lies in its advanced IoT Device Management, device-to-cloud, and cloud-to-device communication offerings with different communication patterns. Bosch IoT Device Management can be used in conjunction with Bosch IoT Edge Agent to provide clients with an even more powerful digital twin experience. A digital twin aids in the orchestration of all aspects of an IoT device, as well as providing a single and simpler model and API for interacting with it. The administration of Bosch IoT Things linked with a device connectivity layer like Bosch IoT Hub is sufficient for simple digital twins, which primarily comprises device status and functionality. Clients can extend this method for more complicated digital twins by combining separate microservices/components that implement specific functions.

Bosch employs 76,100 research and development associates in 128 locations throughout the world, with more than 38,000 of them working as software engineers. In October 2021, Robert Bosch extended its partnership with Carnegie Mellon University and announced an investment of USD 3 million in R&D for investigating real-time digital twins and AR/VR/XR in manufacturing environments. The company is also a member of the Eclipse Foundation International Open-Source Community.

12.1.11.4.2 Strategic choices made

The strategic goal of the Bosch Group is to make connected living easier through products and solutions that either contain artificial intelligence (AI) or were created or manufactured using it. Bosch improves people's lives throughout the world by creating new products and services that elicit excitement. Robert Bosch focuses on collaborating with other companies and participating with leading groups to offer IoT platforms and related software, which play a major role in areas such as connected manufacturing. For instance, Robert Bosch is working together with ANSYS (US), a leading digital twin provider, to address the key trends in automotive industry (Connectivity, electrification, new sharing services, and automation) by merging the finest of physics-based simulation and data analytics-based solutions. With the help of ANSYS and its technology, Bosch is developing digital models that utilize artificial intelligence (AI) and machine learning (ML) capabilities.

Additionally, Bosch partnered with IBM (US) for IoT suite that would help in developing a large ecosystem for connected devices. The company also entered into an innovation partnership with Software AG (Germany) for the IoT cloud. Strategic alliances of the company include the German Platform Industry 4.0 and the US-based Industrial Internet Consortium.

12.1.11.4.3 Weaknesses and competitive threats

Robert Bosch is currently focusing on its most anticipated digital twin solution—digital building twin. The company should also focus on other industries apart from infrastructure. Though the company offers digital twin for several industries, its inclination toward one industry may lead to the loss of opportunities from other industries.

12.2 OTHER KEY PLAYERS

12.2.1 SWIM AI

Founding Year	2015
Headquarters Country	US
Headquarters City	Campbell, California
Ownership Type	Private
Business Overview	SWIM AI was founded in 2015 and is headquartered in California, US. It designs software to solve the fundamental challenges of building intelligent software applications. The company focuses on creating a new generation of streaming applications, unbounded by the constraints of database-centric architectures. The company offers a solution for digital twins. SWIM EDX offers business and operational insights from gray edge data. It autonomously discovers and instantiates stateful digital twins from large edge data, and enables end users to analyze, learn, and predict the future of components, assets, or processes in real time. SWIM EDX can alter operations and provide insightful data for a wide range of edge situations, including smart cities, industrial automation, utilities and service provider networks, and IT and business infrastructure optimization. SWIM AI has raised a total of USD 25.4 million in three rounds of investment. Its most recent funding came from a Series B round on December 20, 2019. SWIM strategically invests in inorganic strategies. It has partnered with Microsoft Azure, ARM, and various other system integrators and OEMs. These partnerships give SWIM a competitive advantage, and the company can differentiate itself from its competitors in edge data handling, local analytics, efficient insights, scale of data supported, and quick application development. In December 2018, SWIM AI and the Roads and Transport Authority (Dubai) signed a Memoranda of Understanding (MoU). The MoU helped the logistics business as well as the Enterprise Command & Control Center (EC3) of RTA to establish an autonomous transport system. The successful project was based on autonomous mobility, asset management, twin digital technology, and artificial intelligence.
Products/Solutions	- SWIM AI offers the following solutions – - SWIM EDX
Geographic Presence	North America and Middle East

Source: Company Website, Press Releases, and MarketsandMarkets Analysis

12.2.2 EMERSON

Founding Year	1980
Headquarters Country	US
Headquarters City	St. Louis, Missouri
Ownership Type	Public (NYSE: EMR)
Revenue (2021)	USD 18.2 Billion
R&D Expenses (2021)	USD 485 Million
Business Overview	Emerson operates through two core business platforms—Automation Solutions and Commercial & Residential Solutions. The Automation Solutions segment of Emerson offers digital twin solutions. Emerson's digital twin supports the needs of control system engineering, allowing distributed teams access to system experts, regardless of location. It supports the proven value of using dynamic simulation for designing, implementing, and testing automation solutions. Emerson's digital twin solutions provide clairvoyance through dynamic simulation software that simulates physical surroundings, allowing clients to execute advanced testing and operator training without disrupting ongoing operations. This reduces the expense and risk of enhancing operations for clients while also increasing worker safety and upskilling. The Ovation digital twin technology from Emerson ensures precise operator training, keeps up with process changes, and validates control logic without disrupting plant operations. Ovation's digital twin technology combines high-fidelity simulation with the control system, allowing for real-time enhanced testing of innovative operational approaches in a risk-free environment before they are implemented in the real plant. The virtual platform-based digital twin simulator is an identical reproduction of the plant control system. The difficulty and cost of maintaining separate modeling software are eliminated with ovation-based models included in virtual controllers.
Products/Solutions	▪ The company offers the following solutions: ▪ Emerson Digital Twin Solutions ▪ Ovation Digital Twin Technology
Geographic Presence	Americas, Asia Pacific, Middle East, Africa, and Europe

Note: The financial year of the company ends on September 30.

Source: Company Website, Press Releases, and MarketsandMarkets Analysis

12.2.3 ABB

Founding Year	1988
Headquarters Country	Switzerland
Headquarters City	Zürich
Ownership Type	Public (XNSE: ABB)
Revenue (2021)	USD 28.9 Billion
R&D Expenses (2021)	USD 1.2 Billion
Business Overview	ABB has a comprehensive offering for digital industries. ABB offers various solutions to digital industries through 4 business segments: Electrification, Industrial Automation, Motion, and Robotics & Discrete Automation. The company is working on creating digital twins of sensors, devices, and systems, practically an identical copy of physical devices. ABB mostly adopts both organic and inorganic strategies to develop and expands its digital twin solutions. For instance, in September 2019, ABB introduced the PickMaste Twin software for robots with digital twin technology to reduce commissioning times. In February 2019, ABB and Dassault Systèmes announced a global partnership to provide customers digital twin solutions. This digital twin solution is based on Dassault Systèmes' industry-leading visualization, modeling, and simulation capabilities. Further, in December 2020, ABB and CORYS, a French simulation company, signed a deal to develop digital twin modeling and simulation technology for the energy and process industries. ABB also offers a digital twin that simulates the actual flowmeter in a virtual environment and predicts the performance of electromagnetic (EM) flowmeters. This digital twin, which is based on Multiphysics modeling, allows for better flowmeter design and performance prediction under field settings. ABB offers digital twin solutions for several industries, including energy & utilities, mining, and supply chain.
Products/Solutions	▪ The company offers the following solutions: ▪ PickMaster Twin software with digital twin technology ▪ ABB Ability Stockyard Management System – It is a configurable system that can be used to digitalize the complete material handling chain, acting as its digital twin
Geographic Presence	Americas, Europe, Asia Pacific, Middle East, and Africa

Note: The financial year of the company ends on December 31.

Source: Company Website, Press Releases, and MarketsandMarkets Analysis

12.2.4 HONEYWELL

Founding Year	1906
Headquarters Country	US
Headquarters City	Charlotte, North Carolina
Ownership Type	Public (NASDAQ: HON)
Revenue (2021)	USD 34.4 Billion
R&D Expenses (2021)	USD 1.3 Billion
Business Overview	The company develops technologies that address critical challenges around energy, safety, security, productivity, and global urbanization. The company operates through four major segments—: Aerospace, Honeywell Building Technologies, Performance Materials and Technologies, and Safety and Productivity Solutions. Honeywell is a pioneer in the development of digital twins. In June 2019, Honeywell launched Honeywell Forge, an industrial Internet of things analytics platform that will collect operational data, analyze it, and optimize infrastructure. Honeywell Forge also includes digital twin. In October 2021, Honeywell and Saipem (Italy), a multinational oilfield services company, signed an agreement for the development and commercialization of a digital twin as a part of Saipem's Snamprogetti Urea technology. Honeywell Forge, Honeywell's enterprise performance management solution, will be used to develop the solution, which would allow customers to increase urea production through a virtual simulation of critical operations or key processes. In October 2020, Honeywell and Microsoft (US) teamed up to introduce Honeywell's domain-specific applications built on the Microsoft cloud platform, allowing industrial clients to achieve new levels of productivity. Integrating Honeywell Forge products with Azure services such as Azure Digital Twins and Azure edge capabilities is also being investigated as a means to offer more innovation to clients. In addition, Honeywell is collaborating with iTRACE Technologies (US) to deploy blockchain technology for label printers in order to help supply chains become more secure. This new integration connects the physical printed item to the digital ledger, creating a digital twin for online authentication, adding another layer of security to industries. Few such industries include transportation and logistics, pharmaceuticals and healthcare, shipping and distribution centers, industrial manufacturing, automotive parts suppliers, and applications that require high precision and error-free printing.
Products/Solutions	▪ Honeywell offers the following solutions: ▪ Immersive Field Simulator ▪ UNISIM DESIGN R490 SUITE
Geographic Presence	Americas, Europe, Asia Pacific, Middle East, and Africa

Note: The financial year of the company ends on December 31.

Source: Company Website, Press Releases, and MarketsandMarkets Analysis

12.2.5 SCHNEIDER ELECTRIC

Founding Year	1836
Headquarters Country	France
Headquarters City	Paris
Ownership Type	Public (EPA: SU)
Revenue (2021)	USD 32.7 Billion
R&D Expenses (2021)	USD 1.7 Billion
Business Overview	Schneider Electric leads digital transformation services by integrating world-leading process and energy technologies to help clients achieve maximum efficiency and sustainability. Schneider Electric uses a digital twin to enable lifetime solutions from design to operation and maintenance. The company helps create the first-born digital twin that enables engineers to test a physical asset twin before anything is constructed. This allows for the process, equipment, and operations to be analyzed and optimized for safety, reliability, and profitability. In July 2020, DORIS Group (France), an engineering and project management company in the energy industry, AVEVA (UK), a global leader in engineering and industrial software, and Schneider Electric entered into a strategic partnership to deliver digital twin technology for the upstream oil & gas industry. This solution enables remote operations and production assurance through a fully functional Living Digital Twin that mirrors all aspects of the operating asset. It also improves traceability through a single point of accountability. This solution brings new assets on stream faster through the use of cloud-enabled software that improves collaboration and increases engineering efficiencies
Products/Solutions	The company offers the following solutions: ▪ Electrical Digital Twin Service ▪ EcoStruxure Machine Expert Twin – digital twin software suite to create digital models of real machines
Geographic Presence	Europe, Asia Pacific, and Africa

Note: The financial year of the company ends on December 31.

The currency conversion rate considered for 2021 is EUR 1 = USD 1.1324.

Source: Company Website, Press Releases, and MarketsandMarkets Analysis

12.2.6 NAVVIS

Founding Year	2013
Headquarters Country	Germany
Headquarters City	München, Deutschland
Ownership Type	Private
Business Overview	NavVis is an IT services and IT consulting company based in Munich, Germany, with offices in the US and China. NavVis has customers worldwide in the surveying, AEC, and manufacturing industries. The company offers digital twin solutions to service providers and businesses, bridging the gap between the physical and digital world. NavVis's SLAM-based mobile mapping devices produce high-quality data at speed and scale, with survey-grade precision. NavVis digital industrial solutions allow users to make better operational decisions, increase productivity, optimize corporate processes, and increase the profitability of their business. NavVis, along with an ecosystem of partners, offers unique services and solutions based on its core technology. PTC (US), Amazon Web Services (US), Autodesk (US), and Siemens (Germany) are a few of the leading partners of NavVis. In May 2022, the company with AVEVA (UK) to provide access to reliable as-built 3D data and take the next step of advancement in digital twin technology. The AVEVA Point Cloud Manager manages large amounts of 3D point cloud data to generate a digital twin that is accessible, detailed, and secure. The company is also working with its partners on next-generation spatial intelligence to the built environment. In addition, over the course of eight rounds of investment, NavVis raised a total of USD 96.4 million. Its most recent round of fundraising was on December 7, 2021.
Products/Solutions	The company offers the following solutions: ▪ NavVis Digital Factory Solution ▪ NavVis IVION Enterprise ▪ NavVis IVION Go
Geographic Presence	Europe, Americas, Asia Pacific, Middle East, and Africa

Source: Company Website, Press Releases, and MarketsandMarkets Analysis

12.2.7 DNV

Founding Year	1864
Headquarters Country	Norway
Headquarters City	Høvik, Oslo
Ownership Type	Public
Revenue (2021)	USD 2.4 Billion
R&D Expenses (2021)	~USD 120 Million
Business Overview	DNV operates through six major business segments—Maritime, Energy Systems, Digital Solutions, Business Assurance, Supply Chain and Product Assurance, and The Accelerator. DNV offers digital twin for asset-intensive industries, such as shipping, oil & gas, and energy & utilities. The digital twin solution offered by the company assists these industries in increasing efficiency and lowering costs while successfully controlling operational risk and safety. DNV's digital twin brings together all of the expertise for these businesses together, resulting in strong analyses, insights, and diagnostics for decision-makers. The company aims to define the future of assurance by providing increasingly digitalized services and leading the assurance of digitalization through data, digital twins, and digitized processes. DNV applies its extensive data science knowledge to assist customers in scoping, building, and maintaining their digital twins. It serves a variety of industries, with over 100,000 customers and operations in over 100 countries.
Products/Solutions	▪ The company offers the following services: ▪ Data smart asset solutions - Digital twin ▪ Assurance of digital twin solutions
Geographic Presence	Americas, Europe, Middle East, Africa, and Asia Pacific

Note: The financial year of the company ends on December 31. The currency conversion rates considered for 2021 is NOK 1 = USD 0.11348.

Source: Company Website, Press Releases, and MarketsandMarkets Analysis

12.2.8 AUTODESK

Founding Year	1982
Headquarters Country	US
Headquarters City	San Rafael, California
Ownership Type	Public (NASDAQ: ADSK)
Revenue (2021)	USD 4.4 Billion
R&D Expenses (2021)	USD 1.1 Billion
Business Overview	Autodesk is a multinational software development firm. Autodesk tools such as AutoCAD, Revit, Maya, 3ds Max, Fusion 360, SketchBook, and others are used by over 100 million customers all over the world to tackle major design, business, and environmental concerns. It offers software for both personal computers and mobile devices and uses the cloud's computational capacity to assist customers in collaborating, designing, simulating, and fabricating their ideas in 3D. Autodesk offer its digital twin technology platform Tandem that is based on cloud. It allows initiatives to begin and end digitally, translating rich data into business intelligence. Architectural Engineering and Construction (AEC) firms can construct and give over a digital twin to building owners and operators by utilizing Building Information Modeling (BIM) data throughout the process. The data end users receive is simply accessible, contextual, and informative, resulting in ready-to-go operations.
Products/Solutions	The company offers the following solutions: ▪ Autodesk Tandem ▪ MOICON Digital Twin
Geographic Presence	Americas, Africa, Middle East, Europe, and Asia Pacific

Note: The financial year of the company ends on January 31.

Source: Company Website, Press Releases, and MarketsandMarkets Analysis

12.2.9 ANDRITZ

Founding Year	1852
Headquarters Country	Austria
Headquarters City	Graz, Styria
Ownership Type	Public (VIE: ANDR)
Revenue (2021)	USD 7.3 Billion
R&D Expenses (2021)	USD 114.3 Million
Business Overview	ANDRITZ provides plants, systems, equipment, and services to a variety of industries. The company is a technological and global market leader in hydropower, pulp and paper, metal working and steel, and solid/liquid separation in municipal and industrial segments. ANDRITZ offers Metris Digital Twin technology based on ANDRITZ IDEAS simulation platform, combined with the power of data analytics with process simulations. The Metris Digital Twin provides comprehensive insights for process performance optimization, operational visibility, and proactive decision-making. Many additional automation systems benefit from ANDRITZ's online Digital Twin solutions, which make them more autonomous. ANDRITZ recently customized Metris Digital Twin applications for the pulp & paper, and mineral processing industry sectors. The company also offers Recast Digital Twin for washing plants and evaporation plants. Recast Digital Twin forecasts and alerts plant operators about potential plant failures such as over- and under-liming. It also forecasts essential operator activities such as white liquor filter washing, liquid-to-gas ratio set points, and slaker temperature to achieve maximum causticizing efficiency. Digital twin solutions offered by the company provide data reconciliation for process instruments such as density, temperature, and flow transmitters, and their application provides suggestions for inventory management by suggesting optimal production set points. The digital twin also serves as a virtual sensor for difficult-to-measure parameters, including density at important sites such as the slaker, each causticizer, white liquor filter, or lime mud filter. ANDRITZ serves customers in over 280 locations in over 40 countries worldwide.
Products/Solutions	The company offers the following solutions: ▪ Metris Digital Twin ▪ Recast Digital Twin
Geographic Presence	Americas, Europe, Africa, Middle East, and Asia Pacific

Note: The financial year of the company ends on December 31. The currency conversion rate considered for 2021 is EUR 1 = USD 1.1324.

Source: Company Website, Press Releases, and MarketsandMarkets Analysis

12.2.10 SOFTWARE AG

Founding Year	1969
Headquarters Country	Germany
Headquarters City	Darmstadt, Hessen
Ownership Type	Public (ETR: SOW)
Revenue (2021)	USD 946.3 Million
R&D Expenses (2021)	USD 171.1 Million
Business Overview	Software AG is a global software development company based in Germany. Using its industry-leading suite of products across integration & API management, IoT & analytics and business transformation, Software AG creates a fluid flow of data between people, departments, systems, and devices. Software AG offers IoT digital twin solutions for designing, managing, and visualizing digital twins of complex assets, asset hierarchies, devices, as well as manufacturing production processes. Users can create an IoT digital twin for any purpose using Software AG and can connect as many devices or systems as per requirement. With Cumulocity IoT device management from Software AG, users can streamline onboarding and manage all of their devices from a single platform. Software AG offers process engineers and operators the flexibility to use self-service analytics to search for trends and directly query process data without the assistance of a data scientist. In addition, Cumulocity IoT and Azure Digital Twins complement one other, allowing users to model, analyze, and stream data in order to reveal new actionable insights. Azure Digital Twins enhances the value of a plant's Cumulocity IoT implementation by allowing factory process engineers to create digital models of their real factories, manufacturing lines, and equipment. These digital twin models are brought to life with sensor telemetry data that is physically surfaced by Cumulocity IoT and fed directly to Azure Digital Twins in real time. Software AG raised a total of USD 388.2 million in two rounds of investment. Its most recent round of investment was a Post-IPO Debt round on December 13, 2021. The company offers its solutions and services in more than 70 countries
Products/Solutions	Software AG offers the following solutions: Software AG IoT digital twins
Geographic Presence	Americas, Europe, Africa, Middle East, and Asia Pacific

Note: The financial year of the company ends on December 31. The currency conversion rate considered for 2021 is EUR 1 = USD 1.1324.

Source: Company Website, Press Releases, and MarketsandMarkets Analysis

12.2.11 BENTLEY SYSTEMS

Founding Year	1984
Headquarters Country	US
Headquarters City	Exton, Pennsylvania
Ownership Type	Public (NASDAQ: BSY)
Revenue (2021)	~USD 1 Billion
R&D Expenses (2021)	USD 220.9 Million
Business Overview	Bentley Systems is a global software development company that caters to the clients responsible for designing and operating the world's infrastructure, such as roads, bridges, airports, skyscrapers, industrial and power facilities, and utility networks. Bentley provides solutions for the entire lifecycle of an infrastructure asset, tailored to the needs of engineers, architects, geospatial professionals, planners, contractors, fabricators, IT managers, operators, and maintenance engineers. The company's offerings include MicroStation-based modeling and simulation products, ProjectWise for project delivery, AssetWise for asset and network performance, Seequent's premier geoprofessional software portfolio, and the iTwin platform for infrastructure digital twins. The Bentley iTwin platform is the building block for creating digital twin solutions. It helps clients to accelerate application development to address various issues, including data integration, visualization, change tracking, and security. Project teams, contractors, and owner-operators can use the iTwin Platform's Open APIs and libraries to build digital twin apps that produce, visualize, and analyze digital twins of infrastructure assets.
Products/Solutions	The company offers the following solutions: ▪ iTwin Platform ▪ ProjectWise, PW365 ▪ SYNCHRO ▪ PlantSight ▪ OpenUtilities
Geographic Presence	Americas, Europe, Africa, Middle East, and Asia Pacific

Note: The financial year of the company ends on December 31.

Source: Company Website, Press Releases, and MarketsandMarkets Analysis

12.2.12 RIVER LOGIC

Founding Year	2000
Headquarters Country	US
Headquarters City	Dallas, Texas
Ownership Type	Private
Business Overview	Since 2000, River Logic has been a global leader in prescriptive analytics (optimization). Its platform, which is geared for business users, allows for enterprise-wide optimization, collaborative planning, and performance management. River Logic solutions allow firms to swiftly create and deploy custom applications at the forefront of advanced analytics technology. River Logic offers a supply chain digital twin to its customer to tackle short-, medium-, and long-term decision-making challenges. With a simple cloud interface that offers a collaborative analysis of endless possibilities, the supply chain digital twin allows clients to move away from siloed planning. River Logic's digital twin integration technology combines physics and data-driven twins to help optimize products throughout their lifecycle.
Products/Solutions	The company offers the following solutions: ▪ Supply Chain Digital Twin
Geographic Presence	North America and Europe

Source: Company Website, Press Releases, and MarketsandMarkets Analysis

12.2.13 ALTAIR

Founding Year	1985
Headquarters Country	US
Headquarters City	Troy, Michigan
Ownership Type	Public (NASDAQ: ALTR)
Revenue (2021)	USD 532 Million
R&D Expenses (2021)	USD 151 Million
Business Overview	Altair is a multinational technology business that specializes in simulation, high-performance computing (HPC), and artificial intelligence (AI) software and cloud solutions. Altair's digital twin integration technology combines physics and data-driven twins to help organizations optimize their products throughout their lifecycle. Clients can use Altair's digital twin building blocks to connect diverse development disciplines and enable simulation-driven teamwork. Organizations can utilize digital twins to improve product performance, obtain visibility into a product's in-service life, determine when and where to undertake predictive maintenance, and determine how to extend a product's remaining usable life (RUL). The Altair digital twin integration platform can provide its clients with a view into the complete complexity of their assets' operations: machine learning insights combined with physics simulation to aid in the discovery and correction of hidden inefficiencies. Altair offers Simulation-driven Digital Twins (The Physics Twin), and Data-driven Digital Twins (The Data Twin). To generate low-fidelity models from detailed simulations, the physics-based, simulation-driven digital twin uses standardized, tool-independent interfaces such as the functional mock-up interface (FMI), co-simulation methods with geometry-based 3D CAE tools, and reduced order modeling approaches. On the other hand, the data-driven twin uses ML algorithms and data science to optimize product performance. The organization has over 3,000 engineers, scientists, as well as 150 plus software partners across the world. The company offers its solution and services in more than 25 countries, with 86 offices and more than 15,000 customers globally.
Products/Solutions	Altair offers the following Digital Twin solutions: ▪ The Physics Twin ▪ The Data Twin ▪ Digital Twin Integration Platform (Fundamental Platform)
Geographic Presence	North America, South America, Asia Pacific, Europe, Middle East, and Africa

Note: The financial year of the company ends on December 31.

Source: Company Website, Press Releases, and MarketsandMarkets Analysis

12.2.14 JOHNSON CONTROLS

Founding Year	1885
Headquarters Country	Ireland
Headquarters City	Cork
Ownership Type	Public (NYSE: JCI)
Revenue (2021)	USD 532 Million
R&D Expenses (2021)	USD 220.9 Million
Business Overview	Johnson Controls offers one of the world's largest portfolios of building technology, software, as well as service solutions, in more than 150 countries. Johnson Controls provides artificial intelligence infused OpenBlue Digital Twin, which is designed for buildings and infrastructures. OpenBlue Digital Twin is a managed service that creates a digital clone of assets, processes, people, places, systems, and devices to enable and map smart buildings. While its primary application is data harmonization and connectivity, twins are brought to life by constructing a replica environment by using data in and around the building. The digital twin of a building encompasses the structure, location, assets, people, and past, present, and future events relating to the building and the environment. Digital twins enable organizations consolidate and enrich multiple data silos to deliver centralized context while optimizing value. In addition, AI infusion across the data context and sources help to enable real-time prediction outputs. OpenBlue Digital Twin is built on the foundations of an open architecture using an API-driven approach.
Products/Solutions	Johnson Controls offers the following solutions: ▪ OpenBlue Digital Twin
Geographic Presence	North America, South America, Asia Pacific, Europe, Middle East, and Africa

Note: The financial year of the company ends on December 31.

Source: Company Website, Press Releases, and MarketsandMarkets Analysis

13 ADJACENT AND RELATED MARKET

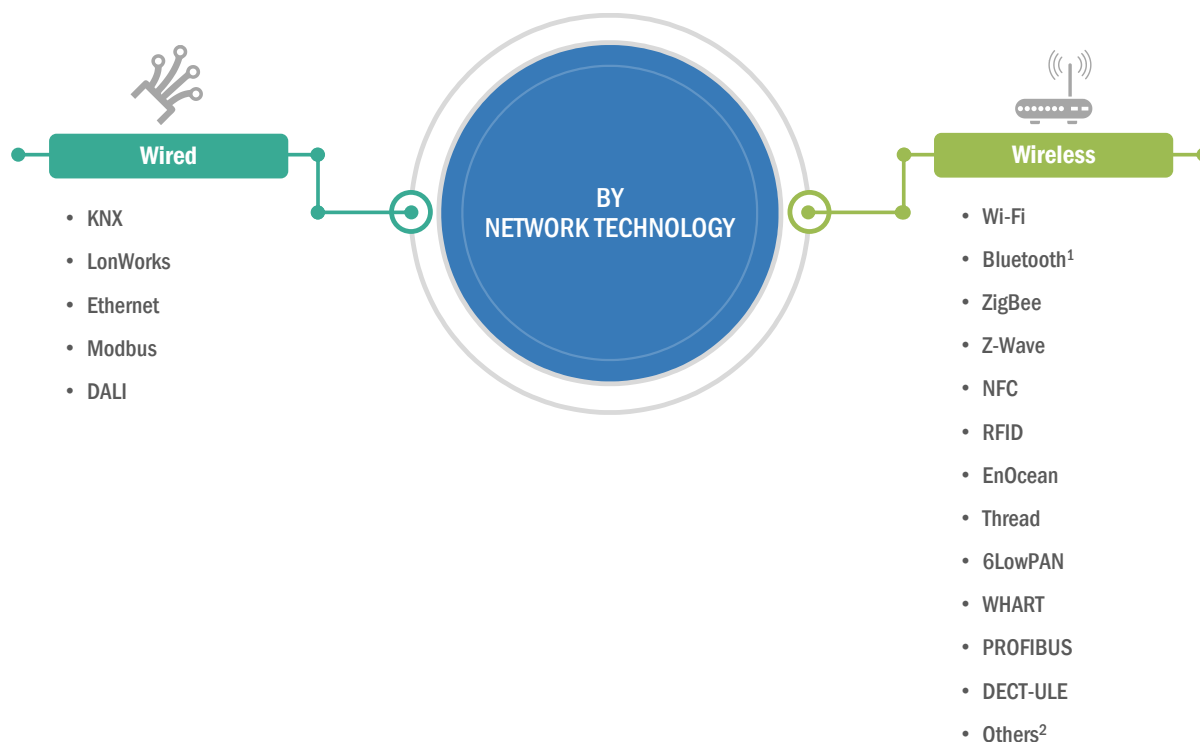
13.1 IOT SENSORS MARKET

Internet of things (IoT) sensors refer to sensors used in smart applications, which require uninterrupted connectivity to carry out real-time data analytics. They require common connecting platforms for collecting and analyzing data. IoT sensors are based on the computing concept that considers every physical object to be connected to the Internet to communicate with each other. As IoT sensors are an integral component of any digital twin solution, therefore IoT sensors are closely associated with digital twin.

13.2 INTRODUCTION

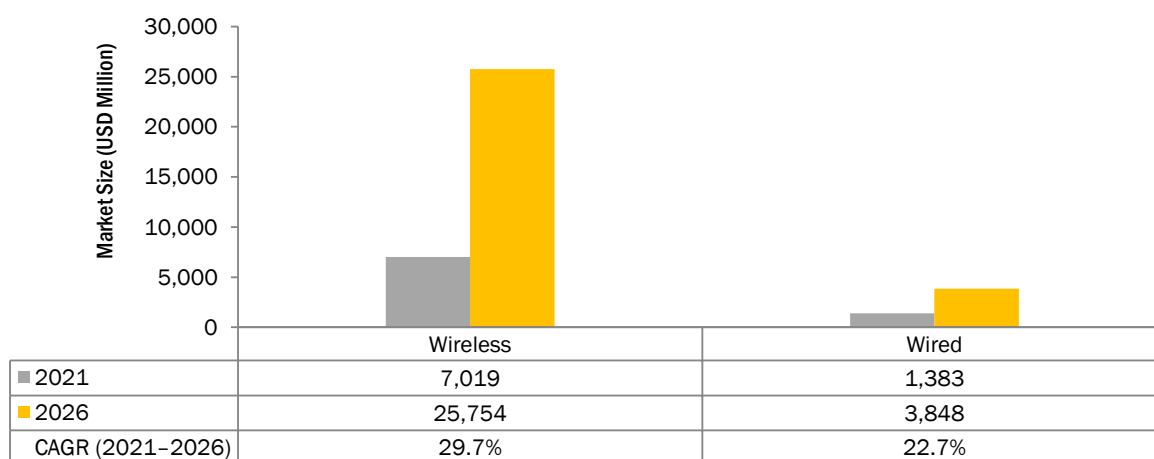
The Internet of Things (IoT) is a web of billions of machines that communicate with each other through various long-range [for instance, cellular (LTE-A and LTE) and satellite] and short-range (for instance, RFID, Wi-Fi, ZigBee, Bluetooth, Z-Wave, and Bluetooth Smart) communication technologies. The growth of IoT is governed by its specialized applications in different markets such as healthcare, automotive, building automation, and industrial. Each of these applications has varied bandwidth requirements, depending on the coverage area, amount of data being transmitted, application, topology, power consumption, and data load.

Connectivity technologies help IoT devices connect with each other as well as with the internet. The market has seen an increase in the number of companies providing low-power wireless network connectivity specifically designed for IoT applications. While introducing any new connectivity protocol and network technology, it has to be mainly checked for short-range wireless communications, low power and energy harvesting, real-time systems, radio-frequency identification, resource-constrained networks, and machine type communication. The application developers and electronic engineers working on systems and products for IoT have an extensive array of connectivity technologies available for their IoT devices. Thus, it is important to understand the different technologies available to ensure the effective utilization of technology for specific solutions or applications most efficiently and cost-effectively.

FIGURE 64 SNAPSHOT OF KEY NETWORK TECHNOLOGIES IN IOT SENSORS MARKET

Note: Bluetooth further includes Bluetooth Smart, Wi-Fi/ Bluetooth Smart, Bluetooth Smart/Ant+, and Bluetooth 5. Other wireless technologies include Ant+, ISA 100, GPS, Sub-Gig, and Cellular (3G and 4G)).

Source: Sensors and IoT Industry Associations, Internet society, SEMI.org, Investor Presentation, Journals and Magazines, Expert Interviews, and MarketsandMarkets Analysis

FIGURE 65 WIRELESS NETWORK TECHNOLOGY SEGMENT TO LEAD IOT SENSORS MARKET FROM 2021 TO 2026

Source: Sensors and IoT Industry Associations, Internet society, SEMI.org, Investor Presentation, Journals and Magazines, Expert Interviews, and MarketsandMarkets Analysis

TABLE 129 IOT SENSORS MARKET, BY NETWORK TECHNOLOGY, 2017–2020 (USD MILLION)

Network Technology	2017	2018	2019	2020	CAGR (2017–2020)
Wired	984	1,052	1,221	1,068	2.8%
Wireless	3,637	4,230	5,366	5,159	12.4%
Total	4,621.1	5,282.4	6,586.5	6,227	10.5%

Source: Sensors and IoT Industry Associations, Internet society, SEMI.org, Investor Presentation, Journals and Magazines, Expert Interviews, and MarketsandMarkets Analysis

TABLE 130 IOT SENSORS MARKET, BY NETWORK TECHNOLOGY, 2021–2026 (USD MILLION)

Network Technology	2021	2022	2023	2024	2025	2026	CAGR (2021–2026)
Wired	1,383	1,757	2,183	2,671	3,227	3,848	22.7%
Wireless	7,019	9,388	12,295	15,897	20,339	25,754	29.7%
Total	8,401	11,145	14,478	18,567	23,566	29,602	28.6%

Source: Sensors and IoT Industry Associations, Internet society, SEMI.org, Investor Presentation, Journals and Magazines, Expert Interviews, and MarketsandMarkets Analysis

The growing demand for wireless data from mobile devices, connected cars, and smart grids, among others, is creating the need for a more robust internet connection. Further, the proliferation of mobile devices and the rapidly increasing adoption of the bring-your-own-device (BYOD) concept as well as IoT within enterprises is expected to drive the growth of the market for wireless network technology. The upcoming 5G mobile technology is expected to provide much faster internet connectivity and coverage, which would also boost the market for wireless network technology.

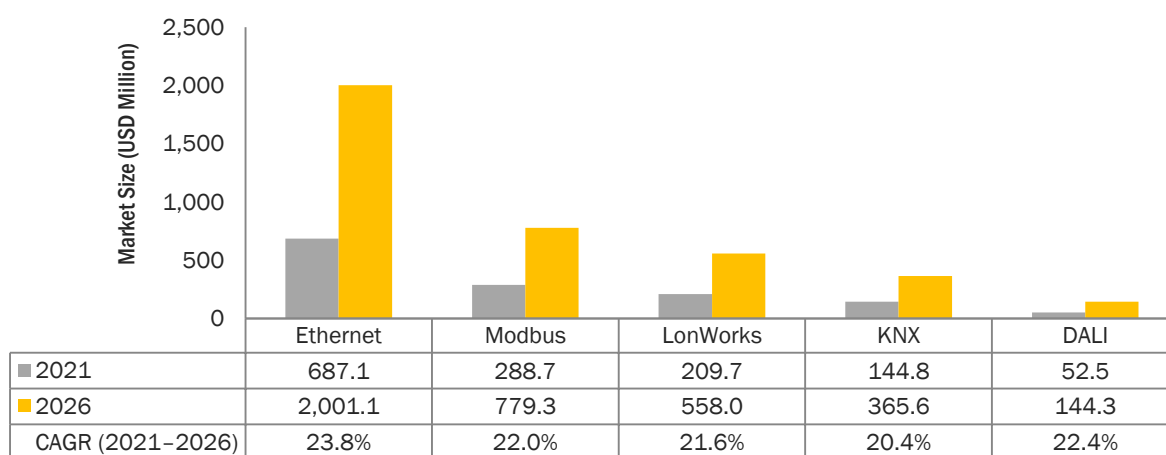
Moreover, factors such as the growing adoption of cloud platforms and lower installation and maintenance costs compared to wired devices are also likely to propel the adoption of wireless network technologies. One of the major factors contributing to the growth of the market for wireless technology is the continuous increase of internet users globally, which would strengthen the overall market for IoT. The use of cloud computing is another factor fueling the growth of this market, which has advantages such as cost savings in business operations. Organizations can save substantial capital costs required for in-house server storage and other application requirements they maintain.

The huge adoption of Ethernet technology is a major contributor to the growth of the market for wired technology. In 2021, Ethernet held the largest share of 49.7% of the wired technology market. The high market share of this technology is attributed to its universal acceptance, growing use in industrial and commercial buildings, and low cost. Moreover, it is cost-effective to produce controllers and I/O modules that reside directly on Ethernet, providing the benefits of Ethernet connectivity without the need for gateways, terminal servers, or other miscellaneous hardware.

13.3 WIRED

Wired network technology is a base to install, troubleshoot, and configure computer operating systems. The network technology works on bus, star, token ring, and mesh topologies. The networks can be characterized based on the distance into local area networks (LANs), metropolitan area networks (MANs), and wide area networks (WANs). The network technologies can also be connected to IoT without using 3G or Wi-Fi and instead by using LoRaWAN, which offers advantages including long range, and low power consumption.

FIGURE 66 ETHERNET SEGMENT TO HOLD LARGEST SIZE OF IOT SENSORS MARKET FOR WIRED TECHNOLOGY FROM 2021 TO 2026



Note: Values have been represented in 1 decimal to show minute differences.

Source: Sensors and IoT Industry Associations, Internet society, SEMI.org, Investor Presentation, Journals and Magazines, Expert Interviews, and MarketsandMarkets Analysis

TABLE 131 IOT SENSORS MARKET FOR WIRED TECHNOLOGY, BY TYPE, 2017-2020 (USD MILLION)

Type	2017	2018	2019	2020	CAGR (2017-2020)
KNX	116.5	120.4	134.9	117.5	0.3%
LonWorks	158.5	166.6	190.0	163.4	1.0%
Ethernet	457.3	498.5	589.7	520.6	4.4%
Modbus	214.1	226.2	259.4	224.3	1.6%
DALI	38.0	40.4	46.6	42.2	3.6%
Total	984.3	1,052.1	1,220.7	1,067.9	2.8%

Note: Values have been represented in 1 decimal to show minute differences.

Source: Sensors and IoT Industry Associations, Internet society, SEMI.org, Investor Presentation, Journals and Magazines, Expert Interviews, and MarketsandMarkets Analysis

TABLE 132 IOT SENSORS MARKET FOR WIRED TECHNOLOGY, BY TYPE, 2021–2026 (USD MILLION)

Type	2021	2022	2023	2024	2025	2026	CAGR (2021–2026)
KNX	144.8	180.6	220.1	264.1	312.8	365.6	20.4%
LonWorks	209.7	264.2	325.2	394.3	472.2	558.0	21.6%
Ethernet	687.1	881.3	1,104.7	1,364.0	1,662.9	2,001.1	23.8%
Modbus	288.7	364.6	450.2	547.5	657.4	779.3	22.0%
DALI	52.5	66.5	82.4	100.6	121.3	144.3	22.4%
Total	1,382.7	1,757.2	2,182.5	2,670.6	3,226.6	3,848.2	22.7%

Note: Values have been represented in 1 decimal to show minute differences.

Source: Sensors and IoT Industry Associations, Internet society, SEMI.org, Investor Presentation, Journals and Magazines, Expert Interviews, and MarketsandMarkets Analysis

The market for Ethernet is expected to witness the highest CAGR during the forecast period. Ethernet connections are more reliable as they are less prone to dropped connections and are more reliable without constant debugging. Wired connections are less affected by local factors such as walls, floors, cabinets, length of the room, interference from other electronic devices, etc. This enables wired connectivity to be much faster than wireless. Wired data transmissions are not sensitive to distances and the placement of devices does not have any adverse effect on the performance of the connection

13.4 WIRELESS

The devices in IoT are connected via different forms of networks. This section focuses on wireless technologies such as ZigBee, Bluetooth, EnOcean, Z-Wave, Wi-Fi, NFC, RFID, ANT, and cellular technologies, which are rising to meet the new challenges faced by the IoT landscape in which information and communication systems will be seamlessly embedded in the surroundings.

TABLE 133 IOT SENSORS MARKET FOR WIRELESS TECHNOLOGY, BY TYPE, 2017–2020 (USD MILLION)

Type	2017	2018	2019	2020	CAGR (2017–2020)
Wi-Fi	2,091.3	2,409.7	3,027.4	2,882.6	11.3%
Bluetooth	953.7	1,097.3	1,376.3	1,308.4	11.1%
ZigBee	223.5	266.0	345.0	339.1	14.9%
Z-Wave	14.4	17.5	23.1	23.1	17.1%
NFC	115.4	127.7	153.7	139.9	6.6%
RFID	83.7	102.3	136.0	136.8	17.8%
EnOcean	20.4	28.6	42.4	46.7	31.9%
Thread	12.8	17.7	26.0	28.4	30.4%
6LowPAN	25.9	35.7	52.3	57.1	30.1%
WHART	23.5	35.2	54.5	61.8	38.0%
PROFIBUS	34.0	47.1	69.2	75.7	30.6%
DECT-ULE	17.8	24.3	35.4	38.4	29.2%
Others	20.3	21.4	24.4	20.8	0.9%
Total	3,636.8	4,230.4	5,365.8	5,158.8	12.4%

Note: Values have been represented in 1 decimal to show minute differences.

Source: Sensors and IoT Industry Associations, Internet society, SEMI.org, Investor Presentation, Journals and Magazines, Expert Interviews, and MarketsandMarkets Analysis

TABLE 134 IOT SENSORS MARKET FOR WIRELESS TECHNOLOGY, BY TYPE, 2021–2026 (USD MILLION)

Type	2021	2022	2023	2024	2025	2026	CAGR (2021–2026)
Wi-Fi	3,902.9	5,194.8	6,770.3	8,710.2	11,089.3	13,971.3	29.1%
Bluetooth	1,770.0	2,354.0	3,065.4	3,940.4	5,012.4	6,309.6	28.9%
ZigBee	466.3	630.4	834.4	1,090.1	1,409.3	1,802.7	31.1%
Z-Wave	32.0	43.7	58.2	76.7	99.9	128.8	32.1%
NFC	184.9	240.1	305.0	382.1	473.3	579.5	25.7%
RFID	190.2	259.8	347.4	458.4	598.4	772.6	32.4%
EnOcean	67.6	95.8	132.5	180.4	242.5	321.9	36.7%
Thread	41.0	58.0	80.0	108.7	145.8	193.2	36.3%
6LoWPAN	82.3	116.2	160.2	217.6	291.7	386.3	36.3%
WHART	90.6	129.8	181.3	249.0	337.3	450.7	37.8%
PROFIBUS	109.2	154.4	213.2	289.7	388.7	515.1	36.4%
DECT-ULE	55.3	77.9	107.3	145.5	194.7	257.5	36.0%
Others	26.5	33.1	40.2	47.9	56.1	64.4	19.4%
Total	7,018.8	9,387.8	12,295.4	15,896.6	20,339.4	25,753.5	29.7%

Note: Values have been represented in 1 decimal to show minute differences.

Source: Sensors and IoT Industry Associations, Internet society, SEMI.org, Investor Presentation, Journals and Magazines, Expert Interviews, and MarketsandMarkets Analysis

The wireless nature of Wi-Fi networks allows users to access network resources from nearly any convenient location within their primary networking environment. Further, wireless networks can serve an increased number of clients with the existing equipment. In a wired network, additional clients would require additional wiring. Connecting new users is easier and quicker. Hence, the market for Wi-Fi has the largest market share.

14 APPENDIX

14.1 INSIGHTS OF INDUSTRY EXPERTS



The healthcare industry would propel the growth of the digital twin market in the coming years. With the help of personalized organ modellings, doctors can treat patients according to their individual health conditions with targeted medications.



North America currently holds the largest share of the digital twin market. The region is expected to provide immense opportunities for players in the market.



The changing face of maintenance, repair, and operations (MRO) is a major driving factor for the growth of the digital twin market. Digital twin offers the ability to calculate maintenance-related key performance indicators. In addition, digital twin also forecasts the behavior of machines under different circumstances.

14.2 DISCUSSION GUIDE

Q. 1. Could you provide a brief introduction of your company and its role in the digital twin market?

Primary's viewpoint _____

Q. 2. What are your views on the growth of the global digital twin market? How is the current scenario, and how is the market expected to grow in the future?

Primary's viewpoint _____

Q. 3. How is your company positioning itself in the market to take advantage of the growing adoption of digital twin across various industries across the world?

Primary's viewpoint _____

MARKET DYNAMICS

Q. 4. What are the drivers, restraints, opportunities, and challenges influencing the digital twin market?

PARTICULARS	DESCRIPTION
Drivers	
Restraints	
Opportunities	
Challenges	

MARKET SEGMENTATION, SIZING, AND GROWTH

Q. 5. Based on application, what were the percentage contributions of the following segments to the overall size of the digital twin market in 2021? Also, mention the CAGR for each segment during 2022–2027.

APPLICATION	MARKET SHARE (%), 2021 CAGR (2022–2027)
Product Design & Development	
Performance Monitoring	
Predictive Maintenance	
Inventory Management	
Business Optimization	
Other Applications	

Q. 6. Based on industry, the digital twin market has been segmented into the following segments: what were the percentage contributions of the following industries to the digital twin market in 2021? Also, mention the CAGR for each segment during 2022–2027.

INDUSTRY	MARKET SHARE (%), 2021	CAGR (2022–2027)
Automotive & Transportation		
Energy & Utilities		
Infrastructure		
Healthcare		
Aerospace		
Oil & Gas		
Telecommunications		
Agriculture		
Retail		
Other Industries		

Q. 7. According to you, what were the percentage contributions of the following enterprises to the digital twin market in 2021? Also, mention the CAGR for each segment during 2022–2027.

ENTERPRISE	MARKET SHARE (%), 2021	CAGR (2022–2027)
Large Enterprises		
Small & Medium Enterprises (SMEs)		

Q. 8. Please list the shares of the regions identified below in the global digital twin market in 2021. What do you expect their CAGRs to be for the next five years?

REGION	MARKET SHARE (%), 2021	CAGR % (2022–2027)
North America		
Europe		
Asia Pacific		
Rest of the World		

Primary's viewpoint_____

Q. 9. According to you, who were the top players having maximum share of the digital twin market in 2021?

Primary's viewpoint_____

14.3 KNOWLEDGE STORE: MARKETSandMARKETS' SUBSCRIPTION PORTAL

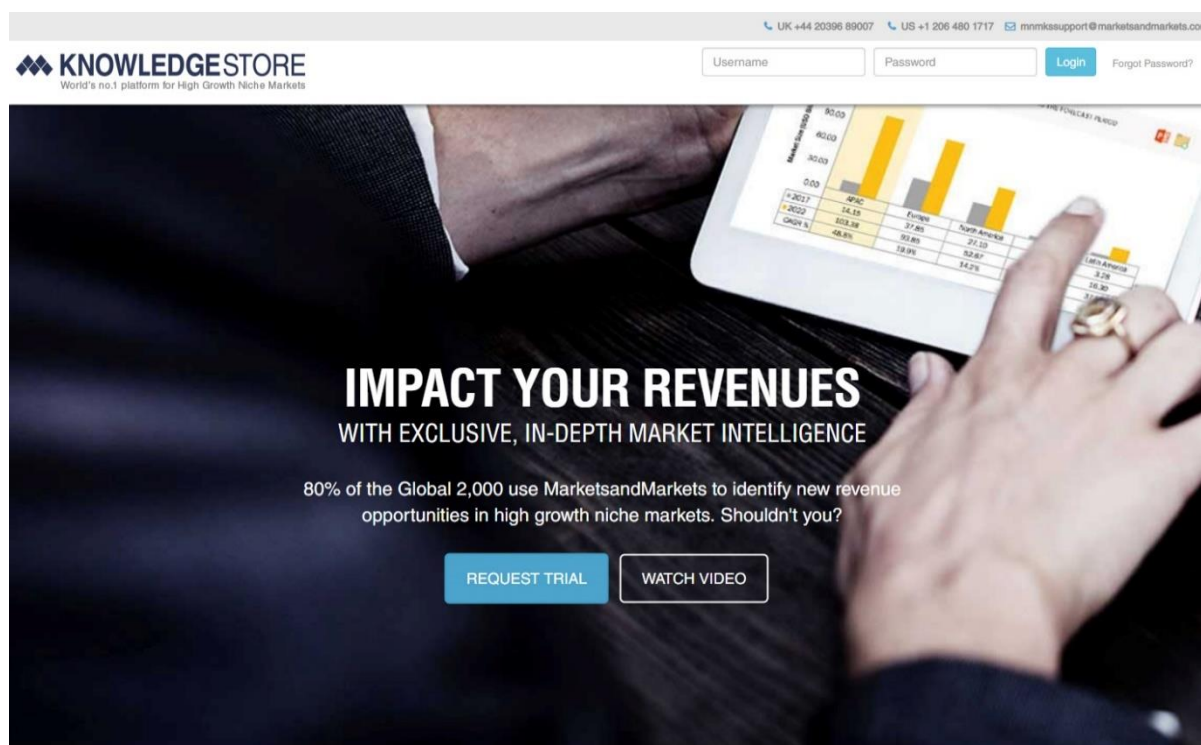
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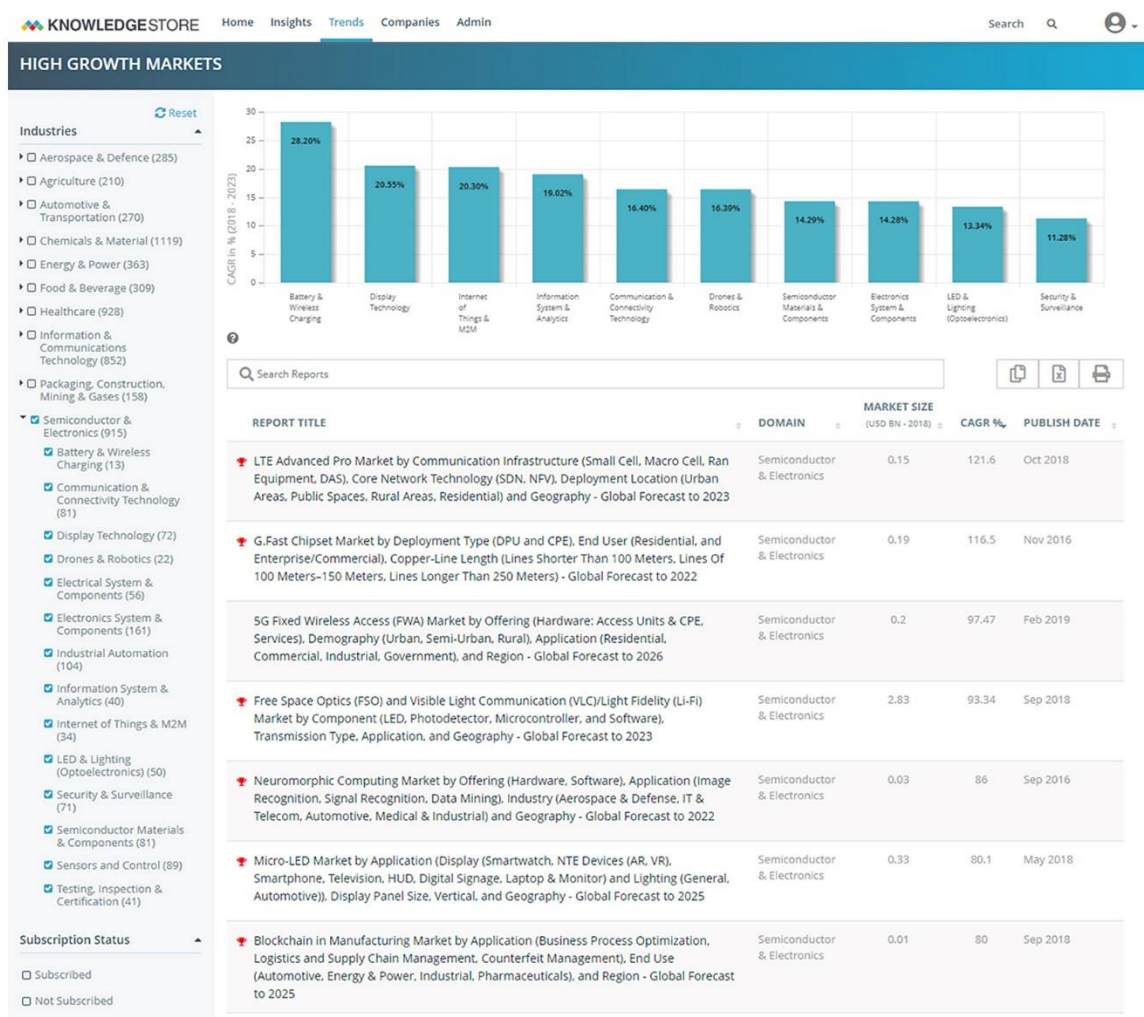
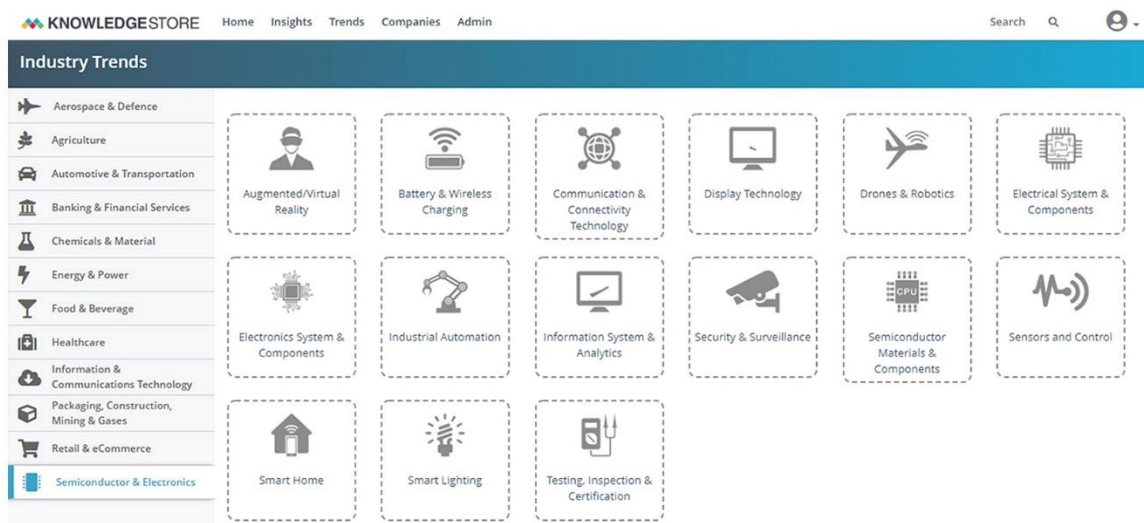
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14.4 AVAILABLE CUSTOMIZATIONS

With the given market data, MarketsandMarkets offers customizations according to a company's specific needs. The following customization options are available for the report:

PRODUCT ANALYSIS

- Product matrix that gives a detailed comparison of the product portfolio of each company

COMPANY INFORMATION

- Detailed analysis and profiling of additional market players (up to eight)

14.5 RELATED REPORTS

SR. NO.	REPORT TITLE	PUBLISHED DATE
1.	IoT Sensors Market - 2021-2026 By Sensor type, Network Technology, Vertical, Application, and Geography (North America, Europe, APAC, RoW) https://www.marketsandmarkets.com/Market-Reports/sensors-iot-market-26520972.html	March 2021
2.	INDUSTRY 4.0 MARKET - 2021-2026 By Technology (Industrial Robots, Blockchain, Industrial Sensors, Industrial 3D Printing, Machine Vision, HMI, AI in Manufacturing, Digital Twin, AGV's, Machine Condition Monitoring) and Geography https://www.marketsandmarkets.com/Market-Reports/industry-4-market-102536746.html	June 2021

14.6 AUTHOR DETAILS

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Apurva is a Management and Engineering professional with more than 14 years of experience in TMT with a focus on core tech areas such as Comms, Compute, Networking and Storage. He is currently focusing on the impact of Emerging Technologies such as IoT, AI & 5G on the global technology value chain. He has previously worked with STMicroelectronics and Samsung and as a TMT equity analyst for investment banks.

Apurva has done his Engineering at Birla Institute of Technology & Science (BITS), Pilani and has completed his post-graduation in management from the Indian Institute of Management, Bangalore.

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